

The impact of technology on data collection: case studies in privacy and economics

Ali Farzanehfar

September 2021

Department of Computing
Imperial College London

Submitted in part fulfilment of the requirements for the degree of
Doctor of Philosophy in Computing of Imperial College London

Abstract

Technological advancement can often act as a catalyst for scientific paradigm shifts. Today the ability to collect and process large amounts of data about individuals is arguably a paradigm-shift enabling technology in action. One manifestation of this technology within the sciences is the ability to study historically qualitative fields with a more granular quantitative lens than ever before. Despite the potential for this technology, wide-adoption is accompanied by some risks.

In this thesis, I will present two case studies. The first, focuses on the impact of machine learning in a cheapest-wins motor insurance market by designing a competition-based data collection mechanism. Pricing models in the insurance industry are changing from statistical methods to machine learning. In this game, close to 2000 participants, acting as insurance companies, trained and submitted pricing models to compete for profit using real motor insurance policies — with a roughly equal split between legacy and advanced models. With this trend towards machine learning in motion, preliminary analysis of the results suggest that future markets might realise cheaper prices for consumers. Additionally legacy models competing against modern algorithms, may experience a reduction in earning stability — accelerating machine learning adoption. Overall, the results of this field experiment demonstrate the potential for digital competition-based studies of markets in the future.

The second case studies the privacy risks of data collection technologies. Despite a large body of research in re-identification of anonymous data, the question remains: if a dataset was big enough, would records become anonymous by being “*lost in the crowd*”? Using 3 months of location data, we show that the risk of re-identification decreases slowly with dataset size. This risk is modelled and extrapolated to larger populations with 93% of people being uniquely identifiable using 4 points of auxiliary information among 60M people. These results show how the privacy of individuals is very unlikely to be preserved even in country-scale location datasets and that alternative paradigms of data sharing are still required.

Statement of originality

All work within this thesis is my own. Contributions by others are either referenced or highlighted at the top of each chapter.

Additionally, the work cited here has led to one journal publication and other manuscripts in preparation:

- **Publication:** A. Farzanehfar, F. Houssiau, and Y.-A. de Montjoye, “The risk of re-identification remains high even in country-scale location datasets,” *Patterns*, vol. 2, no. 3, p. 100 204, 2021
- **In preparation:** Multiple entries using results from chapter 2. Possible contributions for these include:
 - **Effect of machine learning on consumer prices:** A study comparing consumer outcomes in markets populated with advanced algorithmic models compared to those in data model markets.
 - **Effect of gender in insurance pricing models:** A study examining the effect of the gender variable in pricing models produced by experts.
 - **Effect of mixing pricing model types in a competitive market:** A study investigating market dynamics when pricing models of different types are placed in competition with each other.

Copyright Declaration

The copyright of this thesis rests with the author. Unless otherwise indicated, its contents are licensed under a Creative Commons Attribution-Non Commercial 4.0 International Licence (CC BY-NC). Under this licence, you may copy and redistribute the material in any medium or format. You may also create and distribute modified versions of the work. This is on the condition that: you credit the author and do not use it, or any derivative works, for a commercial purpose. When reusing or sharing this work, ensure you make the licence terms clear to others by naming the licence and linking to the licence text. Where a work has been adapted, you should indicate that the work has been changed and describe those changes. Please seek permission from the copyright holder for uses of this work that are not included in this licence or permitted under UK Copyright Law.

Acknowledgements

Turns out a PhD is hard. It's almost impossible to complete without help — my PhD was no exception. First and foremost I have to acknowledge the immense help provided by the lady I live with, Radhika (Ravioli) Ravi. Through no fault of her own she found herself with someone pursuing a research degree within her industry. She was instrumental, not only in some of the technical aspects, but also in providing the emotional support demanded by any PhD student.

One of the many reasons I ended up pursuing and completing this path is my family. My mother, brother, sisters, and father are the safety net for any adrenaline-seeking extreme sport I engage in, be it this PhD, or future activities. Without their unconditional support I would not be taking the risks necessary for early career success. You cannot buy this kind of support.

Science is more collaborative than ever. Florimond Houssiau deserves special mention for walking down the road of two very challenging projects with me. I must also thank Arnaud, Andrea, Shubham, Ana-Maria, Luc, Ben, and Arran for their invaluable support and feedback on both the work and this thesis. Typos (or lack thereof) are attributed to them.

I undertook a large scale data collection effort in my PhD. This was made possible in part by Sharada Mohanty and the rest of the AICrowd team. I would like to thank Jyotish, Shivam, Rabi, and Vrushank in particular. I will never cease being amazed by the continued efficacy of this team.

Learning is difficult without good teachers, learning well is impossible. Firstly, I would like to thank my high-school maths teacher Mojtaba Shahsavan without whom, I would have never realised I was not bad at maths. Secondly, without the help, early work, and insights of Prof. Arthur Charpentier an entire chapter would be missing from this thesis. Finally of course, I would like to acknowledge the early support of my advisor Dr. Yves-Alexandre de Montjoye who recruited me as his very first PhD student.

Last but very far from least, I would like to give special thanks to Dr. Amani El-Kholy and the team at Imperial College London. Without her support and guidance, I would be leaving this PhD with many more battle scars.

I am sure I have missed some very important people. Please get in touch and I will buy you a drink instead.

Dedication

To my brother, Mo Farzanehfar. He is the kindest person I know, with a heart of gold, and unshakeable priorities. Without him I would not be me — even if he still can't take me on in a fair fight.

To my mother, Sarvenaz Mafi. If someday someone calls me intelligent, kind or, tenacious, it can be almost entirely attributed to her.

To my sister, Nina Stone. She's taught me how to be curious about the under-noticed. Without her my world would be inconceivably small.

To my father, Jahangir Farzanehfar. I have never seen anyone personify grit and charm as much as this man — properties I hope to have inherited (along with some modesty).

Contents

Abstract	3
Acknowledgements	7
1 Introduction	21
1.1 Technological advancement and scientific paradigms	21
1.2 Structure of this thesis	25
2 Insurance markets & machine learning	27
2.1 Introduction	27
2.2 Background	30
2.2.1 Limitations in current market-based experiments	31
2.2.2 An overview of existing market-based experiments	32
2.2.3 Data-models Vs. algorithmic modelling	43
2.3 Motivation	48
2.4 Methods	50
2.4.1 Analysis goal: <i>impact of modelling cultures on prices</i>	51
2.4.2 Competition dataset	53
2.4.3 Participant experience flow	56
2.4.4 Participant recruitment & fundraising	56
2.4.5 Participant code templates	59
2.4.6 Insurance pricing game market design	61
2.4.7 Feedback to participants	69
2.4.8 Communication with participants	81
2.4.9 Available online resources	84

2.5	Results	86
2.5.1	Participants & submissions	86
2.5.2	Categorising data and algorithmic models	90
2.5.3	Risk-model quality and exclusion criteria	91
2.5.4	The impact of algorithmic modelling on consumer prices	93
2.5.5	The impact of algorithmic modelling on market entry	97
2.5.6	Sensitivity analysis: <i>beyond a two-stage profit metric</i>	99
2.6	Discussion	100
2.6.1	Motor insurance pricing: <i>algorithmic Vs. data-models</i>	101
2.6.2	Limitations & assumptions	104
2.6.3	Future work	109
2.7	Conclusion	116
3	Privacy & reidentification risk	117
3.1	Introduction	118
3.2	Background	121
3.2.1	An overview of privacy attack models	122
3.2.2	Data sparsity and homogeneity	124
3.2.3	Privacy in heterogeneous (small) data	126
3.2.4	Privacy in homogeneous (big) data	131
3.3	Methods	134
3.3.1	Dataset description	135
3.3.2	Unicity: <i>difficult to compute, easy to estimate</i>	135
3.3.3	Building the unicity model	136
3.3.4	Mathematical lower bound of unicity	139
3.4	Results	145
3.4.1	Results on the decrease of unicity	145
3.4.2	Sensitivity analysis	147
3.5	Discussion	152
3.5.1	Assumptions underpinning the simple unicity model	153

3.5.2	Previous estimates of re-identification risk in large datasets	154
3.5.3	Direction for future work	155
4	Conclusion	157
4.1	Future directions	159
4.1.1	Algorithmic pricing models in a market with real actors	160
4.1.2	Impact of individual conversations on political polarisation	160
	Bibliography	161
	Appendices	192
	Appendix A Insurance markets & machine learning	193
A.1	Full participant feedback report example	193
A.2	Ethics approval	196
	Appendix B Privacy & reidentification risk	197
B.1	A naive unicity model: <i>the combinatorial approach</i>	197
B.2	Convergence of the unicity model	199
B.3	Estimating uniqueness by subsampling a larger dataset	199

List of Tables

2.1	The dataset dictionary provided to competition participants.	57
2.2	Summary table of model type.	91
3.1	Summary of privacy-protecting legal mechanisms.	122
3.2	A fictional example sample from a Call Detail Record dataset.	126
3.3	An illustration of sensitive information, and quasi-identifiers.	128
3.4	$\overline{P}_F$ parameter values	149
3.5	$\overline{P}_A$ parameter values	149
3.6	Summary of unicity results at $N = 20M$ as per the sensitivity analysis.	150

List of Figures

2.1	Summary results from the work of Van de Rijt et al.	40
2.2	Illustration of statistical modelling split between two cultures.	44
2.3	Plot illustrating one potential analysis to be run at the end of experiment. . . .	52
2.4	Distribution of non-zero claims in the dataset.	54
2.5	Scatter matrix with diagonal KDE plots summarising dataset variables.	55
2.6	Participation flow diagram.	58
2.7	A screenshot of the advertising webpage (pricing-game.com).	60
2.8	Illustrative example of a cheapest wins market	62
2.9	Dataset distribution among leaderboards	63
2.10	Diagram view of the profit metric.	66
2.11	Profit Vs. market share feedback	71
2.12	Participant feedback in the form of a claims heatmap	72
2.13	Summary table of driver attributes	74
2.14	Financial summary table of market performance.	75
2.15	The effect of claim caps and dataset size on leaderboard ranking consistency. . .	78
2.16	Example of feedback prior capping.	79
2.17	Competition timeline.	80
2.18	A sample of one of the weekly emails sent to participants.	83
2.19	Participant code traceback upon submission.	84
2.20	Display of how the leaderboards can be accessed and viewed by participants. . .	87
2.21	Participant and submission growth over the competition timeline.	88
2.22	Summary of participant demographics.	89
2.23	World map view of competition participation.	90
2.24	KDE plots of risk-estimation Vs. competitive profit.	92

2.25	Impact of algorithmic modelling on consumer prices.	94
2.26	Between-model correlation and profit comparison of the two modelling cultures.	95
2.27	Comparison of margin values between data and algorithmic models.	96
2.28	Boxplots of profit and market share in mixed markets.	98
2.29	Rolling comparison of rankings when the profit metric is run for multiple rounds.	100
2.30	Weekly boxplots of mean model margin.	112
3.1	An illustrative diagram of the Sweeny attack.	127
3.2	The four inputs to the unicity model are displayed.	137
3.3	Empirical unicity compared to the estimated unicity is displayed.	146
3.4	Unicity expanded to a population of 60M.	147
3.5	The range of distributions studied in the sensitivity analysis are displayed.	150
3.6	Unicity of alternative models compared to observed unicity.	151
A.1	Example of a full participant feedback report.	195
A.2	Ethics approval form.	196
B.1	The combinatorial unicity model	198
B.2	Convergence of the unicity model as the size of the unicity sample increases.	200

List of Abbreviations

AI Artificial Intelligence

GLM Generalised Linear Model

FCA Financial Conduct Authority

VR Virtual Reality

PTSD Post-Traumatic Stress Disorder

SVM Support Vector Machine

RMSE Root Mean Squared Error

UCT Coordinated Universal Time

KPI Key Performance Indicator

KDE Kernel Density Estimation

QBS Query Based System

DP Differential Privacy

GDPR General Data Protection Regulation

LPPM Location Privacy Preserving Mechanism

1

Introduction

In this thesis I will present two case-studies, one related to a novel data-collection mechanism and the other focusing on the privacy risks of data collection. In order to place both of these contributions in context, I will briefly describe the role of new technology in science with a view on paradigm shifts. This is done as both contributions in this thesis are exploring the potential impact of large scale data collection and processing on both the ability to ask new questions with new data, and how this ability carries certain risks, such as to personal privacy.

1.1 Technological advancement and scientific paradigms

One historical view of scientific progress is through the lens of paradigm shifts. In this thesis I will focus on those paradigm shifts that are related to technological advancements, resulting in expanding the ability to measure or control a scientific environment. I use the word “*paradigm*” to mean an exemplar of how to do research in a particular field, as described in the famous work by Thomas Kuhn [2]. One of the many definitions used by Kuhn is stated as:

“[Paradigms are] universally recognised scientific achievements that, for a time, provide model problems and solutions for a community of practitioners.”

Although there are many ways to describe a scientific paradigm in detail¹, I will focus here on two key properties of paradigms that also describe scientific events that lead to a change in existing paradigms — paradigm shifts. The first is a notion of novelty, and the second is open-endedness [2]. Both of these features are inextricably linked with technological advancement which allows new types of questions to be asked and answered.

The first criteria, novelty, is very complex to define, however, for the purposes of this thesis, is meant “*previously unmeasurable*” rather than any notion of intellectual or cultural novelty². Examples of this kind of novelty are usually tied to technological advancements or inventions. The optical and radio telescopes [5], optical, electron and atomic microscopes [6], laser and maser technologies [7], medically utilised X-rays and fMRI scans [8], and satellite technology [9] are example catalysts for the birth of entire fields of research. The very first of these would often be connected to an ongoing paradigm-shift while the others may lead to new discoveries of their own. They are novel in the sense that they allowed for an entirely new kind of observation to be made and reproduced by anyone with access to the right resources.

The second criteria, and arguably the more important one, is open-endedness. In order for a piece of work to be considered an exemplar it must address a scientific question that begs further investigation, either in parallel by applying the method to a different question, or in series by further narrowing the breadth of the initial question. In addition to this contextual open-endedness, from a technological perspective, this criteria requires the technology to be accessible enough. This is so that a significant portion of the community can use the same methodology to first, reproduce the exemplar and second, use the same method to launch new avenues of research. Other than the invention of new measurement devices, the most straightforward examples of open-ended paradigms are found in theoretical scientific discoveries that then require experimental confirmation. Many theories, such as General Relativity or the Standard Model of High energy Particle Physics, provide very general descriptions which can lead to falsifiable predictions. The fact that they can lead to falsifiable predictions is one way that theoretical scientific discoveries become open-ended to some degree and invite future work.

¹And very few ways to obtain consensus on the matter [3].

²For a much more in-depth treatment of scientific novelty see the work of Prof. Michael North of UCLA [4].

Paradigm shifts up until the mid 20th century occurred at a relatively low frequency³ for two key reasons [10]. Firstly, many of the primary natural sciences were not professionalised until the 18th and 19th centuries which is required for a community to share a paradigm and rapidly test hypotheses centering around falsifying predictions using that paradigm [10]. Without a professional science there is no shared paradigm and thus, there is no paradigm-shift. One symptom of non-professional science is seen in their choice of medium for knowledge communication. Typically a field of science begins knowledge communication in the form of demonstrations and then into lengthy books aimed at a general public audience. It finally becomes a professional science (in the modern sense) when the audience is shifted from the general public to an expert community of fellow scientists. The medium appropriately also changes from lengthy books aimed at the general public to, short and dense journal articles and letters, often incomprehensible to the non-expert audience [2].

The second reason for the increased frequency of paradigm shifts is related to the density and frequency of communication within a scientific field. In order for those who share a paradigm to be able to fully understand what their field, often a global community, is doing, they need to be in communication with each other. Before the widespread adoption of the telephone and the internet, it was very difficult for scientists to know exactly what other scientists were doing to the level of detail that the internet can allow. Today this is very different [11] with advances in any sub-field of science often overwhelmingly fast for any individual scientist to fully grasp. In 1950, there were a total of 21 publications in computer science while in 2020 there were in excess of 400,000 publications. Is it really a surprise that paradigm shifting discoveries are happening more often?

Today, the scientific community is experiencing a rich injection of new experimental methods and procedures directly as a result of new large datasets, or the ability to collect data in novel ways [12]. Almost every existing science today has a computational sub-field. Computational biology, computational chemistry, computational medicine, and many fields of computational physics are a few examples. In addition, many entirely new sub-fields are born due to either

³Depending on the historical view taken on, there are many potential reasons why scientific inquiry has exploded in the late 20th and early 21st centuries. Reasons such as wider availability of healthcare, education and general well-being are a few candidates.

the existence of these new datasets or the ability to collect new datasets. Examples for the first include computational social sciences while examples for the second include large scale digital field experiments in economics and social psychology. This is potentially a multi-field scientific paradigm-shift in action⁴. Today data-driven methods relying on granular, machine readable information are ubiquitous in many fields of science, while many fields are on the way to adoption. This could lead to new ways of doing science.

These new computational tools can re-frame many age-old scientific questions by either providing new answers, or better, providing a way to ask new questions. Examples of the former are those studies that use existing data to answer old questions with more accuracy. Such as using mobility data to estimate the local spread of diseases into the future [13] rather than typical manual contact tracing in the past. Examples for the latter are those studies that employ a new data collection mechanism to ask questions in new ways. For example, the advent of digital field experiments today allow social scientists to better quantify their findings than ever before.

However, the ability to collect and process this vast amount of data is accompanied by a host of new risks related to privacy, fairness, and safety [14].

In this thesis I will revisit two of these existing questions using either a new data collection mechanism, or new data itself. For the first, I will present the design and execution of a digital field experiment designed to study the motor insurance market while overcoming three of the most significant limitations in any study of economic markets: scale, participant expertise, and participant time-commitment. For the second, I will present a case-study on privacy using existing mobility data demonstrating that the size of a mobility dataset cannot sufficiently protect the privacy of the data subjects. In other words, we do not get “*lost in the crowd*” when it comes to the anonymisation of mobility data — the increased size of mobility datasets is unlikely to “*naturally*” protect individual privacy.

⁴Not that the use of new tools themselves is a paradigm shift, but rather that their expanded use can lead to new insights.

1.2 Structure of this thesis

This thesis contains 4 chapters in total, these are set out as follows:

1. **Introduction:** This chapter. It includes a brief discussion of scientific paradigms and the impact of technology on how scientific discoveries are made. An argument is made that technological advancement is a key catalyst for many scientific paradigm shifts historically. It is argued that today, the main technological advancement that is catalysing numerous paradigm shifts within multiple fields of science is the ability to collect and process vast amounts of data. With many fields becoming computational, existing questions can be revisited in a new light and entirely new, previously unaskable questions, can be answered. The two case studies within this thesis are then briefly introduced. The first relating to data collection in a study of insurance markets, and the second, touching on the privacy risks of this new paradigm-shif enabling technology.
2. **Insurance markets & machine learning:** A description of the motivation, design, and execution of a large-scale global digital field experiment is the focus of this chapter. This experiment was designed to assess the impact of machine learning on a realistic, cheapest-wins motor insurance market. This is one of the new ways that the answer to previously unaskable questions can be attempted through novel data collection. The experimental procedure is described in detail, with preliminary results analysed and interpreted. Detailed future next steps are also included both with the data collected in this digital field experiment, and future designs on new experiments.
3. **Privacy & reidentification risk:** The second case study shifts away from the uses of data collection and towards one of the main risks, individual privacy. Large scale data collection is allowing the movements of entire populations to be recorded with unprecedented granularity. Existing evidence suggests that this type of high-dimensional homogeneous time-series data is very difficult to anonymise safely, although some suggest that the scale of datasets could provide a sort of “*natural*” privacy protection. In this published study [1], we show that although dataset scale does reduce privacy risk,

this reduction is very small even in the scale of datasets with populations larger than entire countries. This chapter also includes a brief sensitivity analysis, a discussion of the implication of the results and finally, detailed suggestions for future work.

4. **Conclusion:** Finally, some concluding remarks about the two case-studies are included. Beyond an overall interpretation of the results in the thesis, new directions of research, more broad than the previous two chapters, which attempt to exploit the potential of the on-going adoption of this paradigm-shift enabling technology. These avenues of research are quite broad explicitly and attempt to show the wide-ranging applications that this new technology is enabling in science and society.

2

Insurance markets & machine learning

A case-study in simulating an expert-based market

The work contained in this chapter was led by myself in collaboration with Florimond Houssiau, Arthur Charpentier, and Yves-Alexandre de Montjoye. As a rule of thumb, where I use the personal pronoun “*I*”, it is to reflect my contribution and the pronoun “*We*” is reserved for cases which I contributed to along with others.

2.1 Introduction

Scientific fields are distinct from other areas of knowledge in that they ground their findings in empirical validation in a way that strives to be inarguable at the time [15]. Over time, through technological and scientific advancement, the experimenter gains more control over their environment and is able to revisit old questions [16]. This technological advancement typically leads to sub-fields of science being born which may eventually diverge and become their own field of study. The best example of this is maybe the field of computer science which arguably did not exist before the 17th century (regarded as an obscure sub-field of mathematics [17]) and certainly did not exist in its modern form until the 20th century when the term computer started to refer to machines rather than to job titles [18].

Within the natural sciences the correlation between the ability of the experimenter to control their environment and new scientific discoveries is very clear [19]. The confirmation of general relativity theory (a reconstruction of classical mechanics) would never be possible without advancements in optics and construction of lenses [20], at least not in their current form. Of course this increased environmental control can lead to asking entirely new questions as well. An example here would be the relationship between high-energy particle physics and advancements in computer science which allow processing of vast amounts of sensory input in nanosecond time [21].

Unlike the natural world, empirical studies that focus on the behaviour of individuals or small groups are more challenging. Psychologists for decades have come up with ingenious experimental setups to monitor and understand human behaviour. One of area of advancement here is Virtual Reality (VR) technology. This allows researchers to control the environment presented to the human mind by offering some control over visual and auditory input. There is already some application in clinical psychology in treating Post-Traumatic Stress Disorder (PTSD) [22]. Similarly, using virtual environments, experimenters have been able to compare and contrast what participants report they would do in a hypothetical setting to what they would actually do in a virtual environment [23]¹. Aside from VR, there are other technological advances that allow for new experiment designs such as the experience sampling method pioneered by Mihaly Csikszentmihalyi [25] to study the relationship between productivity of experts and their mental state outside of a laboratory setting.

Beyond the individual, the next frontier for the experimentalist is the control of the environment of industries or entire societies. Historically, the study of societies is the remit of the social sciences. Due to the inherently complex and uncontrollable environment that are human societies, social sciences are at their best when they combine both qualitative and quantitative methodologies [26]. However, in recent years this balance between qualitative and quantitative methods is tilting towards the latter.

Although it can be argued that the social sciences will always require a degree of qualitative

¹For those interested, a fascinating account of VR in psychology can be found here [24]

analysis [27], modern computational resources are changing how social science research can be conducted [28]. Most famously this has led to the birth of the sub-field of computational social science, breathing new life into the rich literature of social physics. Generally these new tools are used in two ways: 1) to leverage existing datasets in new ways and 2) to collect new datasets. In the first category, examples span subjects from understanding human behaviour in small groups [29], to mapping movement patterns on a national scale [30]. This line of research builds on the well established research methodology, most common in economics research, of finding appropriate “*natural experiments*”.

The second category is concerned with the ability to collect data in new ways that were previously impractical. Broadly speaking there are 3 ways that data collection is done concerning the study of societies: surveys, live experiments, and simulations. Firstly, one can conduct surveys. The most prolific example of this is perhaps the famous Fragile Families dataset that has been under construction for the past 2 decades through painstaking manual followups with a set of families in the United States [31], while other long-terms studies also exist [32]. However, not only is manual data collection in this way very costly, but also, it may be highly inaccurate as what subjects say they would do is not always aligned with what they actually do in a realistic environment [23]. Secondly, data can be collected in a live laboratory experiment by placing the subjects in a tightly controlled setup and studying their behaviour to stimuli [33]. The infamous Stanford Prison Experiment where participants assigned roles as “prisoners” or “guards” began to exhibit extreme behaviour, falls into this category [34]². However, this kind of data collection is usually limited in scale due to practical reasons as well ethical considerations. The third way that data collection can be done is through designing potentially interactive simulations that mimic certain real-world environments.

Simulated interactive environments fall within the scope of the emerging field of Digital Experimentation that studies experiments in digitally mediated social and economic environments. Specifically studies in Digital Experimentation allow for 2 key features that are difficult to replicate offline [36]: scale, and tight environment control. For example, the work by Salganik et al. in understanding the impact of social influence on performance in a music market [37]

²For the original experimental notes please see here [35]

relied on designing an artificial “*Reddit-like*”³ environment where subjects could vote on the quality of a set number of songs. By manipulating the rankings and splitting the experiment cohort into multiple “*worlds*”, the scientists are able to de-tangle the effect of music quality from that of social consensus when it comes to rankings in a market.

Despite their success, online digital experiments suffer from two key weaknesses. First, many studies require only minimal interaction and time-commitment from experiment subjects as more long-term interactivity cannot be reliably controlled for or incentivised. Second, to achieve scale, these studies usually aim to reach a very wide and diverse pool of experimental subjects globally, with minimal control over participation. In fact, in the absence of appropriate incentives, increased time-commitment may be accompanied by decreased scale as expected intuitively. These two limitations, minimal time-commitment and low levels of shared participant expertise, prevent studying a broad range of targeted scientific questions.

In this work I will present a case-study⁴ of a digital experiment that overcomes these two limitations in studying the impact of machine learning on risk-estimation and pricing in the motor insurance industry. The experiment, executed as an online data science competition with a \$12,000 prize pool, simulated a cheapest-wins insurance market where 2000 participants acted as insurance firms competing for profit. Close to 350 of the participants played weekly for 11 weeks with an average of 2 hour or more invested in the competition per week. Approximately 60% of those reported working professionally within the insurance industry. However, before diving into the experimental design, I will present previous related work to place this case-study in context.

2.2 Background

The work presented here is split into two categories. The first will focus broadly on the design of market-based experiments while the second will cover previous work that studies the impact

³Reddit is an online social network [38].

⁴This experimental method had been used in one failed attempt by myself in 2018, and at a smaller scale, with a much more simple design by Prof. Arthur Charpentier, though no publications ever resulted, due to fatal errors in the design, or very small scale in participant uptake.

of model type (e.g. Generalised Linear Model (GLM) Vs Random Forest) in a given domain. Both pieces will have a focus on the insurance industry where possible. I will touch upon each of these two themes separately, highlighting the relevant pieces of work that connect with the case-study to be presented. Prior to that, I will also describe the principles that were used in the design of the experiment overall, principles that are quite broad and can be found in each of the lines of work that will follow.

2.2.1 Limitations in current market-based experiments

Historical social science research concerning large groups, such as economics, has relied on four types of studies when split methodologically: natural experiments, laboratory experiments, simulations and more recently, digital experiments which are an offshoot of simulations and will be treated separately here. Neither of these, with the exception of simulations, have a direct parallel to the natural sciences as it is usually impractical, unethical, or very expensive to constrain a large-scale social system (e.g. a city) into a lab. However, access to and control over the experimental environment is becoming possible in large-scale social science research due to advances in technology and the flourishing of new methods in digital field experimentation.

In the study of markets, access to and control over the experimental environment is particularly challenging due to three reasons. First, markets are usually large scale ecosystems that involve many individuals. Second, many markets require, or typically involve expert domain knowledge from their participants⁵. Third, in many real-world markets, the expert participants invest a significant amount of time engaging with the ecosystem. Due to the difficulty in replicating these three pieces simultaneously, research on dynamic markets can be quite limited in scope.

Focusing on market-based experiments, the section below will delve into each of the four research methodologies highlighted above and will draw attention to the fact that few existing methods to date are able to capture the three elements of scale, participant expertise, and participant time-commitment within a single design⁶ — something that the case-study that follows arguably

⁵Capital markets, insurance markets, credit markets, etc.

⁶Empirica, a platform designed to address these issues is one very recent example that attempts to solve these issues [36]

achieves.

2.2.2 An overview of existing market-based experiments

The four methodologies touched upon above will be expanded on here. The first are natural experiments where a shift in the real world allows for a unique pre- and post- control over the environment. The second are laboratory experiments, usually small scale studies with very targeted research questions. The third are fully synthetic simulations, usually with some form of real data that first attempt to replicate real-world dynamics and then change some parameters that allow testing of a specific hypothesis, usually in an explanatory capacity. Finally, there is the category of digital field experiments which are usually large scale studies that allow the experimenter to actively manipulate the experimental environment and measure the reaction of the subjects.

Natural experiments: *a tool for scale, with little control*

Natural experiments are a type of field experiment where, the researcher, using a real-world scenario can assign experimental subjects to treatment and control groups [39]. These are one of the most common ways for studying specific markets of a scale that is impractical to replicate in a lab. However, they do have certain limitations mainly related to the inability of the experimentalist to control the environment either for confounding factors [40] or to test alternative hypotheses. In the rest of this section I will present 3 case studies that answer interesting questions but may be victim to the limitations listed here.

Perhaps the most famous modern natural experiment to date is the Oregon Health Insurance Experiment studying the impact of access to health insurance for low-income individuals. The authors used the fact that the state of Oregon used a lottery⁷ to provide health insurance to approximately 30,000 people. The researchers compared the outcomes for these individuals to a second randomly picked sample that had applied for the lottery scheme but did not

⁷Something that would never pass the bar for an ethics committee in my opinion.

win [41]. Notably, although this study has great scale, the researchers would not be able to test any hypotheses by changing the underlying environment, instead they have to contend with studying the real environment as is and hope that their data is applicable. For example, had Oregon decided to provide fewer lottery tickets but targeting them to even lower income areas, perhaps the overall state-level health outcomes would be better.

Another famed market-based natural experiment, this one by Robert Jensen, concerns testing the “*law of one price*”⁸ using the introduction of mobile phones to the fisheries industry in Kerala India [43]. For fishermen in the region, knowing the prices of fish during the day at the many markets along the coast was not possible due to the fact that they can really only visit a single market per day, resulting in large variations in price. In 1997, mobile phone service began to be introduced throughout Kerala with 60% penetration of fishing boats and retail traders by 2001. By conducting a survey of 300 randomly selected fishing boats from before and after 1997 it was revealed that mobile phone introduction led to a stabilisation of sardine prices immediately, providing strong evidence for the law of one price. However, due to the nature of the experiment, further followup studies showed that, price stabilisation only occurred for select fishing boats in select sub-markets and did not occur in general [44]. If Jensen in the original study could control which fishermen to give a phone to, he may have unveiled this nuance as did the followup study.

In another example, Assad et al. study the impact of algorithmic pricing on competition in the German retail gasoline market [45]. In this example, the authors use the 2017 introduction of algorithmic-pricing software into the retail gasoline market to study how gas-stations in the area reacted to the introduction of machine learning algorithms in pricing. In summary, the software adjusts gasoline prices based on geography and current market prices of other stations to maximise revenue⁹. What the authors find is that, in a competitive setting, Artificial Intelligence (AI) adoption leads to a 9% increase in margins of the adopting firms. Due to the multi-year nature of the study they also find evidence of tacit algorithmic collusion by showing that AI-adopted markets experience a higher increase in margin than other markets.

⁸This is the law that suggests, barring any trade limitations and frictions, a given product should sell at the same price everywhere [42].

⁹This aspect was never directly observed but was inferred by the authors based on the results.

As with many natural experiments, in both the Kerala fisheries and German gasoline market studies, the results of the experiments cannot be generalised to other markets, or even to the specific markets they study. In the latter, a stark weakness of the study is that there is no discussion of the pricing software being introduced and whether a diversity of models in the market would impact the findings. Therefore, the mere introduction of a competing algorithmic pricing software vendor could invalidate the results. In the case of the Kerala fisheries market, critics of the study have noted that the observations do not necessarily support the hypothesis and we should not be too quick to confirm a well-known law of economics without due consideration of the experimental results, especially with new tools of measurement at hand [44].

Laboratory experiments: *high degrees of control but small in scale*

If natural experiments are on one side of the spectrum of market studies in terms of experimental control, laboratory experiments are on the other. In these experiments the scientist usually has full control over the environment being examined. Participants are typically invited to a laboratory setting and surveyed or observed engaging with carefully constructed scenarios. That being said, in the context of market studies, laboratory experiments usually take the form of classroom exercises.

One of the key strengths of laboratory experiments is the extreme control they provide the experimenter. An early example of this in the context of markets is the famous work by Ho et al. who empirically study the p -Beauty contest¹⁰ that perhaps epitomises economic research in a lab [46]. For reference this contest rewards any participant in a group who guesses an integer closest to a fraction p of the mean guess, where repeated games often lead to different results with the same participants¹¹. The results of these experiments are often compared to how investors behave in other (notably capital) markets as both relate to the “*what I think you think I think*” type of strategies. In the case of Ho et al., the experimenters physically surveyed ≈ 300

¹⁰Once more in a classroom setting.

¹¹Typically repeated games result in mean estimates closer to 0 (the optimal guess). The original p -beauty contest was introduced by the famed John Maynard Keynes and involved real people voting on who they think is the most beautiful in a newspaper contest. See chapter 12 of the original work [47].

undergraduate students from the same class in a series of 55 highly supervised experiments. In each experiment participants were given the same instructions by the researcher, were observed and completed the study in the matter of a few hours at maximum. Although the study has reasonable scale, it lacks significant time-commitment by the participants, or any expertise. In fact, expertise is a key differentiate in performance in these p-beauty contests, for example see table II in the work of Camerer et al [48].

Among the more explicit market studies in a laboratory setting are those that focus on the effect of information asymmetry on market outcomes. Chen and Jaiprakash simulate an insurance market in a classroom [49] with the aim of investigating how adverse and advantageous selection¹² can interact with each other in the presence of real human agents. Similarly Ali¹³ et al. study the same effect but do not explicitly mention the insurance context [50]. In both cases the vast majority of participants are students recruited from within a classroom. Again the entire experiment takes place within a very short time span of less than 1 day and participants are not experts of any type.

Another laboratory study simulating competition in markets is by Dufwenberg and Gneezy, who investigate the effect of market size on the fierceness of price competition [51]. In this study, once more with student subjects, the researchers simulate a multi-round cheapest-wins market where the student subjects compete for profit¹⁴. Although small in scale once more, the results of this study are interesting.

As seen from this selection of studies, there are three key weaknesses in prior laboratory-based market experiments: scale, participant expertise, and time-commitment. In all of the studies noted here the experimental procedure involved only university students, spending a few hours in one day, playing a game with a handful of other students. As a result, the effects of expertise, sustained effort, and emergent dynamics due to scale cannot be studied¹⁵. This is particularly

¹²Adverse selection is when an insurer under-prices the risk of a policy holder. Advantageous selection is the reverse.

¹³This is a different Ali needless to say.

¹⁴Perhaps the most interesting feature of these experiments is that, with more than two players, once an equilibrium is reached in prices (i.e. profit ≈ 0), it does not change, confirming the theoretical expectation — something that was not observed in their duopolistic markets.

¹⁵Or at least they have not been studied in a lab to date.

relevant for studying markets which require domain expertise, as indicated by one famous study that showed the performance of domain experts is vastly different to others, only when engaged in tasks within their domain [52] and similar to the base population otherwise.

Full market simulations: *a useful tool for explanatory studies*

Full market simulations without any human subjects is one of the ways that a study can scale without many practical obstacles. There are broadly two categories of research in this area. First are fully synthetic agent-based simulations where the focus of the work is designing an appropriate agent to replicate emergent behaviour observed in markets. The second are quasi-simulations where the researchers anchor their simulations with real datasets collected from the markets under investigation. Depending on the model used, these can be seen as observational and explanatory studies. Although literature that uses either of these methods can lead to very interesting and insightful results, neither can replicate dynamically how real-world agents (i.e. human subjects) would behave. Therefore, from the three challenges of scale, participant expertise, and time-commitment, they can only address the first.

The work by Heinrich et al. to understand the risk introduced by insurer model homogeneity [53] is an example of a quasi-simulation. In this work the authors attempt to simulate the entire household catastrophe insurance ecosystem, including customers, insurance companies, reinsurers, shareholders and catastrophe bonds¹⁶. Using micro-behaviours of the agents and historical financial data, they can reproduce the insurance cycle¹⁷ [55]. They find that in their hypothetical world, risk model diversity improved market outcomes for both customers and firms, expanding the diversity of risks covered while reducing market exposure to adverse and unknown events. However, as the authors state, this work lacked any empirical data or real agents, casting some doubt on how much of their results can be applied to the real world.

However, when the agents in a market are not expected to be human subjects at all, full

¹⁶These are risk-based securities which can be used to pass on catastrophe risk to capital markets so that a reinsurer does not have to expose themselves to the full brunt of a catastrophe [54].

¹⁷These are intermittent periods of rapidly rising premiums and reductions on the capacity of companies to write new business on a market-wide scale.

simulations can offer interesting insights in highly stylised environments without a real dataset or real agents. Much of the literature on algorithmic collusion using reinforcement learning relies on fully automated agents which fall into this category [56–59]. Although these may offer interesting insights into specific economic or computer science questions, fully synthetic simulations might not translate to any real world scenario which would involve human actors in the loop at some stage. Even in the domain of algorithmic collusion where fully synthetic simulations are the norm, one could argue that the work by Assad et al. on the German retail gasoline market mentioned earlier [45], shines a more convincing light on the topic than any synthetic simulation; despite its weaknesses in actually expanding on any mechanistic process behind the possible collusion observed.

One of the solutions to addressing the issue of realism in fully synthetic simulations is to anchor any modelling in real data¹⁸. Studies of the dynamics of ride-hailing and ride-sharing platforms fall into this category [60, 61]. As an example, Vazifeh et al. study the ride-sharing market and, using an agent-based simulation, show that these markets can operate as efficiently with 40% fewer vehicles through increased trip information sharing [62]. While other empirical studies actually demonstrate that the current way ride-sharing works increases congestion within cities [63]. What makes these results more compelling than those of a fully synthetic simulation is that the researchers based and tested these results on a real dataset of 150 million taxi trips in the city of New York in the year 2011.

In fact, combining simulations with real-world data mining can be potentially very powerful. An example of this is the work done by Owadally et al. First, the research team, using real home insurance data from the UK, test the hypothesis of whether individual-level, psychological behaviour of actuaries and underwriters can contribute to the widely observed insurance cycles [64]. By modelling an underwriters evaluation of a customer in a graph-based model, the team is able to reproduce elements of the insurance cycle, providing evidence that individual level-behaviour can contribute to these cycles. Once that is accomplished the team then adapted this model to the automobile insurance industry [65]. Here, by combining the model with data-mining approaches which can recognise repeated sub-sequences or anomalies, they are able to

¹⁸The case-study on privacy in chapter 3 in this thesis is one example of how this can be done [1]

produce a potentially, future-forecasting look into the automobile insurance sector, grounded in real data. Such a tool would be very helpful for financial regulation in reducing volatility in the market by allowing regulators to preempt severe crises.

Although quasi-simulations like the work of Vazifteh et al. and Owadally et al. are very insightful, they do not allow the researcher to observe knock-on effects of decisions within the simulation as the data would not have been collected under those conditions. For example, Vazifteh et al. will never be able to know if there are some hidden features unaccounted for by their model that would completely overhaul their results — they do not have data on how the market would react to their proposed solution. For example, it is possible that reducing vehicle numbers would lead to increased demand due to passengers preferring not to share vehicle. In another hypothetical study design, the authors could have surveyed riders to tease out the reason why the current number of vehicles is so large. Similarly, Owadally et al. are not able to test the knock-on effect of certain regulatory decisions as the dataset for that “*what-if*” scenario does not exist.

However, for reasons such as these, many studies of this type formulate their results as explanatory findings, as is the case in all three of the studies noted. Such is also the work by Moro et al. who use a combination of census and US Bureau of Labour Statistics datasets as well as the O*NET¹⁹ database to explain the impact of skills and job opportunities on migration patterns [66]. They frame their findings as the economic resilience of cities²⁰ and quantify their results by modelling the skills of the different labour markets. In this kind of work, the researchers do not need any control over the environment to explain the results. While in the work by Vazifteh et al., some control would improve their findings.

Using simulations to study markets can be insightful, however, they are limited in what kinds of questions they can reliably answer as they cannot account for real-time reaction of human subjects. The work of Vazifteh et al., and other studies noted above, exemplify these limits. Although these studies provide very useful insights to the industry, and can potentially be

¹⁹The Occupational Information Network (O*NET) is a comprehensive database of worker attributes and job characteristics.

²⁰For example a city with only one type of manufacturing job is not resilient economically.

future forecasting as well, they are somewhat untestable as the environment under study (i.e. the NYC taxi-cab ecosystem) cannot retroactively respond to any solution proposed. Needless to say, studies of this kind act as a necessary first step to any full-scale empirical investigation. Unsurprisingly, markets where human subjects are not the agents lend themselves more concretely to simulated environments, however, much like the literature in algorithmic collusion, without deploying these systems in some sort of realistic environment, the results remain highly theoretical.

Digital field experiments: *the potential future of market-level empirical studies*

To address two of the three issues²¹ highlighted in economic research, we can turn to digital field experiments that typically have a large scale, and tend to ask questions which do not require any participant expertise. These experiments are conducted online and usually focus on questions most pertinent to online markets such as the impact of recommender systems, social media, or other forms of influence on overall market dynamics. However, most studies in this category seem to trade time-commitment and participant expertise for scale, thus limiting the kinds of research questions that can be investigated.

Among digital experiments, one of first questions put to the test was the relationship between publicly available rankings and actual product quality. Is the high ranking of products actually justified by their quality, or are they just riding the benefit of a rich-get-richer market? This is the question that Salganik and Watts set out to investigate in 2008 as one of the first digital field experiments of its kind [37]. The researchers set up an online music market with 48 songs that could be ranked and downloaded. Once the rankings of their markets were stable, they created multiple “*worlds*” where they would flip rankings and observe any regression back to the original rank. Most impressively, they recruited more than 12,000 participants globally and engaged in a “*collect and exclude*” strategy by identifying malicious or non-standard behaviour from participants and excluding them post-hoc. Although the expertise or demographics of these participants are not controlled for in a detailed way.

²¹The three issues being: scale, participant expertise and, time-commitment.

Studying a very similar question (i.e. “Does success breed success?”) but with a less controlled design is the work by Van de Rijt et al [67]. In this case the researchers conducted 4 different studies in funding ([kickstarter.com](https://www.kickstarter.com)), review ratings ([epinions.com](https://www.epinions.com)²²), author awards ([wikipedia.org](https://www.wikipedia.org)), and signatures ([change.org](https://www.change.org)). In each case they sampled a large group of website assets (e.g. projects on [kickstarter.com](https://www.kickstarter.com), or petitions on [change.org](https://www.change.org)) and performed a Randomised Control Trial by splitting the assets into experiment and control and only providing positive public feedback to the experimental group. Their hypothesis being that the socially mediated success-breeds-success dynamics operating in each website would then go on to build on the initial success they have endowed the experiment group more than the control group, a finding confirmed by their work (see figure 2.1). Despite showing a measurable effect in multiple categories, this work is not fully able to control for some of the internal measures that may exist within each of the studied platforms. For example, some websites are known to have their own internal recommendation systems that the authors could be inadvertently measuring as opposed to a social phenomenon alone²³.

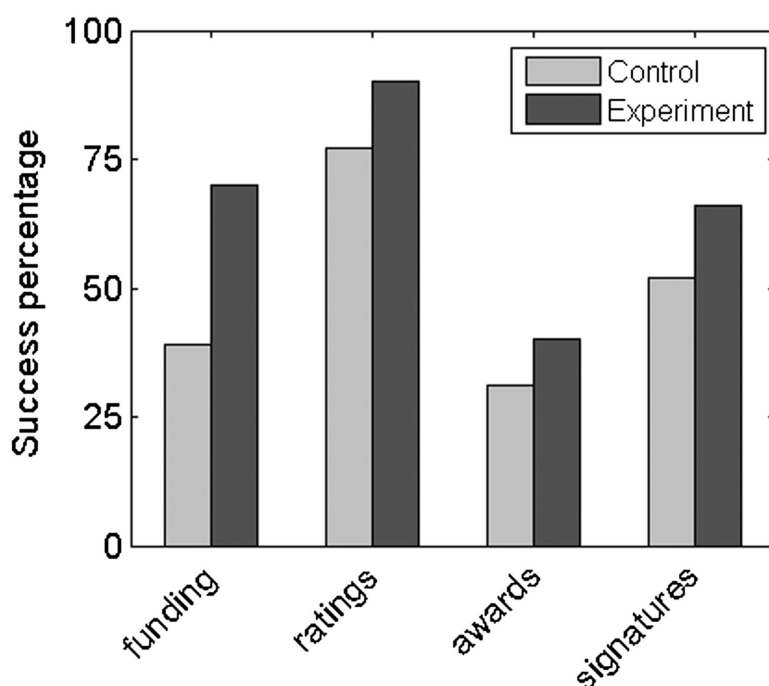


Figure 2.1: Summary results from the work of Van de Rijt et al. [67] highlighting the impact that initial success has on overall success in the four domains of funding ([kickstarter.com](https://www.kickstarter.com)), review ratings ([epinions.com](https://www.epinions.com)), author awards ([wikipedia.org](https://www.wikipedia.org)), and signatures ([change.org](https://www.change.org)).

²²Now deprecated.

²³[netflix.com](https://www.netflix.com) is one example service that has a complex recommendation system in place [68].

Another question that lends itself very well to online digital field experiments is the impact of recommendations and social advertising on consumer behaviour in a market. Two studies with very similar designs across a different range of domains show that recommendation systems have the ability to create less diversity among the habits of each user while simultaneously increasing the overall product diversity [69, 70]. In both of them, researchers were able to conduct a split-world experiment where the control group would receive generic recommendations while the experiment group would receive more personalised suggestion. A third study by Huang et al. [71], investigates a related question, effectively measuring the strength of advertising in different product categories (e.g. status goods, experience goods etc.) by partnering with a large online social network. It is worth noting that a key strength in all three of these studies (and others such as Bapna and Umyarov [72].) is the size of the experiments: they each have in excess of hundreds of thousands of participants with the latter two having millions. This class of field experiments are only possible if the industry incentives align very well with the academic research goals, and even then, are often only used to answer questions where participants are not required to have any expertise.

As seen from these studies, digital field experiments are able to address the issue of scale when studying markets, while the challenges of participant expertise, and time-commitment remain unsolved. As a result, many of the field experiments noted here formulate their research questions so that time-commitment or participant expertise is not a limiting factor. In fact all the studies noted here treat participants as subjects who need not interact with the experimental apparatus for more than an hour or so throughout the lifetime of the study. In addition, none of the studies noted here require participants to have any level of expertise in any subject, with the notable exception of Van de Rijt et al. [67] who use the top 1% of wikipedia contributors in one of their studies.

Finally, a note on the importance of expert participants. In many real markets, the quality of market participants has a significant impact on the overall dynamics, specially if those markets are typically dominated by a small number of players. As an example in the insurance sector, in 1999 one of the largest insurance companies in the US, Reliance Insurance Company went into insolvency in part due to the poor underwriting ability of an underwriting manager,

Unicover [73]. This had market-wide ramifications as many other businesses had to go under as a result of the bad financial practices stemming from the Reliance Insurance problem. Such market-level key-person-risk²⁴ could be studied if there was a simulated market with expert players that then had one bad actor playing the role of Reliance Insurance.

Aside from studies of resilience, any market that requires one party to carry out a value assessment requires expert participants. Examples include insurance markets, markets of highly specialised equipment, and perhaps less obviously, most employment markets and university admissions. In each of these, one part of the market must use their expertise to assess and price the value of the other side of the market, and then compete with others to realise that value, or better, win a bargain. To study these markets, experiments could be designed to investigate the different ways that one side of the market carries out assessments. Example research questions include the effect of different admission processes on long-term alumni outcomes in a competitive market for students, the long-term competitive value of algorithmic pricing to online merchants, or even the impact of banks competing for loan clients in a marketplace.

Principles to achieve scale, time-commitment, and participant-expertise

To overcome the limitations of scale, participant expertise, and participant time-commitment there are three guidelines that should be considered simultaneously²⁵. The first is to embrace a digital experiment design. The second is to ensure the task itself is engaging for participants as a stand-alone (i.e. without any reward) and finally, the third is to target the task so as to attract the appropriate participants naturally.

Firstly, digital experiments have enormous potential in achieving scale and enabling more significant time-commitment. Due to the ease in participating in these field experiments, combined with a global pool of potential participants, they allow participant pools to grow into the thousands or even millions as we have seen in the previous section. In addition, online experiments have the freedom to run for long periods of time with multiple interactions from a global

²⁴The risk resulting from over-reliance on one small part of an ecosystem.

²⁵Other frameworks are possible to achieve these goals. However, these are the guidelines that were used in this case-study.

participant pool while minimising the cost.

Secondly, having the standalone task be engaging is crucial to maximising the time-commitment to the experiment. Existing research in management science suggests a strong link between task engagement and productivity with more engaging tasks resulting in more productive employees [74]. Similar findings are available in the education literature [75]. After all, it is no surprise that more time is spent on tasks that are more fun.

Finally, in order to have expert participants, any experiment design needs to be geared towards those experts. However, a general rule of thumb is to advertise the experiment to those communities exclusively. Although it is possible to specify expertise in certain online participant recruitment platforms such as MTurk [76], this is not a scalable solution due to the payments required. Secondly these platforms typically only control for high-level demographic features and do not control for explicit expertise. For example, to study how a music market entirely populated with professional musicians would differ from an open market, the experimenter would first need a large pool of willing professional musicians.

2.2.3 Data-models Vs. algorithmic modelling

As the case-study in this chapter will concern the impact of machine learning on a competitive insurance market, we first need to understand what is meant by “*Machine Learning*”. In this work I will split modelling into two different categories of data-models and algorithmic models. I will then provide a brief summary of the work done in this field, including some example studies of interest.

The two modelling cultures: *data and algorithmic models*

The difference between data-models and algorithmic modelling is more a cultural distinction than a technical one. The late Prof. Leo Breiman described any statistical modelling task as finding an association between an input x to an output y as illustrated in figure 2.2. The

entirety of modelling then relies on how the scientist decides to view the central rectangle. He then describes two cultures in modelling: data modelling and algorithmic modelling [77].

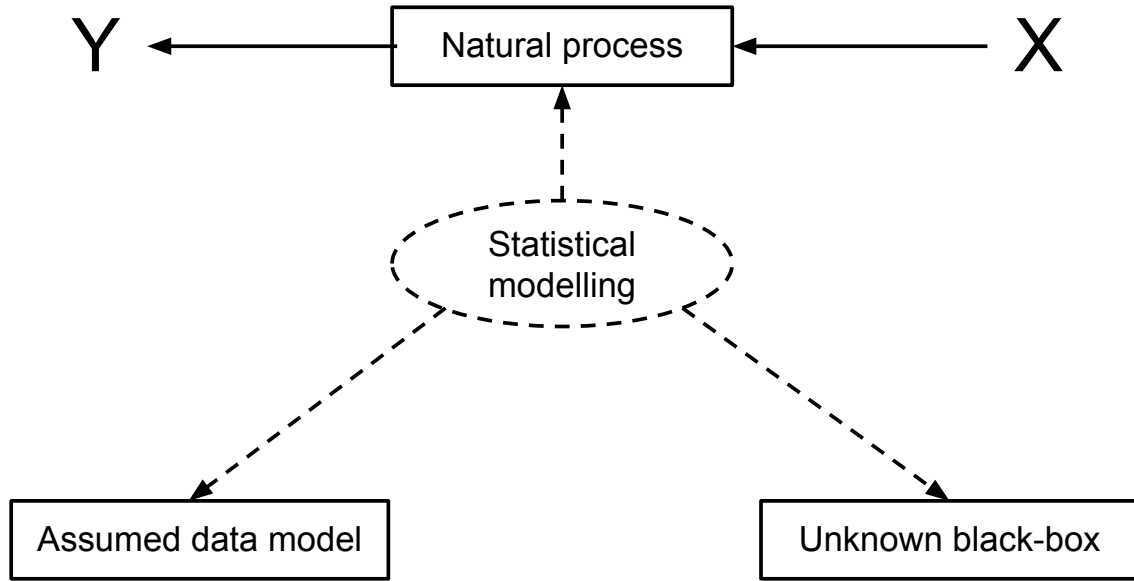


Figure 2.2: Illustration of statistical modelling adapted from [77]. The input data X undergoes some change via a process and results in the output Y . Statistical modelling can then take one of two approaches. The data modelling culture being represented on the bottom left with the algorithmic modelling culture being represented on the bottom right.

Within the data modelling culture, the approach is to assume some form of stochastic model for the inner workings of the box, and then generate predictions based on that. Examples of this include linear and logistic regression, any generalised linear or additive model, and any form of statistical modelling that makes assumptions on how the input data is generated [77]. In the context of the case-study to come, the main strength of this approach is the transparency of the box²⁶ which can allow explanatory analysis to be easily conducted. Furthermore, many of the model validation methods in this culture are based on statistical testing which themselves have concrete and well-established interpretations and assumptions. Especially, within the context of model explainability requirements in the insurance sector [78, 79].

In contrast to this culture is the algorithmic modelling culture where all the approaches of deep learning and many of the recent advances in machine learning would fall. In this culture the inside of the box that actually generates the output is of no interest, and is often considered unknowable. Instead, the aim is to be able to predict the outputs as accurately as possible

²⁶Although one could argue that transparency of the box itself an unverifiable assumption in this case.

using pattern-recognition techniques. These methods can typically capture nuances and non-linearities not captured by data modelling methods at the cost of their explanatory abilities. In fact there is a growing sub-field of machine learning dedicated to the interpretability of these algorithmic modelling techniques to fill this explanation [80].

Purpose of models within the insurance sector

Before delving into comparing these two cultures more deeply, a brief discussion on the purpose of modelling within the insurance sector can be helpful. Within insurance the purpose of models falls into three categories broadly speaking. These are, risk estimation, hypothesis tests, and optimisation and control. These typically occur at different stages of the insurance value chain and are not entirely divorced from each other.

The first category, risk estimation, is what most of this thesis is concerned with. It is the aspect of modelling which allows insurance firms to produce the “pure premium” [81] which is the expected cost of a given policy. Much of insurance modelling is focused on accurately estimating this value so as to reduce the risk of unexpected costs.

The second category, hypothesis tests is a form of explanatory modelling which aims to glean insights from insurance and financial data. Although this type of modelling can focus on contract level data, it is mostly used on a portfolio (or part of a portfolio) as a whole. For example, if a firm needs to stress-test their capital market position they may run hypothesis tests where they artificially alter the interest rates, portfolio mix, the possibility of catastrophic events and so on.

Finally, modelling is used in optimisation and control within insurance. This category of modelling focuses on ensuring that the profit for shareholders over time is maximised. Similarly it is used to ensure maximum utility for policy holders is achieved, or at least, accounted for. Here the modelling would have to account for the goals of the insurer which could be loss-leading in the short term, or greedy profit-maximisation without room for loss.

However, within this thesis I focus on the first category of models when discussing pricing

models.

Studies comparing data Vs. algorithmic modelling

The existing literature comparing these two modelling methodologies focuses on task performance (e.g. “*How well can we predict...?*”) in specific domains. Although the existing work spans a range of domains from clinical predictions, to primary insurance, to risk prediction in capital markets, none of them are designed to study the market wide impact of using one methodology over the other in a competitive environment.

Within motor insurance there are several pieces of work that compare and contrast multiple claim estimation models with each other outside of a competitive setting. Notable is the study by Dugas et al. in 2003 where the authors study linear regression, GLM, mixture models, decision trees, artificial neural networks and Support Vector Machine (SVM)s using real motor insurance data and find that nonlinear methods do offer some advantages in claim estimation [82], as supported by other studies [83–85]. Although this result is not replicated by all other research engaging in this question. For instance Noll et al. from the Swiss Actuarial Association compared Neural Networks, GLMs and tree based models in performance without finding any significant differences. They do mention that the slight under-performance of GLMs may be explained away due a lack of sufficient feature-building typically required for GLMs²⁷ [86]. Similarly Rasa and Mano have found that there is no significant benefit to using non-parametric (roughly algorithmic modelling) versus parametric (roughly data modelling) approaches [87]. They mention:

“The main benefit of using GLMs over neural networks or decision tree analysis is that the models are formulated within a statistical framework. This allows standard statistical tests (such as χ^2 and F tests) to be used for comparing models, as well as providing residual plots for the purpose of model diagnostic checking. On the other

²⁷This could also be the bias of the legacy school of thought that hopes GLMs are ultimately superior to other modelling techniques.

hand, artificial neural networks have a number of advantages mainly because they can learn to solve a problem that is not even precisely defined.”

Outside of applications to insurance, similar comparisons have been made in predicting stock performance or using machine learning for clinical prediction. In stock prediction, Gu et al. apply a series of 12 different predictive algorithms on a rich historical dataset of 30,000 stocks tracked over 60 years [88]. Generally they find that the pricing using models in the algorithmic modelling culture performs better than data-models, with the notable exception of data-models performing better at predicting the performance of the bottom 1000 stocks in their dataset, perhaps due to some hidden linear variable within the dataset. Within the clinical setting, Christodoulou et al. compare the simple Logistic Regression (data modelling) to a variety of nonlinear machine learning techniques (algorithmic modelling) [89] across a range of 71 clinical studies. In their review of the literature, they compare studies with the same research question that have used either logistic regression or “*Machine Learning*”²⁸ and find that there are no significant differences between the two modelling cultures within specific domains.

Lastly, there is the work of Fuster et al. on US mortgage data, comparing the two cultures and delving into the wider impact of shifting from data-models into algorithmic modelling [90]. They use a dataset of approximately 42 million home mortgages in the US that have at least 3 years of history. They then apply a range of modelling techniques to predict loan delinquency and most interestingly compare the sociological impact of each of the modelling techniques on the resulting output scores. They find that with their dataset, the algorithmic modelling methods perform marginally better as they are able to capture non-linearities that data-models may not. However, the authors state that as a result, these models will give worse credit-scores to historically disadvantaged communities, thus risking further marginalisation in a vicious poor-get-poorer phenomenon. Other work, notably by Jon Kleinberg, suggests that knowing the true threshold with better models and applying non-technical mitigation strategies to prevent such a vicious circle is a better approach, than to simply deploy “*worse*” models to protect the vulnerable [91] all the while “*inclusion of variables such as race can increase both equity and*

²⁸Here defined as Decision Trees, Random Forests, Support Vector Machines or, Artificial Neural Networks. All of which fall within the algorithmic modelling culture.

efficiency".

Overall, the current literature in comparing the two modelling cultures is entirely focused on performance as seen by the sample of studies noted here. This is because as many data-rich domains are moving away from data-models towards algorithmic modelling, they are predominantly concerned with being able to achieve similar or better results with these new techniques. While this philosophy may be sensible in a clinical setting, it should not apply everywhere. For example, in the work of Fuster et al., the authors attempt to highlight broader market-wide effects that moving towards algorithmic models may have. Within the insurance sector²⁹ there may be unknown effects on consumers or businesses due to a shift away from data-models. These may relate to the severity of errors, or the impact of mixing modelling cultures, or other unknown factors. Such a question is not answerable with studies of the design discussed thus far.

2.3 Motivation

The price of any insurance product has to account for two factors, first is the risk associated with the insurance product and second, is accounting for any competitive forces from the market. The first step is typically undertaken by underwriters and actuaries who, using data and domain expertise, will propose a given “*break-even*” price³⁰ [81]. The next step or steps will involve adjusting that price to remain competitive in a market environment. These two steps fall under the umbrella of underwriting and pricing [92]³¹. For clarity, I will be referring to these as “*risk estimation*” and “*pricing*”.

The role of data in risk estimation is crucial in determining the potential cost of a particular insurance product [92]. Although the modern role of risk in determining price can be traced back to the Lloyds of London insuring trade ships [93]³², risk-based cost estimation based on

²⁹Or any sector where statistical modelling is a crucial part of a competitive ecosystem.

³⁰Sometimes referred to as the “*pure premium*”.

³¹Depending on the resource these processes may be named differently. In the reference resource “*pricing*” is referred to as “*rate-making*”.

³²Older examples are likely to exist as well.

historical data came into standard practice in the early 20th century [94]. Today, anything from standard motor insurance, or health insurance, to bespoke pension scheme pricing in life insurance will include detailed mathematical modelling based on data collected on individuals. This trend is accelerating as “*big data*” such as telematics, Internet of Things metadata, or smartwatch streams begin to mature [95].

However, knowing the base price is only half the problem, the second half relates to pricing the product competitively. Although, there is some evidence to suggest that no optimal strategy may exist in pricing an insurance product [96]³³, intuitively, going into a market, pricing at or below cost, will not be a long-term profitable strategy. Insurance markets, like any market will have bespoke customer dynamics³⁴ and idiosyncratic behaviours such as customer loyalty, and the impact of marketing and product differentiation. In fact, in the industry, each specific market (e.g. motor, health, life, cyber etc.) will require in-depth domain expertise and familiarity the market landscape to price products in a profitable way, and they may still get it wrong³⁵.

Many state of the art models for risk estimation in the industry today aim to model the data based on inherent assumptions [98]. This data modelling approach consists of a group of statistical techniques that start with assumptions about how the data and target variable are related, and then attempt to fit a function to that relationship. Examples of this kind of approach include logistic and linear regression or a Cox model [77]. Generally speaking, they are highly interpretable models that show a very clear relationship between the input and output of an algorithm, which in the case of insurance, may be a legal requirement [99]. In fact, the vast majority of risk-estimation models in the United Kingdom are reported to be GLMs according to a recent feedback statement from the Financial Conduct Authority (FCA) [100].

Despite this, there is growing evidence that the industry is shifting from the data modelling approach towards an algorithmic modelling approach. Analysis in the algorithmic modelling approach considers the actual relationship between the data and target variables unknown. Instead, the aim is to find an algorithm to predict the target variable using the given data. In

³³Note that this work uses a very different setup than the case-study in this chapter.

³⁴Customer modelling is indeed an entire field of its own (see for example the work by Ansari and Riasi [97]).

³⁵In an interview with an actuary in the Colombian motor insurance market, they shared with me that under-estimating the risk of a specific model of Toyotas has previously cost them dearly in that market.

that same feedback statement from the [FCA](#) a small number of firms have already reported using more advanced modelling techniques such as Decision Trees and Random Forests, both of which fall within algorithmic modelling. Similarly, there is a growing literature of studies that specifically compare data models to algorithmic modelling approaches within the insurance industry signalling that algorithmic modelling will become more popular in the near future (see section [2.2](#)).

Despite the wealth of literature comparing algorithmic modelling to data models, their impact on a competitive market is still unknown. In other words, studies have almost exclusively focused on risk-estimation with very little attention paid to pricing. One notable exception is the work by Heinrich et al. where the authors provide a full simulation of the insurance industry using agent-based modelling to study the effect of risk-model homogeneity [\[53\]](#). However, although their model reproduces some broad industry level features (e.g. the insurance cycle) it fails to capture the real competitive dynamics that can only emerge through the action and reaction of real experts in a competitive environment (see section [2.2](#)).

In this work, I will present the design, detailed methodology, and initial findings of a digital experiment that aims to study the impact of algorithmic modelling in competitive insurance markets. The experiment takes the form of an online data science competition hosted on the [AICrowd](#) platform. It used real motor insurance data in a cheapest-wins market and was targeted to a global community of insurance modelling professionals. The experiment ran for 11 weeks, with 2000 participants, submitting close to 10,000 risk-estimation and pricing models.

2.4 Methods

This section will detail the design of a digital field experiment in the form of a code-based data science competition simulating a cheapest-wins insurance market. The competition used 500,000 real insurance policies from a European country in a simulated market. The close to 2000 participants (with 60% working within the insurance industry ^{[36](#)}) engaged in price

³⁶See figure [2.22](#).

competition within the simulated markets over an extended 11 week period as noted above. After each week they would receive market level personalised feedback about their performance to simulate real-life market feedback. Participants engaged in a broad set of discussions with the community throughout and finally, the top 4 most profitable models shared a prize pool of \$12,000 USD.

In summary, this competition can be thought of as a quasi-simulation where real data is used. However, the agents in this game are competition participants who act as insurance companies. They are given 4 years of history for a set of motor insurance policies. They are then tasked with using the knowledge from this dataset as well as experience from playing in a competitive market with others over 10 weeks, to price incoming policies for the 5th year. The aim of the competition is to maximise competitive profit.

The sections that follow will first, outline the original analysis goals relating to the modelling cultures within insurance pricing. The next sections then describe the case-study design in detail with a focus on the resulting markets and feedback. A discussion of participant recruitment and fundraising, both crucial aspects to this study, is also included.

2.4.1 Analysis goal: *impact of modelling cultures on prices*

Pricing an insurance product, as explained earlier, can be split up into two parts: risk estimation, and pricing. Since much of the existing work concerns upgrading risk estimation models from the data modelling approach to an algorithmic modelling approach, this investigation focuses on the impact of algorithmic modelling on a competitive insurance market. However, the design of the competition allows a study of both risk estimation with room for some pricing to also take shape due to competitive dynamics.

The impact of new modelling can be seen through either the lens of the consumer, the business or the regulator within the insurance market. However, the impact on the business or regulator will be tightly related to the consumer impact³⁷. Therefore, the main question of concern here

³⁷For example, the feedback statement from the [FCA](#) is one way how a regulator can observe how current

will be how does an average consumer fare in a market populated with algorithmic modelling compared with one filled with data models. Figure 2.3 demonstrates one example plot that can shed light on this question from a consumer perspective.

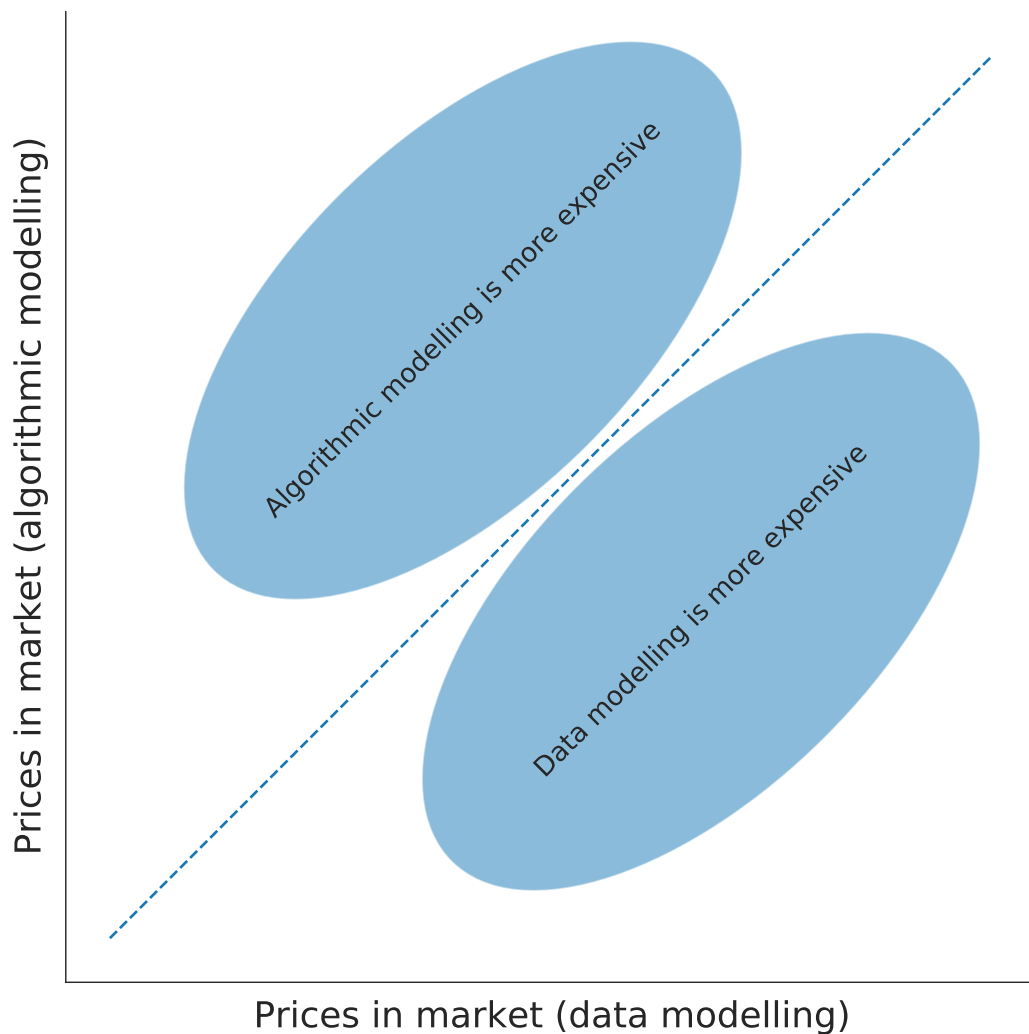


Figure 2.3: An illustrative plot highlighting one key analysis to be run once data collection is complete from a consumer perspective. The horizontal axis represents prices for insurance policies that were paid for in a market filled with pricing models using the data modelling approach. The vertical axis represents those priced using the algorithmic modelling approach. Being on the diagonal signifies no differences between the two approaches from a consumer perspective.

businesses practices are impacting consumers. Which in turn can lead to new regulation which will impact businesses [100].

2.4.2 Competition dataset

The dataset used in this competition comprises of 500,000 motor insurance policies from a real insurer in a European country. The 500,000 policies include 100,000 unique drivers who have renewed their insurance for 5 consecutive years within the last 2 decades. It has never before been used in a public online data science competition. The dataset is sourced from a major insurer in a European country operating nationally, and we obtained approval for using the dataset in this capacity from them. Finally, note that this dataset was collected prior to the anti-gender discrimination regulation in insurance pricing in Europe [101].

The final dataset provided to participants included 26 columns. Of these, 9 are policy level columns such as the level of coverage, or the intended use of the vehicle. Demographic information such as age and gender³⁸ make up 7 additional columns. Another 7 columns are related to the vehicle itself, such as the make and model, the age or the top speed of the vehicle. Two columns describe aggregate information about the location of the registered policy holder. Table 2.1 provides a full data dictionary.

Of particular importance is the target column `claim_amount` which indicates the cost of a policy to an insurance provider — the cost of a claim. A distribution of this variable can be seen in figure 2.4. In panel A, we can see the distribution of claims made with a Kernel Density Estimation (KDE) estimate, while in panel B the proportion of policies with and without a claim are displayed.

The majority of the claims within this dataset are categorised as third party liability claims which is defined as the cost incurred by others as a result of an accident. That is to say, if a policy holder has an accident with Jane causing her \$1000 worth in damages, the value of `claim_amount` for the policy holder would indicate 1000. In the target column, the dataset provider has indicated that catastrophe claims have been removed. This includes claims related to loss of life or medical expenses. The largest claim within the dataset does not exceed €300,000. This value is still quite low and infrequent (less than 1000 claims of this size or

³⁸European law forbids using gender in insurance pricing [102]. However, for research purposes it is included here.

greater exist in the dataset), and thus will allow for tail risk modelling within the competition.

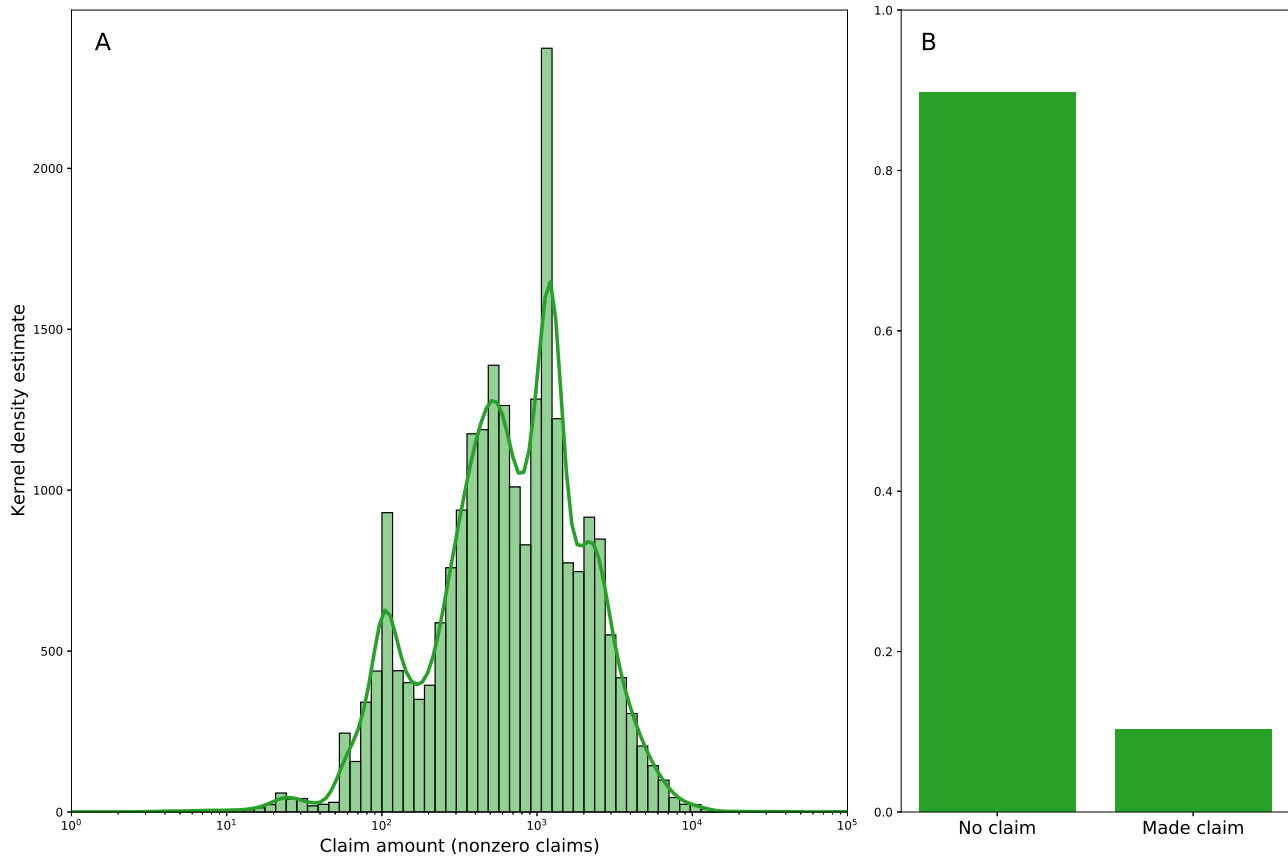


Figure 2.4: Distribution of the claim amount column in the dataset. **Panel A:** Displays a histogram of the non-zero claims appearing in the dataset. The solid line is a KDE estimate of the histogram values. Note the logarithmic scale on the horizontal axis. **Panel B:** A bar plot comparing the proportion of policies that make a claim, to those that do not.

In figure 2.5 we can see a scatter matrix of the continuous variables within the dataset plotted against each other desegregated by whether they made a claim. It seems that there is no first-order and obvious discriminating feature within the dataset.

Like many data science competitions, the raw dataset had undergone 3 changes prior to publication. Firstly, the dataset provider removed any extreme claims as well as medical claims from the set of policies. Second, we used the postal code information originally available in the data to combine it with an external source to increase the available information for modelling. Finally, despite agreement from the data provider to publish the data, we employed steps to reduce the risk of re-identification further. In line with these efforts, we employed a combination of techniques encompassing generalisation, suppression, and pseudonymisation which resulted in the final competition dataset.

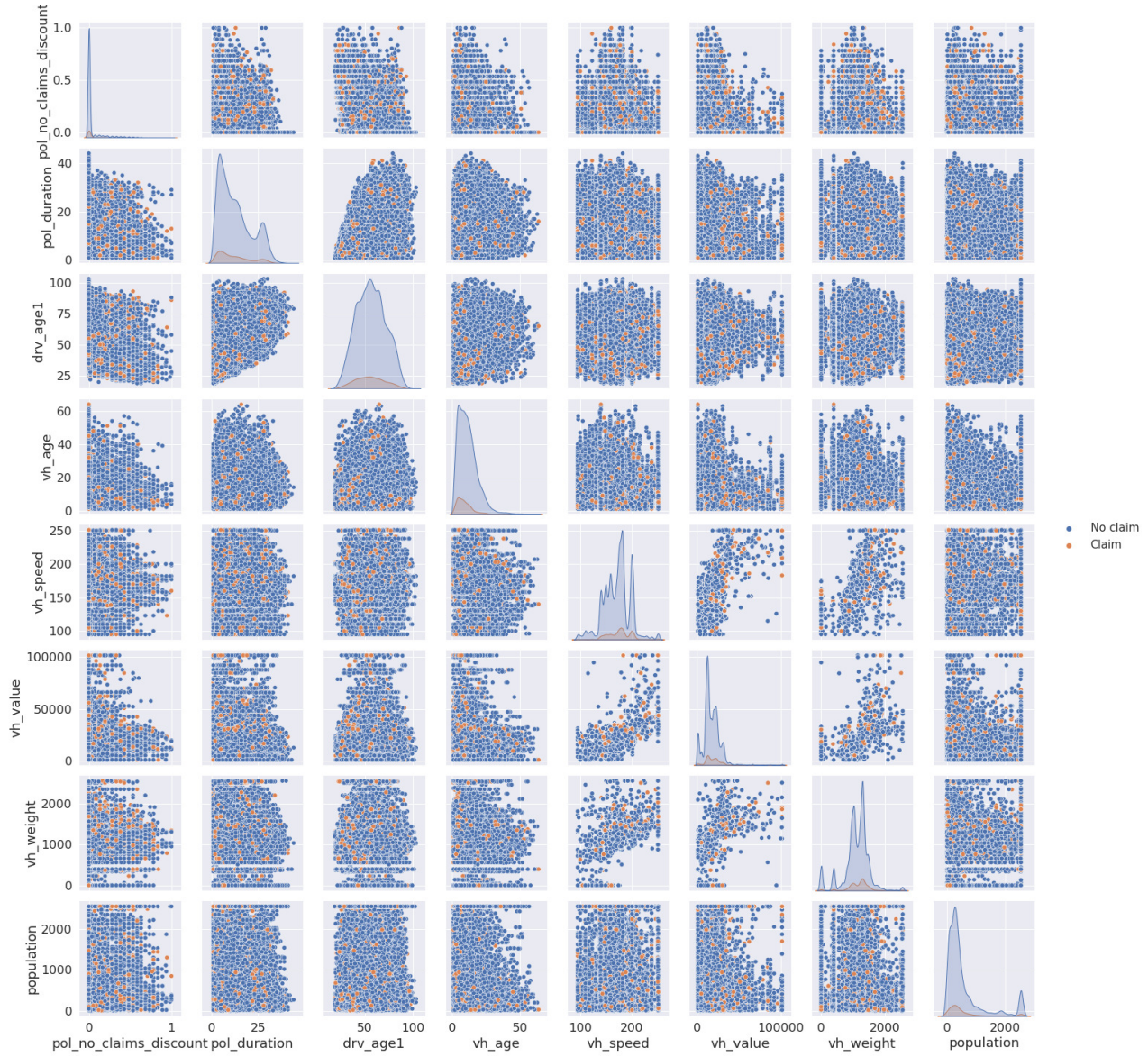


Figure 2.5: Scatter matrix with diagonal KDE plots summarising the continuous variables within the dataset. Each data point corresponds to a single policy with the orange and blue indicating whether the policy resulted in a claim or not. The diagonal is filled with kernel density estimation plots of the histograms.

Finally, this dataset may contain two types of bias. The first is a statistical side-effect of the fact that all of the 100,000 policies covered are renewed at least 4 times in a row. Typically in personal lines insurance, a policy holder would change their provider in the case of making large claims³⁹. However, since the data provider has given us a sample of only renewed policies, this may suggest that the occurrence of claims in this dataset (≈ 0.1) is an underestimation of the real-world claim frequency. The second source of potential bias is the result of the

³⁹This effect is partially explained by the fact that premium prices are increased after large claims [103]

entire portfolio being sampled from the same provider which will include any geographic or demographic preferences that the company may have for writing business.

2.4.3 Participant experience flow

The diagram in figure 2.6 provides an overview of how a participant experienced the flow from sign up to submission. Starting from the top right (blue) participants recruitment is described. In orange is resource download and allocation which then leads to modelling in purple. Participants get feedback (green) on their models and perhaps engage in communication with the community (yellow). Finally, they submit their final model at the end of the competition (red).

With this overview in mind, the next sections, broadly following the direction of figure 2.6, will expand on each of the design components of the pricing game.

2.4.4 Participant recruitment & fundraising

The success of this experiment partially depends on the quality and scale of participants recruited. To have expert participation we need to first reach the experts, and second offer an incentive to participate, beyond the educational and entertainment value that is inherent to most data science competitions. In addition to these two pillars (audience reach and incentives), as with any human subject experiment, it was necessary to obtain ethics approval.

To tap into a global expert participant pool I brokered partnerships with four national actuarial associations. The global insurance community benefits from a highly organised series of professional organisations that have access to almost the entire population of qualified actuaries in the world⁴⁰. By partnering with the four national actuarial associations I was able to engage a further 17 national associations that have a combined $\approx 80,000$ members subscribed to mailing lists⁴¹. These mailing lists served as the primary advertising vehicle for the pricing game.

⁴⁰The total number of actuaries globally is estimated around 60,000 [104] with approximately 80% belonging to a professional institution [105]

⁴¹Many of these will not be actuaries, but students who are studying to become actuaries.

Column	Description
<u>id_policy</u>	Unique policy identifier
<u>year</u>	This dataset covers a 5 year range. This indicates the year of the contract 1-5.
<u>pol_no_claims_discount</u>	This is a compulsory system in most European countries and globally. The coefficient is attached to the driver and indicates previous claims. It starts at 0.631 for new drivers (i.e. first year of insurance). Then, every year without claim, the value decreases by approximately 0.05 until it reaches its minimum of 0. Without any claim, the value evolution would then be: 0.63, 0.58, 0.53, 0.48, ... , 0.03, 0. Every time the driver causes a claim (only certain types of claims are taken into account), the coefficient increases by approximately 0.2, with a maximum of 1. Thus, the range of <u>pol_no_claims_discount</u> extends from 0 to 1 in the dataset.
<u>pol_coverage</u>	There are four types of coverage: Min, Med1, Med2 and, Max, in this order. Min policies cover only liability insurance claims , whereas Max policies covers all claims, including Damage, Theft, Windshield Breaking, Assistance, etc. The two Med policies represent intermediate coverage.
<u>pol_duration</u>	Policy duration represents how old the policy is. It is expressed in years, accounted from the beginning of the current year. There may be very old policies in the dataset for loyal customers.
<u>pol_sit_duration</u>	Represents how old the current policy characteristics are. This is different from <u>pol_duration</u> , because the same insurance policy could have evolved in the past (e.g. coverage, or vehicle, or drivers, ...). The <u>pol_sit_duration</u> is a metric for how up-to-date the characteristics are.
<u>pol_pay_frequency</u>	The price of the insurance coverage can be paid annually, bi-annually, quarterly or monthly. Be aware that you must provide a yearly quotation in your final submission.
<u>pol_payd</u>	PAYD or "Pay As You Drive" is a boolean (i.e. a string with Yes or No), which indicates whether the client has subscribed a mileage-based policy or not.
<u>pol_usage</u>	This indicates how the driver uses the vehicle with 4 possible values: WorkPrivate, Retired, Professional (which denotes a professional usage of the vehicle), and AllTrips which is quite similar to Professional (including professional tours).
<u>drv_sex1</u>	European rules force insurers to charge the same price for women and men. But gender can still be used in academic studies, and that's why <u>drv_sex1</u> is still available in the datasets. For our purposes, let's assume that this can be used as variable in this pricing game. In this dataset the values are M or F.
<u>drv_age1</u>	This is the age of the first driver in years.
<u>drv_age_lic1</u>	The age of the first driver's driving license in years.
<u>drv_drv2</u>	This boolean (Yes/No) identifies the presence of a secondary driver on the contract. There is always a first driver, for whom certain characteristics are provided, but a secondary driver is optional.
<u>drv_sex2</u>	The gender of the second driver if present in the data.
<u>drv_age2</u>	When <u>drv_drv2</u> is Yes, then the secondary driver's age is present. When no second driver is registered, this is 0.
<u>drv_age_lic2</u>	The age of the second driver's driving license in years, if present.
<u>vh_make_model</u>	A hash representing the make and model of the vehicle. These hashes (random strings) represent text such as Ford Focus or Honda Civic.
<u>vh_age</u>	This variable is the vehicle's age, the difference between the year of production of the vehicle and the current year. One can consider values of 1 or 2 to correspond to new vehicles.
<u>vh_fuel</u>	Diesel, Gasoline or Hybrid. This is the fuel type of the vehicle.
<u>vh_type</u>	Tourism or Commercial. You'll find more Commercial types for Professional policy_usage than for WorkPrivate.
<u>vh_speed</u>	This is the maximum speed of the vehicle, as stated by the manufacturer.
<u>vh_value</u>	The vehicle's value (replacement value) is expressed in Euros.
<u>vh_weight</u>	The weight (in kg) of the vehicle.
<u>population</u>	The population of the town where the policy holder is registered multiplied by a constant scale.
<u>town_surface_area</u>	Approximate surface area (in hectors multiplied by a constant scale.) of the town where the policy holder is registered.
<u>claim_amount</u>	The amount of a claim expressed in euros for the year (a value of zero indicates that no claim was made).

Table 2.1: The dataset dictionary provided to competition participants.

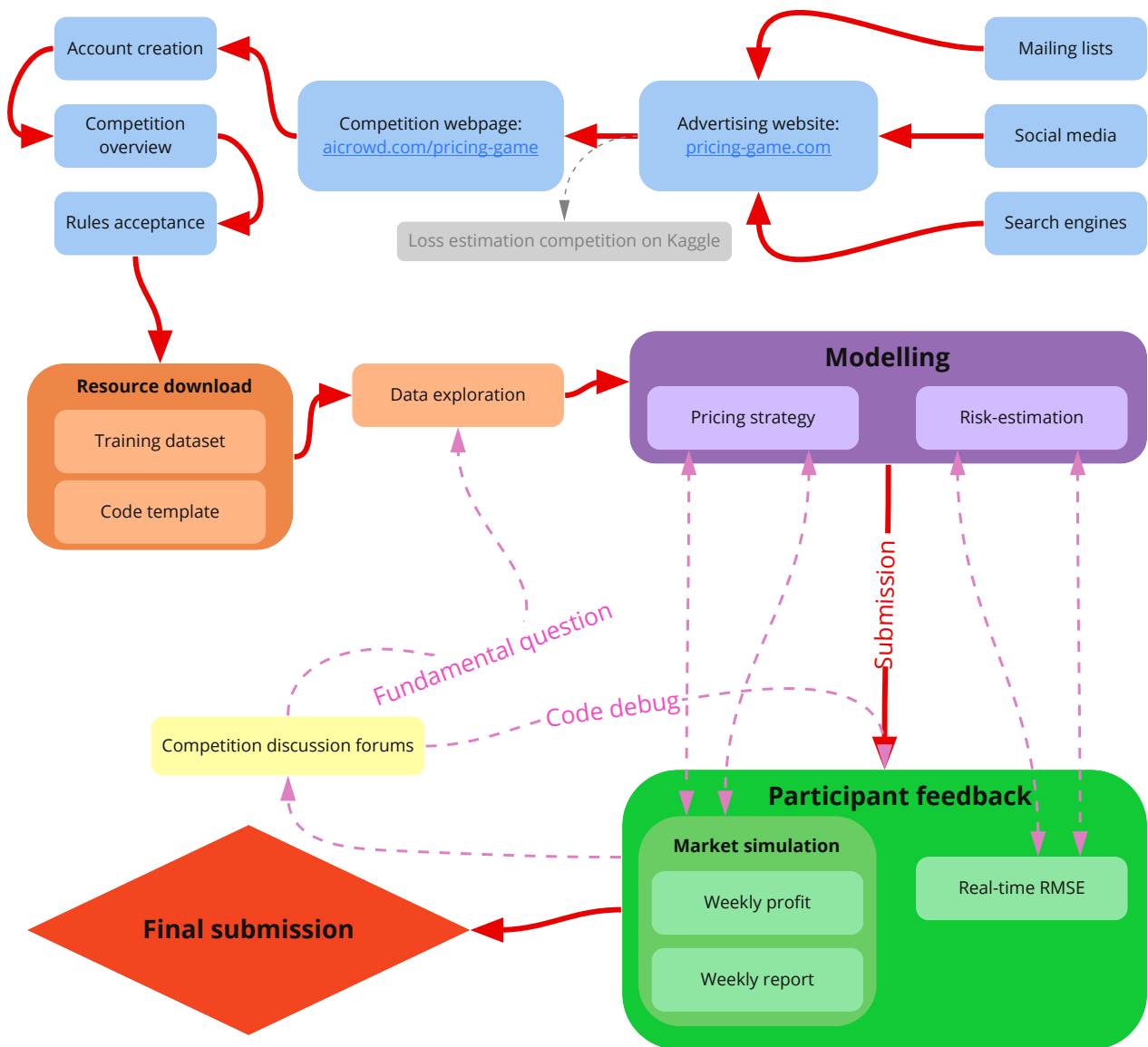


Figure 2.6: Diagram illustrating the participant flow from recruitment to final submission. The participant flow starts from the top right with recruitment (in blue) and ends in the bottom left with their final submission (in red). The bold red arrows indicate the participant flow direction. The dashed pink arrows indicate possible feedback loops which include the modelling - feedback loop as well as the discussion - modelling loop.

Although many of these mailing lists are predominantly targeted at qualified and experienced actuaries, there will be participants that are not qualified actuaries. Some will be working within the actuarial industry as consultant actuaries, or actuaries in training. Others will be students in the midst of their education and some will be professionals outside of the actuarial industry within other areas of financial mathematics. However, a majority of participants reported working within the industry themselves (figure 2.22).

The advertising process was all done by sharing the URL to a website designed to summarise

and gather interest for the game. [Pricing-game.com](https://pricing-game.com) (see figure 2.7) was designed to advertise two data science competitions. The primary competition, and the overall website design, was focused on the insurance pricing game hosted on Alcrowd with a market simulation using real data. There was a second sister competition using synthetic data that was included due to the requirements of the actuarial society partners which was unrelated to the main Alcrowd game.

The insurance pricing game incentivised participation with a \$13,000 USD prize pool that I helped raise through 9 different sources. These funding sources were either actuarial institutions themselves, or private companies that are linked to the actuarial institutions. The funds were raised only through the promise of having the logo of the funding organisation appear on the promotional material. Additionally a series of community engagement prizes⁴² were included to encourage participant discussions and engagement.

Finally, as this digital field experiment requires the prolonged study of human subjects, we obtained ethics approval from Imperial College London prior to launching the study. The study was given ethical approval by the Head of the Department of Computing and Research Governance and Integrity Team (RGIT) at Imperial College London and has been reviewed by the Joint Research Compliance Office at Imperial College London (see figure A.2).

2.4.5 Participant code templates

This data science competition, unlike many others⁴³, only accepted code submissions for 2 reasons. The first, is related to realism of the markets as it is rare for a primary insurer to know exactly the make-up of their incoming portfolio ahead of time. Therefore, by requiring code-submissions the markets can execute participant pricing models on data that the participants never see ahead of time.

The second reason for requiring code submissions is to explicitly separate the pricing and risk-estimation aspects of insurance pricing for analytical purposes. As the analysis goal is to

⁴²Personalised t-shirts and mugs.

⁴³The standard submission format of data science competition is to submit the results directly, not the code that generates the result.

Insurance Pricing Game
Active tracks
Timelines
Organizers

a global
data science competition
with real motor insurance data


Real motor insurance data


Build a prediction model for claims


Play in a simulated marketplace

#Active tracks

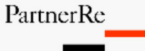
#track A

Motor insurance market simulation

Play as an insurance company, using **real historical data** in a competitive market with other players. See if you can make a profit with realistic market conditions.

Register for post-competition town-hall event →



	\$6000 (USD)	
	\$3500 (USD)	
	\$1500 (USD)	
	\$1000 (USD)	

→ Learn more about PartnerRe

#track B

Worker Compensation Claim Prediction

Predict the claim amount of workers compensation claims using a **synthetic dataset** generated for this competition!

Join Competition! →



sponsored by

→ Learn more about Actuaries Institute Australia

Figure 2.7: A screenshot of a portion of the advertising webpage pricing-game.com is displayed.

understand the impact of algorithmic modelling on consumer prices, it may be necessary to be able to separate any pricing aspect from risk-estimation as much as possible, in order to investigate the cause of any discrepancy observed. Since this separation is natural within the insurance world we induce it through the strict code templates provided to participants.

The templates provided to participants included 3 key functions:

1. `fit_model`: fit model to data
2. `predict_expected_claim`: estimate the claim for incoming data
3. `predict_premium`: offer price for incoming data

The full templates are available on the competition page online⁴⁴.

2.4.6 Insurance pricing game market design

The markets in this game fundamentally have 3 components that could be designed and controlled. These are the customer model, the leaderboard design, and finally, the rules of the market. The decisions surrounding these three components are driven firstly by what is considered to be most realistic, and secondly by whether the decisions will positively impact the participant community, thus encouraging them to continue playing, increasing the chance of addressing the participant time-commitment and scale limitations noted earlier (see section 2.2).

The customer model: *cheapest-wins*

The markets in the pricing game are all “*cheapest-wins*” markets, meaning that they are populated by customers that are extremely price-sensitive. Figure 2.8 illustrates how this can work. Every participant in this game represents one insurance company. They are tasked with providing a premium quote for every policy (or customer) they encounter. So does every other company. The customer will pick the cheapest price offered to them (figure 2.8A).

⁴⁴aicrowd.com/pricing-game

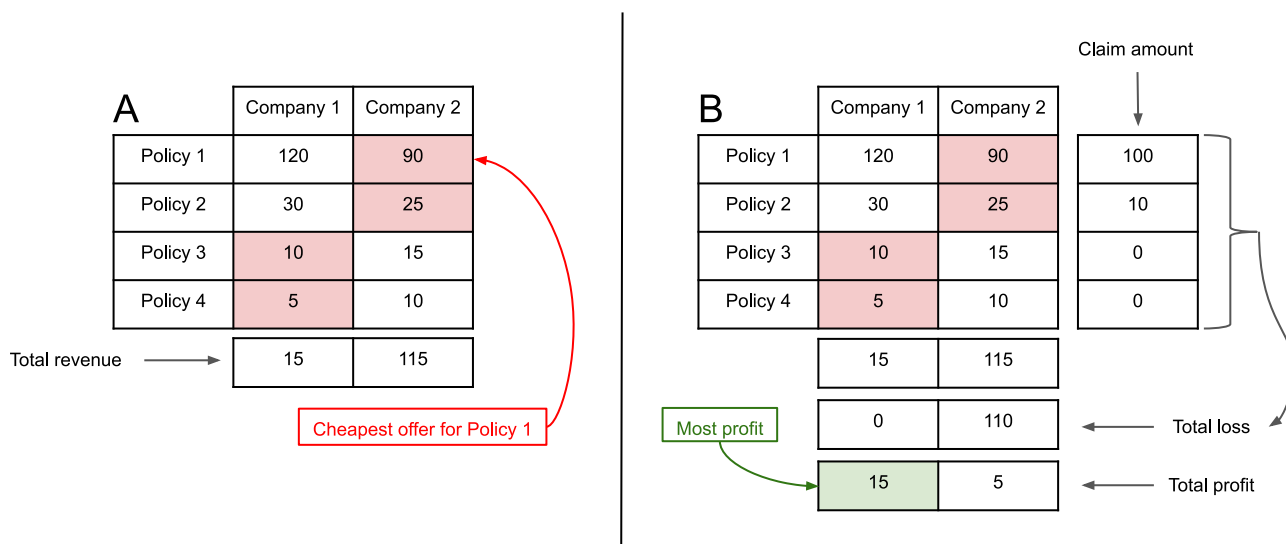


Figure 2.8: Illustrative example of a cheapest wins market. **Panel A:** First we see 2 companies (columns) competing for 4 policies (rows). The cheapest prices offered by each company to a particular policy is highlighted. **Panel B:** Once the policies have been allocated to companies, their revenues can be computed by summing the premiums paid by each policy to each company. Next the claims are tallied up and the profits for each company are computed (in the final row).

In figure 2.8B, we can see that companies 1 and 2 each have a set amount of revenue — now they have to pay out the cost of the claims associated with policies 1-4. In this case, once the claims are taken into account in the cheapest-wins market, we can see that Company 1 wins as it has the most competitive profit⁴⁵.

Leaderboard design

Since insurance pricing is fundamentally split into two tasks, the core gameplay loop⁴⁶ is also split into two. This is realised through the inclusion of two different types of leaderboards: risk-estimation measured by Root Mean Squared Error (RMSE), and pricing measured by competitive profit. Once participants download the training data (green in figure 2.9) they are then free to begin using the two leaderboards.

The profit leaderboard was considered to be the primary leaderboard in this game. In real markets, it is rare that a policy can be priced repeatedly by the same company. To replicate

⁴⁵In practice the markets in the competition use a more complex profit metric through randomisation of competitors and a set of rules. See section 2.4.6 for a complete description.

⁴⁶Terminology borrowed from the video game design literature. This is the repetitive activity that is performed by a player to advance in any game [106]

this dynamic, each iteration of the profit leaderboard used a disjoint sample of the dataset. This dataset was updated weekly⁴⁷ for a total of 10 weekly profit leaderboards. Each leaderboard dataset is a sample of the first 4 years of policies with the time progressing from week 1 to week 10. That is, the week 1 leaderboard will have almost exclusively policies from year 1 and the week 10 leaderboard will consist of almost entirely year 4 policies, although some yearly mixing was allowed in general due to dataset constraints. This data is displayed in blue in figure 2.9.

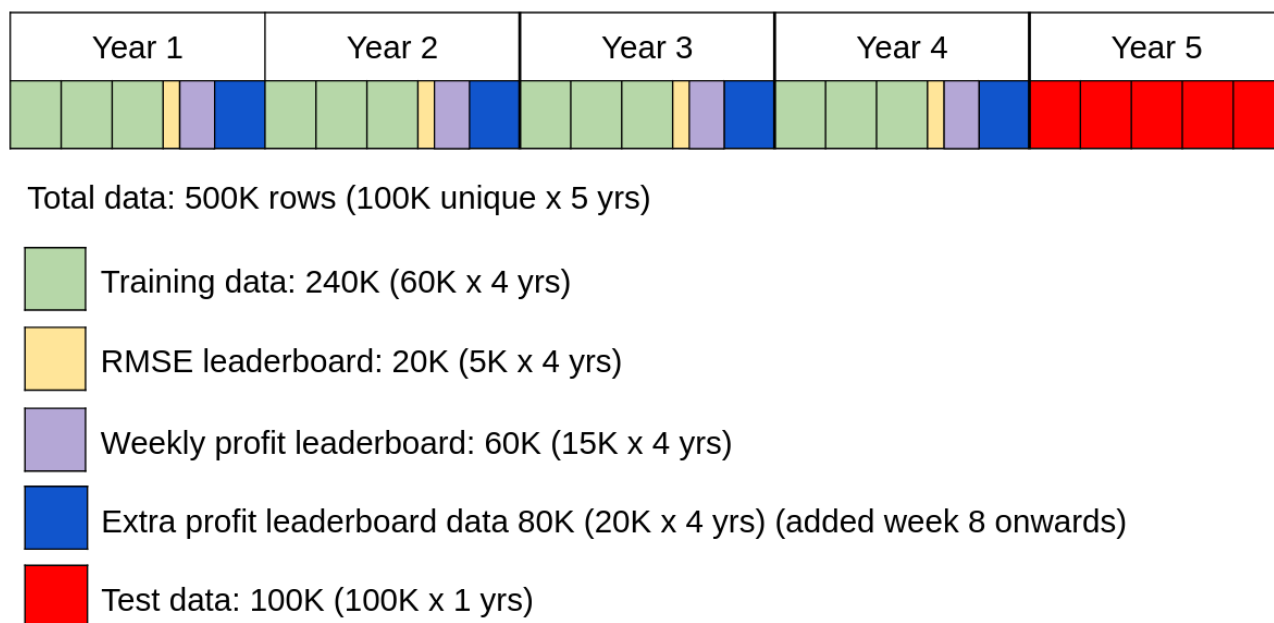


Figure 2.9: Illustration of how the dataset is split among the two different leaderboards, training, and final evaluation. The data provided to participants is highlighted in light green. In yellow the the data for the [RMSE](#) leaderboard is displayed. Purple indicates the profit leaderboard data. Red is the final year test data upon which the profit competition is held. Finally, the blue is the data that was not provided to participants initially. This data was eventually released to participants in an attempt to bolster the profit leaderboards from week 8 onward (see section 2.4.7 for more details on the leaderboard expansion).

The [RMSE](#) leaderboard is designed to measure the risk-estimation ability of models based on their [RMSE](#) — a crucial part of the underwriting process in insurance. This leaderboard satisfied two requirements for the game: risk-estimation benchmarking and real-time community feedback. The first, risk-estimation benchmarking, refers to the fact that participants are able to see what the benchmark for [RMSE](#) is in the competition, allowing them to update their models if necessary. The second, real-time community feedback, acts as an incentive for partic-

⁴⁷Profit leaderboard deadline was set at 10PM Coordinated Universal Time ([UCT](#)) every Saturday. This was to maximise the ability of professionals to spend time on the competition during Saturday, see their results, and then make any changes on Sunday.

ipants to remain engaged between the weekly profit leaderboards based on previous evidence of the positive effect of real-time feedback on engagement [107, 108]. This leaderboard used a consistent sample of the dataset over the entire course of the game with 4 years of history, unlike the weekly profit leaderboards which use disjoint sets of the data (see figure 2.9).

Leaderboard metrics: *risk-estimation and competitive profit*

The leaderboard rankings are each based on their own metric. The key metric of interest is competitive profit. This metric is based on a cheapest-win customer model in repeated randomised markets of size 10 in a two stage metric. This design accounts for realistic market-sizes, filters out bad models using the two stages, and finally through randomisation of markets, ensures that no participant is given undue advantage⁴⁸.

The aim for the competitive profit metric is to ensure that the final leaderboard is representative of “*realistic*” models. Three criteria were identified as critical in classifying models as realistic. These are:

1. **Training profitability.** Profitable on their own portfolio in a monopoly (i.e. training data)
2. **Nonzero market-share.** Have a non-zero market-share (i.e. win at least one policy)⁴⁹
3. **Hold their own.** Reasonable performance against other models that perform well

The first two are required of most companies active in a competitive market, while the third must follow to sustainably remain in the market. In real markets, the third requirement, “*holding their own*” must be true in a stable market populated with historically profitable models⁵⁰. In other words, any model in a stable market must perform relatively well against other models.

⁴⁸Alternative metrics such as an elimination-style process, akin to football world cup competition, was considered. However, this was decided against, as these metrics tend to be highly sensitive to initial pairings [109].

⁴⁹In practice this requirement is more complex and is tied to costs. Here we use non-zero market-share as a proxy for those requirements.

⁵⁰Or almost profitable, since insurance markets are sometimes, surprisingly loss-making [110]

Requirements (1) and (2) are controlled for in these markets in a straightforward way. The first is checked pre-submission and the second is measured during evaluation and flagged if not satisfied for at least 5% of markets⁵¹.

To satisfy the third requirement we employ a two stage randomised profit metric. First, average competitive profit is computed by placing models in randomised markets of size 10 repeatedly and ranking all models based on their mean profit in the market. Once this ranking converges we obtain the initial global rank G_1 .

Once we have the initial global rank we rerun the markets against the top of G_1 to filter out the poor performing models. We pick the top 10% of G_1 (or top 15 models, whichever is larger) and re-generate randomised markets once more. The difference is that this time, every submission is placed in a market with 9 other models that are sampled from this top 10% of G_1 . This results in a final global rank G_2 which is used as the final leaderboard. Figure 2.10 illustrates this computation process. A sensitivity analysis conducted to see if a 2-stage metric is sufficient supports this decision (see section 2.5.6).

By ensuring that the final global rank, G_2 is a result of competition against already profitable models we come closer to simulating an environment where models are competing only against other models that perform well (requirement 3).

The two-stage profit metric can be computed analytically due to the randomised nature of the markets in combination with the cheapest-wins customer model. Given N competitors offering prices $P_{M \times N}$ competing for M policies, whose claims are represented by y , we can compute the “conversion rate” C_{ij} of each policy, i , for each competitor j . The conversion rate is the fraction of markets in which a given policy is offered the cheapest price by a given competitor and can be understood as the probability of model j winning policy i . In other words, to compute the profit earned by company j resulting from a given policy i we have:

⁵¹In practice a very small number of competitors are disqualified due to failing the market-share rule.

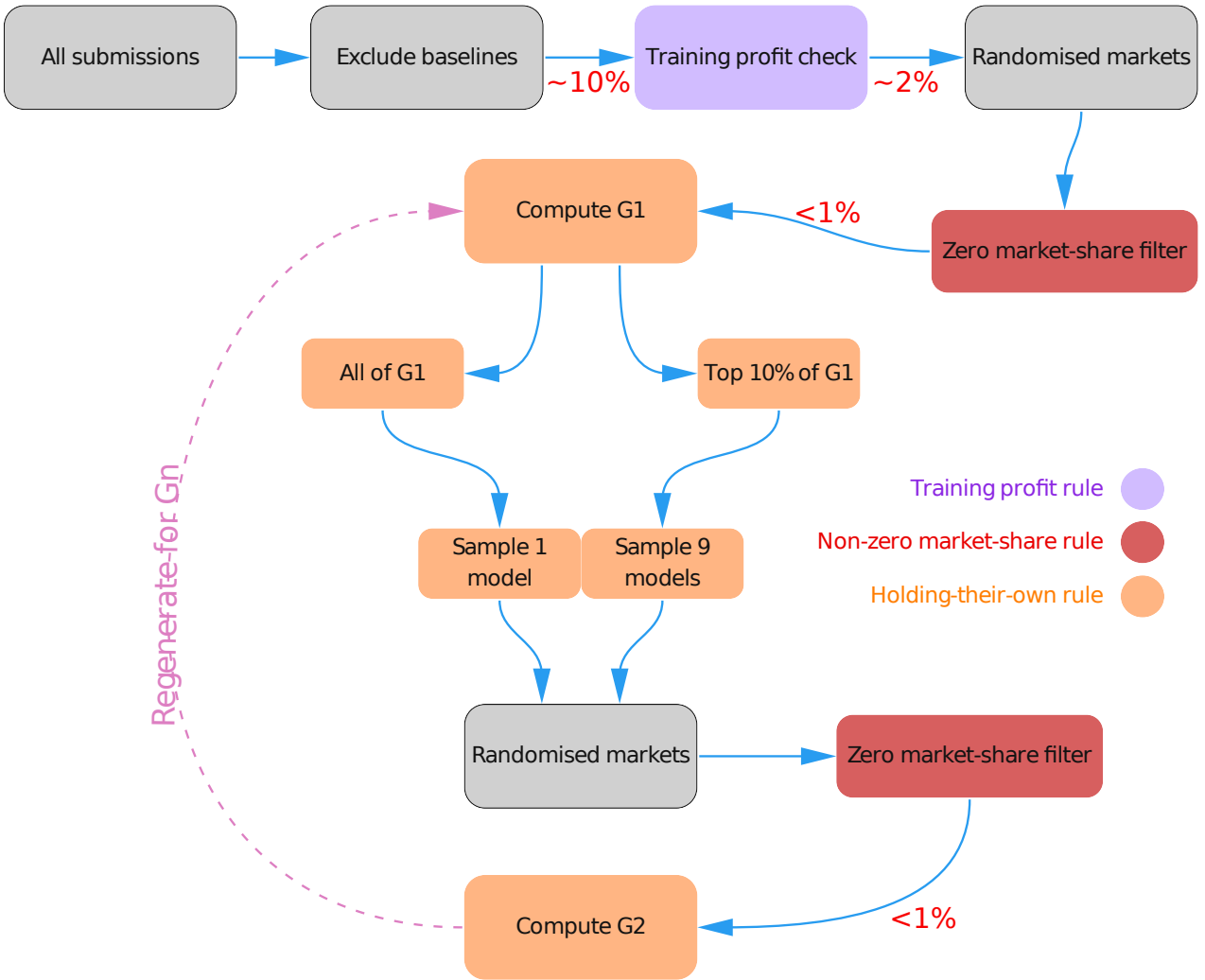


Figure 2.10: Diagram view of the profit metric. Starting from the top left, submissions are fed into the pipeline with baseline submissions (e.g. those who uploaded the template) excluded. The first check is related to the training profitability which is done upon submissions (in purple). Upon the weekly deadline all submissions are then placed in randomised cheapest-wins markets of size 10. At this stage the models that do not win any policies are excluded (in red). Once this is complete the first global rank G_1 is computed and used to compute the second global rank G_2 which is the final ranking. Finally, if further repetitions of the metric are required they can be done as indicated by the pink dashed arrow (see section 2.5.6 where $G_{1..21}$ are computed as well). The percentage numbers in red indicate what fraction of submissions are dropped after each filter stage.

$$prof_{ij} = (P_{ij} - y_i) \cdot C_{ij} \quad (2.1)$$

The key to calculating G_2 here is then to compute the conversion rates C_{ij} for the second stage

of the metric. To do so first let us focus on a particular model j competing for a particular policy i whose claim is y_i . Say there are k_{ij} models that are more expensive than j and $N - 1 - k_{ij}$ models that are cheaper, with N representing the total number of submissions in play.

To compute G_1 we first must compute all possible markets that could involve model j and then the fraction of those in which j is the cheapest price for i . That would give us the conversion rate in G_1 , labelled C'_{ij} . Therefore we have:

$$Q_{ij} = \text{nCr}(N - 1, 9), \quad \text{Possible markets of size 10 with } j \text{ in them} \quad (2.2)$$

$$q_{ij} = \text{nCr}(k_{ij}, 9), \quad \text{Possible markets where } j \text{ is the cheapest} \quad (2.3)$$

$$C'_{ij} = \frac{q_{ij}}{Q_{ij}} \quad (2.4)$$

Using the conversion rates C'_{ij} , we can compute the initial global rank G_1 through equation (2.1). Once that is computed then we must simply repeat the process keeping in mind that now the computation splits into two, one case where j is in the top 10% of G_1 and the other where it is not. In either case, let us assume that there are ℓ_{ij} models in the top 10% that are more expensive than our model j .

If j is in the top 10% of G_1 the computation is as follows. Assuming there are a total of R models that are in the top 10%:

$$Z_{ij} = \text{nCr}(R - 1, 9), \quad \text{Possible markets of size 10 with } j \text{ in them} \quad (2.5)$$

$$z_{ij} = \text{nCr}(\ell_{ij}, 9), \quad \text{Possible markets where } j \text{ is the cheapest} \quad (2.6)$$

$$C_{ij} = \frac{z_{ij}}{Z_{ij}} \quad (2.7)$$

Similarly if j is not in the top 10% we have:

$$Z_{ij} = \text{nCr}(R, 9), \quad \text{Possible markets of size 10 with } j \text{ in them} \quad (2.8)$$

$$z_{ij} = \text{nCr}(\ell_{ij}, 9), \quad \text{Possible markets where } j \text{ is the cheapest} \quad (2.9)$$

$$C_{ij} = \frac{z_{ij}}{Z_{ij}} \quad (2.10)$$

At the end of this calculation we will have computed the final conversion rates which allow for the competitive profit ranking to be generated (illustrated in orange in figure 2.10).

The randomised metric plays three crucial roles. Firstly, it prevents the undesirable situation where models make it to the top of G_1 due to good performance against other models that do perform badly (e.g. have terrible risk-estimation). In the scenario where most models submitted are not realistic, it is likely that those appearing near the top of G_1 are only there because, on average, they beat all the unrealistic models. In such a situation, the two stages filter out these models and ensure that the final rank G_2 , is based on competition against these good models.

Secondly, this metric design also mitigates the effect of any single player that behaves poorly. A common example is a player who may be pricing near zero or negatively despite the rules above. If markets were not limited in size, and not randomised, such a player would effectively “win” all policies in the market, leaving none to be won by others — effectively destroying the market. In our markets, such a model will only affect a small part of the overall average profit due to the randomisation.

Finally, it allow for realistic dynamics, such as the winner’s curse to emerge and be observed. The winner’s curse is a phenomenon which involves an insurer under-pricing the risk of a given policy, thus winning that policy in a market, and being “cursed” with the responsibility of paying out the cost of the risk when it eventually materialises. The profit metric allows for dynamics such as the winners curse to emerge as it requires participants to estimate the risk of each contract and prevents the market from being dominated by one or two players entirely through the randomisation mechanism. However, the idea of winner’s curse extends beyond the world of insurance (e.g. to many forms of auctions) with an informative discussion found

elsewhere [111].

For the risk-estimation leaderboard the metric is the [RMSE](#) of the output of the `predict_expected_claim` function compared to the ground truth of the risk-estimation leaderboard data (i.e. the value in the `claim_amount` column). More formally:

$$RMSE = \sqrt{\sum_i^n \frac{1}{n} (y_i - \hat{y}_i)^2} \quad (2.11)$$

where y_i indicate the ground truth values of claims and $\hat{y}_i$ are the predicted expected claim values. There are risk-estimation metrics other than [RMSE](#), such as the mean squared error, the mean absolute error, and other, more industry standard metrics. However, to increase accessibility to the risk-estimation leaderboard [RMSE](#) was picked.

2.4.7 Feedback to participants

Real-world insurance markets involve feedback that allows market players to react. For the public these reactions are either observed in the capital markets or in the insurance markets themselves⁵². However, for insurance firms the feedback is more precise and data-heavy. Usually it includes information about how a company has performed in the recent past. This feedback is also typically split into two categories corresponding to risk-estimation and pricing. For risk-estimation, it would include detailed claims histories of the portfolio for that year, who was sold what product and how much it cost — usually a summary of what the portfolio looks like. For competitive pricing feedback, there is less agreement on a market standard. Some firms obtain conversion rates from brokers⁵³, while others can collect the information themselves based on online interactions of users or a history of quotes provided to a potential customer in-store or on the phone [113].

⁵²Examples include a shift in prices, product offerings, or changes in quarterly results.

⁵³An insurance broker is an a service provider who sells, or negotiates insurance on behalf of a client [112].

Description of market feedback provided to participants

In this game, we approximate these two kinds of feedback using a weekly feedback report that is made accessible to participants after the weekly profit leaderboard. However, this feedback was not provided to participants in one piece. Instead the level of feedback complexity was increased from the beginning of experiment until halfway through to allow the markets to stabilise and to avoid any volatility due to being flooded with new information, something that is hypothesised to happen in real markets [114].

The final form of the feedback included information about risk-estimation and pricing intertwined into four sections. First are two plots that describe the profit, market-share and claim distribution of the participant model. The second is a list of Key Performance Indicator (KPI)s for the competitor. The third and fourth are made of tables containing either summary statistics about the policies encountered in the markets, or summary statistics about the financial performance of the participant in the markets. A full example report of the feedback is included in the appendix (see figure A.1). Note that the information within the report and the financial summary tables are gathered using a limited number of market simulations since they cannot be computed analytically, unlike the rest of the feedback report. However, agreement between the analytical and simulation-based methods could be achieved to arbitrarily high accuracy, we only continued the simulation-generation process until the final G_2 rankings from both methods were indistinguishable.

The first part of the feedback report includes two figures. The first (figure 2.11) displays the market-share against profit in a sample set of simulated markets using the participant models to help with the pricing aspect of market feedback. Figure 2.11A and B show this plot for a profitable and non-profitable model respectively⁵⁴. The plots also highlight three sample markets in red, yellow and, blue which correspond to markets within the least, middle and most profitable 5% quantiles of markets they participated in, as a guide to what direction the participants should aim to move. In this case for example, the model in figure 2.11A has room

⁵⁴These are KDE plots to avoid providing raw data to participants who may be able to game the system to learn information about others in the market.

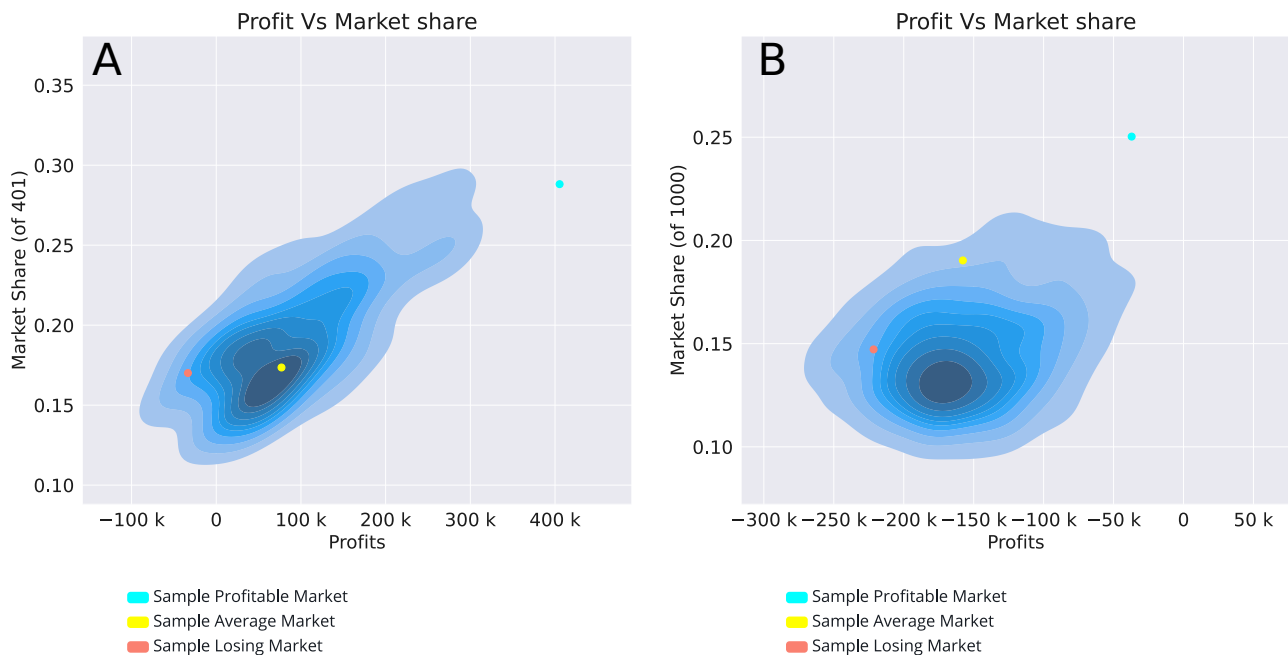


Figure 2.11: Profit Vs. market share plots of two submissions in the final leaderboard are displayed. On the left (A), a model in the top 10 of the rankings, it is clear that there is no distinct relationship between market share and profit while on the right (B) there may be a positive correlation between the two — a model with a ranking of 50+. Note that even the sample profitable market here is loss-making as this model likely never made any profit in any of the markets they competed in.

to slightly reduce their prices and gain market share, which may lead to higher profits based on the apparent trend, while for the model in figure 2.11B, margin optimisation may not be the main issue as they are loss-making in all markets they competed in. Note that even the sample profitable market here is loss-making.

The second figure (figure 2.12) displays a heatmap which indicates to a participant whether they are picking up many of the existing large claims or if they are able to avoid them using their risk-estimation model. On the horizontal axis the total claim amount has been stratified into 4 categories. On the vertical axis there are 4 categories based on how often a policy is won in a market by the participant. The top 3 categories are set by first computing conversion rates for every policy and then splitting them into 3 quantiles. The final “never” category is then filled with all policies that have a conversion rate of zero (i.e. the participant never offers the cheapest price). Participants were made aware of the meaning of the 3 quantiles as being the same as that of the summary tables (see below), based on the conversion rates. The heat axis is vertically normalised and counts the fraction of claims within a certain range that fall into

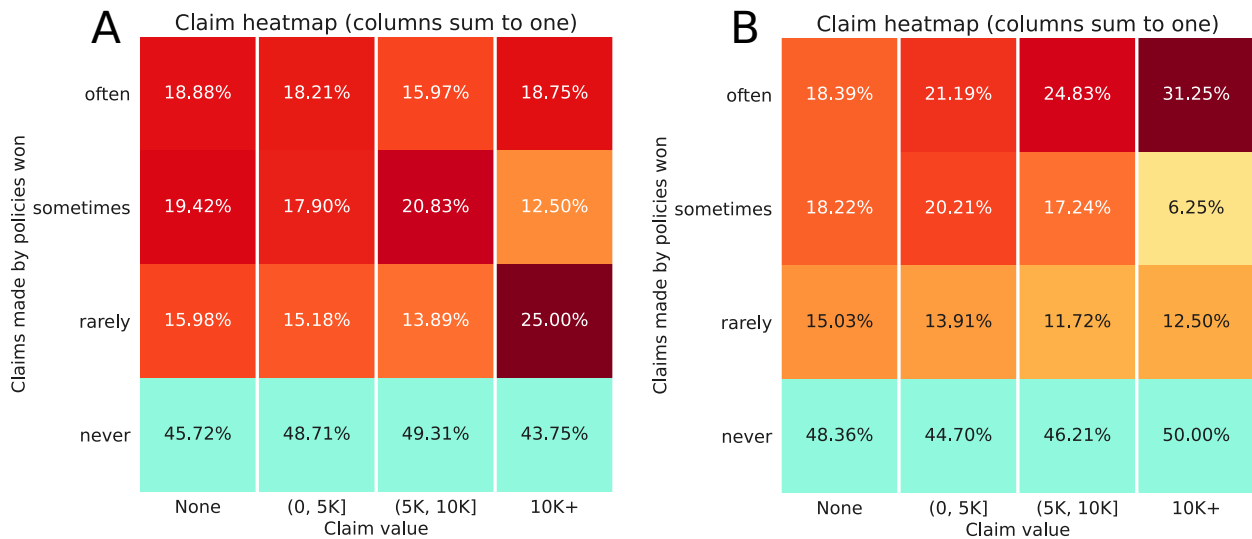


Figure 2.12: Claim heatmap plots of the same two submissions as in figure 2.11. On the horizontal axes policies are split into 4 claim groupings. On the vertical axis, the top three rows correspond to quantiles of conversion rates for policies won by the participant model — the last row is reserved for policies that were never won by the participant. The value (and heat) in each cell represents the fraction of all policies within that column grouping, which has a particular conversion rate. In (A), the feedback for a model in the top 10 final ranking can be seen. Here the model is minimising exposure to very expensive claims, while the opposite is true for the model on the right-hand side with a ranking of 50+ (B).

a particular quantile. For example in figure 2.12A, we can see that the model captures 25% of claims above €10,000 “rarely” which translates a real probability based on the conversion rate distribution of that particular model⁵⁵.

Comparing our two example models again, in the case of figure 2.12B it is clear that the participant is picking up almost a third of very expensive claims in the dataset “often”, while this is not the case in figure 2.12A. Therefore, this suggests that the participant receiving the former feedback most likely needs to hone their claim estimation model so that they can better identify and price these very expensive claims as that seems to be the biggest contributing factor to their losses.

The next piece of feedback is a series of 7 KPIs, 2 of which are publicly available on the leaderboards⁵⁶. These serve as a very high-level summary that captures the performance of a participant’s model at a glance. They are included to allow those participants that are

⁵⁵Note that an ideal heatmap will have all the heat in the top left of the plot with the bottom right indicating 100%.

⁵⁶These are the profit, and market share.

pressed for time, the ability to continue active participation using their expertise without being overwhelmed by the more detailed feedback report⁵⁷. The average market-share, loss, profit, revenue, and the total number of markets are relatively easy to understand as they are simple mean value computations over markets. The average loss-ratio⁵⁸ and average profit-per-policy are slightly more nuanced and are defined mathematically here in a simulation for a total of N markets with a K policies each. The claims, revenue, and market-share within each market is indicated by c_i , r_i , m_i . The loss ratio and average profit-per-policy are defined as:

$$\text{Average Loss-Ratio} = \frac{1}{N} \sum_i^N \frac{c_i}{r_i} \quad (2.12)$$

$$\text{Average profit-per-policy} = \frac{1}{N} \sum_i^N \frac{(r_i - c_i)}{K m_i} \quad (2.13)$$

Note that the average profit-per-policy is an average over markets rather than policies. This is noted since it is entirely possible for the average loss ratio to be less than 1, yet the average profit per policy to be positive⁵⁹.

The next significant part of the feedback provided is in the form of 4 summary tables, including aggregate statistics about policies that were won in the markets by the participant. These aggregate statistics align with the 4 categories of information within the dataset⁶⁰. An example table summarising demographic statistics about the driver is included in figure 2.13. Note that the rows correspond to the same policy groupings as those indicated in figure 2.12 with the addition of three extra rows which highlight performance in the 3 sample markets that are displayed in figure 2.11. Note that the actual conversion rate grouping limits on each row are also enumerated here unlike in figure 2.12.

The final piece of feedback are 3 tables which include summary financial information about market performance. The first of these tables includes summary statistics relating to revenue,

⁵⁷See chapter A for full report.

⁵⁸The loss ratio is defined as the ratio of claims to revenue

⁵⁹For example see this [online discussion](#) on the pricing game forums.

⁶⁰These are information about the vehicle, policy type, location, and demographics of the drivers.

Driver

Policies won	drv_sex1	drv_age1	drv_age_lic1	drv_drv2	drv_sex2	drv_age2	drv_age_lic2
often: 26.5 - 100.0%	M (64%)	57.09	35.30	No (64%)	F (60%)	50.50	28.60
sometimes: 2.5 - 26.5%	M (61%)	57.60	35.68	No (65%)	F (59%)	51.08	29.23
rarely: 0.1 - 2.5%	M (61%)	57.81	35.75	No (67%)	F (62%)	51.04	29.27
never:	M (59%)	60.22	37.69	No (69%)	F (62%)	51.21	29.30
in sample profitable market	M (62%)	56.37	34.76	No (62%)	F (58%)	50.34	28.60
in sample average market	M (64%)	56.56	34.80	No (62%)	F (60%)	50.14	28.13
in sample losing market	M (63%)	57.01	35.21	No (62%)	F (60%)	50.12	28.27

Figure 2.13: One sample summary table about driver demographics that is provided in the weekly participant feedback. The top three rows represent quantiles of conversion rates of policies won by the participant in simulated markets. Each cell then represents either the mean value of a continuous variable or the most common category of a categorical variable. The bottom three rows correspond to the three sample markets also visible in the profit Vs. market share plots (see figure 2.11).

claims and losses based on conversion rates. The last two focus on the three sample markets, firstly on policies that were won, and next on policies that were lost, providing some information to participants about their competitors in those markets. Figure 2.14 displays examples of these three tables.

Addressing market volatility

Uncertainty of outcomes is an inherent feature of any stochastic process and insurance markets are not immune. In the real world, the uncertainty comes from 3 sources. First, is the realisation of claims that have an inherent random element in any real insurance dataset — they are accidents after all. Second, capital markets can be notoriously unpredictable [115], something that almost every insurance market is exposed to, due to their reserving practices. Third, uncertainty comes from external events such as a new actor disrupting the market, new regulations, or even something as insignificant as a new marketing campaign from a competitor.

The markets in the insurance pricing game are much simpler and have a different set of 5 sources that lead to volatility:

1. Claims uncertainty

Financials By Conversion Rate

Policies won	claim_frequency	premiums	conversion_rate	profit per policy	profit_margin
often: 40.1 - 100.0%	0.09	104.81	0.79	1.58	0.16
sometimes: 7.2 - 40.1%	0.08	112.25	0.22	3.11	0.18
rarely: 0.2 - 7.2%	0.09	115.47	0.03	0.26	0.20
never:	0.10	213.63	0.00	0.00	0.39

Financials In Sample Markets

Policies won	claim_frequency	premiums	conversion_rate	profit	revenue	loss	market_share	profit_margin
In sample profitable market	0.09	116.37	0.56	405456.53	3188236.88	2782780.35	0.29	0.17
In sample average market	0.08	95.96	0.73	77030.17	1583377.03	1506346.86	0.17	0.17
In sample losing market	0.09	106.17	0.76	-33441.67	1716621.07	1750062.74	0.17	0.16

Financials Of Policies Lost In Sample Markets

Policies lost	premiums (mean)	premiums (median)	premiums (std)	distance to winner (mean)
In sample profitable market	104.70	93.63	70.80	68.26
In sample average market	105.34	100.33	64.42	64.05
In sample losing market	102.86	97.16	65.45	64.13

Figure 2.14: Examples of the three financial summary tables are included here. The top table includes financial information about policies won disaggregated by the conversion rate quantiles associated with those policies for the participant model. The next two tables provide information about the sample markets that are highlighted in the profit Vs. market share segment of the report (see figure 2.11.). These tables are designed to be self-explanatory and easy to understand for participants and as such, the rows are identical to the attribute summary tables (see figure 2.13).

2. Claim value to portfolio size ratio
3. Price transparency
4. No cumulative profits
5. No customer loyalty

The first is shared with the real-world, claims uncertainty. This is due to the fact that the competition uses a real-world dataset. The second source of uncertainty is related to the relative portfolio size with respect to claim amounts, expanded on below. The other three

reasons are related to 3) a lack of transparency of prices that can prevent price convergence, 4) no cumulative profits over leaderboards which means week by week the markets essentially restart, and 5) a customer model that does not account for customer loyalty which means every single policy on the books of a company is up for grabs every week.

For reasons of accessibility (4) was not addressed, while addressing (5) would add unnecessary degrees of freedom to the overall game-design. Finally, addressing (3) would introduce an unpredictable attack landscape to the game which could allow participants to engage in malicious behaviour. This not only would undermine the scientific questions but also, would be unfair to the rest of the participants.

To highlight why the second source of uncertainty, namely the relative size of claims and portfolio sizes, does not exist in the real world compared to the pricing game, consider what happens in the real world. There, an insurer may have 1 million policies⁶¹ while in our markets a 100% market-share would correspond to a maximum of 18,000 policies in the weekly leaderboards⁶². This means that one very large claim can wipe out the entire revenue of many insurers in our markets, while this does not happen in the real world. Even in that case, insurers in the real world tend to have a complex set of safe-guards to protect their bottom line from very large kicks. Either in the form of re-insurance, or by refusing to write any businesses for a segment of the market they deem too risky.

There are two components to addressing the issue of relative portfolio size. First, we can increase the size of the weekly leaderboards or, second, we can cap the maximum value of the claims within the portfolios. Any decision would be constrained by the fact that we have a limited amount of data that can be injected into the leaderboards while ensuring no policy appears twice, and that capping the value of claims could have serious implications for modelling.

To measure volatility in our markets along these two dimensions, we used models from week 5 on the final test data where the size of the leaderboard was altered with different cap levels. Using models from week 5 on the test data, I start by simulating markets with an increasing

⁶¹Admiral, Axa, Direct Line, and Aviva, each have 15%, 13%, 12% and 11% of policies in the UK motor insurance market, with a total market-size of 31 million policy holders [116]

⁶²Note that before week 8, the leaderboard portfolio size was approximately 8,000 policies.

portfolio size, from 10,000 policies to 100,000 policies. I then measure the consistency between the leaderboards of these different markets. Ranking consistency for a particular market-size is defined as the fraction of models in the top 10 of a leaderboard at that size which remain in the top 10 when compared to the ranking at the largest market-size. This assumes the latter to be closest to the “true” ranking as this is the final leaderboard dataset.

Formally, I define the ranking consistency as $\mathcal{RC}_{i,j}^n$ between two leaderboards i and j for the top n models:

$$\mathcal{RC}_{i,j}^n := J(G_2^i(:n), G_2^j(:n)) = \frac{|G_2^i(:n) \cap G_2^j(:n)|}{|G_2^i(:n) \cup G_2^j(:n)|} \quad (2.14)$$

Where $G_2^j(:n)$ indicates the set of top n models for leaderboard j in the second stage of the global rank computation⁶³. $J(A, B)$ is the Jaccard index [117] of the two sets A and B .

In figure 2.15A we can see that the leaderboard consistency, defined as the fraction of models from the top 10 that remain in the top 10, steadily increases with dataset size. As a result of this investigation, we increased the size of the leaderboards from approximately 8,000 policies to 18,000 policies from week 8 onward.

Surprisingly, by investigating the cap values between 0 and 400K⁶⁴ we can see no discernible effect on ranking similarity in aggregate. In figure 2.15B, we can see the leaderboard consistency (X-axis) is broadly similar to each other for different cap values (Y-axis). Each boxplot here represents the distribution of leaderboard consistency values for a given cap⁶⁵. In other words, the cap of the claim values does not seem to impact the consistency of the leaderboard positions in an extreme way.

Although we did not observe any tangible effect on leaderboard consistency due to claim amount, we did cap the claims at €50,000 to improve the readability of the weekly feedback

⁶³See section 2.4.6 for details on how the global rank is computed.

⁶⁴Maximum claim size is at 300K.

⁶⁵For example the boxplot of the €50K cap displayed in figure 2.15A would consist of the distribution the Y-value of the points displayed. This can be seen as the fourth boxplot from the top in figure 2.15B.

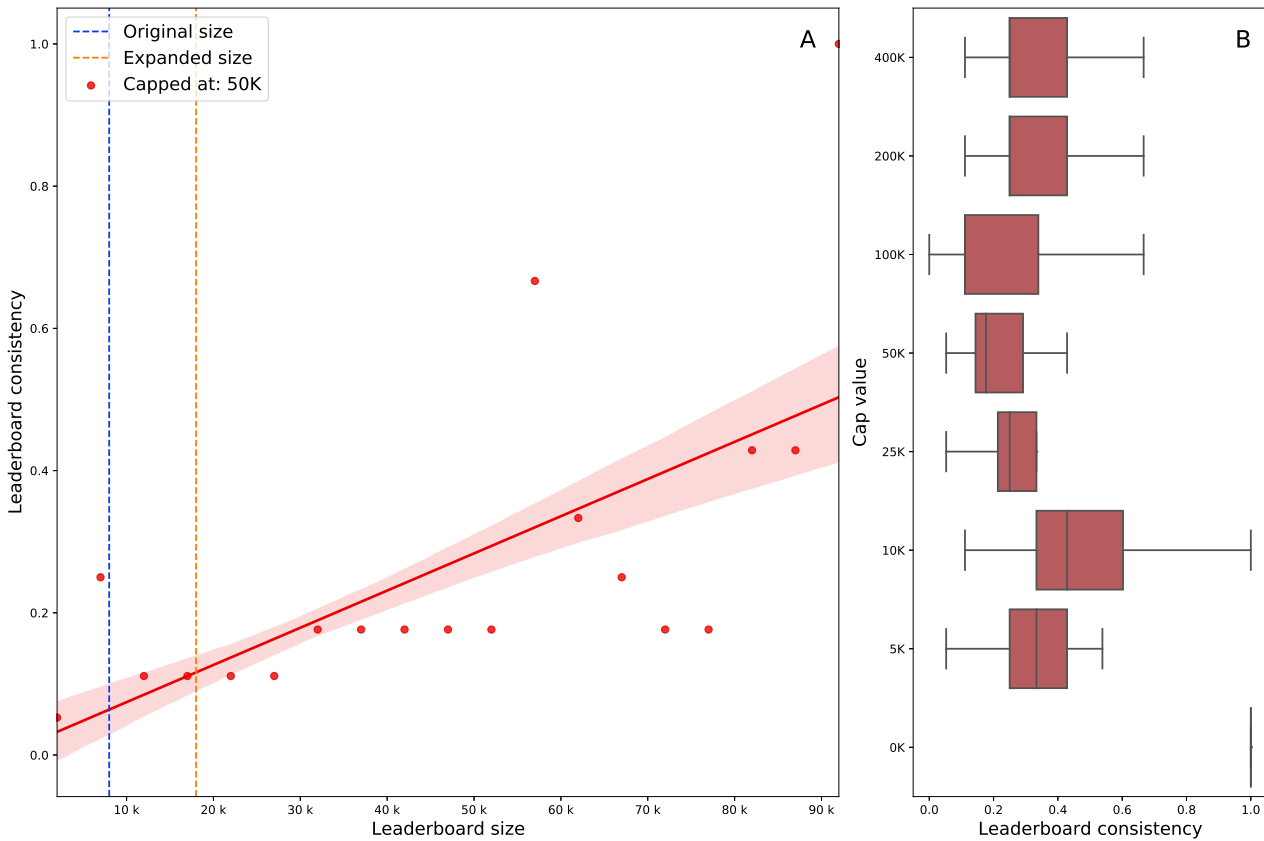


Figure 2.15: The effect of claim caps and dataset size on leaderboard ranking consistency is displayed. **Panel A:** On the horizontal axis are samples of increasing size of the final test leaderboard. On the vertical axis are values of leaderboard consistency defined as the fraction of models in the top 10 of the leaderboard that will remain in the top 10 with the full dataset. In red, the line of best fit for leaderboard consistency against dataset size is displayed along with the 50% confidence interval. The blue and orange vertical lines represent the original and expanded dataset size. **Panel B:** Boxplots representing the leaderboard consistency distributions for various levels of claim caps are displayed. Each boxplot represents the distribution of consistency values when comparing a different leaderboard size to the maximum size of 100K policies. Consistency values of 1 with a cap at zero indicates that without any claims (i.e. capped at zero), the leaderboard positions will not change as expected. This is a sign that the dataset samples are representative of the final leaderboard as the leaderboard rankings in this case only reflect revenues.

plots. A common pattern in the participant feedback plots, clearly demonstrated the impact of very large claims. For example, in figure 2.16 we can see two clear regions, in one the participant did not win a policy with a very large claim while they did so in the other, effectively swinging their leaderboard ranking from the top to the bottom.

The value of the cap was picked based on consultations with industry practitioners. It was set at 2% of the total market revenue based on a market with 100,000 policies. This corresponded

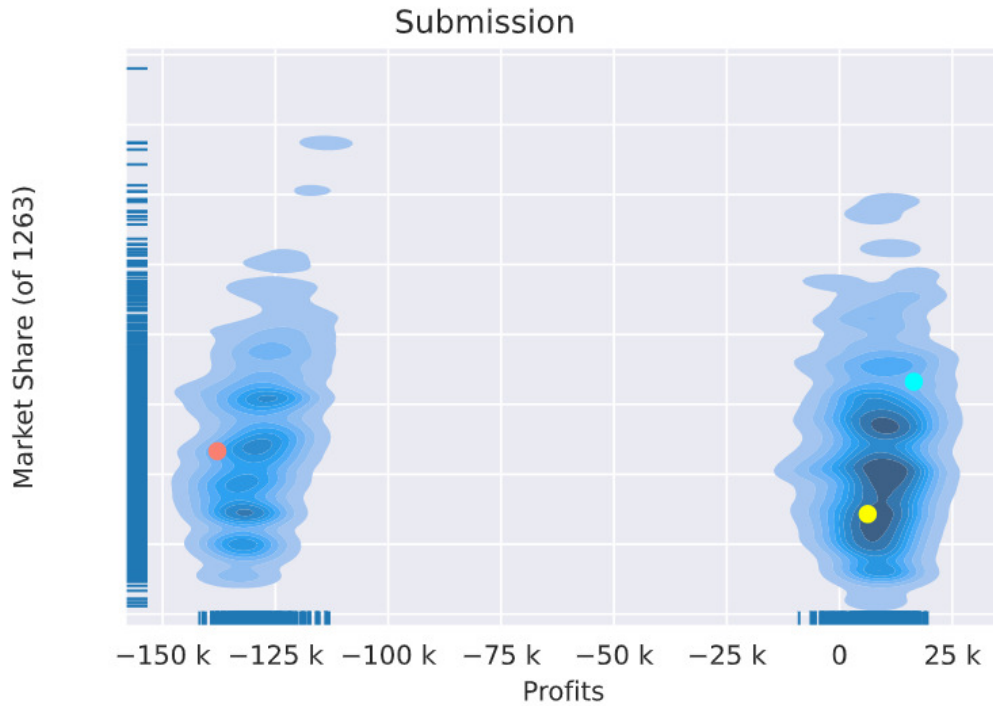


Figure 2.16: An example piece of participant feedback prior to capping claims. The two distinct regions correspond to markets where a participant did suffer one large loss or did not. This would inject a large level of volatility in the leaderboard rankings of a given participant with the bad luck of winning one large claim dictating whether they are near the top, or the bottom. Such issues were far less common after the introduction of a claim cap at €50K.

most closely to a cap of €50,000.

Capping the claims in such a way was an unavoidable middle ground which had to occur at week 5. The reason for this was twofold. Firstly, the relative claim value to portfolio sizes means that a single claim can often wipe out the revenues from some of the top firms in the leaderboard, and second, due to the randomised profit metric, the feedback provided to participants in such situations can be very difficult to read (see figure 2.16 for example). A broader discussion on the effects of the introduction of this cap is included in section 2.6.2.

Evolution timeline of the competition

On the one hand, real markets involve feedback, while on the other hand, providing even slightly unrealistic feedback can sway the markets in an unrealistic way. However, there is evidence that markets also have inertia [118], that is to say, if participants first use their expertise to

replicate realistic models on the platform, and are then exposed to imperfect feedback, the risk of large variations due to this feedback may be minimised.

With this assumption in mind, the feedback in the insurance pricing game started out minimal and was gradually transformed into the form presented in the previous section. The evolution timeline of the feedback, and the overall competition, can be seen in figure 2.17.

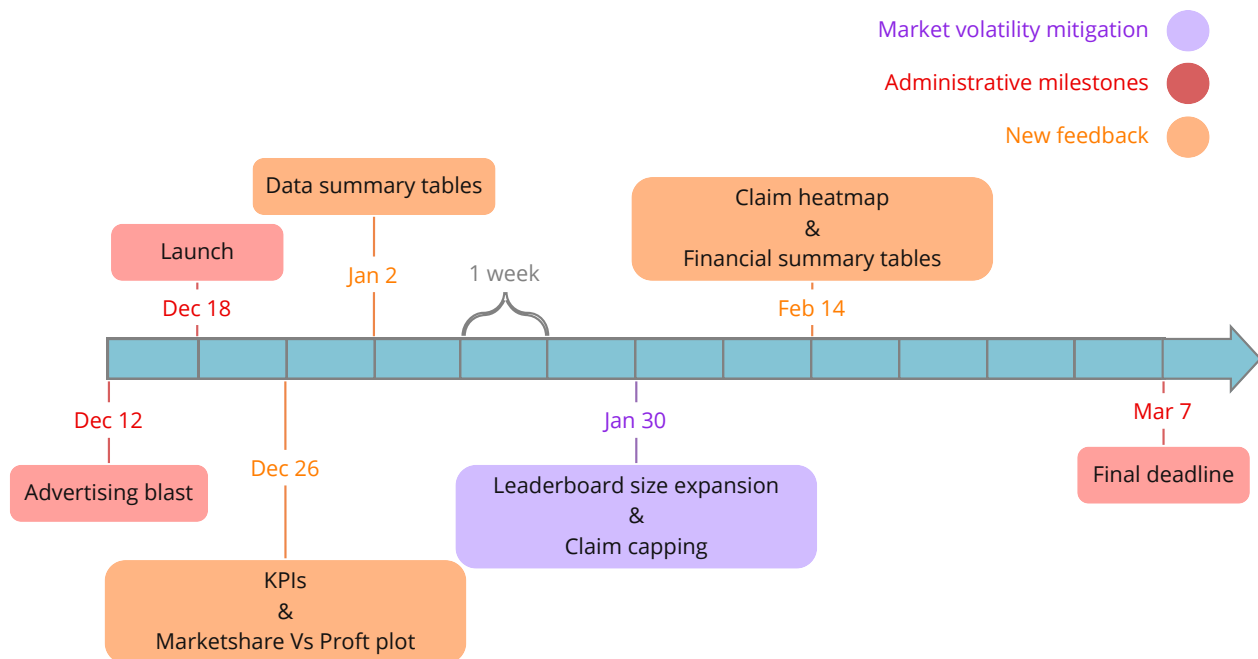


Figure 2.17: The competition timeline is displayed. The timeline starts on the left with the advertising blast and is split into weekly sections. There are three categories of events highlighted. These are administrative (in red), introduction of new feedback (in orange) and, introduction of competition changes (in purple).

As seen in figure 2.17, the feedback provided to participants only came to its final form in week 8 of the competition. The core pieces of the feedback were already finalised by January 3rd (week 3 of the competition) while the claim heatmap and a few financial summary tables were not added until February 14th (week 8).

The timeline also highlights when capping of the claim values and the increasing of leaderboard dataset size was done to reduce volatility.

2.4.8 Communication with participants

This experiment required consistent and high-quality levels of communication to maintain active participation. This communication occupied 3 broad categories. The first and most common type of communication revolved around diagnosis and resolution of issues relating to submission. These are bug fixes, clarification of how the leaderboards work and so on. The second most common category is more broad discussion posts, with technical questions relating to the game or the industry. For example, describing why a particular metric was chosen⁶⁶, why a particular choice in leaderboard design was made, what the standard industry practice is in pricing⁶⁷, or why a particular feedback report may look like what it does. The third category is communication initiated by the organisers and focused on either informing participants of important updates to the competition, or nudging them to continue to play. Each of these will be described in detail here.

The most important⁶⁸ part of communicating with participants is making sure they can submit without issue. A typical data science competition would involve downloading the training dataset and the test dataset (without the target). Then the participant would train a model and generate a file with predictions on the test dataset. In this competition the process is much more complex and as such, there was a significant amount of questions that needed answering, and issues needing resolution during the competition, solely related to being able to submit the code that participants intended to submit.

The most interesting category of posts which indicated high levels of expert engagement with the competition were participant discussions posts. These are participants sharing their own analyses or thoughts on the competition, often drawing parallels to their own experiences in the industry. There are two general forms these conversations take. First is when a participant shares a full-fledged analysis of a problem, typically inspired by their weekly feedback, and opens the floor for discussion with other participants and the organisers⁶⁹. The second is when

⁶⁶For example see [this discussion](#).

⁶⁷Another discussion is [linked here](#).

⁶⁸And perhaps least interesting.

⁶⁹A great example is [included here](#).

a participant asks a highly technical question targeted to the expert community⁷⁰. These could be to prompt others to share their experience of the industry, or thoughts about whether they can help solve a problem unrelated to the submission process.

Finally there is the communication that is initiated by the organisers falling into two categories. First are weekly updates⁷¹ on the discussion forums focusing on the profit leaderboard outcome and game updates. The second category is in the form of weekly emails that include some high-level information about that week. A portion of one such email can be seen in figure 2.18.

Since participants within this competition are actively involved in a research study, we followed four main guidelines while communicating with them. First, all communications were neutral to our hypothesis. That is, nowhere do we suggest that data modelling is more appropriate than algorithmic modelling in any form⁷². Second we strive not to reveal the strategies and models of one participant when addressing technical questions that may relate to debugging. In fact, we worked with the AICrowd team to launch a new debugging feature allowing participants to view private traceback of their code (see figure 2.19 for examples of failed and successful submissions). Third, we do not reveal any details of the dataset source or details of the privacy risk-mitigation steps so as to reduce the chances that a participant can obtain an undue advantage. Finally, the team responded to all participants in a timely, friendly, and comprehensive manner. By putting our best foot forward we can ensure that participants remain interested — a key to maintaining high levels of engagement. Since this is a code-based competition, the participants had to learn how our submission process worked in addition to the data science work required. This put participant engagement at risk if they encountered a problem they could not solve. Therefore, being present and available on the discussion forums was a top priority activity during the competition. In fact, of the 300+ participant posts on the online forum, only a handful were unresolved for longer than 24 hours.

⁷⁰An example about loss models is [linked here](#)

⁷¹One example update in week 8 [is linked](#).

⁷²For example, we do not suggest decision trees might be better than logistic regression



Hey ali_farzanehfar!

It's been some time since we sent you a mail. 👁️👁️

By now, we have 👤 1,600+ participants from 🌍 40+ countries making 🚀 7,500+ submissions!

There are **3 more weeks** till this competition comes to an end, and we have some important ✨ **updates** and 🎁 **surprises** for you! But before that, we are very happy to report that **almost a quarter of the top 20** entries on the week 7 leaderboard are from participants that joined the competition very recently!

Newcomers can still compete with the experienced players, and also beat them! 🏆



Check out Week 7 Leaderboard

And if you are new to the competition, check out our resources for newcomers and get started!



[Starter Kit\(Python\)](#)



[End-to-end Guide on Making a Submission](#)



[Forum](#)



[More Notebooks](#)



[Starter Kit\(R\)](#)



[Discord](#)

Figure 2.18: A part of the email sent to participants in week 8, after the week 7 leaderboard.

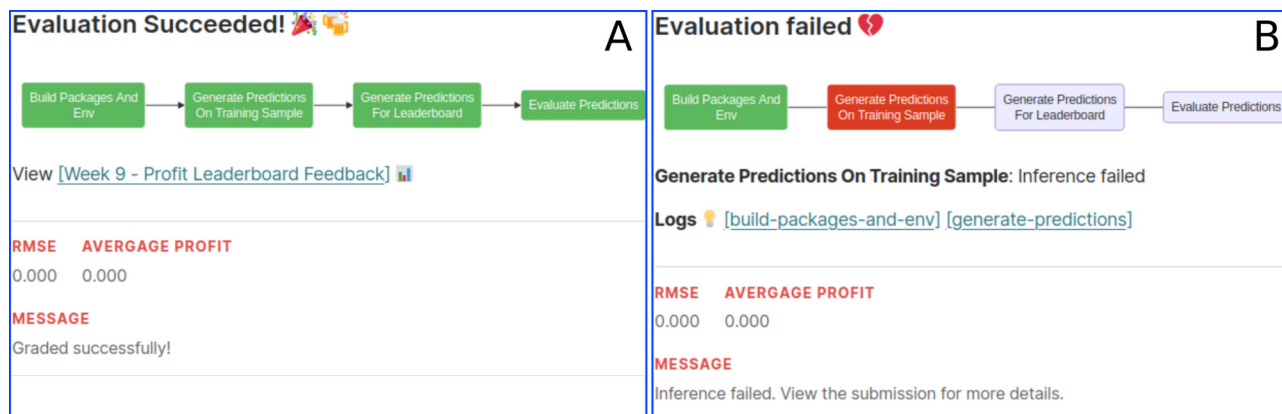


Figure 2.19: The live participant traceback is displayed here. When a participant is successful (A), they eventually receive weekly feedback if this submission is picked by them to be included in the final leaderboard. If they fail in the submission (B) they are given feedback (see links in B) on why their code failed so that they can correct the mistake. Alternatively they have links to their weekly private feedback (see A).

2.4.9 Available online resources

As this competition was entirely online and publicly accessible, all resources and design elements noted above, barring the dataset will remain online. In this section I will share a selection of archived online resources that can demonstrate the public-facing elements of the competition in more detail. These will include the main competition webpage that explains the rules of the game, a sample of forum discussions, and finally, the various leaderboard pages and the participant private view of them.

The main competition webpages

There are three main pages that participants visit to gather information about the competition. The first is the competition homepage located at aicrowd.com/pricing-game⁷³. This page describes the game in lay terms assuming only basic knowledge of insurance pricing. It also explains the profit metric as well as how to submit, including links to submission templates.

The second important webpage is the official [challenge rules](#)⁷⁴ which must be accepted before dataset download is permitted. Most importantly, these rules explain to participants that this

⁷³For an archived version please visit the [the archived link](#)

⁷⁴Web archived version is [available here](#).

is part of a digital field experiment, that they provide consent to use their data for this purpose and how their data will be stored and used. It also covers issues relating to privacy, tax, and prizes.

The third important page available is the [resources page](#)⁷⁵ that linked to the dataset dictionary and the dataset itself. However, the dataset itself is no longer available online.

The online discussion forum

Perhaps more frequently visited than the competition webpages are the discussion forums for this competition. Discussion forums for all AICrowd challenges are hosted on the [discourse service](#). The forum for the insurance pricing game is entirely public and available online at discourse.aicrowd.com/c/pricing-game-challenge/373. I will highlight three sample discussions here that each fall into one of the categories of “*submission issue*”, “*participant-initiated discussion*”, and “*organiser announcement*”.

1. *Submission issue example*: Participants with a common issue asking for a resolution ([online link](#)).
2. *Participant-initiated discussion example*: Participants discussing the claim distribution experienced in week 9 ([online link](#)).
3. *Organiser announcement example*: increasing of leaderboard size and capping claims at €50,000 ([online link](#)).

Leaderboards and personal submission pages

This challenge included a total of 14 leaderboards. The first 12 are split into 11 profit leaderboards, 1 [RMSE](#) leaderboard. The final 2 are also an extra profit leaderboard and an extra [RMSE](#) leaderboard that were created after the final submission and prior to the release of the results. The aim of these final two leaderboards was primarily to keep the community engaged

⁷⁵Access to this page will require an AICrowd account.

before the final results are released, in case they needed to be reached for any reason to discuss their submission⁷⁶.

Figure 2.20 demonstrates what the weekly leaderboard page looks like for week 5. It can be seen that participants also have the ability to navigate to any other leaderboard if need be. In addition to that, by clicking the “view” button on their own submission, they can view the status of the submission (see figure 2.19) as well as any weekly feedback that is available.

2.5 Results

This digital field experiment ran for 11 weeks with ≈ 2000 participant sign-ups, submitting $\approx 10,000$ models. Using manual code inspection of only the leaderboard eligible submissions, they are categorised into either data models or algorithmic models (see table 2.2 for categorisation). After this division, markets with only one type of model are generated and consumer prices are compared for the same policies in the two market types. The next question explored focuses on the advantages of data modelling Vs. algorithmic modelling with respect to market entry by simulating markets where the number of algorithmic models increases from 0 to 10 (the full market). The presentation of results is preceded by a brief discussion on the quality of models submitted. Finally a sensitivity analysis on the profit metric is conducted.

2.5.1 Participants & submissions

Over the 11 week lifetime of the insurance pricing game, there were 10,009 submissions made, growing steadily from launch to the deadline. Figure 2.21A displays the daily number of website visits to the challenge page. The peak at launch is due to the global email advertising campaign to 21 national actuarial communities. In figure 2.21B the cumulative growth of participants is displayed. Participants contributing to this plot are only those who successfully submitted

⁷⁶Competition on AICrowd usually involve a “Town-Hall” event. In our case, this event was held at the end of the competition. However, in order to ensure participants do attend the event, and to satisfy the sponsoring partners, we withheld the reveal of the final leaderboard until this event. As such, this bonus round was included. The video stream from the town-hall event is [available online](#).

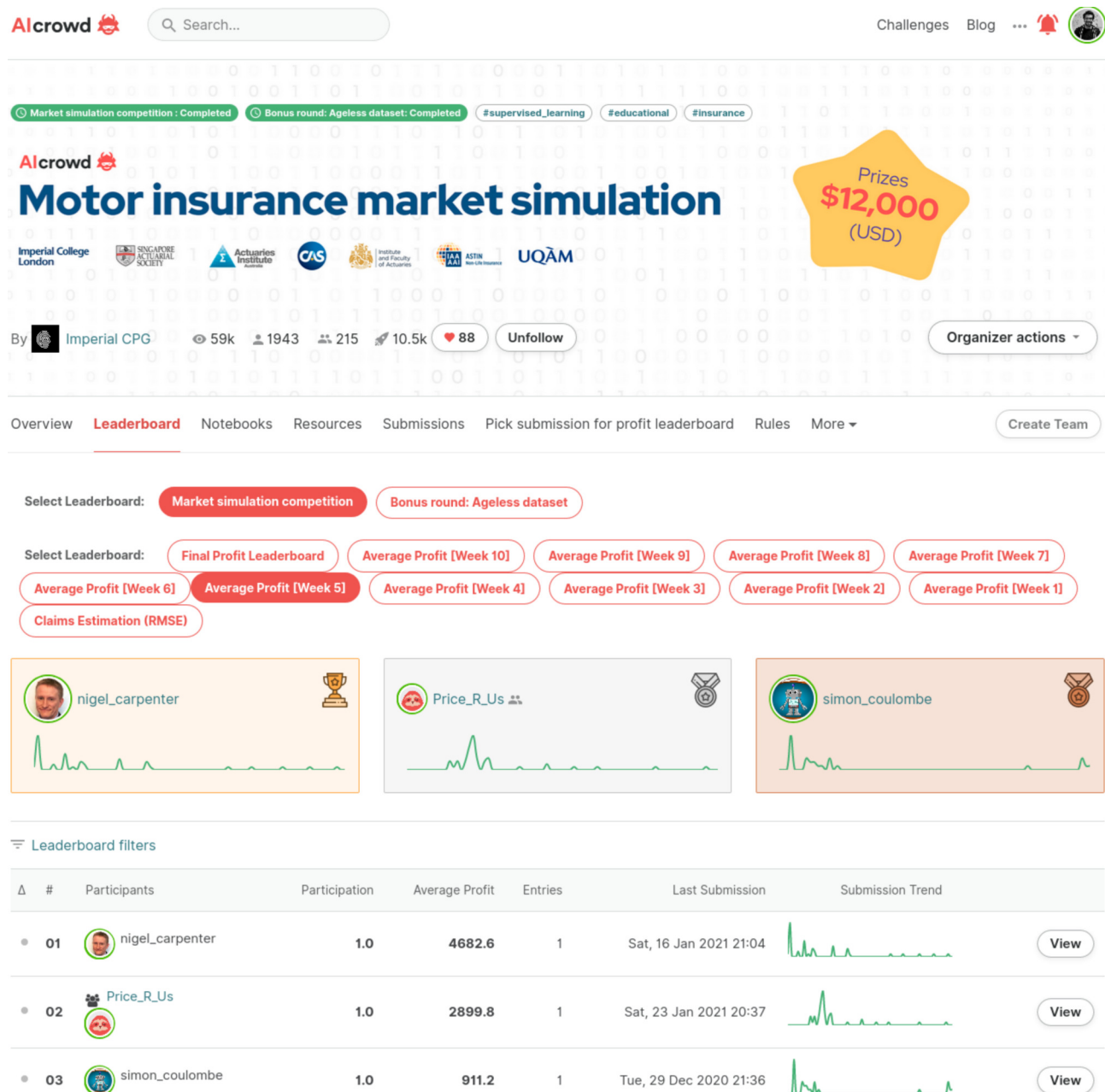


Figure 2.20: The page for the week 5 leaderboard is displayed. Note that other leaderboards are also accessible. Each participant can click on the “view” button next to their own submission to view the status and any available feedback for their submission.

at least once during the competition (total - orange line) or at least once in the last 7 days (active - blue line). From week 3 to the deadline the platform averaged 100 submissions per day. Most interestingly, week 3 can be considered as the point where the market participation stabilises, with active participants remaining at approximately 100 (pink dashed line figure 2.21B). Similarly, from week 3 onward, the number of successful submissions outstrips the number of failed submissions (figure 2.21C), which indicates the active participants, making up the stable

markets, have learned how to submit successfully and consistently.

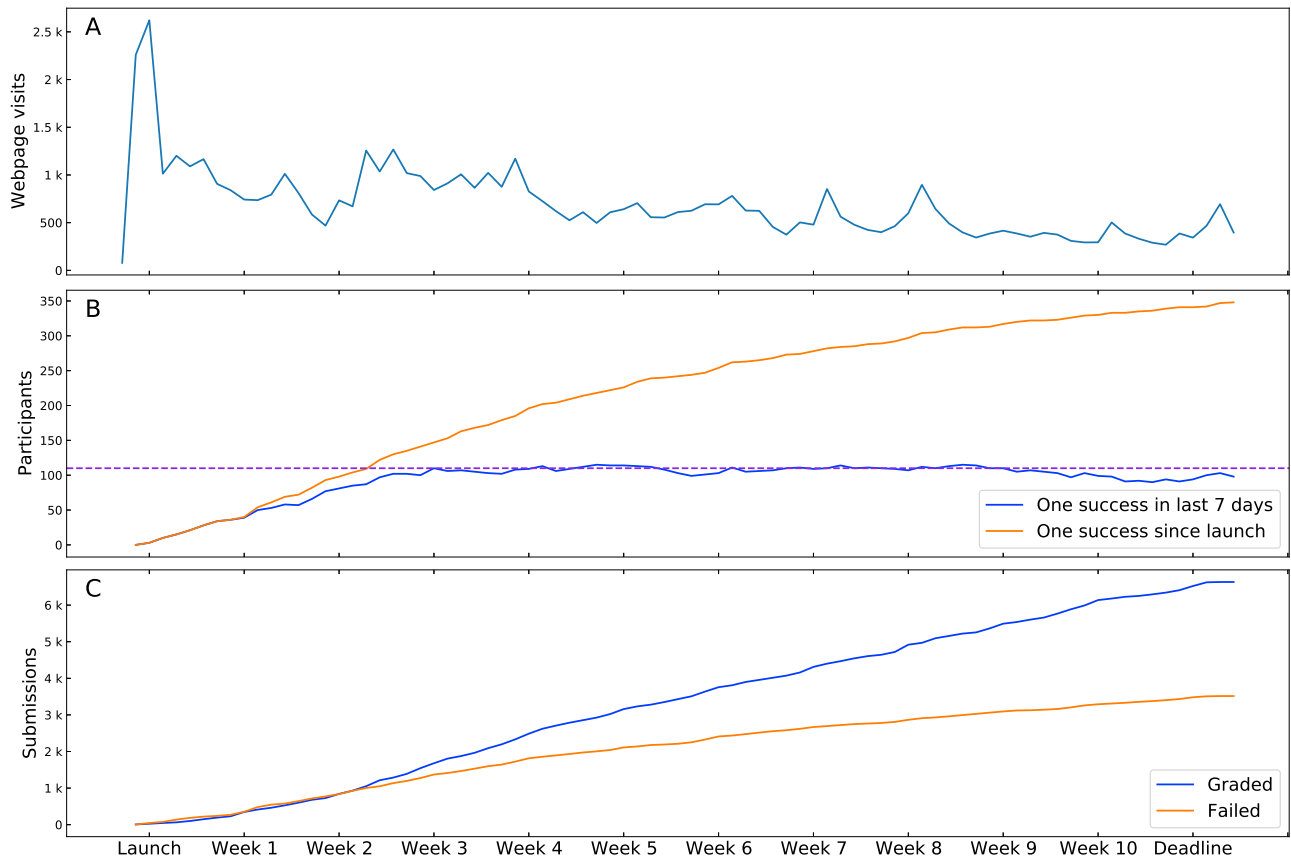


Figure 2.21: A time series view of webpage visits (A), active participant growth (B), and submission growth (C). **Panel A:** Daily webpage visits are displayed. Note the peak at launch due to the advertising campaign. **Panel B:** A cumulative plot of active participants in blue (i.e. made one successfully graded submission attempt in the past 7 days). Those who had at least one successful attempt since launch are highlighted in orange. Note that after week 2, the number of weekly active participants remains at 100. **Panel C:** The cumulative growth of successfully graded (blue) or failed (orange) submissions over time. After week 2, participants seem to better grasp the submission ecosystem, with graded submissions, outstripping failed attempts.

Participants were surveyed weekly, with a response rate of 40% up to week 3. It was found that close to 60% are insurance industry professional (*"Ins. ind. prof."* - figure 2.22B) with more than a third in the 30 - 40 year range of age (figure 2.22A). However, there will be a subset of participants that are not actuarial experts. They may be actuaries in training, underwriters, data scientists, investment professionals and so on. However, the majority will be insurance industry professionals with some experience in modelling. Even if that experience is within other areas of insurance, such as life and health, or catastrophes.

Most importantly, it was found that more than 80% of participant spent upwards of 2 hours

on the competition in the early weeks, some more than 8 (figure 2.22D). High-levels of time-commitment by expert participants is a required component of an experiment like the pricing game. Additionally, the participants were queried about the frequency of modelling within their professional job. More than 40% reported modelling using either Python or R with a frequency of monthly or higher (figure 2.22C).

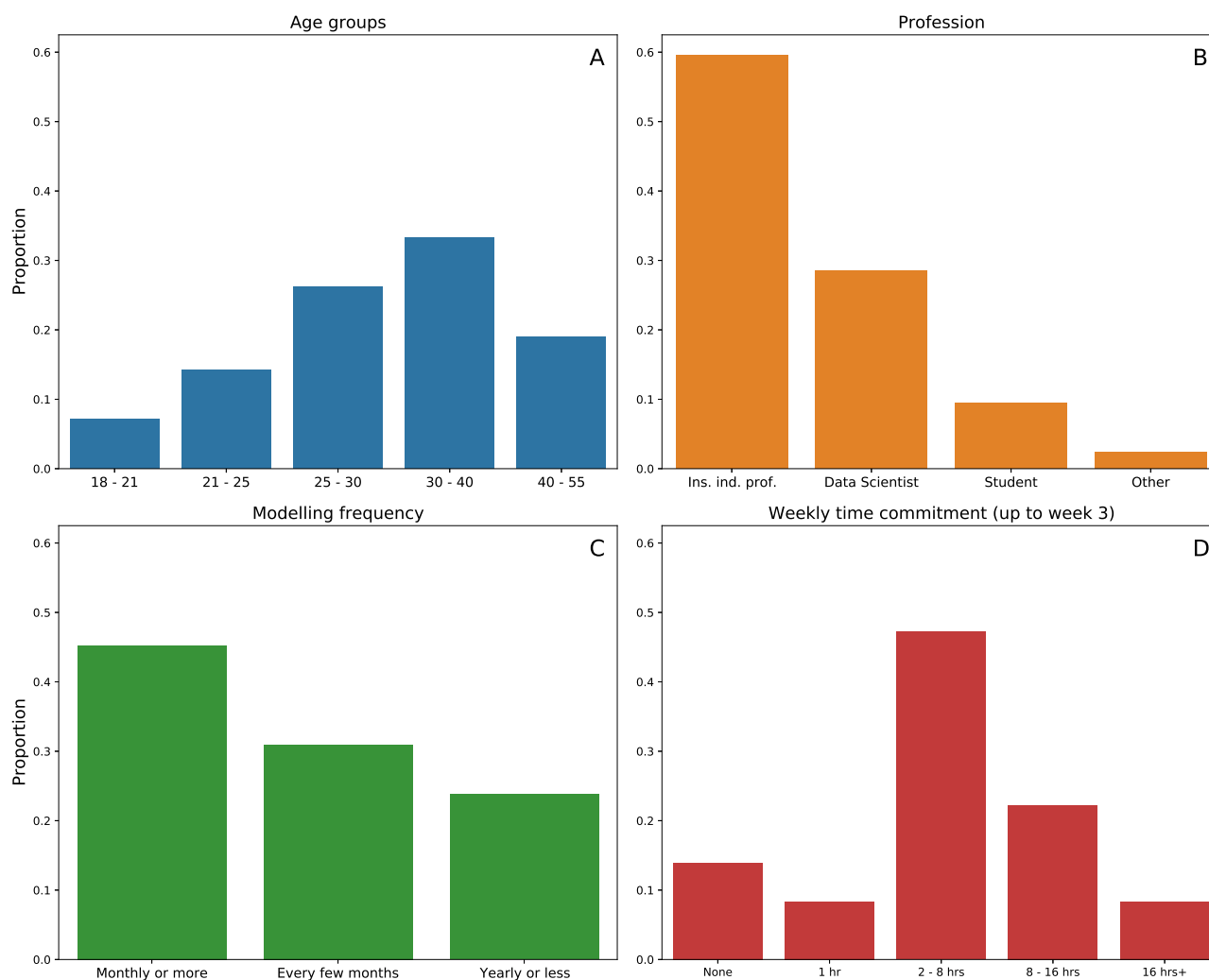


Figure 2.22: A summary of participant demographics and time spent in the competition is displayed. **Panel A:** Proportion of participants broken down by age groups is displayed. **Panel B:** Participants' self-declared professions are compared in proportion. With the majority reporting as insurance industry professionals. **Panel C:** Displays the frequency with which the participant engages in Python or R modelling in their day-job. **Panel D:** The mean amount of time spend weekly by participants up to week three, with 80% spending upwards of 2 hours per week.

Finally, to demonstrate the global nature of the participant pool, the breakdown of nationalities (based on IP addresses) is shown in figure 2.23. It is clear that there were 5 major hubs of activity. India had the most number of participants (256) followed by the United states with

209. France, Canada, and the United Kingdom follow with 185, 130 and 106 participants respectively. Note that no participants were registered from countries that are crossed out. This map shows all sign-ups to the competition⁷⁷.

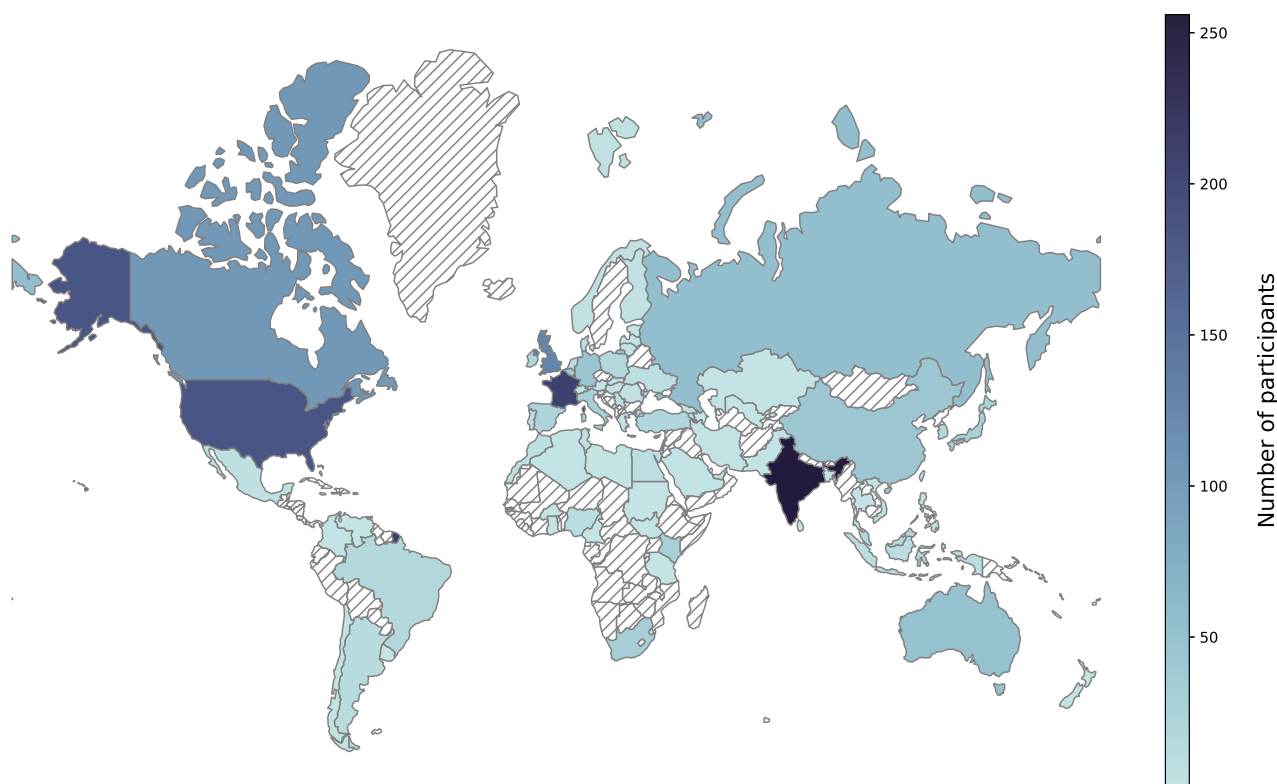


Figure 2.23: A world map view of competition participation. With more than 50 countries represented, this digital field experiment can be considered to have a global sample of the actuarial expertise in the world. The crossed out countries did not participate as recorded by IP addresses.

2.5.2 Categorising data and algorithmic models

In order to study the impact of algorithmic modelling on consumer prices, we first must categorise submissions into either data models, or algorithmic models. Table 2.2 displays the taxonomy of model types that were encountered in the leaderboards. In the first column on the left-hand side we can see a list of the most common model types while on the right-hand side we can see whether they are categorised as data models (DM) or algorithmic models (AM). In practice, submissions were split into three categories. Many models incorporate a combination

⁷⁷Not everyone that signed up, attempted to submit see figure 2.21.

Algorithm type	Model type	AM	Algorithmic model
Tree-Based	AM	DM	Data model
Artificial Neural Network	AM		
Gradient Boosting Machine	AM		
K-means	AM		
Support Vector Machine	AM		
Expert Rules-based	DM		
Generalised Linear Model	DM		
Generalised Additive Model	DM		
Spline-based	DM		

Table 2.2: Table providing examples of how categorisation is done, distinguishing between algorithmic models, or data models. Note that mixed models are those that combined one or more of each model type in the final pricing model.

of these two types and when one model contains more than one type it is categorised as a mixed model and is excluded from the analysis that follows.

2.5.3 Risk-model quality and exclusion criteria

Here I will present some analysis on the relationship between risk-estimation and profitability of the submitted models. I will investigate the relationship between leaderboard rank and risk-estimation. This also will involve stratifying these findings based on model type. Finally, I will provide the criteria used to exclude models from the analysis.

In figure 2.24 the risk-estimation ability of models is displayed, and disaggregated across model type. As clearly demonstrated in these KDE plots, both data-models and algorithmic models have localised risk-estimation ability within the range of $\text{RMSE} \in (560, 640)$. Moreover, the risk estimation ability of models that fall within this range seems entirely uncorrelated to the their total competitive profit. Perhaps the most stark feature of this figure is the wider spread of algorithmic models in risk estimation, which may be due to their tendency to overfit due to a higher level of decision boundary complexity. The story is almost identical when considering profitability (i.e. profit per policy).

The results presented here excludes a small set of models. The overall exclusion criteria is based on what is considered “*very unrealistic*”. In practice these models can appear in the

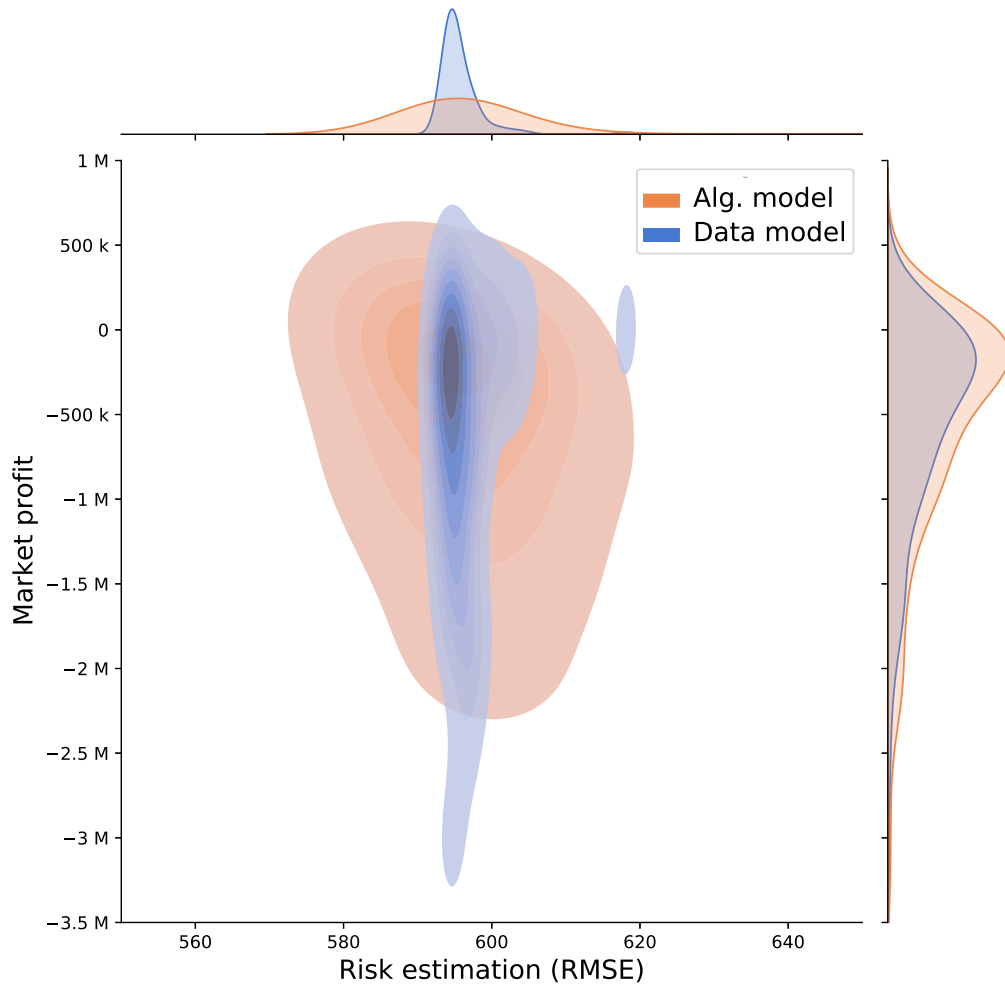


Figure 2.24: KDE plots of risk-estimation Vs competitive profit are displayed. On the horizontal axis the risk-estimation ability of models is displayed as measured by their claim-estimation function against the test data. On the vertical axis their competitive profit is displayed using their premium prices. This plot is made up of 211 models after excluding mixed models, and the three exclusion criteria: baseline models, constant models, and models with negative prices.

leaderboard and, by the design of the randomised metric⁷⁸, will not significantly impact the overall leaderboard ranking.

Models are excluded based on the following three criteria:

1. **Baseline models.** They are one of the two baselines with minimal changes.
2. **Constant models.** They are a constant, or an “almost”⁷⁹ constant model (i.e. always

⁷⁸See section 2.4.6 for a description of the metrics.

⁷⁹An “almost” constant model is one where a handful of prices are very large while the rest are close to zero or negative. This is a workaround for one of our market entry requirements that states models must be profitable on their own training data while still providing very unrealistic prices. See section 2.4.6 for a detailed discussion.

predict the same price).

3. **Negative prices.** They have a large proportion ($\geq 10\%$) of their prices as negative.

In practice however, only 3 models were excluded out of the 211 that are classed as either an algorithmic model, or a data model. Of these three, 2 were excluded on the basis that they are near constant models with identical prices for all but one policy. The third was excluded as 13% of the prices offered were negative.

2.5.4 The impact of algorithmic modelling on consumer prices

Figure 2.25A compares prices paid by policies in fully data-model based markets of size 10 to fully algorithmic model filled markets. Although the effect is small, this preliminary result suggests that prices in markets with algorithmic models are slightly cheaper than in data model based markets (see centre quantile line in blue). Additionally the vast majority of policies have very similar prices in both market types indicating that they are in general agreement with relation to the risk levels identified in the dataset. Expanding further, figure 2.25B displays the distribution of prices offered to policies respectively in each market type. The solid and dotted lines indicate the inter-quartile range for each distribution of offered prices⁸⁰. Note that a two-sided independent t-test, as well as a two-sample Kolmogorov-Smirnov on these two distributions of prices offered returns a p-value of 0⁸¹.

This preliminary result suggests that markets filled with algorithmic models could result in cheaper prices for consumers. More interestingly, it seems that the variance of prices offered by algorithmic models is not significantly higher when compared globally⁸² (see figure 2.25B). Therefore, the cheaper market prices does not seem to be due to a higher variance in offered prices overall — a higher variance combined with a cheapest-wins market could have resulted in cheaper prices.

⁸⁰Note that the claims distribution (i.e. margin-free prices) exhibits the same pattern displayed in figure 2.25, only with reduced values that accounts for a margin-free market.

⁸¹Although it is not exactly zero, it is close to zero within more than 230 significant digits.

⁸²“Globally” meaning across all markets. That is, the variance between the largest and smallest prices offered across all markets.

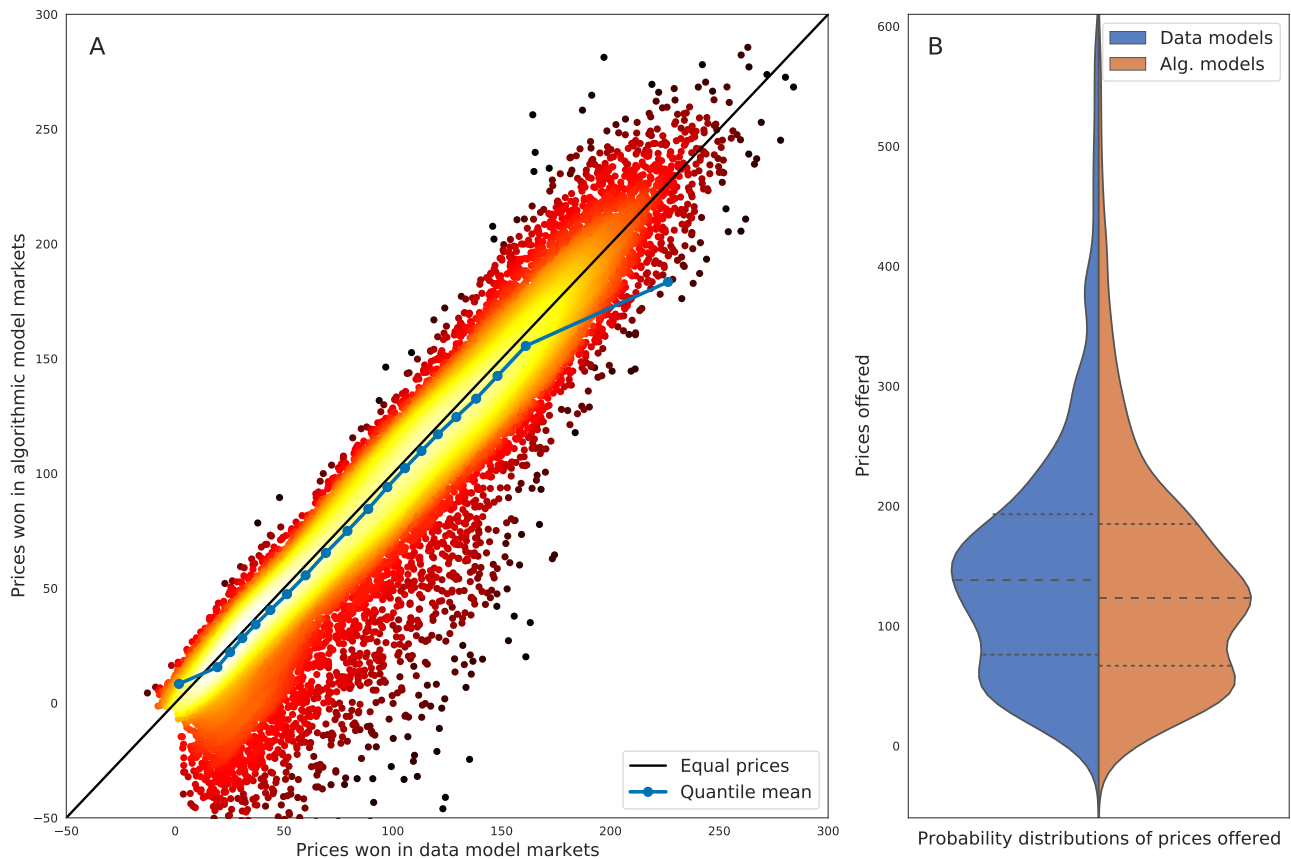


Figure 2.25: A comparison of consumer prices in markets with data models or algorithmic models only. **Panel A:** Each dot on the plot represents a single policy with the price won in a data model market on the horizontal axis. The vertical axis indicates the algorithmic model price won. The thick blue line marks the centre point of 5% quantiles of the prices won. Yellow and white areas indicate higher concentration of points than in orange and red areas. Note that the prices of data models are slightly more expensive than algorithmic models. **Panel B:** The distribution of prices offered to policies by the two modelling types. Note that the prices offered by algorithmic models are slightly lower with the values in the orange distribution being slightly lower than the blue (the dashed and dotted lines on the violin plot indicate the inter-quartile range).

However, there are three other possible contributing factors. First, it could be that the lower prices offered as seen in figure 2.25B is enough to explain the discrepancy of prices. Second, given that the the distributions of prices offered are note very dissimilar, it could be that the prices of algorithmic models are more heterogeneous than those of data models. In other words, they have less correlation between them for a given policy, even if overall this effect is not observed. This would result in lower prices being realised on average. Finally, it is possible that data-models simply have a higher margin loaded into their prices.

In figure 2.26A we can see that algorithmic models indeed have less correlation between them

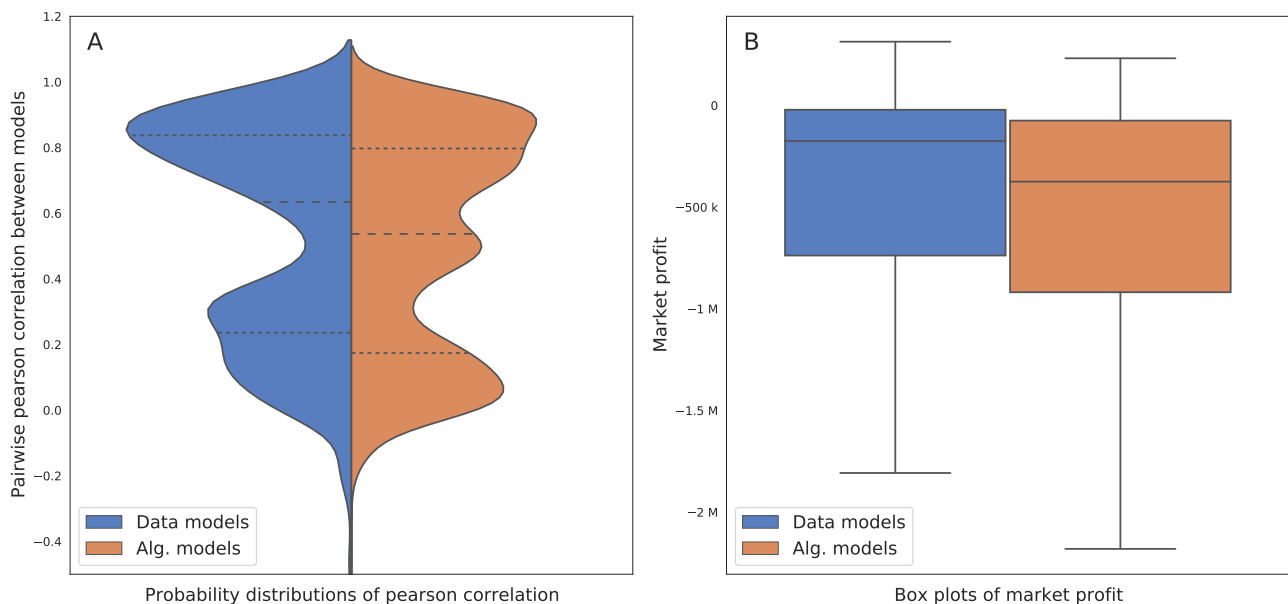


Figure 2.26: This figure displays a comparison of model correlation as well as profits between algorithmic and data models. **Panel A:** A violin plot comparing the distribution of Pearson’s R when computed pairwise for all algorithmic models (orange) and all data models (blue). The dashed and dotted lines indicate the inter-quartile range of each distribution (Kolmogorov-Smirnov and T-test p-values < 0.05). Note the larger correlation coefficient values for data-models. **Panel B:** Comparing boxplot distributions of profit in algorithmic (orange) markets to those of data (blue) markets (Kolmogorov-Smirnov test p-value = 0.16, T-test p-value = 0.08). Note the lower profits on average of the algorithmic models in orange.

than do data models. This suggests that in a cheapest-wins market, they are more likely to exhibit a higher price-variance for any particular policy, resulting in a cheaper overall price. In figure 2.26B on the other hand, we can see that the profits are overall lower for algorithmic models (orange). This could be due to cheaper prices offered as displayed previously, however, neither the t-test of the means or the Kolmogorov-Smirnov performed on these boxplots indicate significant results in this preliminary analysis.

Finally, figure 2.27 demonstrates that the two model types do not have significantly different margins loaded into their prices. The violin plots each indicate the distribution of margins for each model type with the inter-quartile ranges highlighted in dashed and dotted lines. Neither a two sample independent t-test nor, a two-sample Kolmogorov-Smirnov test yielded significant results, thus all but ruling out the possibility that the differences in realised market prices observed in figure 2.25 is due to a difference in profit-loading. This is especially true, since if there were an effect due to profit loading it would be in the opposite direction as the margins

of algorithmic models seem to be higher.

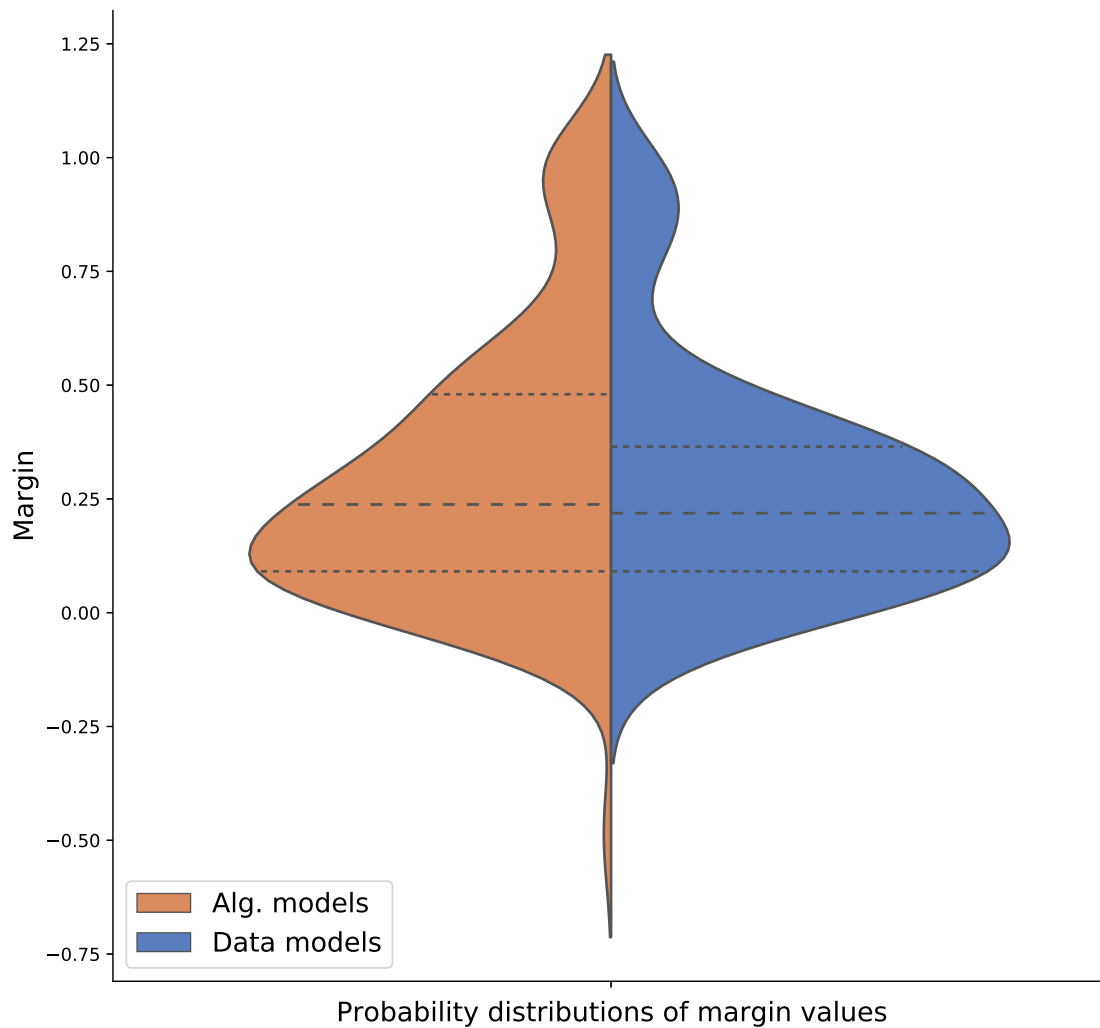


Figure 2.27: Comparison of margin values between data and algorithmic models. The violin plot displays distributions of the median margin value for algorithmic (orange) and data models (blue) in the final leaderboard. Note that the distribution of models are extremely similar for the two modelling camps, with algorithmic models, having slightly higher margins. A two sided independent t-test resulted in a p-value of 0.61, while a two-sample Kolmogorov-Smirnov test yielded a p-value of 0.52. Both not significant.

2.5.5 The impact of algorithmic modelling on market entry

If indeed individual firms are moving towards algorithmic modelling, then it begs the question, how does such a mass movement start? Is there a competitive advantage to early adoption of machine learning in pricing? The second question studied here revolves around market entry as the population of data models slowly gives way to algorithmic models. The profits and market shares of these firms are investigated throughout this shift.

To study this, I start with a market filled with data models into which I inject one algorithmic model while removing one data model — monitoring the profit and market-share distribution of the market throughout this shift. Whether the algorithmic model benefits from any competitive advantage can then be measured by computing the profit and market share of randomly picked algorithmic models injected into randomly generated data-model markets, comparing them to the average profit of the remaining data models before and after the injection of the algorithmic model. I then incrementally replace data models with algorithmic models until the entire market is algorithmic. In practice, this process is repeated many time so that all combinations of markets with X algorithmic models and Y data models are run and evaluated. The results are then tallied and placed in boxplots.

The results of this evolution can be seen in figure 2.28 for both profits and market share. Note that this analysis is limited to the top 10 models of each type to control for performance.

Although no clear pattern is visible, the two sided t-tests suggest that data models in figure 2.28A are consistently more profitable than algorithmic models. Additionally, even though more variance is observed here, this seems to be mostly upside variance. A similar pattern is visible in figure 2.28B with larger market shares and variances attributed to data models.

This preliminary observation is in need of further study to confirm this possible effect and understand its root causes. For example, the increased volatility of data models is likely due to their increased homogeneity (see figure 2.26A). However, at this early stage, it suggests that firms may have a counter incentive to switch to algorithmic modelling if current markets are indeed filled with data-models only. However, the possibility of maintaining high market shares might

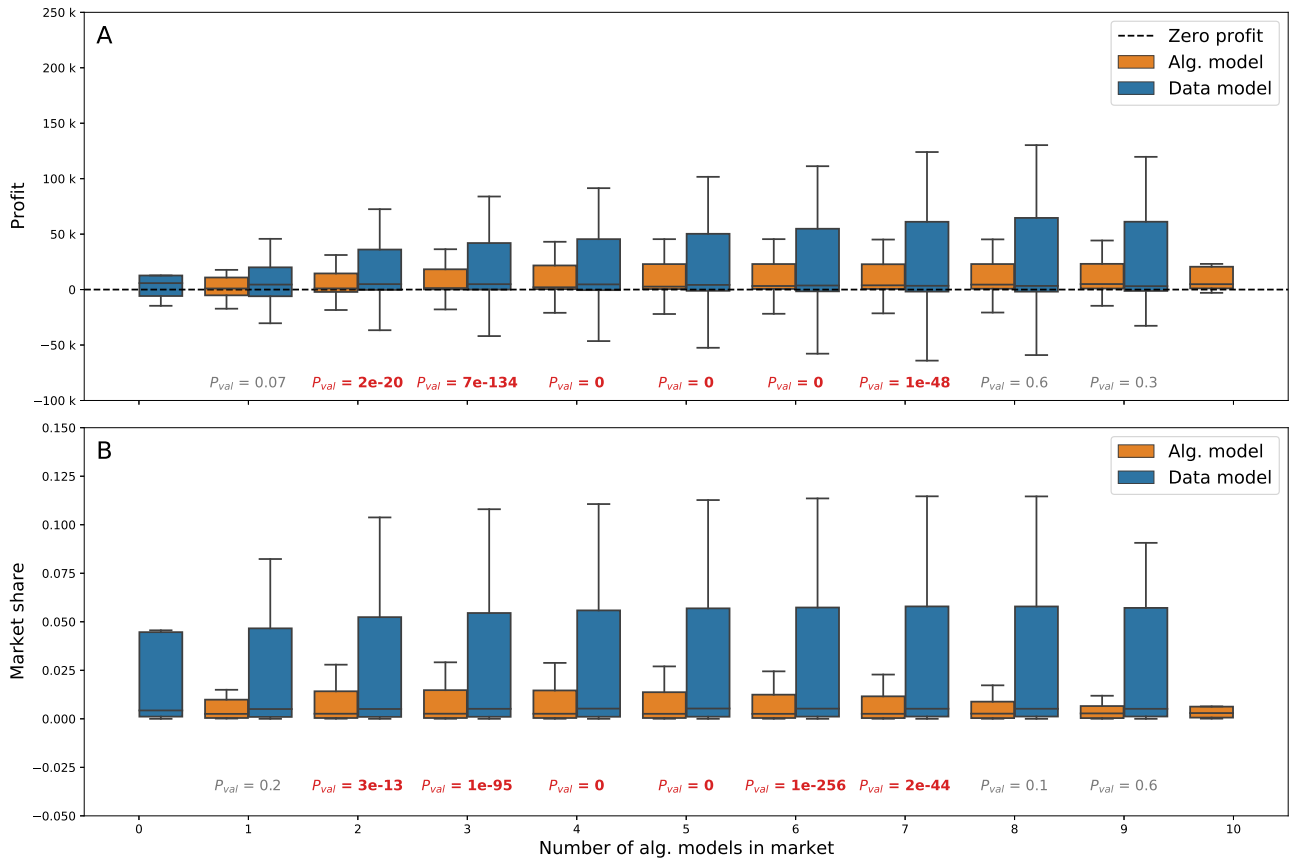


Figure 2.28: Boxplots of profit (A) and market share (B) for markets with both model types in competition are displayed. **Panel A:** The profit boxplots of algorithmic (orange) Vs. data models (blue) are plotted against the number of algorithmic models in a market of size 10. Note the increased stability of profits for algorithmic models (orange). **Panel B:** Boxplots of market shares are displayed in the same format as panel A. On both panels A and B, the value of two sample t-test is displayed which compares the two boxplots. The T-test here is comparing the probability distributions of profit and market shares resulting from each configuration of models listed on the horizontal axis. In bold and red are t-tests with a p-value below the customary threshold of 0.05. The boxplots in both these cases display the inter-quartile range. The are the median, and the 25% and 75% quantiles.

necessitate remaining in the camp of data models, despite some evidence of firms switching. Regardless, if even a few algorithmic models in the market can inject uncertainty to data-model performance, being an algorithmic model may be beneficial if the profit and market share effects observed here are countered through other dynamics in the market⁸³. The implications of this possibility combined with the results found relating to prices for consumers is expanded on in the discussion in section 2.6.

⁸³For example, in this study, all firms start with the same dataset and compete for the same policies. Brand value does not exist and capital and reserving effects are absent. It is possible that some of these effects can counter this preliminary observation.

2.5.6 Sensitivity analysis: *beyond a two-stage profit metric*

A significant degree of freedom that may have an impact on the markets in this study is the levels of depth required from the profit metric. One of the reasons for the two stage design of the profit metric was to reduce the probability that a model climbs the ranks by mostly performing well against other bad models⁸⁴. What happens when we go beyond two stages?

To study the effect of going beyond a two stage profit metric, instead of running the market for only 2 rounds, I run it for 21 rounds and compare the rankings at each round i to the round prior $i - 1$ ⁸⁵, measuring leaderboard consistency (equation (2.14)). Here the consistency is measured as previously by the fraction of models that appear in the top n of the leaderboard in both iteration i and $i - 1$. I alter n from 1 to the maximum value of 252⁸⁶ and compute their distribution of consistency constructed using different values of n . In other words, I first compute G_1 , and continue iterating until G_{21} , and for each G_i , G_{i-1} pair, record consistency values for the top n ranging from 1 to 252.

Figure 2.29 displays the result of this sensitivity experiment. Figure 2.29A displays the distributions of the consistency values while in figure 2.29B, three of the samples within each distribution are picked and displayed.

From figure 2.29B we can see that the ranking tends to either converge after round 2 or weakly oscillates from round 2 onward depending on what fraction of the top of the leaderboard is investigated. Values stabilising after round 2, indicates that the similarity of round 3 to round 2 is similar to round 4 to 3 and so on. However, from figure 2.29A it can be seen that the volatility between round 1 and 2 is much higher than between rounds 2 onward.

Based on these results, the choice of 2 stages is justified as the consistency of rounds 3 with 2 tends to be very similar to any further rounds as evident from the boxplots in panel A.

⁸⁴This is requirement (3) listed in section 2.4.6.

⁸⁵See figure 2.10.

⁸⁶This includes all mixed models but still excludes models based on the exclusion criteria.

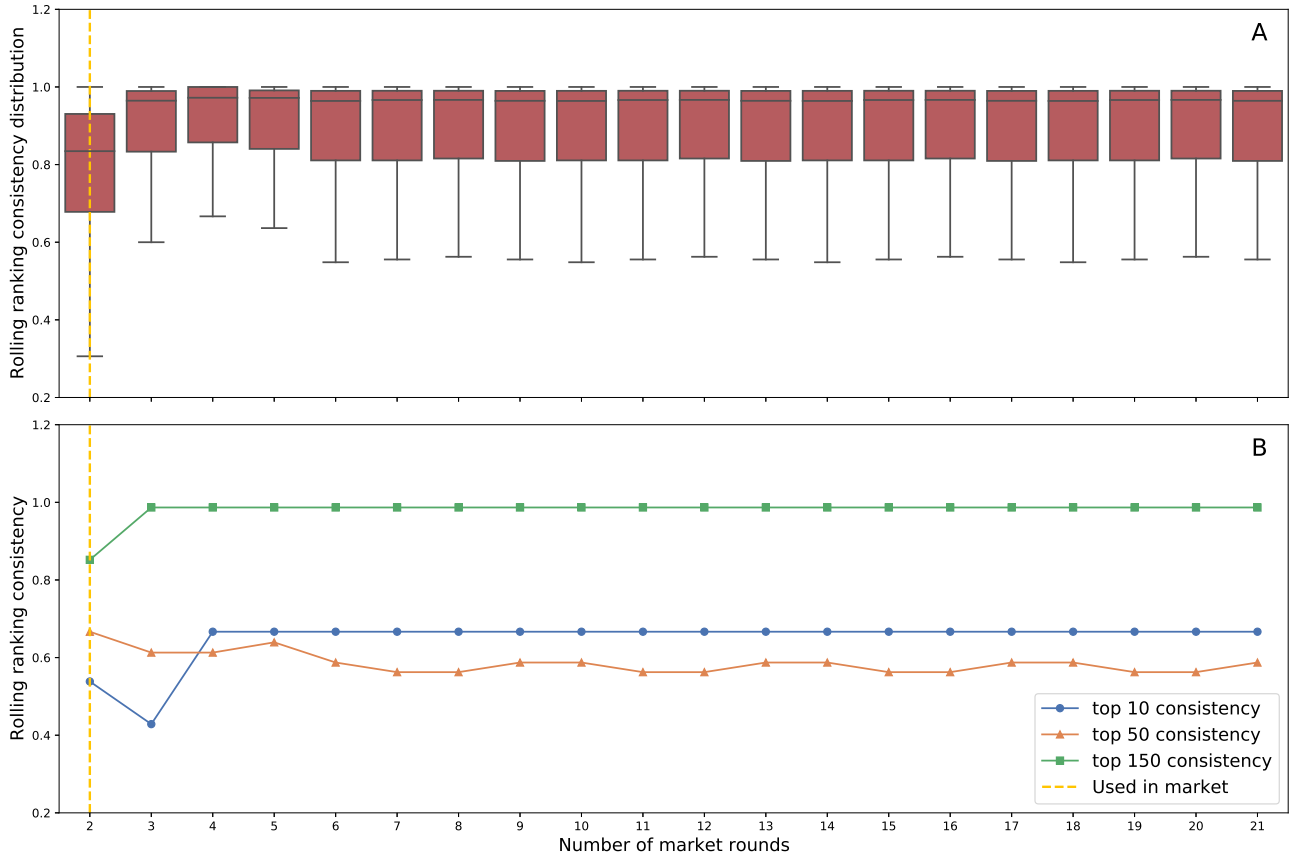


Figure 2.29: Rolling comparison of rankings when the profit metric is run for multiple rounds. **Panel A:** The distribution of rolling ranking consistency (i.e. what fraction of models in the top of round i were also in the top of round $i - 1$) is displayed for rounds 2 to 21. The dashed yellow line represents the metric used in the competition. Each box contains 252 values corresponding to increasing values of n in the consistency computations (equation (2.14)). Note that after stage 2, the differences between stages becomes immaterial. **Panel B:** 3 of the 252 samples within each boxplot in the top panel are displayed as a sample, to show their evolution over different market rounds.

2.6 Discussion

The preliminary results here suggest that competition based digital field experiments can unlock a new class of research questions. In this case-study I demonstrate that it is possible to simultaneously overcome the triple limitations of scale, participant expertise, and participant time-commitment by gamifying the experimental design, advertising it properly and in a highly targetted fashion, and ensuring the online environment is intuitive and engaging. Advertising and online engagement was only possible due to strategic partnerships with institutions representing the expert participants⁸⁷ and the online competition hosting platform [AICrowd](#).

⁸⁷National Actuarial Institutions of the United Kingdom, Australia and Singapore, along with 17 other national actuarial institutions that followed.

Over a course of 11 weeks we engaged with 2000 participants who registered for the competition from upwards of 50 countries (figure 2.23). More than 300 of the participants actively engaged with the online competition (figure 2.21). Although this accounts for only 15% of registrants, it is useful to remind ourselves that the total number of active general insurance companies in the UK is close to 340 [119]. Of these participants, approximately 60% self-report as experts in the insurance industry with an additional 30% identifying as data scientists. This split is also indicative of the paradigm-shift within the industry discussed before (see section 2.3). Finally, over 11 weeks, close to 80% of participants reported spending more than 2 hours on the competition weekly (see figure 2.22). This level of commitment from experts would normally not be possible within a laboratory setting, or within a typical digital field experiment.

2.6.1 Motor insurance pricing: *algorithmic Vs. data-models*

The case-study itself focused on studying the impact of algorithmic modelling on the motor insurance market. The key reason for picking the insurance market as the experimental environment was three-fold. First, we had access to real historical motor insurance data. Not only does having real data make the competition much more attractive to experts, but also, there is reliable evidence that third-party liability motor insurance is a highly commoditised product [120], ensuring that our cheapest-wins customer model is not too far from reality — minimising the complexity introduced by any assumptions on customer modelling. Second, taking inspiration from a wide body of work in natural experiments (see section 2.2), this study is performed during an ongoing paradigm-shift in pricing of general insurance. While previously data-models were the norm, the industry is slowly moving towards algorithmic modelling (see section 2.3). This made it possible to engage the community and be relatively confident that participant submissions will fall somewhat equally into these two camps. In fact the final split was approximately 60/40 tilted towards algorithmic modelling by the end of the competition. Thirdly, the insurance industry benefits from a very well-organised global network of professional institutions. This allowed us to gain access to a global participant pool through a select few strategic partnerships. These three factors, dataset availability, a potential on-going

paradigm-shift, and access to a global participant pool of experts, contributed to picking the motor insurance market as the subject of the case-study.

Preliminary analysis of the models studied suggests that consumers will pay less in cheapest-wins markets as algorithmic models become more prevalent. However, this effect is small (figure 2.25A) and requires confirmatory analysis. Regardless, given that the two models types have similar margins within the final leaderboard (figure 2.27) and are similar in their risk-estimation capability (figure 2.24) this difference in prices cannot be merely due to better risk-estimation or lower average margins.

Three of the possible sources for this discrepancy are lower offered prices, lower between-model correlation, or lower margins for algorithmic models. The first possibility is not entirely excluded and could explain the full effect (figure 2.25B). Additionally, less price correlation between different algorithmic models for the same policy (figure 2.26A) can translate into cheaper prices as well. This effect would be more likely if the differences in the margins of offered prices is negligible, which seems to be the case (figure 2.27).

However, these preliminary analyses do not delve deep enough to tease out the true source of the discrepancy due to limitations of time. In other words, why are algorithmic models more price-variant, and Does this fully explain the possible cheaper prices observed? The answer is most likely related to model-complexity and decision boundaries. This line of thought is expanded in “*Future work*” below.

The analysis above focuses on the difference between a fully stable algorithmic market, and a fully stable data-model based market — the before and after states of the on-going paradigm-shift. However, there is a transitory period in between⁸⁸ where the majority of models are data-models with a few selected algorithmic models in play. To simulate this environment, I conducted an experiment where a full data-model based market is slowly transformed into an algorithmic market (see section 2.5.5).

Preliminary results from this analysis suggest that in markets with both model types, data-

⁸⁸This is the phase we are most likely inhabiting right now.

models experience slightly better profit and market share outcomes, while both model types experience significantly larger levels of volatility, compared to algorithmic models (figure 2.28). In a cheapest-wins market, where data-models are more closely correlated than algorithmic models (figure 2.26) this volatility is partially expected, though further study is required. Further, the better outcomes for data models should be investigated as the current industry momentum is towards adoption of algorithmic models.

Today, if such a result is applicable to the real world, it can provide an disincentive for firms to swap to algorithmic modelling in this transitory period. Additionally, the work of Henrich et al. on risk-model diversity suggests that this volatility could be due to the higher levels of correlation between data-models [53]. This is despite a growing body of research that sometimes finds algorithmic models to be more accurate than data-models [82–85], which combined with the possible cheaper overall consumer prices observed here is contributing to the adoption of algorithmic models in the industry.

In our modern societies, large insurance and re-insurance firms are the great shock absorbers of the economy [121], not only by measuring risk and providing a safety net, but also by acting as institutional investors and keeping the financial pillars of the markets relatively risk-free [121]. However, this industry is experiencing a paradigm-shift in the most fundamental aspect of their service — risk pricing. In order to ensure the transition from data-model based markets to algorithmic markets is done smoothly, both insurance firms, and policy makers must ensure that they account for the competitive dynamics that may arise from mixing these model types in a market environment, and not analyse the impact of modelling on their bottom line in isolation.

The next two sections of this discussion will focus on the limitations of the case-study presented, and some suggestions for future work.

2.6.2 Limitations & assumptions

Competition-based experiments in general and, this case-study in particular, suffer from a number of limitations and were executed based on a set of assumptions. These limitations and assumptions fall into two categories: those related to the overall digital experiment design, and those relating to the research question of this case-study.

Limitations of competition based experiments

Competition based experiments generally suffer from two key limitations. Firstly, online data-science competitions typically have a very standard form and a set audience that may not intersect with our target demographic fully, introducing a potential selection bias overall. In other words, the Venn diagram of the group of people that always compete in data-science competitions, and the desired global participant pool for these experiments, is indeed not a circle. Therefore, it is likely that the expert participants in any data science competition will not necessarily be representative of the industries under study. However, it is unlikely that their expertise and domain knowledge will be differ significantly from their non-competing peers given the scale and global nature of the experiment.

Secondly, code-based competitions in particular have a set of unique challenges concerning the ease of participation, resulting in a second layer of bias. Typical data science competitions require only the submission of a solution file to participate. This would simply require the participant to generate a solution file on their local machine with their own customised environment. They then upload that file onto the competition webpage using a web interface. The insurance pricing game on the other hand, required a complex set-up process for successful submission to occur.

We mitigated the potential bias introduced by this submission difficulty in six ways. First, we allowed submissions in the two leading languages used by our target participant pool — Python and R. Second, we provided a very simple, yet flexible code template to allow participants to quickly submit their models, removing the overhead of structuring their code. Third, we

provided a fully browser-based submission pipeline through Google Colaboratory notebooks in both R and Python in addition to submissions through compressed files. Fourth, we provided two baseline models that participants could use to familiarise themselves with the submission pipeline. These were very simple out-of-the-box models⁸⁹ and are removed from the analysis. Fifth, we provided detailed error feedback to participants while ensuring that the error feedback cannot be used to gain insights into the unseen leaderboard data. This feedback mechanism (figure 2.19) allowed participants to rapidly resolve their issues, further reducing the overhead for participation. Finally, the research team, along-side Alcrowd had to adapt dynamically to the requests of participants by making technical changes to the platform, and ensuring that any issue is resolved as fast as possible when raised on the forums.

Limitations of the market simulation design

I identify 6 categories of limitations within this case-study, though more may exist. These are related to the data, customer model, profit metric, market incentives, model classification, and finally, planned analysis. Each of these will be discussed below and any mitigation strategies employed will be highlighted. If mitigation strategies were not possible, then possible improvements are discussed in the next section on “*future work*”.

Data limitations. First, there are limitations with respect to the dataset related to bias and relative dataset size. The policies in the data are a uniform sample of policies held by a particular insurance company such that each policy has 5 continuous years of history. This means that, most-likely, no policy holder in this sample decided to change their insurance company within these 5 years, this is despite evidence that customers who do make a claim tend to be more likely to switch providers [103] — an effect which is related to the claim size. Unfortunately, as we did not have access to more data, we could not feasibly correct for this effect without sacrificing the size of the leaderboards. However, this effect is understood to not be extremely large in industry.

⁸⁹One is a constant mean model while the other is a 3 variable logistic regression. See the template sections at the bottom of the [main webpage](#).

In addition to dataset bias, the number of policies in the dataset is small relative to the largest claim. As discussed in section 2.4.7 on market volatility, the ratio of the largest claim to the total revenue of a firm can be unrealistically high in our markets as the total market only has 100K policies, compared to many millions in the real world. As a result, the markets can exhibit high volatility. To mitigate, but not remove, this effect we capped the largest claims from 280K to 50K and increased the portfolio size of the leaderboards.

Although capping the claims improved participant feedback readability and reduced volatility, it also includes a set of concerns.

Firstly, the cap was introduced in week 5 of the competition which could be cause for concern, especially were this competition one where the results of one leaderboard impacted the next. However, due to the fact that the only leaderboard which resulted in prizes is the final one, 6 weeks after this change, the impact of this cap introduction was minimised.

Second, this issue could not be reliably identified earlier without imposing severe assumptions on the game outcome, or by delaying the game launch significantly. For example, one way that this issue could be identified would be through play-testing the game with a large participant pool — a time prohibitive task. Another way that this could have been identified is by access to detailed market share and catastrophe events data from the dataset provider, information that is confidential and was not provided to us. Lastly, we could have a-priori capped the claims at a low value to ensure this issue does not arise. However, that option would require the research team to make assumptions on the “undesirable” market dynamics that would emerge in order to implement a preventative cap.

Third, modelling very large claims is one of the key aspects of any insurance risk modelling task, and capping them seems to circumvent this issue. However, in this dataset, due to the relative claim to portfolio size, “very large” claims still exist. As an example, consider that position number 5 on the leaderboard has a total average profit of $\approx 50K$, meaning that a single large claim could still wipe them out. What has been essentially removed is an unrealistic of catastrophe level events in the 5 year history of the dataset through the capping of claims which would wipe out many insurers in the leaderboard. However, it is also possible that some of the

work required to model these extreme (50K+) events has been removed, even if their entire frequency in the dataset was fewer than 20.

Fourth, removing the cap can also impact the modelling decisions of participants mid-challenge. For example, they could have modelled the claim distribution in some way before the cap introduction. However, this is unlikely to be hugely impactful for two reasons. One, in figure 2.15 we can see that the cap value does not change leaderboard consistency values significantly, meaning that competitor models are somewhat agnostic to this cap; And two, the frequency of extremely large claims (that were capped) was very very small and they were considered to be unpredictable, and essentially random by the participants.

Fifth, regardless, there were two other options that were considered before introducing a cap. One, we considered scaling the entire claim portfolio down. However, this could not be done since a) some participants had hard-coded claim values in their models and b) it could simply have no effect at all since the prices could all adapt accordingly and shrink. Two, we considered creating portfolio copies, so that when one firm wins a particular contract, they actually win several copies of it. However, distributing the claim amount among these copies and deciding how many copies each contract type should have would necessitate too many assumptions and mitigate the benefit we hope to derive from using real data.

Finally, we did also consider introducing solvency rules, or a cost to participate in the markets. This could be done in two ways, first, they could be consistent across the leaderboard, and second, they could refresh with each leaderboard. Since the leaderboards in this game refresh each week⁹⁰ we could not reasonably implement the first. For the second, we could either introduce a base cost to each policy, or we could include features such as capital or reserves. We did not introduce a base cost as it would simply reduce the values on the profit leaderboard uniformly. Capital or reserves were also not introduced as, if they were to be shared identically by all participants and refreshed weekly, they would also not have any discernible effect other than moving the values on the leaderboard by a constant value.

It is possible that more complex capital reserves, or solvency requirements would allow for real-

⁹⁰a decision made to boost participation numbers and not give late joiners a disadvantage.

world dynamics to emerge, however, we did not have the resources at the time to investigate this feature.

Customer model limitations. Second, in these markets, customers are assumed to be fully price elastic. In practice, there is evidence that customers in insurance markets suffer from the “*loyalty trap*”. For example, millions of British drivers are automatically renewing their car insurance every year and overpaying as a result [122]. This will undoubtedly factor into the pricing strategies of existing firms. However, for the purposes of this case-study, we treat every policy as a new customer. This is supported by findings that motor-insurance is becoming increasingly commoditised, with a large fraction of customers searching for insurance on aggregator websites that approximate a cheapest-wins market [123]. This assumption is justified insofar as the actual competition is only executed on a single year of data and does not account for customer loyalty.

Profit metric limitations. Third, the profit metric designed in the markets may not fully allow for controlling market-size for future studies in a post-hoc analysis. In the real world, when a market-size is small, the players within those markets are also typically consistent. As a result, the behavioural dynamics that emerge are a by-product of learned behaviour by the players about their competitors [46]. However, in our case-study due to the randomised nature of the metric, competitors are effectively continuously competing in a market-size of 100-300 participants (i.e. the full leaderboard size) in a mitigated fashion. However, the market-size limitation and randomisation were unavoidable as they mitigated the effect of potential malicious behaviour by competitors. By randomising markets and limiting their size, we could ensure that no small participant group could effectively “*destroy*” our markets. While in a free-for-all cheapest-wins market, if a single participant decided to price at a marginal cost (or lower), no policies would be left for other competitors. The metric protects the overall leaderboard rankings from this type of possible bad behaviour.

Market incentive limitations. Fourth, our markets incentivise maximising profits for a single year of incoming policies. In realistic insurance markets, some firms may aim to maximise market shares in order to gain historical data to improve their risk-estimation or market

presence. Others may be maximising profitability⁹¹ rather than pure profits, while many will likely rely heavily on their investment arms to balance out their financials. However, none of these nuances are captured here. This was done to focus the research question on purely the risk-estimation and pricing aspects divorced from any other complications.

Model classification limitations. Fifth, the specific research question of data-models Vs algorithmic models requires a reliable classification of these two modelling approaches. This classification is inherently non-binary and requires expert judgement to be made on a case-by-case basis, making the process potentially error-prone. To mitigate these effects, the classification of models reported here were performed twice by different members of the team independently with points of disagreement discussed⁹², and were excluded from the analysis relating to modelling type.

Planned analysis limitations. Finally, the core question that the experimental design revolves around is about comparing two modelling approaches separately, while the experimental markets themselves mix these modelling approaches. It is possible that participants who adapt their models according to experience in purely data-model based markets will behave differently to those with experience in algorithmic-model based markets. Although we cannot control for this aspect in this study, preliminary evidence suggests that the effect of this mixing is not enough to counteract the observations I have reported here(see section 2.5).

2.6.3 Future work

The main aim of this work has been to show the potential of competition based digital field experiments — there is yet a rich set of scientific questions to be tackled with this dataset or this experimental methodology. These fall into three categories: those that are directly related to the analysis presented here, those that may be answered with the existing dataset but are not touched upon here, and those that would require a new experimental design.

⁹¹Profit per policy.

⁹²Sometimes with expert actuarial input

Work related to the analysis presented here

Using the existing dataset there are three categories of work that remains. First and foremost the results presented in section 2.5 require additional analysis to be confirmed. This analysis will require more rigorous statistical testing, beyond what is presented here, as well as stratifying the results based on risk-estimation ability, or profitability of the models presented.

After confirmation, the root cause of the observed effects should be investigated. Given that the differences between the data-models and algorithmic models most likely relate to the underlying decision boundary within each model type, investigating the root cause will require exploring these decision boundaries.

One approach to explore the decision boundaries could be through selective dataset noise injection. In practice, comparison of decision boundaries across different algorithms is a difficult task, however, since all the models here are trained to do the same thing, it is possible to investigate their decision boundaries without any knowledge of what the models actually are, through manipulating the input and monitoring the output. For example, by altering, in-turn, each column and measuring the impact on predictions it is possible to build first order visual representations of the decision boundaries. After that, the decision boundaries of the two modelling cultures can be compared, for example by measuring the number of regions that the dataset space is split up into by each model type, on average. Existing theoretical measures, such as the Kolmogorov complexity [124], could be investigated but they are unlikely to be fruitful as complexity measures are either designed for very specific algorithms without broad applicability (e.g. a complexity measure for some neural networks can be found here [125]), or they are intractable to compute (e.g. Kolmogorov itself).

Another approach to measure decision boundaries is to standardise all submitted models using model extraction techniques first, and then employ standard complexity measures for comparison of models. For example, each model, be it a GLM or a neural network, is first approximated by a very large Random Forest. This RF model would be trained on the input of the target models to predict their output. These models (one RF for each submission) can then be com-

pared to each other and their decision boundaries measured and visualised using standard complexity measures designed for random forests [126]. A random forest is used as an example here, the base model could be any one of a number of highly flexible algorithms. In fact several base models should be used and the results should be compared to ensure they all tell the same story and any emerging pattern is not an artefact of the choice of base model.

Remaining work using the existing dataset

There are entirely new research questions that can be addressed with the existing dataset. Here I will give detailed examples of 2 such questions and suggest preliminary analyses for each.

The first question is related to the source of profit margins in the absence of many real-world financial pressures. In figure 2.27 we can see that the average margins loaded onto prices never dip below zero, as would be expected in a cheapest-wins market where all firms have the same data and access to the same resources. The questions that arise are along the lines of, how do markets behave if models with similar risk-estimation ability with the same margins were to compete with each other? Is there a way to link uncertainty in risk estimation to margins?

In reality margins are effected in three ways. First, in a real market, insurers must pay a cost of capital to their shareholders which requires that they be profitable. This requires a non-zero margin to be loaded as most insurance firms have very active investment arms.

Second, the real market is not a commodity market perfectly. In practice, not only do customers care about the brand reputation, and identity, but also customers care about who the recommendation to join a firm is coming from, and so social network effects may play a role as well. If the competition is not totally perfect, then margins will not be exactly zero.

Finally, insurance pricing is inherently related to risk estimation. This risk is estimated differently by different models, much of it potentially unmeasurable. The participants, like real firms, are tasked to adjust for these by an amount that depends on their personal risk aversion. The margin that they land on eventually is made to cover the whole of this combination of risks. This stochastic risk necessitates a non-zero margin.

Since in this competition, capital markets are not simulated, and customers do not exhibit any brand loyalty or non-price related preference, it leaves the possibility that margin focused studies with this dataset would reveal the effect of risk estimation uncertainty on margins, potentially disaggregated by model type.

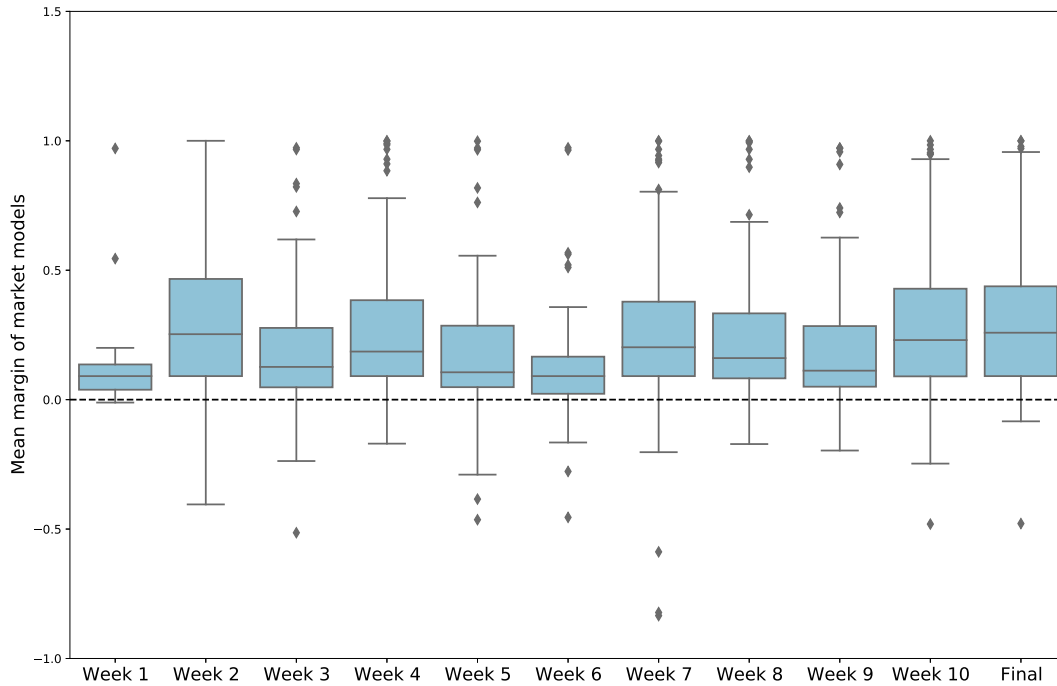


Figure 2.30: Weekly boxplots of the mean margin used by models in leaderboard markets are displayed. The horizontal dashed line indicates a margin of zero, expected in a commodity market. Note that all boxes are hovering above the zero line indicating a nonzero margin throughout the competition timeline.

Overall it would be interesting to see if there exists either a) an implicit consensus about what level of margin is acceptable in a competitive market only concerning risk-estimation. Or b) is the margins for a product that is priced based on risk completely related to the predictability of risk in the dataset. If the level of predictability was higher, would margins erode naturally as part of emergent competitive dynamics? What is the relationship between task predictability and margins? These questions could possibly be answered with the existing dataset.

The second question that this preliminary investigation is equipped to cover is whether our expert participants use gender in their pricing models. This question is of particular interest due

to the recent regulation that prohibits the use of gender in any part of the pricing process [101]. Here I would study whether the use of gender in pricing is related to better risk-estimation or not. If not, then this may suggest that the use of this variable is an artefact of training carried over with the expertise of our participants, rather than a result of the iterative data exploration and modelling process.

To measure the impact of gender on prices paid, the same approach in comparing consumer prices can be used (see figure 2.25). First, we must alter the underlying dataset such that the gender of the primary driver is flipped (e.g. male to female). Next, two versions of the market should be run, once with the gender-flipped dataset and once with the original dataset. Finally, we can plot the prices paid by women when labelled as women, against prices paid by the same women when they are labelled as men. A similar analysis should follow for men labelled as women. The distribution of these scatter plots with relation to the diagonal will shed some light as to the role of gender in pricing by our participants. However, this study must also ensure that the datasets are balanced and that the analysis controls for risk-estimation of the models under investigation. This is so that any observed effect can be attributed to a choice in modelling made by the participants rather than any parameter optimisation that takes place within the model fitting process.

Future directions of experiment design

Collecting new datasets using bespoke experiment designs is another promising direction for future competition based digital field experiments. Here I will outline 2 possible study designs, each focusing on a different research question. I will also propose initial experiment designs for each of these and list the required resources to execute them.

The first design would focus on studying the impact of price aggregators on consumers, companies, and the market. Price aggregators allow consumers to enter their information and find out which company is offering them the most competitive quote, say for car insurance. Although some theoretical studies exist [127], within a market their impact has yet to be studied empirically.

One study design to investigate the impact of aggregators would contain two phases. The primary phase would proceed identically to the original study here with some minor tweaks with relation to the rules regarding profit⁹³. Phase two is characterised by the introduction of a “*Price Comparison Website*”. In practice this would be a portal on the competition webpage which would allow participants to submit the attributes for a limited set of policies of their choice (similar to an actual price comparison website). They would then be able to retrieve price information from a set of competitor models that were deployed during the previous weekly profit leaderboard and adjust their prices accordingly.

Not only would such a design allow for studying the impact of an aggregator, but also it opens the door to an empirical study of algorithmic pricing. By limiting the amount of training data available to participants, the design could incentivise players to automate information extraction from the price comparison website. At the extreme this would involve fitting a model on the prices of others, thus leading to a model that algorithmically adjusts based on the prices of others. Taken to its limit, if everyone decides to abandon their original risk models, this would lead to a somewhat unorthodox market where no one is using a real risk-model. Instead all participants would partially rely on the prices of others.

Additionally, the existence of cumulative profits would allow the study of a host of new research questions. For example, observing how strategies evolve over time, whether initial testing is done or not, or the frequency of strategy change as the game progresses would be interesting. Additionally, investigating questions related to market entry and how firms evolve strategies when a new smaller player enters the market can have insightful policy-level implications.

The second research question of interest could be focused on the utility of high dimensional “*big data*” in insurance pricing. The current design was inspired by the “*Machine Learning*” revolution that is sweeping the insurance industry. However, pricing models only tell half the story. The other half concerns data complexity. In fact some believe that “*big data*” will invert

⁹³Notably, the profits should be computed cumulatively (i.e. the total profit is a function of the profit from the previous weeks). This would be to reduce market volatility (see figure 2.15) and disincentivise bad behaviour such as not submitting real models — submitting real models is a key requirement for the second proposed phase of the study.

the typical information asymmetry within insurance markets⁹⁴ to one where the insurer has more information about the insuree than vice-versa [128]. However, the predictability of claims data can be very limited and it might be the case that traditional claims data combined with existing modelling techniques is already at the limit of this predictability, with the gain from using more complex datasets being marginal.

Investigating the question of inverse selection, similarly lends itself to a two-phased design. The difference here is that the existing required dataset must contain both high-dimensional and legacy style data about the same policies. For example, this could be in the form of traditional and telematics data about the same drivers over a number of years. With this dataset in place, phase 1 of the design would involve only providing the traditional data to participants for a few weeks. In phase 2 the training data would then be enriched with new telematics information with the older legacy dataset potentially removed. At the analysis stage, the profitability of markets with only traditional data, a mixture of both, and only telematics data can be compared. This would be possible if advertising of the competition is done intentionally to boost participant take-up at the beginning of each phase. This would result in three groups of participants, those that joined at the beginning and continued to the end, and two groups that only played during one phase. Such a study would shed a preliminary light on the actual impact of high-dimensional data in the insurance context.

Finally, there are other areas of improvement in future studies. The customer model could be vastly improved by accounting for loyalty, renewal dynamics, and product preferences. Another area of obvious improvement is through expansion of the dataset size. One of the key limitations of this study is the relative size of the claims to the number of policies that can lead to potentially unrealistic competitive dynamics. Lastly, the impact of re-insurance and investment income together could be modelled by designing not one, but three parallel markets. One similar to the markets here, the other, a reinsurance market with the ability to insure the primary market, and finally the third being a synthetic simulation of the capital markets to allow the reinsurers to act according to market conditions. The work of Heinrich et al. [53] can act as a blue print

⁹⁴Where the insuree has more information than the insurer. This is referred to as “*inverse selection*” alluding to the inversion of the typical insuree-insurer information asymmetry.

for how such a game could be designed. However, such a game would require substantive time commitment and financial resources to execute.

2.7 Conclusion

Online digital field experiments vastly expand the ability of an experimenter to control and measure the environment. The primary goal of this case study has been to demonstrate the ability of this experimental format in simultaneously overcoming three challenges: scale, participant expertise, and time-commitment. These “*triple threats*” are often the reason why many market-based experiments are not able to delve into deeper interesting questions. With preliminary success along these three axes, I hope that this case-study can be built upon and expanded, potentially in other domains, to investigate and observe emergent human behaviour when experts are concerned. Other areas where this could be applied is university admissions where universities have to model student success and compete with other universities⁹⁵, bank loans where banks model credit-worthiness of clients, and even job applications — all areas that affect our societies at a very large scale.

⁹⁵My personal suspicion is that without a cumulative “*student success*” leaderboard, university success (measured as student outcome) would be highly volatile year-on year. That is to say, “*good*” universities are only good because they pick good students, resulting in a vicious (or virtuous) cycle. With an appropriate admissions dataset, and a willing participant pool, a design could be developed to test this hypothesis.

3

Privacy & reidentification risk

A case-study on the effect of dataset size

Many technological advancements carry risks — the advent of “*big data*” and machine learning is no different. In this chapter, I will touch upon the relationship between the right to privacy and technological advancement. I will focus on how technology often expands our ability to record private information about ourselves, and thus, necessitates a more clear definition of what is considered private. I will then dive into a specific case-study on mobility data which touches on how this new technological advancement (i.e. expanded data collection) has lead to a new conversation on privacy once more.

The main work in this chapter is focused on a case-study on the scalability of re-identification risk in mobility data. As this work was published, much of what I present here will be an expanded version of the published work. My specific contributions to the technical work here are a) designing and performing the experiments, b) building the models, c) helping with the mathematical results. These contributions are also reflected at the end of the publication itself [1].

3.1 Introduction

Many of our rights and freedoms began in much more basic forms than their modern day incarnations. For example, the right to liberty meant to protect against actual restraint and the right to life meant to protect against physical harm. Today the right to life is understood as the right to enjoy life, not just live it. The right to liberty is to be able to engage in many of the civil freedoms of modern society, such as voting, rather than just not being in chains. The right to privacy has undergone a similar metamorphosis over time [129]. Yet it is somewhat unique in that its evolution is closely tied to the advancement of technology. Arguably more so than other basic rights.

The advancement of technology has accelerated the means to gather and publish information about a person without their consent. Starting with private written correspondence which separated a person from their thoughts [130], concerns about the census [130], all the way to the monitoring of internet browsing habits [131] which can reveal things about us that even we are not aware of such as our personality, and other features [132–136]. The advent of photography had clear implications for the invasion of privacy, first appearing in the case of a young woman in the early 1900s whose photo was used without her consent for advertising purposes [137]. The right to one’s own likeness would never have been tested to such a degree had the tools to capture the likeness been nonexistent. Similar arguments have been made about wiretapping [138], lie detection [139] and many other forms of data collection.

As technology advanced it became possible to collect more homogeneous datasets [140]. Heterogeneous datasets contain clear rows and columns where each column contains a different type of information and each row contains information about a particular individual. Examples include census data, medical records at a typical doctor’s office, student grades and so on. Homogeneous datasets on the other hand normally contain event-level data about each individual being monitored. Here a “row” of the data would contain information about a particular interaction with a system. For example, Call Detail Records collect information about each text, call or internet connection request made by any phone on their network [141]. To construct the trace of one individual one has to search the database and construct a time-series from the events

that contain interactions from a particular phone. Most often, when there are references to “*big data*”, it is this homogeneous, often large scale and time-series format that is being discussed which is collected, passively, at scale, with very high frequency.

As collecting data became easier, so did analysing it, resulting in many communities realising and utilising the great potential of “*big data*”¹ [142, 143]. This potential has been realised in many different areas spanning academia and research, industry, and governmental organisations. In academia, new fields have emerged such a computational social science, bioinformatics, computational medicine and many others that are built on the back of large datasets [144]. Studies exploring large scale human migration patterns [145] all the way to those that investigate body language within an organisation to measure team efficacy [146] are the result of this kind of data. In industry, not only have new datasets led to entirely new business models such as online advertising [147], but they have revolutionised more traditional businesses such as the insurance [100], music [69], and legacy advertising [71] industries. Finally, government bodies and regulators are also using this new resource, from fraud detection algorithms by regulators [148] to automated methods to combat online criminal activity by the police [149]. In short, this data is useful.

As in the past, the information contained within these data silos is highly sensitive. Some of this sensitivity is obvious — many heterogeneous data sets, such as medical data, contain highly personal information including a few very sensitive attributes (e.g. what disease a person has). However, with homogeneous data, this sensitivity might be less obvious due to the fact that no single very sensitive attribute exists. The sensitivity here is contained within the patterns in the data as a whole. For example, online browsing history can reveal your personality [135] and mobility data has been shown to be deceptively rich in information. It can be used to predict an individual’s income [150, 151], their home and work locations [152–157], when they sleep and wake up [158–162], their gender and age [163], their personality [164], who their friends are [165, 166], and where they tend to socialise [167]².

So how do we use this information without putting the privacy of data subjects at risk?

¹Also see the discussion on paradigm shifts in chapter 1.

²Many more possibilities, not cited here, likely exist.

Anonymisation has been the historical answer. Anonymisation is the process of changing a dataset such that the data subject is not or no longer identifiable [168]. One of the more widely accepted definitions³ of anonymous data is that used by the General Data Protection Regulation (GDPR) [168]:

“[Anonymous data is] information which does not relate to an identified or identifiable natural person or to personal data rendered anonymous in such a manner that the data subject is not or no longer identifiable.”

In practice, this is achieved either through a combination of legal means such as the OpenSafely system [170], by modifying the data that is published such as k -anonymity based systems, or by limiting the level of control lost over the dataset through a query based system [171] such as Diffix [172].

Although standard anonymisation techniques have had moderate success on heterogeneous data, there has been growing evidence that these methods are not as successful on high-dimensional, homogeneous, big datasets as evidenced by a large volume of literature on re-identification attacks. In fact, supposedly anonymised data of this type has been re-identified in the case of Netflix movie ratings [173], numerous studies on mobility data [160, 174], cryptocurrency ledgers [175], credit card transaction histories [176], mobile phone applications [174], public transport smart-card data [177], genomic data [178], social network data [179], and many other cases.

Many of these re-identification case-studies report very high levels of risk. Yet, despite the high levels of risk highlighted by the literature, some criticism has pointed out that due the relatively small size of the studied datasets, the risk could be over-estimated [180–182], arguing that with a larger dataset, the risk would be acceptably small [181]. Intuitively, it is harder to be unique in a larger population compared to a small one. In fact, revisiting a 2015 study, one criticism suggested that the risk of re-identification can be reduced by a factor of 95 when

³Note that this is a “*negative definition*” in that anonymous is only defined by its absence as a property of data. For an interesting discussion on the informativeness of these kinds of definitions see the early work of 20th century philosopher Wilmon Henry Sheldon [169].

increasing the population from 1.5M to 20M [181]. However, until very recently, the impact of size on the re-identifiability of these datasets remained unknown.

In this chapter I will present a published case-study that shows the effect of dataset size does not significantly counteract re-identification risk. The importance of this is highlighted by summarising the existing work in the area of technical data privacy with a heavy focus on attacks on data considered anonymous. The case is then made that dataset size is not sufficient protection using 3 months of location data of 1M people in a European country. It is shown that 1) the empirically measured risk of re-identification decreases slowly with dataset size, 2) that this decrease can be approximated with a simple model taking into account 3 population-wide marginal distributions, and 3) it can be proven that unicity (a re-identification risk metric) is convex with a linear lower bound. The estimates here show that 93% of people would be uniquely identified in a dataset of 60M people using 4 points of auxiliary information, with a lower bound at 22%. This lower bound increases to 87% when 5 points are available. Taken together, these results show how the privacy of individuals is very unlikely to be preserved even in country-scale datasets.

3.2 Background

The origins of the modern notion of a right to privacy can be traced back to “*the right to be let alone*” [129] — this is still the core of what privacy protection strives to achieve. This idea evolved from not having the internal domestic on-goings of a family published by the press. Over time, with the advent of new technology the fear that “*what is whispered in the closet shall be proclaimed from the house-top*” started to inch closer to reality. Perhaps the best way to see this link between technology and privacy is by observing the arms race between privacy-invading technologies and legal remedies meant to protect this right.

In line with changing technology, the law has had to adapt over the years. Table 3.1 lists a few key steps that show how technology has required a more precise definition of the legal right to privacy. For example, when technology was able to listen in on a private conversation inside

Technology	Early example of legal case or law	Date
Modern mail	Postal Service Act (USA) [186]	1792
Analogue Photography	Roberson v. Rochester Folding Box Co. (USA) [140]	1902
Telephone	Olmstead v. United States [140]	1928
Mobile phones	Privacy Act of 1974 (USA) [187]	1974
Internet	EU Data Protection Directive (EU) [188]	1995
AI and Machine learning	European regulation on AI [189]	Pending

Table 3.1: An illustrative summary of legal mechanisms put in place to respond to the privacy invading aspects of new technology.

a house through wiretapping, the law responded by restricting that ability [138]. Similarly, debates surround internet browsing history [183], the use of key loggers [184], and famously a case involving Google collecting WiFi hot-spot information at a very large scale [185], all re-ignited interest in what society should consider private.

With the introduction of GDPR [168], the law began to take a more, future-proof definition of what is private information. It defines “*anonymous*” data as one where “*the data subject is not [...] identifiable*”. Other documents, notably the opinions of the Article 29 working party on anonymisation techniques [185] go further to provide guidance on assessing and testing for anonymity.

3.2.1 An overview of privacy attack models

The opinions of the Article 29 working party set out 3 principles for assessing anonymity that are still used today [185]. These are:

1. **Singling out.** The risk of singling out an individual
2. **Linkability.** The ability to link multiple records relating to an individual, without necessarily identifying an individual
3. **Inference.** The ability to infer information about an individual

Almost every privacy attack on a dataset will include at least one of these features as we will see in this overview. However, the vast majority of the literature on privacy attacks focuses

on the ability to single out an individuals' record. The reason for this is that, barring a few special cases (e.g. group privacy), in order to reveal private information about an individual, that individual first must be found.

To be clear, these principles are focused on the privacy of individuals, and have less focus on the notion of group privacy (i.e. where one discovers information about a group) as well as privacy from an institutional standpoint which covers everything from business to state secrets. An expanded discussion of group privacy [190] can be found elsewhere, as can a treatment of privacy in institutions [191].

To be able to assess the effectiveness of any privacy attack, the notion of an “*attack model*”, borrowed from the cryptography literature, is necessary [192]. In privacy, an attack model is a framework used to ask questions about a scenario and structure our thinking around a given privacy attack. In privacy, an attack model has 3 components:

1. **Dataset.** A description of the target dataset under question including all the information it contains.
2. **Goal.** Which outlines what the sensitive information is, and what the attacker wishes to achieve.
3. **Auxiliary information.** An assumption on what tools and data the attacker has access to in order to achieve the goal.

A stylised toy example to highlight the usefulness of having an attack model is a comparison of the two statements below:

- **No attack model.** Allan will aim to discover the address his previous boss (*goal*).
- **With attack model.** Allan will aim to discover the address of his previous boss (*goal*) using published voter registration information (*dataset*) in combination with the name and age of the target (*auxiliary information*).

Although without an attack model, the broad aim of the privacy attack is understood, technically there is little that can be said about the likely success of Allan. This is clarified by the second statement which satisfies the three key requirements (i.e. dataset, goal, and auxiliary information) needed for an attack model. One can then go on to ask more detailed and technical questions to assess the actual risk of re-identification of Allan's previous boss.

Given that most empirical attacks focus on singling out, their attack models have some similarities. Namely many share the following:

1. **Goal:** To single out any individual record
2. **Auxiliary information:** Access to the dataset as well as partial information about any individual record.

The specifics of each attack will differ depending on the use-case and context in which the dataset is published.

Finally, before delving into examples of privacy attacks it is worth remembering that all of this analysis is conducted to enable data to be used safely, and sometimes, publicly. Although in this chapter, I will focus on privacy attacks rather than uses of data, any analysis of the risks of sharing data should always consider the use-case in question and the benefits that could come from that use. Regulations such as the [GDPR](#) aim to account for various ways under which anonymous data can be used, yet current criticism of the regulation focuses on both the difficulty to implement and use the data under such strict requirements. Some go so far as to suggest that privacy regulation hampers innovation and a more equitable trade-off is necessary [193]. After all, utility is why we even attempt to anonymise data to begin with.

3.2.2 Data sparsity and homogeneity

In much of the privacy literature there are references to “*big data*” and being vulnerable to re-identification attack. Here I will briefly expand on 2 features of datasets which will be used

to categorise two types of privacy attacks. These features are dataset sparsity and dataset homogeneity.

Sparsity

Given that singling out is one of the most commonly executed privacy attacks, the ease of singling out is naturally a very important dataset feature. The ease of singling out attacks relies on the ability of easily accessible auxiliary information to uniquely identify a record. In turn, this is related to the “*sparsity*” of the dataset.

With singling out in mind, dataset sparsity can be defined as the inverse of the fraction of the possibility space that is occupied by records within a dataset. More formally, an intuitive measure of sparsity of a dataset (D) can be $\mathcal{S}(D)$:

$$\frac{1}{\mathcal{S}(D)} \propto \frac{\text{Number of actually unique records}}{\text{Number of possible records in the entire space of records}} \quad (3.1)$$

There exist a few intuitive examples of sparse and non-sparse datasets. Examples of highly sparse datasets include, any dataset with a free text field⁴, highly granular mobility datasets, genome datasets, web browsing data and so on. Examples of highly non-sparse data include large datasets of, first names and gender or one with age and city.

Homogeneity

Dataset homogeneity in this work refers to the type of information contained within each column of the dataset. If each piece of information within the record of an individual has a different “*type*” than other pieces, then it is not homogeneous. Otherwise it is. Typically homogeneous datasets are large-scale, time-series, and highly sparse datasets.

⁴Here the number of possible unique texts in a dataset is far smaller than the number of possible ways that the alphabet can be arranged in a free text field.

ID_INIT	ID_REC	CALL_TYPE	DURATION	TIME	DATE
065978198280	0612283725	MOC	90	12:00:00	01/01/2012
065978198280	0613069656	MOC	82	12:02:27	01/01/2012
065978198280	0613481951	MOC	78	12:04:41	01/01/2012
065978198280	0600000001	MTC	92	12:07:13	01/01/2012
065978198280	0600000002	MTC	94	12:09:40	01/01/2012
065978198280	0612063352	MTC	114	12:12:40	01/01/2012
065978198280	0613103364	SMS-MO	0	12:13:42	01/01/2012
065978198280	0613751973	SMS-MO	0	12:14:44	01/01/2012
065978198280	0613672843	SMS-MO	0	12:15:44	01/01/2012
065978198280	0612769488	SMS-MT	0	12:16:42	01/01/2012
065978198280	0613164676	SMS-MT	0	12:17:39	01/01/2012
065978198280	0613399901	SMS-MT	0	12:18:39	01/01/2012

Table 3.2: A fictional example sample from a Call Detail Record dataset. This sample contains interactions by a single subscriber with the network within a 20 minute period. In other words, this entire table would only account for 20 minutes of one individual, data about whom is usually limited to a single row in heterogeneous datasets.

For example, in a dataset of medical records, the columns (i.e. a piece of information) could describe the age, gender, disease, marital status and so on, of the data subject. This means the information about a data subject is heterogeneous. Similarly, census data, most survey datasets, and tax records are all heterogeneous.

Examples of homogeneous datasets include histories of transactions, mobility, web-browsing or even movie ratings over time. Many other time-series datasets fall into this category. In addition, other examples include genomic data, a list of downloaded mobile phone applications or even a list of most commonly purchased items by a grocery shopper. As a visual example consider the data displayed in Table 3.2. Here each row represents one interaction of the same subscriber with their mobile network. The entire table, in turn is the record of one individual. This would correspond to a single row in a heterogeneous dataset.

3.2.3 Privacy in heterogeneous (small) data

In the late 1990s, MIT PhD student Latanya Sweeney performed a famous privacy attack on a publicly available dataset. At the time, the governor of Massachusetts had decided to publicly release as dataset of medical records with identifiable information retracted (i.e. pseudony-

mous). Realising that the combination of age, zip code⁵, and gender [194], singles out the vast majority of records⁶ within the dataset, Latanya, after having warned the Governor of the risks, mailed him a copy of his medical record having been singled out from the dataset. Figure 3.1 demonstrates the intuition behind the attack.

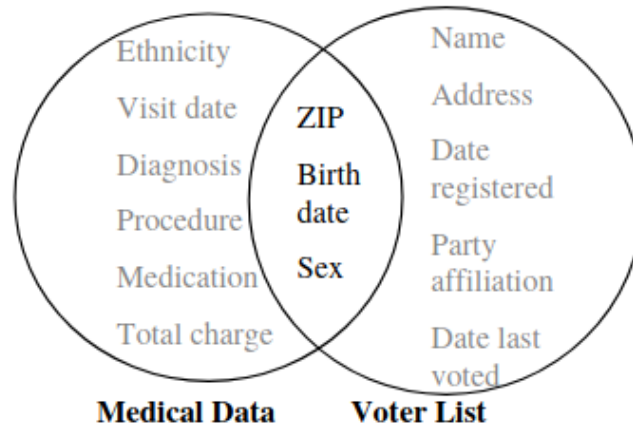


Figure 3.1: An illustrative diagram of how Sweeney performed the original linkability and singling-out attack on the governor using publicly available voter registration data. Figure from original publication [194].

In the immediate aftermath of this attack, Sweeney formalised the concept of k -anonymity as a property possessed by some anonymous datasets, that protects individuals against being singled out [194]. This property, defined most intuitively for heterogeneous data⁷, relies on a distinction between what is considered auxiliary information available to the attacker, and what is considered the secret sensitive information. The auxiliary information are typically knowledge that is publicly available or easy to obtain such as the age, gender and zipcode of a target. The secret information could then be the type of disease if the dataset in question is a medical dataset. In technical terms, auxiliary information is considered “*quasi-identifiers*” and the secret is considered “*sensitive*” information. In a k -anonymous dataset, no combination of quasi-identifiers can lead to fewer than k records in the dataset, thus masking their sensitive information within groups of minimum size k . Table 3.3 summarises these concepts in a stylised example dataset containing medical information.

⁵Age and zipcode were obtained from public voter registration lists.

⁶A combination that was later estimated to single out 87% of the US population at the time.

⁷In fact, Charu Aggarwal has shown that k -anonymity does not translate to high-dimensional heterogeneous data due to the issue of sparsity noted before [195].

ID	Quasi-identifiers			Sensitive information
	sex	dob	zipcode	disease
800f48...	female	28-09-1955	43221	hypertension
f1f333...	female	06-01-1955	43210	hypertension
3b3c6...	male	08-07-1959	89127	prostate cancer
93db7...	male	10-06-1959	89127	gastritis
e1d54...	male	27-09-1959	89127	skin cancer

Table 3.3: An illustration of sensitive information, and quasi-identifiers. Note that in this particular example, if the date of birth is coarsened to the nearest year, and the zipcode to the first three digits, the dataset would become 2-anonymous.

Shortly after the introduction of k -anonymity, weaknesses within the framework were discovered that led to further refinements of the concept, namely ℓ -diversity [196] and t -closeness [197]. The protection that k -anonymity offers is against singling-out. However, it is entirely possible for a particular equivalence class⁸ in a k -anonymous dataset to all share the same auxiliary information. In this case, although no individual record is singled out, no protection is offered against an inference attack. This can happen when all members of a particular equivalence class share the same sensitive information⁹. The property of ℓ -diversity, protects against this type of information leakage, by ensuring that no record within an equivalence class will have fewer than ℓ unique values for the sensitive information [196].

However, even if ℓ -diversity is respected, it is still possible that the distribution of sensitive information within an ℓ -diverse equivalence class leaks significant information. For example if the chance of a record having heart disease is 3 in 4 given their equivalence class, this information is significant if the 3 in 4 chance is substantially different from the rest of the dataset¹⁰. The principle of t -closeness was introduced as a measure to ensure the distribution of sensitive information is within a distance t (to be defined on a case-by-case) to the dataset-wide distribution of sensitive information [197].

⁸This is a group of k records that share the same quasi-identifiers, and so, occupy the same anonymity set.

⁹For example if there two men living in the same zipcode, with the same age are both gay (sexuality being the sensitive attribute), although they could be 2-anonymous, an attacker finding these two would know their sexuality without singling them out. This is an example of an inference attack without any aspect of linkability or singling out as described by the Article 29 Working Party guidance [198].

¹⁰Intuitively, the distribution of the base-population, not just the dataset, should also be considered. However, this is no longer t -closeness.

k -anonymity, among other generalisation methods, are designed to secure privacy when the dataset is eventually released (potentially to the public); this is not without risks. For example, none of these methods protect against future data releases. If, after a dataset is made k -anonymous, the data is updated either directly, or by proxy through the publication of a second dataset, new quasi-identifiers are released which can potentially break the k -anonymity property [199, 200].

Despite demonstrated weaknesses, when used properly and in combination with other methods, k -anonymity today can be reliably used as part of the standard toolkit to anonymise certain datasets. In fact, k -anonymisation is one of the few techniques that is explicitly discussed by the Article 29 Working Party opinion on anonymisation techniques [198].

Instead of altering the data and releasing it, another paradigm is to keep the data and give controlled access to designated users. This is the principle of most Query Based System (QBS). Generally speaking, query based systems use one or more of the following methods, often in combination with legal safeguards as well. These methods are:

1. Pseudonymised access
2. Differential privacy
3. Ad-hoc noise addition

Pseudonymised access relies heavily on user authentication and is the framework most commonly used by query based systems. This is where the data controller provides access to a pseudonymous version of the dataset to users that have been authenticated via technological and legal means. The OPAL platform for using large-scale mobility data [201], or the Open-Safely platform for medical research [170] which has led to 9 peer-reviewed high-impact medical publications [170, 202–209] in 1 year are examples of these query based platforms. The framework used by the famous Swedish NGO Flowminder also falls into this category, with their most recent use of the technology focusing on the current COVID-19 pandemic [210]. Flowminder famously used Call Detail Records during the disastrous Nepal earthquake of 2015 to help the

United Nations organise an emergency response [211]. Although pseudonymised access can work very well in combination with legal safeguards, it is not an easily scalable framework and will always rely on detailed due diligence and pre-existing vetting processes which cannot be automated today.

Differential Privacy (DP) is another framework that is designed for use in query based systems. Using mathematical guarantees, DP protects a particular definition of individual privacy, namely, membership in a dataset. As a framework, studied elsewhere extensively [212], DP describes how to add noise to a dataset so that the result of any computation on the dataset would be indistinguishable if any one record were to be removed. In other words, the noise is added in such a way to make it impossible for an attacker to discover information about any one data subject through creative uses of the QBS.

In practice, DP faces issues along three dimensions: a) utility, b) broad applicability and, c) deployment. Concerning utility, DP uses a tunable parameter (i.e. the privacy budget) that measures the strength of the DP guarantee [212]. Although many private and public bodies such as Apple, Google, LinkedIn, Microsoft, and even the US census [213–217] use DP, existing criticism suggests that either the results are either not usable due to a very low privacy budget [218] or effectively not private due to a high privacy budget [219]. The second issue, broad applicability is related to the fact that most DP frameworks define privacy as the effect of a single record being observable within the dataset. If no single record can be identified then the dataset is considered differentially private. However, this does not account for many normative privacy violations that may not be violations of differential privacy [220]. For example, a bank publishing the fact that 80% of people living in a specific neighbourhood have defaulted on one loan payment in the past could be entirely differentially private while some would argue that it is still a normative privacy violation. Finally, the deployment of DP systems is notoriously difficult. The US census, noted above, famously incorporated DP within their 2020 data gathering efforts and faced many issues with operational details such as a lack of trained personnel in both theory and operations, and a lack of verified DP toolkits [221].

The third type of query-based systems are those that use ad-hoc noise addition. This is where

data analysts are given access to a system which can be used to run analyses on the data and retrieve noisy answers, (similar to DP frameworks). An example of such a system is Diffix [172]. Diffix used a noise addition mechanism which could be exploited to infer private information about the records within the system [222]. Although Diffix later fixed this vulnerability [223], such attacks highlight the need for any query-based system to either be based on mathematical guarantees or be supplemented by access control mechanisms that can be legally enforced.

Overall, anonymisation techniques have had modest success in de-identifying small heterogeneous datasets. They have been in use historically, sharing data with limited risk when done properly¹¹. However, if de-identification is not done properly, there are risks. Data initially considered anonymous has been subject to re-identification attacks. From improper pseudonymisation [177], to poor noise addition [173], or even data made available through a query based system [222]. However, these risks seem to be much larger when we move from heterogeneous datasets, to large, sparse, homogeneous datasets.

3.2.4 Privacy in homogeneous (big) data

Homogeneous “*big data*”, in the context of privacy, refers to datasets that are sparse, often collected as a side-effect, and usually contain many pieces of information about an individual. In this category contains many time-series datasets such as mobility data, movie ratings, transaction information, browsing histories and others. Genetic data also falls into this category as well. Similarly much of the datasets collected as a result of Internet of Things technology is considered big and homogeneous [224].

Unlike small heterogeneous datasets, these larger homogeneous datasets have a rich history of being re-identified. Narayanan and Shmatikov famously showed that close to 90% of people could be re-identified in the Netflix movie ratings dataset [173], while Riederer and Chaintreau used the unicity of traces to match the same individual across multiple datasets [225]. More broadly unicity (a metric for re-identification risk) has been used to quantify re-identification risk across a number of domains, including the mobility of vehicles [226], apps downloaded by

¹¹Although, often times what is considered “*proper*” today may not be considered as such in the future.

smartphones over time [227, 228], smart cards used in public transport [160], credit card transaction histories [176], and location data from mobile phones in a number of countries [174, 229, 230]. Beyond that, genetic data has been subjected to successful re-identifications attacks [231], as has data from seemingly anonymous cryptocurrency ledgers [175].

To measure the high levels of exposure to singling-out attacks for these datasets, the metric, unicity, was introduced in 2013 [174]. In the original case-study, the authors, using a dataset of 1.5M location trajectories over 1 year, showed that knowledge of only 4 events (e.g. calls or texts) can single out approximately 95% of the data-subjects. More formally, unicity (ε_p) is defined as the fraction of trajectories that are unique based on knowledge of p randomly chosen points in a given trajectory¹².

Studies such as the original unicity case-study, led to a wider conversation about the efficacy of anonymisation on high-dimensional, unstructured datasets. In fact the Article 29 Working Party opinion on anonymisation techniques [198] cites both the unicity [174] and Netflix [173] studies. Beyond that, in the US in 2014, the President’s Council of Advisers on Science and Technology [232] stated that

“Unfortunately, it is increasingly easy to defeat anonymization by the very techniques that are being developed for many legitimate applications of big data. As the size and diversity of available data grows, the likelihood of being able to re-identify individuals (that is, re-associate their records with their names) grows substantially.”

Although, it should be stated that at the time, the distinction between pseudonymisation and anonymisation was not as clear as it is today. That clarity came later, most notably with the GDPR in 2018 [168] which explicitly recognises the risks of pseudonymised data above and beyond those associated with anonymised dataset.

Although almost, every example above deals with simply anonymised, or often pseudonymised datasets, the ability to anonymise and publish homogeneous big data is at best, an open research question, and at worst, highly unlikely. In location data for example, studies have

¹²See equation (3.7) for the full definition.

shown that reducing the time and space resolution of the data did not meaningfully decrease re-identification risk [174]. In fact, as noted earlier, in the work by Charu Aggarwal, it is proven that achieving k -anonymity style privacy is highly unlikely in high-dimensional datasets due to a lack of a desirable privacy-utility trade-off in such paradigms of privacy protection [195]. Similarly, other studies such as Gramaglia et al. have shown that mobility datasets do not grant k -anonymity and conclude that mobile phone data require extreme spatial and temporal generalisation in order to slightly improve user privacy [233]. Similarly Song et al. experiment with resolution reduction, and do not find a substantial decrease in unicity [234]. In credit card transaction data as well, there is preliminary evidence that there is no desirable balance between privacy and utility [176], with highly perturbed datasets with arguable utility still being associated with very high levels of re-identification risk. In addition, Boutet et al. have conducted an extensive study on Location Privacy Preserving Mechanism (LPPM)s on mobility data and have found [235]:

“Similarly to previous studies, we show that reducing the uniqueness through generalisation requires a very coarse-grained information which drastically limits the utility of the data. Lastly, we show that a de-anonymisation attack re-identifying anonymous users using only a limited background information on the mobility habit of users provides a very high success rate, even if the raw data are obfuscated by classical LPPMs.”

Although the evidence suggests that many homogeneous, high-dimensional datasets are re-identifiable, some researchers and industry practitioners have, argued that these high unicity numbers are an artefact of the small size of the datasets considered, and are overestimating the risk of re-identification [180–182]. This is understandable as the probability of being singled out must decrease as the pool of possible matches increases, controlling for the available auxiliary information. In one of the previous studies for instance, Riederer et al. [225] relied on a location dataset of 1.7k people, while other case studies report unicity on dataset sizes ranging from several thousands (resp. 12k, 55k) [226, 227] to over one million people (1.5M) [174]. It can be argued that some data controllers (notably large telecommunication companies) hold datasets

of tens of millions of individuals [236]. Maybe their size effectively renders their datasets anonymous, or so goes the hypothesis. In fact, examining a published study [176], El Emam et al. estimated that the unicity of a dataset of $\approx 20\text{M}$ trajectories will be as low as 1% given four points of auxiliary information; the conclusion being that privacy is preserved in such large datasets [181].

Answering the question: “*how does unicity scale with dataset size?*”, is the key aim of this chapter. To answer this we 1) study three months of location data and show empirically that unicity decreases slowly with the size of the dataset, 2) approximate this decrease with a simple statistical model taking into account three population-wide marginal distributions along with the underlying geography, and 3) prove that the decrease of unicity is a convex function of the dataset size and obtain a linear lower bound on unicity. Finally, I perform a sensitivity analysis suggesting that the decrease in unicity is agnostic to broad perturbations to the input distributions. These results disprove previous claims, instead showing that unicity is likely to remain high even in country-scale datasets.

3.3 Methods

To answer study the effect of size on re-identification risk, here I focus on a published case-study using mobility data [1] which is a type of homogeneous sparse data. For heterogeneous data, a study by Rocher et al. attempts to answer the question elsewhere [237].

The results of the scaling experiment are constructed using 3 different methods:

1. Empirical measurement of unicity
2. Building a unicity model to extrapolate to larger populations
3. Computing a theoretical lower bound on the unicity

In this section, after describing the dataset, I will describe how each of these steps were performed before demonstrating the results in the next section.

3.3.1 Dataset description

The dataset in this study consists of the location traces of close to 1M individuals who have visited at least 10 distinct locations over 3 months for one telecommunication operator. Each record in the data consists of a timestamp with hourly granularity, the GPS coordinates of the antenna the user was connected to the most during the hour, and a unique pseudonym. The records are generated when an individual interacts with the mobile phone network (i.e. makes/receives a call or sends/receives a text). Further, the data contains nearly 6000 antennas that span the whole of the country. Antennas in our database are distributed throughout a country with a heterogeneous geography. The median area covered by an antenna is 16 km² with smaller antennas covering areas closer to 0.05 km². More than 50% of antennas cover areas that are less than 10 km² and the distribution of subscribers across the country roughly follows its population density. Finally, for each trajectory in the data, multiple antennas can be associated to one hour (e.g. multiple calls made within the same hour). In such a case, the most popular antenna within the hour is picked, with ties broken randomly.

3.3.2 Unicity: *difficult to compute, easy to estimate*

Unicity is a measure for the risk of re-identification in homogeneous anonymous datasets [174]. It has previously been used to show that four points of auxiliary information (time and location points) are enough to uniquely identify 95% of people in a large-scale location dataset [174]. These four points of auxiliary information could be in the form of geo-tagged “*tweets*”, online check-ins or information obtained by more traditional means, such as observing someone making a call. Unicity (ε_p) is defined as the fraction of trajectories that are unique based on knowledge of p randomly chosen points in a given trajectory.

Computing the exact unicity of a dataset D is extremely computationally expensive (see equation 3.7). Therefore, unicity is computed empirically in line with previous work [174]. Here the original algorithm is adapted to compute unicity over a range of population sizes, thus measuring how re-identification risk scales with dataset size. To do so, unicity is estimated by

randomly sampling a smaller data set of users $Q \subset D$ and computing the fraction of trajectories in Q that are unique in D given p randomly selected points from each user in Q . Namely, from each trajectory $T^{(i)} \in Q$, p points are picked uniformly at random, labelled $A_p^{(i)} \subset T^{(i)}$. The unicity estimate ε_p is then the fraction of trajectories in Q that are unique in the entire data set D , given $A_p^{(i)}$ as auxiliary information (see algorithm S2 of [176]). With a large enough Q , this process can result in a very accurate estimate of ε_p in D [174].

3.3.3 Building the unicity model

In order to extrapolate beyond the size of the existing dataset a simple statistical model of unicity is built. This model, $\mathcal{M}$ takes into account three population-wide distributions: activity (P_A), circadian (P_C), and frequency (P_F). It then samples location traces for each user independently of other users to estimate unicity of a dataset of size N . These location traces are then grouped together to compute unicity.

Figure 3.2 displays the information extracted from the dataset, three distributions and the antenna network. (P_C) characterises the circadian cycle, the overall likelihood of a record to occur at a given time in a week. The existence of circadian cycles are well-documented in the computational social science literature [158, 159, 161, 162] and their empirical form is used in the model. The frequency distribution, (P_F), is the relative overall likelihood of a location to be visited. This distribution too has been studied before and has been widely shown to be well approximated by a power-law distribution [238–242] as is also the case here (figure 3.2B, $R^2 = 0.99$). Note that the top 10 most popular antennas (inset of figure 3.2B) captures 85% of the activity for the average trajectory. The activity distribution, (P_A), captures the number of records $\ell_{(i)} = |D_i|$ that appear in each user trace. It is approximated here with a β distribution ($a = 1.72$, $b = 14.7$, $R^2 = 0.98$). Finally, S_i is the set of locations visited by person i . It is a sub-graph sampled from the Delaunay tessellation¹³ of the antenna coordinates ($\mathcal{L}$) in the dataset.

¹³The Delaunay tessellation of a set points, splits the space into triangles with the points as vertices such that no point is inside the circumcircle of any triangle. See the original work for more details [243].

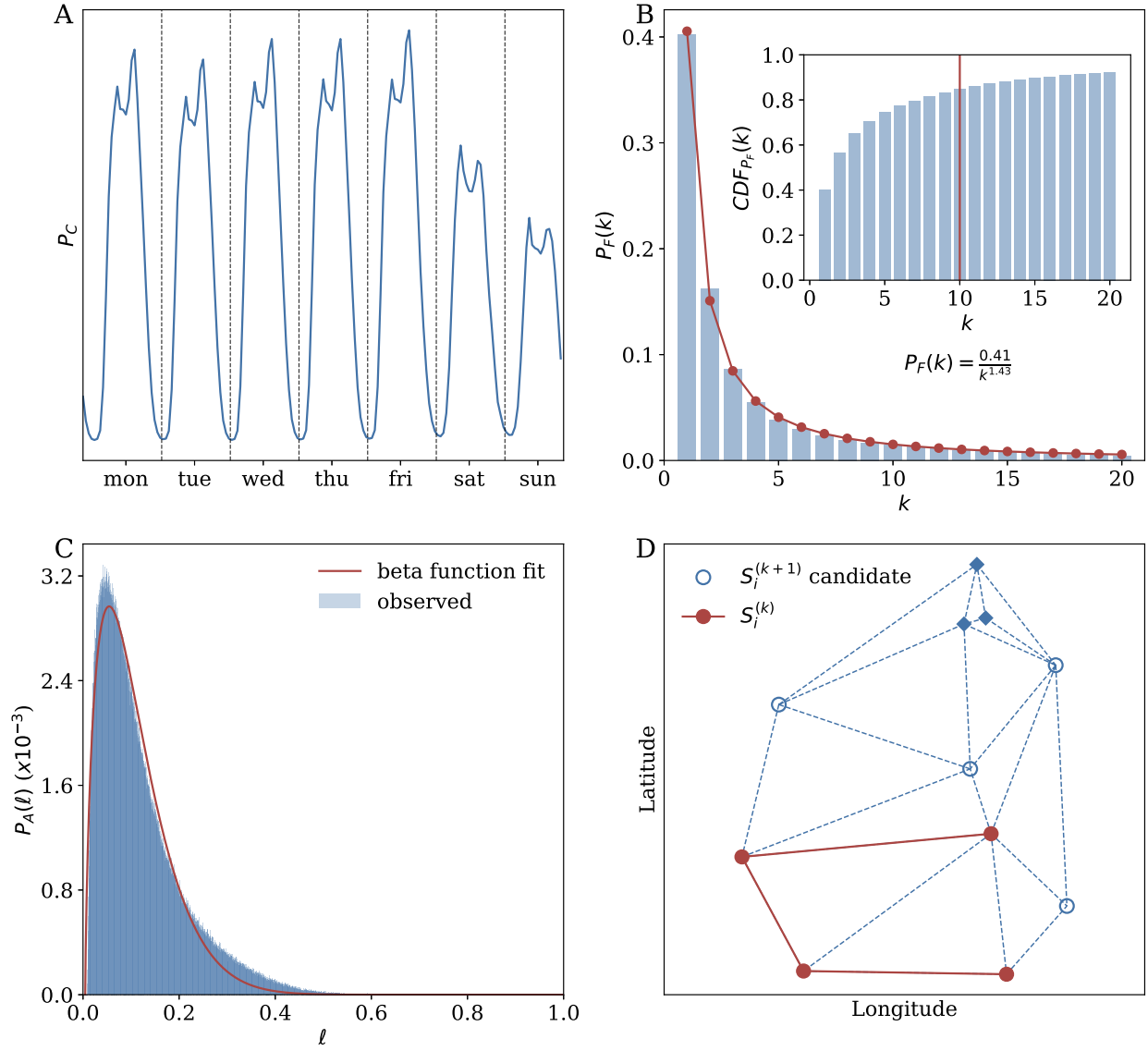


Figure 3.2: The four inputs to the unicity model are displayed. **A:** The circadian distribution, P_C . **B:** The frequency distribution, P_F , along with a power law fit (solid line - $R^2 = 0.99$). The inset displays the cumulative distribution with 85% of activity captured by the top 10 locations. **C:** The activity distribution, P_A , indicating the distribution of the number of records per trajectory along with a β distribution fit (solid line - $R^2 = 0.98$). **D:** Illustration of the sub-graph sampling method used to generate an antenna set S_i where $S_i^{(k)} \in S_i$. The underlying antenna network is represented by dotted lines. The filled nodes (circle) correspond to locations already selected, while the hollow nodes are potential locations that could be selected next ($S_i^{(k+1)}$ candidates). Remaining locations are represented by filled diamonds.

In short, for each user, our model samples a list of 10 connected antennas ($S_1, \dots, S_{10}$) on the network and an activity (number of records in the user's trace) $A \sim P_A$. Each record's timestamp C and position X is then sampled according to the circadian distribution $C \sim P_C$ and $X = S_K$, $K \sim P_F$.

Formally, the model $\mathcal{M}$ can be written as:

$$\mathcal{M}(P_A, P_C, P_F, \mathcal{L}, N) = D = (D_1, \dots, D_N) \quad (3.2)$$

Where $\mathcal{L}$ and N represent the set of all antennas and the dataset size. Each $D_i \in D$ is a location trace for a unique user, represented as a list of L_i records $\left(X_i^{(j)}, C_i^{(j)}\right)_{j=1}^{L_i}$. The length L_i of trace D_i is sampled from the empirical activity distribution P_A :

$$\mathbb{P}[L_i = \ell] = P_A(\ell) \quad (3.3)$$

The timestamps of each record in a trace, $\left(C_i^{(j)}\right)_{j=1}^{L_i}$, are sampled independently from the empirical circadian distribution P_C :

$$\mathbb{P}[C_i^{(j)} = c] = P_C(c) \quad \forall j \in \{1, \dots, L_i\} \quad (3.4)$$

For the spatial component, for each user, a connected sub-graph S_i of size 10 is first sampled from the Delaunay tessellation of the antenna coordinates $\mathcal{L}$. This sub-graph is then randomly ordered as a list, which is denoted by $S_i = \left(S_i^{(k)}\right)_{k=1}^{10}$ with a slight abuse of notations. Finally, the locations of the records $X_i^{(j)} \in X_i$ are sampled independently from S_i according to the empirical frequency distribution P_F :

$$\mathbb{P}[X_i^{(j)} = S_i^{(k)}] = P_F(k) \quad \forall j \in \{1, \dots, L_i\} \quad (3.5)$$

Further, the set of 10 distinct antennas S_i visited by the trajectory i (figure 3.2D) is obtained following the procedure outlined below:

1. Use the geographic coordinates of all antennas in the location set $\mathcal{L}$ to generate the Delaunay tessellation $\mathcal{G}$ (full diamonds in figure 3.2D).
2. Pick a starting antenna $S_i^{(j)}$ uniformly at random from $\mathcal{L}$.

3. Add $S_i^{(j)}$ to the set $S_i = \{S_i^{(j)}\}$ (full red circles in figure 3.2D).
4. Find all the neighbouring antennas to $S_i^{(j)}$ according to $\mathcal{G}$, denoted by $Neighbours(\mathcal{G}, j)$.
5. Add all the locations in $Neighbours(\mathcal{G}, j)$ to a choice set U_i (hollow circles in figure 3.2D).
6. Pick, uniformly at random, an antenna $S_i^{(k)}$ from $W_i = U_i \setminus S_i$, and add it to S_i .
7. Repeat steps 3 - 6 until $|S_i| = 10$.

This process effectively samples a connected sub-graph denoted by nodes S_i from the Delaunay tessellation $\mathcal{G}$. This sub-graph will then be used as the set of locations visited by the user i in the unicity model $\mathcal{M}^{14}$.

Finally, when the size N of the dataset sampled by our model $\mathcal{M}$ increases, this corresponds to sampling more individuals from the same underlying geography. This is what is meant throughout this work when the dataset size is increased: the dataset is considered to be a growing sample from the same underlying population.

3.3.4 Mathematical lower bound of unicity

Through a mathematical analysis of unicity, it can be shown that unicity decreases as a convex function with dataset size, as a result, a linear lower-bound on unicity can be computed. This linear lower bound on unicity relies on the relationship between unicity and the size of the dataset to be convex. Here, it is proven that as long as trajectories are independent and identically distributed, the expected unicity of a population, as a function of population size, follows such a form.

Let $\mathcal{T}$ the set of all possible trajectories and assume $\mathcal{T}$ to be finite, an assumption which agrees with the empirical setting as the number of possible points (number of hour-location combinations) that could constitute a trajectory is finite, as is the set of all points. We define a

¹⁴Note that the above algorithm describes the procedure through which the location set S_i is produced. It does not imply that this is how users actually move from node to node.

data set D as a collection of N trajectories: $D = (D_1, \dots, D_N)$, $D_i \subseteq \mathcal{T}$. A unicity model $\mathcal{M}$ is a randomised mechanism that for a value of N generates a data set D of size N given that:

1. All trajectories $D_i \in D$ are sampled from the *same* distribution T_X
2. All trajectories are sampled *independently* from T_X , meaning $D_i \perp D_j$

The first assumption states that an underlying distribution for population trajectories T_X exists, and it captures correlation between individuals on a large-scale (e.g. commuting patterns, cities, weekends). The second assumption assumes that the correlation between specific individuals is negligible¹⁵ when estimating unicity.

We can thus write that a data set is a random variable $D = (D_1, \dots, D_N)$, with $D_i \sim T_X$ and $D_i \perp D_j$ if $i \neq j$. In the following notation, all random variables are represented by upper-case letters¹⁶ and their realisations or possible values by lower-case letters¹⁷.

Unicity is computed with knowledge of a random set of points in a trajectory. Given a set $A_p^{d_i} \subseteq d_i$ of p points *picked uniformly at random* from a trajectory d_i , we denote the probability that this trajectory $d_i \in d$ is unique in d as:

$$\mathbb{P}_{A_p^{d_i}} [A_p^{d_i} \not\subseteq d_j : \forall j \neq i] \quad (3.6)$$

Unicity $\varepsilon_p(d)$ for a data set d is then defined as this probability (equation 3.6) averaged over all trajectories in a particular data set d . Let $N = |d|$, we thus have:

$$\varepsilon_p(d) = \frac{1}{N} \sum_{i=1}^N \mathbb{P}_{A_p^{d_i}} [A_p^{d_i} \not\subseteq d_j : \forall j \neq i] \quad (3.7)$$

¹⁵While for specific individuals, such as housemates, this may not hold, it does hold on average for the vast majority of individuals within the dataset.

¹⁶ $D_i, X \dots$

¹⁷ $d_i, x \dots$

We are interested in the *expected unicity* of a data set D generated by our randomised mechanism $\mathcal{M}$. This is the expectation of equation 3.7 taken over all possible data sets d .

$$\bar{\varepsilon}_p(N) = \mathbb{E}_D [\varepsilon_p(D)] \quad (3.8)$$

From here we can use equations 3.6 and 3.7 to obtain:

$$\bar{\varepsilon}_p(N) = \mathbb{E}_D [\varepsilon_p(D)] \quad (3.9)$$

$$= \mathbb{E}_D \left[\frac{1}{N} \sum_{i=1}^N \mathbb{P}_{A_p^{D_i}} [A_p^{D_i} \not\subseteq D_j : \forall j \neq i] \right] \quad (3.10)$$

$$= \frac{1}{N} \sum_{i=1}^N \mathbb{E}_D \left[\mathbb{P}_{A_p^{D_i}} [A_p^{D_i} \not\subseteq D_j : \forall j \neq i] \right] \quad (3.11)$$

where in equation 3.11 we have substituted equation 3.6 while also specifying that the trajectories are now random variables as the expectation is taken over the data set D . Next we will expand this to obtain:

$$\bar{\varepsilon}_p(N) = \frac{1}{N} \sum_{i=1}^N \mathbb{E}_D \left[\mathbb{P}_{A_p^{D_i}} [A_p^{D_i} \not\subseteq D_j : \forall j \neq i] \right] \quad (3.12)$$

$$= \frac{1}{N} \sum_{i=1}^N \sum_{d \in \mathcal{T}^N} \mathbb{P}[D = d] \cdot \mathbb{P}_{A_p^{d_i}} [A_p^{d_i} \not\subseteq d_j : \forall j \neq i] \quad (3.13)$$

In equation 3.13 we used the definition of the expected value to obtain the sum over all possible data sets d . Let us expand this term (representing the other factor by Θ):

$$\sum_{d \in \mathcal{T}^N} \mathbb{P}[D = d] \Theta = \sum_{d \in \mathcal{T}^N} \mathbb{P}[D = (d_1, d_2, \dots, d_N)] \Theta \quad (3.14)$$

$$= \sum_{d \in \mathcal{T}^N} \prod_{k=1}^N T_X(d_k) \Theta \quad (3.15)$$

$$= \sum_{d_1 \in \mathcal{T}} \cdots \sum_{d_N \in \mathcal{T}} \prod_{k=1}^N T_X(d_k) \Theta \quad (3.16)$$

$$= \sum_{d_i \in \mathcal{T}} T_X(d_i) \sum_{d_1 \in \mathcal{T}} \cdots \sum_{d_{i-1} \in \mathcal{T}} \sum_{d_{i+1} \in \mathcal{T}} \cdots \sum_{d_N \in \mathcal{T}} \prod_{k \neq i} T_X(d_k) \Theta \quad (3.17)$$

Going from 3.16 to 3.17, the i^{th} component of the sum over the trajectories was pulled out to the front for later use. Going back to 3.11 we obtain:

$$\bar{\varepsilon}_p(N) = \frac{1}{N} \sum_{i=1}^N \sum_{d \in \mathcal{T}^N} \mathbb{P}[D = d] \mathbb{P}_{A_p^{d_i}} [A_p^{d_i} \not\subseteq d_j : \forall j \neq i] \quad (3.18)$$

$$= \frac{1}{N} \sum_{i=1}^N \sum_{d_i \in \mathcal{T}} T_X(d_i) \sum_{d_1 \in \mathcal{T}} \cdots \sum_{d_{i-1} \in \mathcal{T}} \sum_{d_{i+1} \in \mathcal{T}} \cdots \sum_{d_N \in \mathcal{T}} \prod_{k \neq i} T_X(d_k) \mathbb{P}_{A_p^{d_i}} [A_p^{d_i} \not\subseteq d_j : \forall j \neq i] \quad (3.19)$$

At this stage equation 3.19 contains the expression for the expected value over $D \setminus \{D_i\}$ of the probability. We consolidate all sums to reflect this fact:

$$\bar{\varepsilon}_p(N) = \frac{1}{N} \sum_{i=1}^N \sum_{d_i \in \mathcal{T}} T_X(d_i) \sum_{d_1 \in \mathcal{T}} \cdots \sum_{d_{i-1} \in \mathcal{T}} \sum_{d_{i+1} \in \mathcal{T}} \cdots \sum_{d_N \in \mathcal{T}} \prod_{k \neq i} T_X(d_k) \mathbb{P}_{A_p^{d_i}} [A_p^{d_i} \not\subseteq d_j : \forall j \neq i] \quad (3.20)$$

$$= \frac{1}{N} \sum_{i=1}^N \sum_{d_i \in \mathcal{T}} T_X(d_i) \mathbb{E}_{D \setminus \{D_i\}} \left[\mathbb{P}_{A_p^{d_i}} [A_p^{d_i} \not\subseteq D_j : \forall j \neq i] \right] \quad (3.21)$$

The probability expression essentially represents the chances that any randomly picked p points

from d_i , appear in any other trajectory d_j . This can be computed as, by developing the probability over $A_p^{d_i}$:

$$\mathbb{P}_{A_p^{d_i}} [A_p^{d_i} \not\subseteq D_j : \forall j \neq i] = \sum_{a_p^{d_i}} \frac{1}{C_{|d_i|}^p} \mathbb{1}_{D_j \neq i} [a_p^{d_i} \not\subseteq D_j, \forall j \neq i] \quad (3.22)$$

Where $C_{|d_i|}^p$ is the number of ways p points can be picked from the trajectory d_i . Inserting this back into equation 3.21 results in the following.

$$\bar{\varepsilon}_p(N) = \frac{1}{N} \sum_{i=1}^N \sum_{d_i \in \mathcal{T}} T_X(d_i) \mathbb{E}_{D \setminus \{D_i\}} \left[\sum_{a_p^{d_i}} \frac{1}{C_{|d_i|}^p} \mathbb{1}_{D_j \neq i} [a_p^{d_i} \not\subseteq D_j, \forall j \neq i] \right] \quad (3.23)$$

Note that: 1) The expression above is agnostic to the index i which cancels with the leading factor of $\frac{1}{N}$ and, 2) The expected value of the indicator variable of a random event is the probability of that event. Noting that $D \setminus \{D_i\}$ represents the same variable set as the ones impacting the indicator function $D_{j \neq i}$, we move in the expectation operator past the sum and the constant factor and therefore re-write equation 3.23 as:

$$\bar{\varepsilon}_p(N) = \sum_{d_i \in \mathcal{T}} T_X(d_i) \sum_{a_p^{d_i}} \frac{1}{C_{|d_i|}^p} \mathbb{P}_{D_{j \neq i}} [a_p^{d_i} \not\subseteq D_j] \quad (3.24)$$

$$= \sum_{d_i \in \mathcal{T}} T_X(d_i) \sum_{a_p^{d_i}} \frac{1}{C_{|d_i|}^p} \prod_{j \neq i} \mathbb{P}_{D_j} [a_p^{d_i} \not\subseteq D_j] \quad (3.25)$$

$$= \sum_{d_i \in \mathcal{T}} T_X(d_i) \sum_{a_p^{d_i}} \frac{1}{C_{|d_i|}^p} \prod_{j \neq i} \mathbb{P}_X [a_p^{d_i} \not\subseteq X] \quad (3.26)$$

$$= \sum_{d_i \in \mathcal{T}} T_X(d_i) \sum_{a_p^{d_i}} \frac{1}{C_{|d_i|}^p} \left(\mathbb{P}_X [a_p^{d_i} \not\subseteq X] \right)^{N-1} \quad (3.27)$$

In equation 3.25 we used the independence of different trajectories, and we used the fact that all trajectories are identically distributed in going from equation 3.26 to 3.27. Finally, with a

change of dummy variables from d_i to y we obtain:

$$\bar{\varepsilon}_p(N) = \sum_{y \in \mathcal{T}} \frac{1}{C_{|y|}^p} T_X(y) \sum_{a_p^y} \left(\mathbb{P}_X[a_p^y \not\subseteq X] \right)^{N-1} \quad (3.28)$$

Finally note that the form of equation 3.28 is:

$$\bar{\varepsilon}_p(N) = \sum_{y \in \mathcal{T}} \frac{1}{C_{|y|}^p} T_X(y) \sum_{a_p^y} \left(\mathbb{P}_X[a_p^y \not\subseteq X] \right)^{N-1} \quad (3.29)$$

$$= \sum_{y \in \mathcal{T}} \frac{1}{C_{|y|}^p} T_X(y) \sum_{x(y)} \left(f(x) \right)^{N-1} \quad (3.30)$$

$$= \sum_{y \in \mathcal{T}} \frac{1}{C_{|y|}^p} T_X(y) g(y, N) \quad (3.31)$$

Functions of the form $f(m) = b^m$ are convex for all positive m and b . Further, the sum of convex functions is itself convex, and therefore $g(y, N)$ is convex in N in equation 3.31. Using this fact one more time leads to realisation that $\bar{\varepsilon}_p(N)$ must also be convex, since all elements being multiplied out in equations 3.29 to 3.31 are probabilities and are thus non-negative.

Finally, while our analysis considers real-valued functions, a direct consequence of this convexity result is that the sequence $\{\varepsilon(D(N))\}_{N=1,2,3,\dots}$, which represents the population dependence of unicity for discrete population sizes, is also a convex sequence.

Given this result, the unicity of a dataset of size N will always obey the following relation (treating unicity as a function of a real-valued N):

$$\varepsilon(D(N)) \geq \varepsilon(D(N')) + (N - N') \cdot \left. \frac{d\varepsilon}{dN} \right|_{N=N'} \quad (3.32)$$

The discrete equivalent of this expression replaces the differentiation operator by a difference

operator. Re-arranged, this gives a lower bound for unicity:

$$\varepsilon(D(N')) - \varepsilon(D(N)) \leq (N - N') \cdot \left(\varepsilon(D(N' - 1)) - \varepsilon(D(N')) \right) \quad (3.33)$$

Unicity being a strictly decreasing convex function, it is lower-bounded by its linear tangent. This means that the rate of unicity decrease will continue to slow down as the dataset size is increased.

3.4 Results

Experiments are performed on a dataset of call detail records containing the location of 1M individuals over three months. Each record contains a unique user ID, an hourly time stamp, and an antenna ID which relates to a location. We formally model this dataset as a sequence $D = (D_1, \dots, D_N)$, populated with user time/location traces of the type $D_i = (X_i, C_i)$. X_i and C_i are lists of positions (antennas) and times (hours) representing the spatial and temporal components of a user's location trace.

Using this dataset, the decrease in unicity with the dataset size is studied by randomly sampling individuals from the original dataset and measuring the unicity of the sample as we increase its size (see section 3.3 for details). In line with previous work [174], the subscript p in $\varepsilon_p(N)$ is used to indicate the number of points of auxiliary information used in the computation of unicity.

3.4.1 Results on the decrease of unicity

The results of how unicity decreases with dataset size are presented in three ways. First, empirical measurement of unicity decrease. Second a comparison of empirical unicity with the unicity model, which then allows extrapolation of this decrease to larger datasets using the

model. Finally, attention is drawn to the linear lower bound which results from the convexity of unicity (see section 3.3.4).

Firstly, figure 3.3A shows that unicity empirically decreases slowly with the size of the dataset. With three points of auxiliary information unicity (solid orange line) goes down from $\varepsilon_3(100K) = 0.98$ in a dataset of 100k people to $\varepsilon_3(1M) = 0.93$ in a dataset of a million people. With two points (solid blue line) this decreases slightly faster reaching $\varepsilon_2(1M) = 0.69$, while unicity with four points or more (solid red and brown lines) decreases very slowly with $\varepsilon_4(1M) = 0.98$. These results show that, while the size of the dataset impacts unicity, the decrease of unicity is slow.

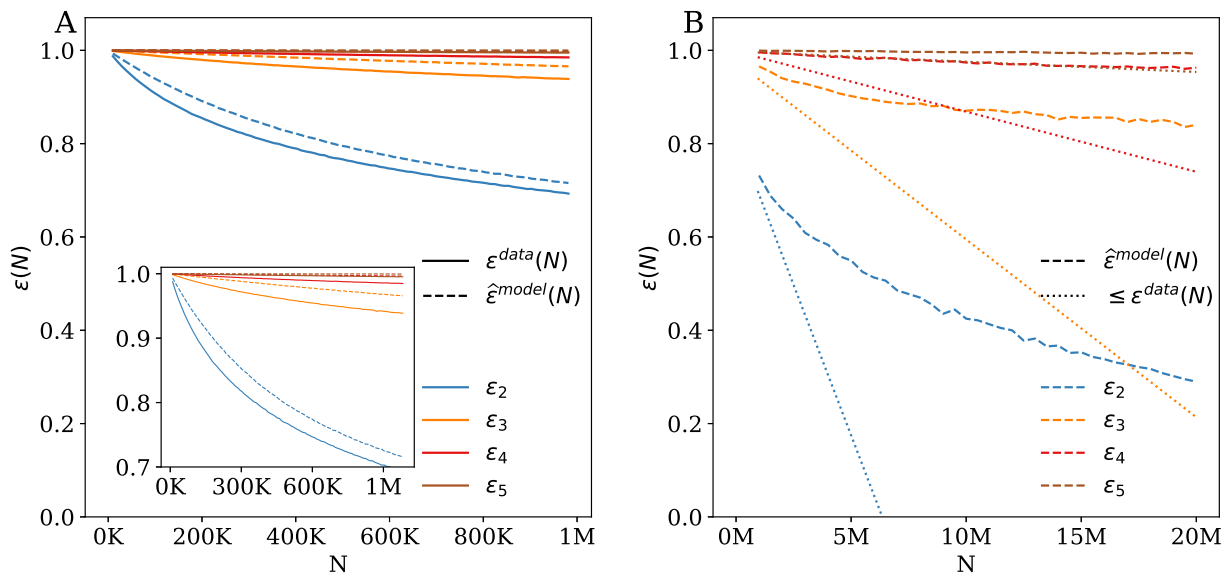


Figure 3.3: Empirical unicity compared to the estimated unicity is displayed. **A:** Empirical (solid lines) and estimated (dashed lines) unicity decrease slowly with the size of the dataset. Inset: closeup of the region $\varepsilon \geq 0.7$. **B:** The estimated unicity remains high even in large datasets. This is confirmed by the lower bound results (dotted lines). Note that the decrease of risk is slow in both the empirical (**A**) and simulated (**B**) cases.

Secondly, figure 3.3A also shows that our simple statistical model closely follows the empirical measure of unicity from 1 to 1M people (dashed and solid lines). Using the model, we then study how unicity is likely to evolve as the size of the dataset increases to 20M people (figure 3.3B). For $N = 20M$, our model estimates unicity with three points to be close to $\hat{\varepsilon}_3(20M) = 0.93$ while knowing one more point would increase this to the region of $\hat{\varepsilon}_4(20M) = 0.99$.

In fact, extending our model to a population of 60M (figure 3.4), 93% of people would be uniquely identified using 4 points of auxiliary information with a lower bound at 22%. This

bound increases to 87% when 5 points are available. These results re-enforce the conclusion that the risk of re-identification remains high even in country-scale location datasets.

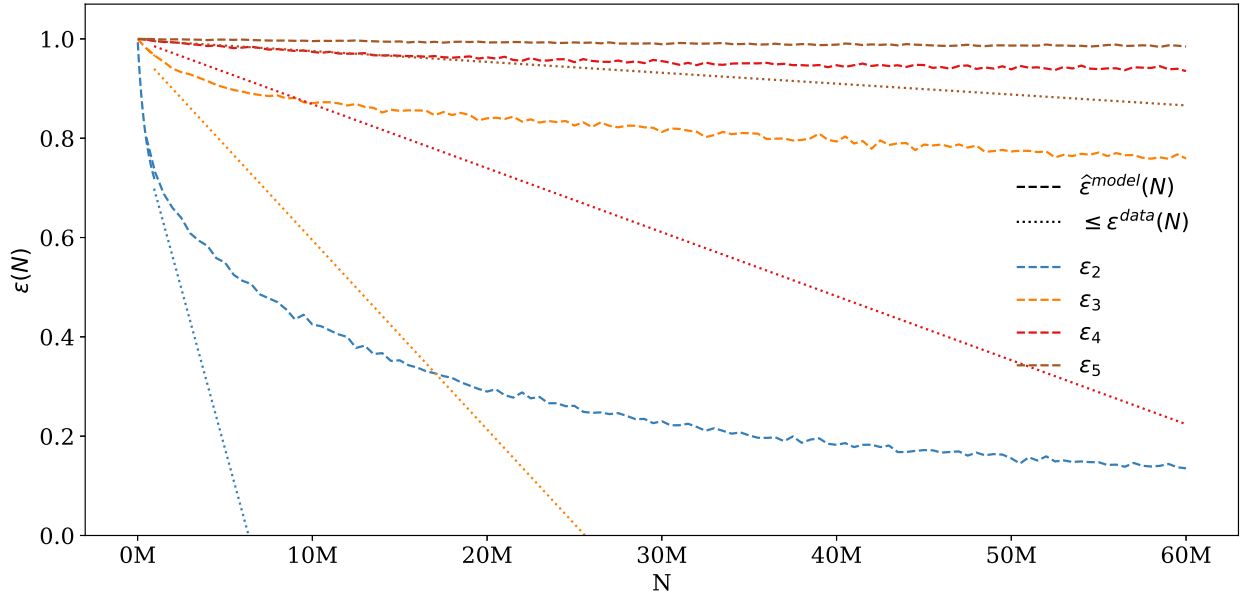


Figure 3.4: Estimated unicity values along a lower bound on unicity. The unicity of the model (dashed) is compared to the lower bound results (dotted). In each case the estimated unicity remains high even in very large datasets with 60M trajectories. This is confirmed by the lower bound results which does not dip below 20% with 5 points even with 60M mobility traces in the dataset.

Finally, note that in both figure 3.3A and B as well figure 3.4, the lower linear bound is displayed as a dotted line. This is an effect of unicity reduction following a convex form as shown in section 3.3.4. Using the tangent to the empirical unicity curves estimated by discrete difference over the range of $N \in [0.9M, 1M]$, we obtain a lower bound of 0.73 for $\epsilon_4(20M)$ and 0.9 for $\epsilon_5(20M)$ (figure 3.3B (dotted)). The bounds for a population of 60M are also displayed in figure 3.4.

3.4.2 Sensitivity analysis

Our simple statistical model for unicity takes as input three distributions, but these distributions may vary depending on specifics of the dataset, such as the country where it was collected or the sources of location information. Here I present a sensitivity analysis to ensure the robustness of our model to even broad changes to the distributions.

We first perturb the P_A and P_F distributions around their empirical forms using a scaled earth movers distance¹⁸ as the guiding metric. The P_C distribution on the other hand has been shown to be very stable across datasets [158–162] and we keep it constant throughout our analysis.

Broadly speaking, concerning the frequency and activity distributions:

- **Frequency distribution.** With increasing values of the exponent of the power law unicity decreases
- **Activity distribution.** Given a fitted β function $\beta(a, b)$:
 1. Unicity decreases with increasing values of a
 2. Unicity increases with increasing values of b

In figure 3.2 we saw that the empirical frequency and activity distributions are each approximated by a power-law and a β function respectively. These are:

$$\overline{P_F}(k) = \frac{c}{k^\gamma} \quad (3.34)$$

$$\overline{P_A}(l) = c \cdot l^{a-1} \cdot (1-l)^{b-1} \quad (3.35)$$

Here, we perturb these distributions around their empirical fit. For each of the two, we change their fit parameters to obtain a set of new distributions $P_q \in \mathcal{H}_q^{all}$ where $q \in \{A, F\}$ corresponds to either the frequency, F , or activity, A , distributions. $\mathcal{H}_q^{all}$ corresponds to the set of all possible distributions. The P_q included in this sensitivity analysis are picked such that the earth movers distance (d_{emd}) between P_q and the observed distribution P_{obs} , is no more than 70% of its maximum distance given. While in general the earth movers distance between two probability distributions is not bounded, in our case, the distributions are defined over a discrete finite space and the distance is thus bounded by a finite number.

¹⁸Also called first Wasserstein distance [244].

The set of distributions included in this analysis are a subset of the elements of $\mathcal{H}_q$ which is constructed as a random sample of $\mathcal{H}_q^{all}$. Therefore, in this analysis, we choose those $P_q \in \mathcal{H}_q \subset \mathcal{H}_q^{all}$ such that:

$$\mathcal{H}_q^{selected} = \{P_q \in \mathcal{H}_q \mid d_{emd}(P_q, P_{obs}) \leq 0.7 \cdot \sup_{P_q \in \mathcal{H}_q} \{d_{emd}(P_q, P_{obs})\}\} \quad (3.36)$$

Our estimate of the parameter values that satisfy this heuristic are included in table 3.4 and table 3.5¹⁹. The resulting distributions are visible in figure 3.5.

Table 3.4: $\overline{P_F}$ parameter values

γ	c
0.390	0.173
0.802	0.281
1.214	0.409
1.626	0.540
2.038	0.655
2.450	0.747
2.862	0.816

Table 3.5: $\overline{P_A}$ parameter values

a	b	c
1.186	5.142	0.0036
1.186	17.094	0.0146
1.186	29.046	0.0275
1.689	5.142	0.009
1.689	17.094	0.0639
1.689	29.046	0.1546
2.193	5.142	0.01927
2.193	17.094	0.2298
2.193	29.046	0.7137

In figure 3.5 the resulting frequency and activity distributions are displayed. These distributions are combined to produce 63 different instantiations of the unicity model. Table 3.6 summarises the unicity values for models using the broad range of distributions in figure 3.5, at a dataset size of 20M trajectories. Note that the lowest unicity values across all instantiations of the model are still high with $Min(\varepsilon_4(20M)) = 43.1\%$ and would still be considered as putting people's privacy at risk.

Further, I study how certain aspects of human mobility contribute to unicity. Starting from empirical user location traces $D_i = (X_i, C_i)$, first, we find that removing the association between times (C_i) and locations (X_i) by shuffling the vectors and recombining them, only slightly

¹⁹The constant values c are normalising factors.

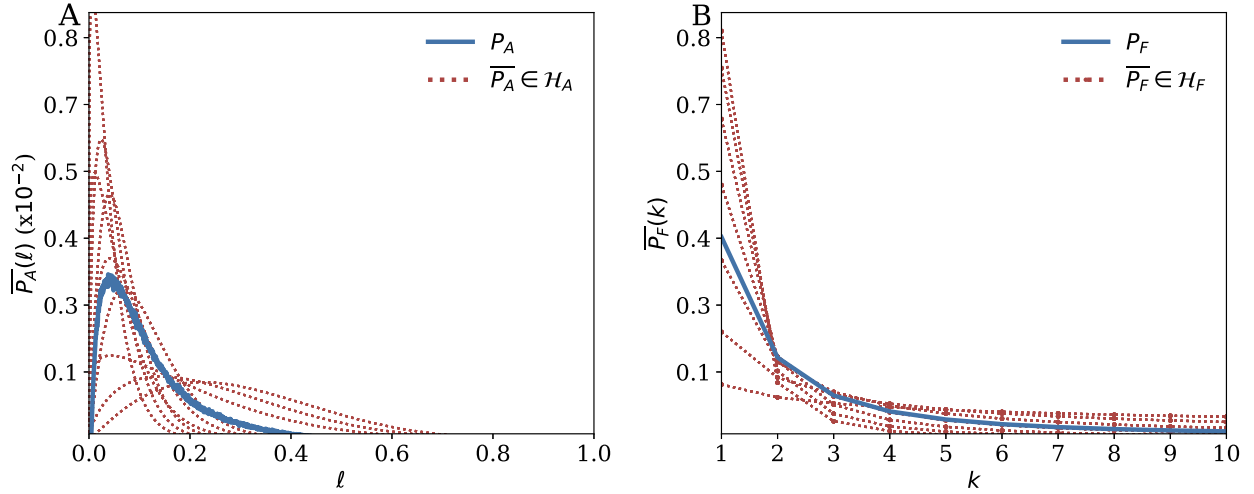


Figure 3.5: The range of distributions studied in the sensitivity analysis are displayed. The range of perturbed activity $\overline{P}_A$ (A) and frequency $\overline{P}_F$ (B) distributions are displayed (dotted) along with their empirical forms (solid). The k variable for $\overline{P}_F$ is the rank of the most popular positions with 1 being the most and 10 the least commonly visited locations. The ℓ variable for $\overline{P}_A$, indicates the number of hours that a trajectory appears in the dataset, from none to 24 hours a day for 3 months.

	ε_2	ε_3	ε_4	ε_5
Mean	0.307	0.735	0.876	0.935
Standard deviation	0.175	0.216	0.159	0.113
Minimum	0.071	0.260	0.431	0.544
Maximum	0.704	0.997	1	1

Table 3.6: Summary of unicity results at $N = 20M$ as per the sensitivity analysis.

impacts unicity values (figure 3.6A). Specifically, consider a dataset D' composed of trajectories $D'_i = (X'_i, C'_i)$ such that:

$$X'_i = \sigma_i(X_i)$$

$$C'_i = \pi_i(C_i)$$

where σ_i and π_i refer to random permutations of the spatial and temporal components of D_i . This only marginally impacts unicity, showing that unicity does not depend on the specific places being visited at specific times, as long as those times and places appear in the trace with

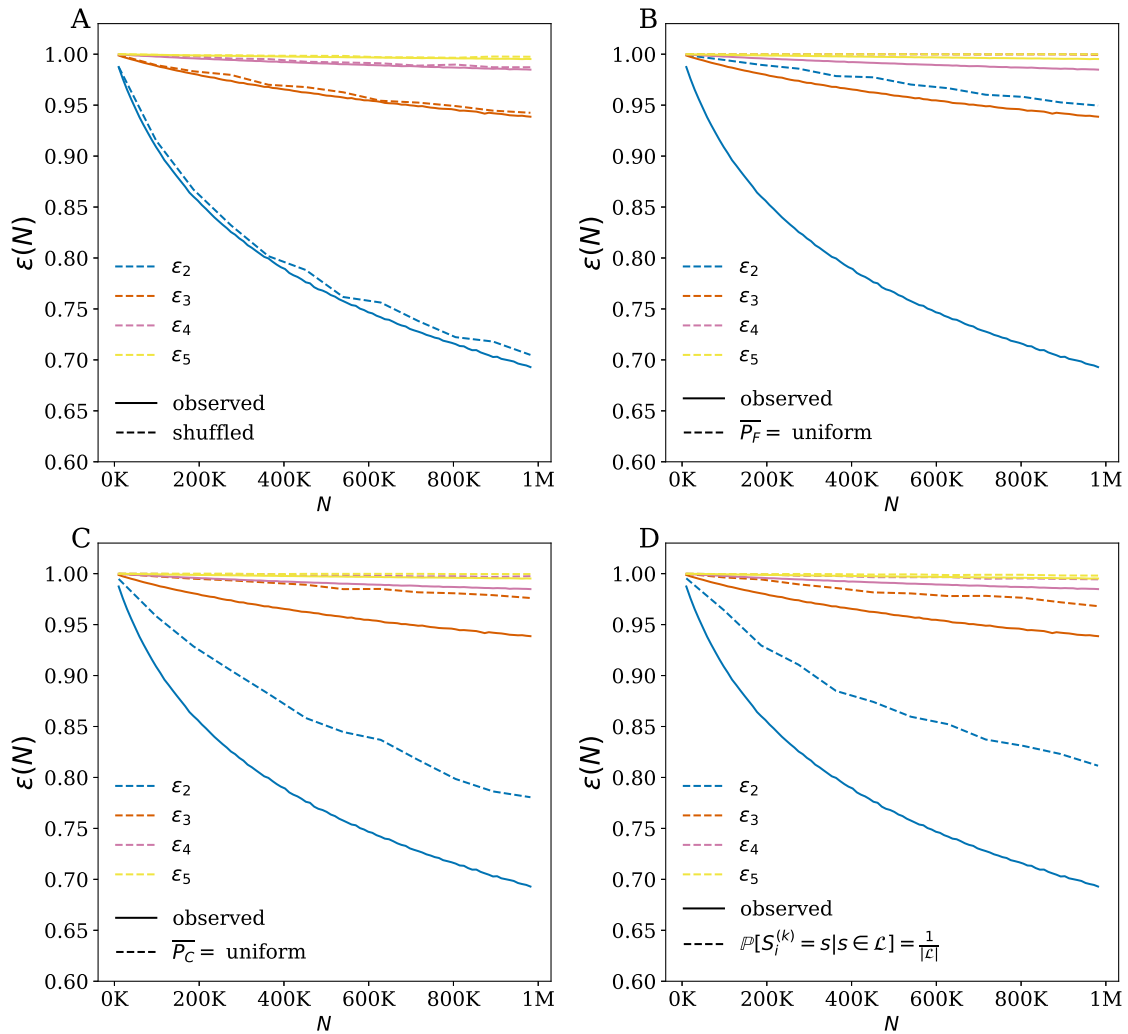


Figure 3.6: Unicity of alternative models compared to observed unicity. **A:** The space and time components of trajectories are shuffled with minimal impact on unicity. The P_F distribution (**B**) and the P_C distribution (**C**) are separately replaced with the uniform distribution, which increases the unicity of these trajectories greatly. Lastly, the antenna set S_i is populated with uniformly picked antennas from $\mathcal{L}$ at random (**D**), instead of using the sub-graph sampling method. Here, unicity is again increased substantially.

their respective frequencies independently (figure 3.6A).

Second, replacing P_C or P_F with uniform distributions (figure 3.6B and C) or attempts to model unicity using a simple combinatorial model (appendix figure B.1) also causes the model to overestimate unicity. These demonstrate the importance of all three distributions, and the underlying geography to correctly capture the unicity of mobility datasets.

Finally, I replace the set of locations in each trajectory with uniformly picked locations. Instead of using the sub-graph sampling method displayed in figure 3.2D, I populate each S_i with

antennas picked from the entire set of locations $\mathcal{L}$ uniformly at random. We find that this leads to unicity being overestimated (figure 3.6D).

This analysis, combined with the relative simplicity and generality of the unicity model, suggest that these results would generalise to many other high-dimensional location datasets. Likewise, the strong underlying combinatorial effect²⁰ that underpins unicity combined with previous research [176, 227, 228] suggests that unicity will similarly decrease slowly in other types of high-dimensional data.

3.5 Discussion

Taken together, these results show that the scale of a dataset does not prevent re-identification. Human mobility, much like a physical fingerprint, is highly unique and can be used to find a person across mobility datasets without appropriate safe-guards.

Legally, the European Union [GDPR](#) sets a high threshold for what constitutes anonymous data, namely that the individual should not be identifiable taking into account both the “*available technology at the time of the processing*” but also future “*technological developments*” (Recital 26). The Article 29 Working Party, the predecessor to the European Data Protection Board, in its guidance sets out three criteria to assess whether a dataset is anonymous: singling out, linkability, and inference [198] with the former two being directly applicable here. As an example, the Centre for Humanitarian Data of the United Nations (UN OCHA) adopted 5% as a threshold for what constitutes an acceptable re-identification risk [245]. Even our lower bound of 22% far exceeds this liberal threshold.

Here we study the unicity of location datasets with a spatial resolution of $\approx 1 \text{ km}^2$ and a temporal resolution of an hour. Fine grained GPS data is likely to lead to even higher values of unicity and previous research has shown that, in general, many existing de-identification methods do not meaningfully reduce the risk of re-identification. For instance, research [174, 227] has shown that reducing the spatial and temporal resolution of the data further, only

²⁰This combinatorial intuition is expanded on in the appendix. See chapter [B](#).

slowly decreases the risk while another study [233] concluded that location data cannot be anonymised without extreme generalisation — reducing utility to unacceptable levels.

Ensuring that this data can be accessed and used broadly is of paramount importance, as is the privacy of individuals. A range of privacy engineering techniques allowing data to be used while giving individuals strong privacy guarantees have been developed and are starting to be used [172, 201, 246]. As standards for anonymisation are being redefined, in the EU and around the world, it is essential for them to emphasise the strong limits of de-identification, and to give guidance on the use of modern privacy-engineering solutions.

In the next 3 sections I discuss the underlying assumptions of the unicity model, a brief critique of a previous method of risk-estimation in the literature, and end with possible directions for future work.

3.5.1 Assumptions underpinning the simple unicity model

Here I evaluate the four assumptions underpinning the simple unicity model.

Model input independence. First, the model treats each of the four inputs in figure 3.2 as independent from each other. Considering them, or some of them, jointly might further improve the model. However, this would also increase its complexity, and therefore sensitivity to small changes in the data. While further exploration would be interesting, we consider that the simple model approximates the decrease in unicity with increasing dataset size well enough to support the conclusion that unicity is unlikely to be low even in population-scale datasets.

Convergence of input distributions. Second, our model uses input distributions extracted from a dataset of 1M people to study the unicity of datasets with up to 60M people. This assumes that these distributions estimated from a smaller sample are representative of the larger sample (i.e., the estimation of the distributions has converged). I show that this is a reasonable assumption by instantiating our model $\mathcal{M}$ with distributions extracted from samples of sizes significantly smaller than 1M, and showing that the unicity results remain largely unchanged

(figure B.2 in appendix). I also perform a sensitivity analysis to evaluate the impact of broad variations on these input distribution on our results.

Standardisation of visited graph size. Third, the model assumes each trajectory to contain at most 10 unique locations. This standardisation of the set of visited locations, allows for the mean frequency distribution (P_F) to be used in the modelling process (figure 3.2B). As seen in the inset of figure 3.2B, more than 85% of the activity in the average trajectory is captured by the top ten locations visited. Furthermore, we find that P_F only changes slightly when the number of unique locations is altered, and that our conclusions are not impacted by this choice.

Locality of visited graph. Finally, our model assumes that the set of locations appearing in each trajectory can be described by a connected planar sub-graph of the underlying antenna network. We believe this to be a reasonable assumption, as previous work suggests that sub-graphs spanned by each trajectory in human mobility are highly localised, with the distribution $P(r_g)$ of the radius of gyration – a metric for how far people tend to travel on average – following a power-law with increasing radius [247].

3.5.2 Previous estimates of re-identification risk in large datasets

The results found here are in stark contrast with the linear extrapolation made by El Emam [181], who reports a unicity of 0.01 with four points at a population of 20M. Here I explore why and replicate the results by this previous estimate (see section B.3 for expansion on this method).

El Emam [181] proposed a method to estimate the uniqueness of a population-size (N) dataset given the unicity $\varepsilon(m)$ of a smaller sample dataset of size m . This method essentially copies records, or parts of records from an underlying dataset, in order to increase its size. Using this method, he estimates that the uniqueness for a population of size $N = 22\text{M}$ is of about 1%, given a uniqueness of 90% of a sample of size $m = 1\text{M}$ of the same dataset. This estimate forms the basis for his claim that uniqueness is low in large-scale datasets.

We here show that this method by El Emam a) is unrealistic, b) provably gives the lowest possible estimate for the risk in the larger dataset, and that c) using our dataset, we observe

that the real empirical unicity is significantly higher than the upper bound given this method.

First, the method is unrealistic as it effectively generates a dataset D of size N where a fraction α of records are unique while all the other records are identical to exactly one and only one other record. The parameter α is selected such that the expected estimated uniqueness on a sample of size m , which we denote by $\nu_D(m)$, is equal to the empirical unicity. This assumes that users in the real mobility dataset are either unique or exact duplicates of another user.

Second, it is provable (see chapter B) that the risk estimated by El Emam will be lower or equal to the risk of *any* other dataset of size N , as this estimate is an *affine* function of m . In other terms, this method will *always* return the absolute lowest possible estimate of the risk.

Third, the El Emam method, applied to our dataset demonstrates that the estimate of the risk is significantly lower than the real empirical value, leading to the risk of re-identification to be strongly underestimated. For a dataset of 200K people, we empirically observe a $\varepsilon_2(200K) = 0.86$. Using this number, El Emam's method would estimate the risk of a larger 1M person dataset to be $\varepsilon_2(1M) = 0.3$ while the correct empirical value is ≈ 0.7 .

3.5.3 Direction for future work

There are 4 main areas of further exploration currently missing in the literature. Firstly, in expanding the modelling to account for dependencies between the input distributions. Second, concerning mobility data and expanding the size of a dataset. Third, replicating these results more broadly on other high-dimensional homogeneous datasets. Finally, expanding the results here to any high-dimensional dataset using broad mathematically robust methods. Below I will expand briefly on how each of these lines could be pursued.

First, although in this work the three distributions used as model inputs are treated independently, in reality, this is not the case, and future work could improve on the estimates here by accounting for these inter-dependencies. Specifically, there is evidence that activity and frequency distributions are related [248]. Additionally, initial evidence from this work showed

that for each trajectory, the larger the activity, the lower the power in the frequency power law fit (i.e. they tend to spread their movements among a more diverse set of locations).

Second, here I focused on a case-study where a data set size grows due to a provider sharing a larger sample, while a dataset can also grow due to expanded collection abilities. The latter, in the context of mobile phone location data, would be a provider increasing their network coverage organically (i.e. by installing more antennas). Modelling this increase is an open research question as it includes modelling the growth of a graph in a general enough way that can be applied to multiple network providers. Further, it would include a study of how many customers each new antenna would capture, although some preliminary analysis on this already exists [249].

Third, this case-study has focused on location data captured from mobile phone metadata. However, as described in the background section (see section 3.2.4), there are many different forms of homogeneous, high-dimensional data being collected today. Replicating these results on a broad set of new datasets would shine a light as to whether the risk reduction behaviour observed in location data is also present in other forms of data when they are scaled. These include Internet of Things data, credit card transaction histories, browsing data, mobile phone data and so on.

Finally and perhaps, most interestingly, there is room for research that can estimate re-identification risk, for any high-dimensional homogeneous dataset. This has already been done for many heterogeneous datasets by Rocher, et al [237]. In that work the authors use generative copula-based methods that can accurately estimate the likelihood of a target being correctly re-identified, even in datasets with missing information. In that case this is possible as many structured datasets will contain roughly similar information about an individual (e.g. age, gender). In the case of homogeneous datasets, this presents a more substantial challenge. Indeed such a model will likely require information about dataset sparsity as well as some domain level information.

4

Conclusion

Technological advancements are one of the most common catalysts for scientific paradigm-shifts. From the telescope to DNA sequencing technology, they allow scientists to peer into previously unexplored avenues of knowledge, thus ushering in a new age of experimental work [19]. Machine learning and the ability to collect and store vast quantities of data, is arguably facilitating these paradigm-shifts in many scientific communities today.

In this thesis I have explored two scientific questions that are effected by machine learning and the increased data collection ability in the science today. The first question is focused on studying the impact of machine learning on insurance markets while the second discusses one of the risks of expanded data collection to individual privacy.

Regarding the first, typically studies of economic markets are constrained either by the scale, participant-expertise, or participant time-commitment. However, recent advances in digital field experiments are removing these constraints. Using a digital field experiment can allow experimenter to collect data in ways previously impossible.

Using an online data science competition, we designed a cheapest-wins, competitive insurance market to study the impact of machine learning on consumers, companies and the entire market. In the insurance pricing game presented in this thesis, 2000 participants (60% insurance industry professionals) signed up, and approximately 350 played over a course of 11 weeks.

They used 500K real historical motor insurance policies to train and submit pricing models, in order to maximise their competitive profit. Over the course of the competition, participants submitted over 10,000 models averaging 100 models a day. The models were then categorised into algorithmic models and data-models which correspond roughly to machine learning and statistical inference based on guidance from the literature [77].

Post-submission categorisation of models resulted in a 60/40 split tilted towards algorithmic models. With submissions split between data and algorithmic models, two experiments were performed on the impact of machine learning (i.e. algorithmic models) on insurance markets. First, markets were split into purely data-model or algorithmic model based and consumer prices were found to decrease slightly in markets with purely algorithmic models, although the effect is small. The potential cause of this was also investigated and found to be likely due to a combination of cheaper offered prices and lower levels of between-model correlation for algorithmic models compared to data-models. Lower correlation would indicate a larger diversity of prices per policy, which, in combination with the cheapest-wins customer model, can result in cheaper prices being realised.

An additional scientific question studied was related to market entry. As today most motor insurance markets globally are almost entirely data-model based, studying the impact of gradually injecting algorithmic models into a market could indicate what the immediate future of the insurance markets will look like. Starting with a market of 10 data-models, and replacing data-models with algorithmic models one by one, shows preliminary signs of increased profit volatility for data-models. Cheaper consumer prices combined with a increased earnings stability would provide an incentive for firms to adapt algorithmic pricing in the future.

The second case-study addresses one of the many possible risks associated with the new data collection ability being exercised in sciences today — the risk to individual privacy. This case-study specifically focuses on the relationship between the size of a dataset, and the risk of re-identification. Intuition suggests that size buys you privacy, that you can get lost in a crowd of many records if the dataset is large enough.

To address this specific question, using 3 months of location data, we 1) show the risk of re-

identification to decrease slowly with dataset size, 2) approximate this decrease with a simple model taking into account 3 population-wide marginal distributions, and 3) prove that unicity (the risk metric used) is convex and obtain a linear lower bound. Our estimates show that 93% of people would be uniquely identified in a dataset of 60M people using 4 points of auxiliary information, with a lower bound at 22%. This lower bound increases to 87% when 5 points are available. Taken together, our results show how the privacy of individuals is very unlikely to be preserved even in country-scale location datasets [1].

The ability to collect and process data is currently transforming the way we work and live. Many fields, and in particular the social sciences, are grappling with how to harness the power of this new technology. As a result, we are faced with three interconnected challenges. First, we must learn how to re-visit many well-established principles and observe them through the lens of this new technology. Second, we must be vigilant to apply this new technology to fields where quantitative paradigms are not the norm (e.g. political science). Finally, all the while, we must keep any potential risks to our fundamental rights such as privacy at the forefront of our inquiries.

4.1 Future directions

Each of the chapters includes an extended future work section related to the digital field experiments and studies of re-identification risk. Here I will expand the scope wider and discuss possible future directions where old societal and scientific questions can be revisited using new machine learning techniques or new large datasets. I will touch upon two specific applications, one in measuring the impact of algorithmic pricing agents in a generic product market, the other in reducing political polarisation in online communities using individual conversations.

4.1.1 Algorithmic pricing models in a market with real actors

Within many product markets, automated algorithmic pricing systems are beginning to see wider use. This has been the case in capital markets with high frequency trading algorithms for many years, however, they are just starting to appear in consumer facing markets such as gas stations [45] or two-sided platforms such as eBay or Amazon [250]. Currently their impact and interaction with real-world actors has yet to be studied. More specifically, do automated algorithmic pricing agents benefit consumers within a market? under what conditions do they engage in anti-competitive behaviour? These questions have been studied in a theoretical setting previously [56–59], however, no controlled, real-world experiment exists.

One study design with real actors would be possible in the form of a digital field experiment in a gamified setting, similar to the case-study in chapter 2. Using real world product-level demand data these questions can be studied by recruiting participants to be sellers in a market competing for fully elastic customers. By then providing the sellers with the possibility of “paying” for one of a range of automated pricing algorithms such a study may be able to observe not only how individuals engage in a market which includes automated agents, but also, whether the prices paid by consumers are negatively effected by the mere presence of these agents.

4.1.2 Impact of individual conversations on political polarisation

Recent research in political polarisation of communities suggests that this is a worsening problem today [251]. Some studies investigating the causes of polarisation have found that homogeneity of a social network leads members to ascribe more extreme views to their opposing political party [252, 253]. Although evidence exists that mere exposure to opposing views does not mitigate polarisation [254], there is a possibility that an increase of group heterogeneity may also reduce polarisation if there is indeed a causal link. In other words, does having a conversation about a polarising topic with a person of the opposing view move the political needle for either party? What if they are each given summary facts about a particular topic

before hand, does being informed change the dynamic?

Applying gamified design to study such an effect would involve recruiting participants and repeatedly pairing them for one-to-one conversations over an extended period of time. By pre-exposing a set of participants to curated facts about their discussion points and surveying all participants at regular time intervals about their political leanings along the axes under consideration, it could be possible to tease out the effect of in-person conversations on political leanings en-mass. Needless to say, such a study would face a number of challenges, namely participant incentives, that would be the topic of future research. However, solving those challenges would leverage the scalability of digital experiments within an emerging sub-field of computational social science.

Bibliography

- [1] A. Farzanehfar, F. Houssiau, and Y.-A. de Montjoye, “The risk of re-identification remains high even in country-scale location datasets,” *Patterns*, vol. 2, no. 3, p. 100 204, 2021.
- [2] T. S. Kuhn, *The structure of scientific revolutions*. University of Chicago press, 2012.
- [3] J. Naughton, *Thomas Kuhn: The man who changed the way the world looked at science*, en, Section: Science, 2012. [Online]. Available: <http://www.theguardian.com/science/2012/aug/19/thomas-kuhn-structure-scientific-revolutions> (visited on 06/14/2021).
- [4] M. North, *Novelty: A history of the new*. University of Chicago Press, 2013.
- [5] H. C. King, *The history of the telescope*. Courier Corporation, 2003.
- [6] W. J. Croft, *Under the microscope: a brief history of microscopy*. World Scientific, 2006, vol. 5.
- [7] J. Hecht, “A short history of laser development,” *Applied optics*, vol. 49, no. 25, F99–F122, 2010.
- [8] B. Kevles, *Naked to the Bone: Medical Imaging in the Twentieth Century*, en. Rutgers University Press, 1997, ISBN: 978-0-8135-2358-3.
- [9] B. A. Smith, L. A. Soderblom, R. Beebe, J. Boyce, G. Briggs, M. Carr, S. A. Collins, A. F. Cook, G. E. Danielson, M. E. Davies, *et al.*, “The galilean satellites and jupiter: Voyager 2 imaging science results,” *Science*, vol. 206, no. 4421, pp. 927–950, 1979.

- [10] A. Mazloumian, Y.-H. Eom, D. Helbing, S. Lozano, and S. Fortunato, “How Citation Boosts Promote Scientific Paradigm Shifts and Nobel Prizes,” *PLOS ONE*, vol. 6, no. 5, pp. 1–6, 2011, Publisher: Public Library of Science. DOI: [10.1371/journal.pone.0018975](https://doi.org/10.1371/journal.pone.0018975). [Online]. Available: <https://doi.org/10.1371/journal.pone.0018975>.
- [11] DBLP computer science bibliography, *dblp: Publications per year*, 2021. [Online]. Available: <https://dblp.org/statistics/publicationsperyear.html> (visited on 06/14/2021).
- [12] Nature Editor, “The scientific events that shaped the decade,” en, *Nature*, vol. 576, no. 7787, pp. 337–338, Dec. 2019, Number: 7787 Publisher: Nature Publishing Group. DOI: [10.1038/d41586-019-03857-x](https://doi.org/10.1038/d41586-019-03857-x). [Online]. Available: <https://www.nature.com/articles/d41586-019-03857-x> (visited on 06/14/2021).
- [13] M. F. C. Gomes, A. Pastore y Piontti, L. Rossi, D. Chao, I. Longini, M. E. Halloran, and A. Vespignani, “Assessing the International Spreading Risk Associated with the 2014 West African Ebola Outbreak,” *PLoS Currents*, vol. 6, Sep. 2014, ISSN: 2157-3999. DOI: [10.1371/currents.outbreaks.cd818f63d40e24aef769dda7df9e0da5](https://doi.org/10.1371/currents.outbreaks.cd818f63d40e24aef769dda7df9e0da5). [Online]. Available: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4169359/> (visited on 09/28/2018).
- [14] S. Zuboff, “Big other: Surveillance capitalism and the prospects of an information civilization,” *Journal of Information Technology*, vol. 30, no. 1, pp. 75–89, 2015.
- [15] M. R. Cohen and E. Nagel, *Introduction to logic and scientific method*. Allied Publishers Private Limited, New Delhi, 1936.
- [16] D. C. Gazis, “Influence of technology on science: A comment on some experiences at ibm research,” *Research Policy*, vol. 8, no. 3, pp. 244–259, 1979.
- [17] M. Tedre, *The science of computing: shaping a discipline*. CRC Press, 2014.
- [18] C. Evans, “Broad band, the untold story of the women who made the internet,” *Internet Histories*, vol. 3, no. 2, pp. 195–198, Apr. 2019, ISSN: 2470-1475. DOI: [10.1080/24701475.2019.1598078](https://doi.org/10.1080/24701475.2019.1598078). [Online]. Available: <https://doi.org/10.1080/24701475.2019.1598078> (visited on 06/11/2021).

- [19] P. DeHart Hurd, "Technology and the advancement of knowledge in the sciences," *Bulletin of science, technology & society*, vol. 14, no. 3, pp. 125–131, 1994.
- [20] L. C. Crispino and D. J. Kennefick, "A hundred years of the first experimental test of general relativity," *Nature Physics*, vol. 15, no. 5, pp. 416–419, 2019.
- [21] S. Stapnes, "Detector challenges at the lhc," *Nature*, vol. 448, no. 7151, pp. 290–296, 2007.
- [22] B. O. Rothbaum, L. Hodges, R. Alarcon, D. Ready, F. Shahar, K. Graap, J. Pair, P. Hebert, D. Gotz, B. Wills, *et al.*, "Virtual reality exposure therapy for ptsd vietnam veterans: A case study," *Journal of Traumatic Stress: Official Publication of The International Society for Traumatic Stress Studies*, vol. 12, no. 2, pp. 263–271, 1999.
- [23] M. T. Schultheis and A. A. Rizzo, "The application of virtual reality technology in rehabilitation.," *Rehabilitation psychology*, vol. 46, no. 3, p. 296, 2001.
- [24] C. J. Wilson and A. Soranzo, "The use of virtual reality in psychology: A case study in visual perception," *Computational and mathematical methods in medicine*, vol. 2015, 2015.
- [25] M. Csikszentmihalyi and R. Larson, "Validity and reliability of the experience-sampling method," in *Flow and the foundations of positive psychology*, Springer, 2014, pp. 35–54.
- [26] D. L. Morgan, "Paradigms lost and pragmatism regained: Methodological implications of combining qualitative and quantitative methods," *Journal of mixed methods research*, vol. 1, no. 1, pp. 48–76, 2007.
- [27] B. Castellani, "Focus: Complexity and the failure of quantitative social science," *Focus*, vol. 12, p. 12, 2014.
- [28] D. Lazer, A. S. Pentland, L. Adamic, S. Aral, A. L. Barabasi, D. Brewer, N. Christakis, N. Contractor, J. Fowler, M. Gutmann, *et al.*, "Life in the network: The coming age of computational social science," *Science (New York, NY)*, vol. 323, no. 5915, p. 721, 2009.
- [29] D. O. Olgun, P. A. Gloor, and A. S. Pentland, "Capturing individual and group behavior with wearable sensors," in *Proceedings of the 2009 aaai spring symposium on human behavior modeling, SSS*, vol. 9, 2009.

- [30] P. Deville, C. Linard, S. Martin, M. Gilbert, F. R. Stevens, A. E. Gaughan, V. D. Blondel, and A. J. Tatem, “Dynamic population mapping using mobile phone data,” *Proceedings of the National Academy of Sciences*, vol. 111, no. 45, pp. 15 888–15 893, Nov. 2014, ISSN: 0027-8424, 1091-6490. DOI: [10 . 1073 / pnas . 1408439111](https://doi.org/10.1073/pnas.1408439111). [Online]. Available: <http://www.pnas.org/content/111/45/15888> (visited on 12/15/2017).
- [31] N. E. Reichman, J. O. Teitler, I. Garfinkel, and S. S. McLanahan, “Fragile families: Sample and design,” *Children and Youth Services Review*, vol. 23, no. 4-5, pp. 303–326, 2001.
- [32] V. J. Felitti, R. F. Anda, D. Nordenberg, D. F. Williamson, A. M. Spitz, V. Edwards, and J. S. Marks, “Relationship of childhood abuse and household dysfunction to many of the leading causes of death in adults: The adverse childhood experiences (ace) study,” *American journal of preventive medicine*, vol. 14, no. 4, pp. 245–258, 1998.
- [33] S. Friedman, D. Friedman, and S. Sunder, *Experimental methods: A primer for economists*. Cambridge university press, 1994.
- [34] P. G. Zimbardo, “On the ethics of intervention in human psychological research: With special reference to the stanford prison experiment,” *Cognition*, vol. 2, 2017.
- [35] P. G. Zimbardo, C. Haney, W. C. Banks, and D. Jaffe, *The Stanford prison experiment*. Zimbardo, Incorporated, 1971.
- [36] A. Almaatouq, J. Becker, J. P. Houghton, N. Paton, D. J. Watts, and M. E. Whiting, “Empirica: A virtual lab for high-throughput macro-level experiments,” *Behavior Research Methods*, pp. 1–14, 2021.
- [37] M. J. Salganik, P. S. Dodds, and D. J. Watts, “Experimental study of inequality and unpredictability in an artificial cultural market,” *science*, vol. 311, no. 5762, pp. 854–856, 2006.
- [38] Wikipedia contributors, *Reddit — Wikipedia, the free encyclopedia*, <https://en.wikipedia.org/w/index.php?title=Reddit&oldid=1028963157>, [Online; accessed 24-June-2021], 2021.

- [39] J. S. Sekhon and R. Titiunik, “When natural experiments are neither natural nor experiments,” *The American Political Science Review*, vol. 106, no. 1, pp. 35–57, 2012, ISSN: 00030554, 15375943. [Online]. Available: <http://www.jstor.org/stable/23275362>.
- [40] T. Dunning, “Improving causal inference: Strengths and limitations of natural experiments,” *Political Research Quarterly*, vol. 61, no. 2, pp. 282–293, 2008.
- [41] A. Finkelstein, S. Taubman, B. Wright, M. Bernstein, J. Gruber, J. P. Newhouse, H. Allen, K. Baicker, and O. H. S. Group, “The oregon health insurance experiment: Evidence from the first year,” *The Quarterly journal of economics*, vol. 127, no. 3, pp. 1057–1106, 2012.
- [42] P. Isard, “How far can we push the” law of one price”?” *The American Economic Review*, vol. 67, no. 5, pp. 942–948, 1977.
- [43] R. Jensen, “The digital provide: Information (technology), market performance, and welfare in the south indian fisheries sector,” *The quarterly journal of economics*, vol. 122, no. 3, pp. 879–924, 2007.
- [44] J. Srinivasan and J. Burrell, “Revisiting the fishers of kerala, india,” in *Proceedings of the Sixth International Conference on Information and Communication Technologies and Development: Full Papers-Volume 1*, 2013, pp. 56–66.
- [45] S. Assad, R. Clark, D. Ershov, and L. Xu, “Algorithmic pricing and competition: Empirical evidence from the german retail gasoline market,” 2020.
- [46] T.-H. Ho, C. Camerer, and K. Weigelt, “Iterated dominance and iterated best response in experimental” p-beauty contests”,” *The American Economic Review*, vol. 88, no. 4, pp. 947–969, 1998.
- [47] J. M. Keynes, “The general theory of employment,” *The quarterly journal of economics*, vol. 51, no. 2, pp. 209–223, 1937.
- [48] C. F. Camerer, T.-H. Ho, and J.-K. Chong, “A cognitive hierarchy model of games,” *The Quarterly Journal of Economics*, vol. 119, no. 3, pp. 861–898, 2004.

- [49] L. Chen and P. Jaiprakash, “An insurance market simulation with both adverse and advantageous selection,” *Risk Management and Insurance Review*, vol. 20, no. 1, pp. 133–146, 2017.
- [50] S. N. Ali, M. Mihm, L. Siga, and C. Tergiman, “Adverse and advantageous selection in the laboratory,” *Available at SSRN 3448457*, 2019.
- [51] M. Dufwenberg and U. Gneezy, “Price competition and market concentration: An experimental study,” *international Journal of industrial Organization*, vol. 18, no. 1, pp. 7–22, 2000.
- [52] E. Dane, K. W. Rockmann, and M. G. Pratt, “When should I trust my gut? Linking domain expertise to intuitive decision-making effectiveness,” en, *Organizational Behavior and Human Decision Processes*, vol. 119, no. 2, pp. 187–194, Nov. 2012, ISSN: 0749-5978. DOI: [10.1016/j.obhdp.2012.07.009](https://doi.org/10.1016/j.obhdp.2012.07.009).
- [53] T. Heinrich, J. Sabuco, and J. D. Farmer, “A simulation of the insurance industry: The problem of risk model homogeneity,” *Journal of Economic Interaction and Coordination*, pp. 1–42, 2021.
- [54] Wikipedia contributors, *Catastrophe bond — Wikipedia, the free encyclopedia*, https://en.wikipedia.org/w/index.php?title=Catastrophe_bond&oldid=1010068497, [Online; accessed 24-June-2021], 2021.
- [55] R. A. Winter, “The liability insurance market,” *Journal of Economic Perspectives*, vol. 5, no. 3, pp. 115–136, Sep. 1991. DOI: [10.1257/jep.5.3.115](https://doi.org/10.1257/jep.5.3.115). [Online]. Available: <https://www.aeaweb.org/articles?id=10.1257/jep.5.3.115>.
- [56] L. Rocher, A. J. Tournier, and Y.-A. de Montjoye, *Adversarial pricing: Algorithmic competition and collusion*, personal communication, Jun. 2021.
- [57] T. P. Peixoto and S. Bornholdt, “No need for conspiracy: Self-organized cartel formation in a modified trust game,” *Physical review letters*, vol. 108, no. 21, p. 218 702, 2012.
- [58] G. Tesauro and J. O. Kephart, “Pricing in agent economies using multi-agent q-learning,” *Autonomous agents and multi-agent systems*, vol. 5, no. 3, pp. 289–304, 2002.

- [59] T. Klein, “Autonomous algorithmic collusion: Q-learning under sequential pricing,” *Amsterdam Law School Research Paper*, no. 2018-15, pp. 2018–05, 2019.
- [60] L. Chen, A. Mislove, and C. Wilson, “Peeking beneath the hood of uber,” in *Proceedings of the 2015 internet measurement conference*, 2015, pp. 495–508.
- [61] E. Bokányi and A. Hannák, “Understanding inequalities in ride-hailing services through simulations,” *Scientific reports*, vol. 10, no. 1, pp. 1–11, 2020.
- [62] M. M. Vazifeh, P. Santi, G. Resta, S. H. Strogatz, and C. Ratti, “Addressing the minimum fleet problem in on-demand urban mobility,” *Nature*, vol. 557, no. 7706, pp. 534–538, 2018.
- [63] M. Diao, H. Kong, and J. Zhao, “Impacts of transportation network companies on urban mobility,” *Nature Sustainability*, pp. 1–7, 2021.
- [64] I. Owadally, F. Zhou, and D. Wright, “The insurance industry as a complex social system: Competition, cycles and crises,” *Journal of Artificial Societies and Social Simulation*, vol. 21, no. 4, p. 2, 2018, ISSN: 1460-7425. [Online]. Available: <http://jasss.soc.surrey.ac.uk/21/4/2.html>.
- [65] I. Owadally, F. Zhou, R. Otunba, J. Lin, and D. Wright, “An agent-based system with temporal data mining for monitoring financial stability on insurance markets,” *Expert Systems with Applications*, vol. 123, pp. 270–282, 2019, ISSN: 0957-4174. DOI: <https://doi.org/10.1016/j.eswa.2019.01.049>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0957417419300557>.
- [66] E. Moro, M. R. Frank, A. Pentland, A. Rutherford, M. Cebrian, and I. Rahwan, “Universal resilience patterns in labor markets,” *Nature communications*, vol. 12, no. 1, pp. 1–8, 2021.
- [67] A. Van De Rijt, I. A. Akin, R. Willer, and M. Feinberg, “Success-breeds-success in collective political behavior: Evidence from a field experiment,” *Sociological Science*, vol. 3, pp. 940–950, 2016.

- [68] C. A. Gomez-Urbe and N. Hunt, “The netflix recommender system: Algorithms, business value, and innovation,” *ACM Transactions on Management Information Systems (TMIS)*, vol. 6, no. 4, pp. 1–19, 2015.
- [69] D. Holtz, B. Carterette, P. Chandar, Z. Nazari, H. Cramer, and S. Aral, “The engagement-diversity connection: Evidence from a field experiment on spotify,” in *Proceedings of the 21st ACM Conference on Economics and Computation*, 2020, pp. 75–76.
- [70] D. Lee and K. Hosanagar, “How do recommender systems affect sales diversity? a cross-category investigation via randomized field experiment,” *Information Systems Research*, vol. 30, no. 1, pp. 239–259, 2019.
- [71] S. Huang, S. Aral, Y. J. Hu, and E. Brynjolfsson, “Social advertising effectiveness across products: A large-scale field experiment,” *Marketing Science*, vol. 39, no. 6, pp. 1142–1165, 2020.
- [72] R. Bapna and A. Umyarov, “Do Your Online Friends Make You Pay? A Randomized Field Experiment on Peer Influence in Online Social Networks - Online E-Companion Appendix,” en, Social Science Research Network, Rochester, NY, SSRN Scholarly Paper ID 2498080, Sep. 2014. [Online]. Available: <https://papers.ssrn.com/abstract=2498080> (visited on 06/13/2021).
- [73] L. D. Cluff, J. Black, D. Brandon, T. Givens III, and R. Healy, “Summary of reinsurance activities and rating actions tied to selected insurers involved in the failed” unicover” venture,” General Accounting Office Washington DC, Tech. Rep., 2001.
- [74] E. R. Fluegge-Woolf, “Play hard, work hard: Fun at work and job performance,” *Management Research Review*, 2014.
- [75] M. Lister, “Gamification: The effect on student motivation and performance at the post-secondary level,” *Issues and Trends in Educational Technology*, vol. 3, no. 2, 2015.
- [76] S. Palan and C. Schitter, “Prolific. ac - a subject pool for online experiments,” *Journal of Behavioral and Experimental Finance*, vol. 17, pp. 22–27, 2018.
- [77] L. Breiman, “Statistical modeling: The two cultures (with comments and a rejoinder by the author),” *Statistical science*, vol. 16, no. 3, pp. 199–231, 2001.

- [78] G. D. P. Regulation, “Regulation eu 2016/679 of the european parliament and of the council of 27 april 2016,” *Official Journal of the European Union*. Available at: http://ec.europa.eu/justice/data-protection/reform/files/regulation_oj-en.pdf (accessed 20 September 2017), 2016.
- [79] M. E. Kaminski, “The right to explanation, explained,” *Berkeley Tech. LJ*, vol. 34, p. 189, 2019.
- [80] Z. C. Lipton, “The mythos of model interpretability: In machine learning, the concept of interpretability is both important and slippery.,” *Queue*, vol. 16, no. 3, pp. 31–57, 2018.
- [81] A. H. Bailey, “The pure premium method of valuation,” *Journal of the Institute of Actuaries and Assurance Magazine*, vol. 21, no. 2, pp. 115–136, 1878.
- [82] C. Dugas, Y. Bengio, N. Chapados, P. Vincent, G. Denoncourt, and C. Fournier, “Statistical learning algorithms applied to automobile insurance ratemaking,” in *Casualty Actuarial Society Forum*, 2003, pp. 179–213. [Online]. Available: <http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.448.5227&rep=rep1&type=pdf#page=183>.
- [83] F. L. Kitchens, “Financial Implications of Artificial Neural Networks in Automobile Insurance Underwriting,” *Int. J. Electron. Financ.*, vol. 3, no. 3, pp. 311–319, Aug. 2009, ISSN: 1746-0069. DOI: [10.1504/IJEF.2009.027853](https://doi.org/10.1504/IJEF.2009.027853). [Online]. Available: <http://dx.doi.org/10.1504/IJEF.2009.027853> (visited on 08/21/2019).
- [84] N. Chapados, Y. Bengio, P. Vincent, J. Ghosn, C. Dugas, I. Takeuchi, and L. Meng, “Estimating Car Insurance Premia: A Case Study in High-Dimensional Data Inference,” *Advances in Neural Information Processing Systems*, Oct. 2001.
- [85] L. Guelman, “Gradient boosting trees for auto insurance loss cost modeling and prediction,” en, *Expert Systems with Applications*, vol. 39, no. 3, pp. 3659–3667, Feb. 2012, ISSN: 09574174. DOI: [10.1016/j.eswa.2011.09.058](https://doi.org/10.1016/j.eswa.2011.09.058). [Online]. Available: <https://linkinghub.elsevier.com/retrieve/pii/S0957417411013674> (visited on 08/20/2019).

- [86] A. Noll, R. Salzmänn, and M. V. Wuthrich, “Case Study: French Motor Third-Party Liability Claims,” en, Social Science Research Network, Rochester, NY, SSRN Scholarly Paper ID 3164764, Nov. 2018. [Online]. Available: <https://papers.ssrn.com/abstract=3164764> (visited on 08/20/2019).
- [87] C. Mano and E. Rasa, “” a discussion of modeling techniques for personal lines pricing,” *Trans 27th ICA*, 2012.
- [88] S. Gu, B. Kelly, and D. Xiu, “Empirical Asset Pricing via Machine Learning,” National Bureau of Economic Research, Working Paper 25398, Dec. 2018. DOI: [10.3386/w25398](https://doi.org/10.3386/w25398). [Online]. Available: <http://www.nber.org/papers/w25398> (visited on 08/22/2019).
- [89] E. Christodoulou, J. Ma, G. S. Collins, E. W. Steyerberg, J. Y. Verbakel, and B. Van Calster, “A systematic review shows no performance benefit of machine learning over logistic regression for clinical prediction models,” en, *Journal of Clinical Epidemiology*, vol. 110, pp. 12–22, Jun. 2019, ISSN: 0895-4356. DOI: [10.1016/j.jclinepi.2019.02.004](https://doi.org/10.1016/j.jclinepi.2019.02.004).
- [90] A. Fuster, P. Goldsmith-Pinkham, T. Ramadorai, and A. Walther, “Predictably Unequal? The Effects of Machine Learning on Credit Markets,” en, Social Science Research Network, Rochester, NY, SSRN Scholarly Paper ID 3072038, Nov. 2018. [Online]. Available: <https://papers.ssrn.com/abstract=3072038> (visited on 08/22/2019).
- [91] J. Kleinberg, J. Ludwig, S. Mullainathan, and A. Rambachan, “Algorithmic fairness,” in *Aea papers and proceedings*, vol. 108, 2018, pp. 22–27.
- [92] M. M. George E. Rejda, *Principles of Risk Management and Insurance*, 12th Edition, ser. Pearson Series in Finance. Prentice Hall, 2013, ISBN: 0132992914,9780132992916.
- [93] G. J. Marcus, *Heart of Oak: A Survey of British Sea Power in the Georgian Era*. Oxford University Press, 1975.
- [94] C. A. Kulp, “The rate-making process in property and casualty insurance. goals, techniques, and limits,” *Law and Contemporary Problems*, vol. 15, no. 4, pp. 493–522, 1950, ISSN: 00239186. [Online]. Available: <http://www.jstor.org/stable/1189883>.

- [95] B. Keller, M. Eling, H. Schmeiser, *et al.*, *Big data and insurance: implications for innovation, competition and privacy*. Geneva Association-International Association for the Study of Insurance, 2018.
- [96] M. Rothschild and J. Stiglitz, “Equilibrium in competitive insurance markets: An essay on the economics of imperfect information,” en, in *Uncertainty in Economics*, P. Diamond and M. Rothschild, Eds., Academic Press, Jan. 1978, pp. 257–280, ISBN: 978-0-12-214850-7. DOI: [10.1016/B978-0-12-214850-7.50024-3](https://doi.org/10.1016/B978-0-12-214850-7.50024-3).
- [97] A. Ansari and A. Riasi, “Modelling and evaluating customer loyalty using neural networks: Evidence from startup insurance companies,” en, *Future Business Journal*, vol. 2, no. 1, pp. 15–30, Jun. 2016, ISSN: 2314-7210. DOI: [10.1016/j.fbj.2016.04.001](https://doi.org/10.1016/j.fbj.2016.04.001). [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2314721016300019> (visited on 06/13/2021).
- [98] S. B. McGrayne, *The Theory That Would Not Die: How Bayes’ Rule Cracked the Enigma Code, Hunted Down Russian Submarines, & Emerged Triumphant from Two Centuries of C*. Yale University Press, 2011.
- [99] D. V. Carvalho, E. M. Pereira, and J. S. Cardoso, “Machine learning interpretability: A survey on methods and metrics,” *Electronics*, vol. 8, no. 8, p. 832, Jul. 2019, ISSN: 2079-9292. DOI: [10.3390/electronics8080832](https://doi.org/10.3390/electronics8080832). [Online]. Available: <http://dx.doi.org/10.3390/electronics8080832>.
- [100] F. C. Authority., *Feedback Statement: Call for inputs on big data in retail general insurance*, 2016. [Online]. Available: <https://www.fca.org.uk/publication/feedback/fs16-05.pdf> (visited on 02/21/2019).
- [101] European Commission, *Sex Discrimination in Insurance Contracts*, en, Text, 2011. [Online]. Available: https://ec.europa.eu/commission/presscorner/detail/en/MEMO_11_123 (visited on 06/13/2021).
- [102] C. V. Schoubroeck, “Protection of consumers against discrimination: The “test-achats” ruling,” in *IJBF Online*, K. G. Saur, 2009. [Online]. Available: https://www.degruyter.com/document/database/IJBF/entry/ijbf.ID_IJBF2014-9591/html.

- [103] A. C. Yeo, K. A. Smith, R. J. Willis, and M. Brooks, “Modeling the effect of premium changes on motor insurance customer retention rates using neural networks,” in *Computational Science - ICCS 2001*, V. N. Alexandrov, J. J. Dongarra, B. A. Julian, R. S. Renner, and C. J. K. Tan, Eds., Berlin, Heidelberg: Springer Berlin Heidelberg, 2001, pp. 390–399, ISBN: 978-3-540-45718-3.
- [104] S. Panda, “We should have 1,000-1,200 actuaries at the earliest: Irdai chairman,” *Business Standard India*, Mar. 2021. [Online]. Available: https://www.business-standard.com/article/pf/we-should-have-1-000-1-200-actuaries-at-the-earliest-irdai-chairman-121031000619_1.html (visited on 06/13/2021).
- [105] J. D. Gribble, “Actuarial Supply and Demand,” en, *The Congress of the International Actuarial Association*, p. 27, 2006.
- [106] E. Guardiola, “The gameplay loop: A player activity model for game design and analysis,” in *Proceedings of the 13th International Conference on Advances in Computer Entertainment Technology*, ser. ACE '16, Osaka, Japan: Association for Computing Machinery, 2016, ISBN: 9781450347730. DOI: [10.1145/3001773.3001791](https://doi.org/10.1145/3001773.3001791). [Online]. Available: <https://doi.org/10.1145/3001773.3001791>.
- [107] Y. Luo, I. Zelenika, and J. Zhao, “Providing immediate feedback improves recycling and composting accuracy,” en, *Journal of Environmental Management*, vol. 232, pp. 445–454, Feb. 2019, ISSN: 0301-4797. DOI: [10.1016/j.jenvman.2018.11.061](https://doi.org/10.1016/j.jenvman.2018.11.061). [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0301479718313264> (visited on 06/13/2021).
- [108] F.-H. Tsai, C.-C. Tsai, and K.-Y. Lin, “The evaluation of different gaming modes and feedback types on game-based formative assessment in an online learning environment,” en, *Computers & Education*, vol. 81, pp. 259–269, Feb. 2015, ISSN: 0360-1315. DOI: [10.1016/j.compedu.2014.10.013](https://doi.org/10.1016/j.compedu.2014.10.013). [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0360131514002334> (visited on 06/13/2021).

- [109] M. P. Kim, W. Suksompong, and V. V. Williams, “Who can win a single-elimination tournament?” *SIAM Journal on Discrete Mathematics*, vol. 31, no. 3, pp. 1751–1764, 2017.
- [110] G. C. Taylor, “Underwriting strategy in a competitive insurance environment,” en, *Insurance: Mathematics and Economics*, vol. 5, no. 1, pp. 59–77, Jan. 1986, ISSN: 0167-6687. DOI: [10.1016/0167-6687\(86\)90009-0](https://doi.org/10.1016/0167-6687(86)90009-0). [Online]. Available: <http://www.sciencedirect.com/science/article/pii/0167668786900090> (visited on 04/29/2020).
- [111] R. H. Thaler, “Anomalies: The winner’s curse,” *Journal of Economic Perspectives*, vol. 2, no. 1, pp. 191–202, Mar. 1988. DOI: [10.1257/jep.2.1.191](https://doi.org/10.1257/jep.2.1.191). [Online]. Available: <https://www.aeaweb.org/articles?id=10.1257/jep.2.1.191>.
- [112] Wikipedia contributors, *Insurance broker — Wikipedia, the free encyclopedia*, https://en.wikipedia.org/w/index.php?title=Insurance_broker&oldid=1029578544, [Online; accessed 24-June-2021], 2021.
- [113] S. Guven and M. McPhail, “Beyond the cost model: Understanding price elasticity and its applications,” in *Casualty Actuarial Society E-Forum, Spring 2013*, Citeseer, 2013.
- [114] E. Liljeblom and M. Stenius, “Macroeconomic volatility and stock market volatility: Empirical evidence on Finnish data,” *Applied Financial Economics*, vol. 7, no. 4, pp. 419–426, Aug. 1997, ISSN: 0960-3107. DOI: [10.1080/096031097333538](https://doi.org/10.1080/096031097333538). [Online]. Available: <https://doi.org/10.1080/096031097333538> (visited on 06/13/2021).
- [115] O. Jeanne and A. Korinek, “Excessive volatility in capital flows: A pigouvian taxation approach,” *American Economic Review*, vol. 100, no. 2, pp. 403–07, May 2010. DOI: [10.1257/aer.100.2.403](https://doi.org/10.1257/aer.100.2.403). [Online]. Available: <https://www.aeaweb.org/articles?id=10.1257/aer.100.2.403>.
- [116] Research and Markets Institute, *United Kingdom Motor Insurance Market Report 2021: COVID-19 will have Long-Term Impacts on the 14 Billion Industry - ResearchAndMarkets.com*, en, Feb. 2021.
- [117] P. Jaccard, “The distribution of the flora in the alpine zone. 1,” *New phytologist*, vol. 11, no. 2, pp. 37–50, 1912.

- [118] D. Miller and M.-J. Chen, “Sources and consequences of competitive inertia: A study of the u.s. airline industry,” *Administrative Science Quarterly*, vol. 39, no. 1, pp. 1–23, 1994, ISSN: 00018392. [Online]. Available: <http://www.jstor.org/stable/2393492>.
- [119] Association of British Insurers, *UK Insurance & Long-Term Savings - Key Facts*, 2015. [Online]. Available: <https://www.abi.org.uk/globalassets/sitecore/files/documents/publications/public/2016/keyfacts/keyfacts2016.pdf> (visited on 06/13/2021).
- [120] L. Wong, “Myths and critical success factors of personal lines insurance,” en, *The Journal of Risk Management and Insurance*, vol. 19, no. 1, pp. 45–57, Jun. 2015, Number: 1, ISSN: 2773-9260. [Online]. Available: <https://www.jrmi.au.edu/index.php/jrmi/article/view/145> (visited on 06/13/2021).
- [121] P. M. Liedtke, “What’s insurance to a modern economy?” *The Geneva Papers on Risk and Insurance-Issues and Practice*, vol. 32, no. 2, pp. 211–221, 2007.
- [122] The Actuary Magazine, *UK drivers caught in £1.2bn insurance ‘loyalty trap’*, en, Page, Feb. 2019. [Online]. Available: <https://www.theactuary.com/news/2019/02/2019/02/26/uk-drivers-caught-ps12bn-insurance-loyalty-trap> (visited on 06/13/2021).
- [123] G. Robertshaw, “An examination of the profitability of customers acquired through price comparison sites: Implications for the uk insurance industry,” *Journal of Direct, Data and Digital Marketing Practice*, vol. 12, no. 3, pp. 216–229, 2011.
- [124] M. Li and P. Vitanyi, “Algorithmic Complexity,” en, in *An Introduction to Kolmogorov Complexity and Its Applications*, ser. Texts in Computer Science, M. Li and P. Vitanyi, Eds., Cham: Springer International Publishing, 2019, pp. 101–199, ISBN: 978-3-030-11298-1. DOI: [10.1007/978-3-030-11298-1_2](https://doi.org/10.1007/978-3-030-11298-1_2). [Online]. Available: https://doi.org/10.1007/978-3-030-11298-1_2 (visited on 06/13/2021).
- [125] H. Karimi, T. Derr, and J. Tang, “Characterizing the decision boundary of deep neural networks,” *arXiv preprint arXiv:1912.11460*, 2019.

- [126] X. Zhao, Y. Wu, D. L. Lee, and W. Cui, “Iforest: Interpreting random forests via visual analytics,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 25, no. 1, pp. 407–416, 2019. DOI: [10.1109/TVCG.2018.2864475](https://doi.org/10.1109/TVCG.2018.2864475).
- [127] D. Ronayne, “Price comparison websites,” *International Economic Review*, vol. n/a, no. n/a, 2021. DOI: <https://doi.org/10.1111/iere.12504>. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/iere.12504>. [Online]. Available: <https://onlinelibrary.wiley.com/doi/abs/10.1111/iere.12504>.
- [128] M. K. Brunnermeier, R. Lamba, and C. Segura-Rodriguez, “Inverse Selection,” en, Social Science Research Network, Rochester, NY, SSRN Scholarly Paper ID 3584331, Apr. 2020. DOI: [10.2139/ssrn.3584331](https://doi.org/10.2139/ssrn.3584331). [Online]. Available: <https://papers.ssrn.com/abstract=3584331> (visited on 06/13/2021).
- [129] L. Brandeis, “The right to privacy,” *Harvard law review*, vol. 4, no. 5, pp. 193–220, 1890.
- [130] R. F. Hixson, *Privacy in a Public Society: Human Rights in Conflict*. Oxford: Oxford University Press, 1988.
- [131] European Commission. (2018). “Proposal for an ePrivacy Regulation,” [Online]. Available: <https://web.archive.org/web/20181203123422/https://ec.europa.eu/digital-single-market/en/proposal-eprivacy-regulation> (visited on 12/03/2018).
- [132] M. Spiliopoulou, C. Pohle, and L. C. Faulstich, “Inferring demographic attributes of anonymous internet users,” in *International WEBKDD Workshop, San Diego, Calif., August*, vol. 15, 1999.
- [133] K. De Bock and D. Van den Poel, “Predicting website audience demographics for web advertising targeting using multi-website clickstream data,” *Fundamenta Informaticae*, vol. 98, no. 1, pp. 49–70, 2010.
- [134] S. Goel, J. Hofman, and M. I. Sirer, “Who does what on the web: Studying web browsing behavior at scale,” in *International Conference on Weblogs and Social Media*, 2012, pp. 130–137.
- [135] M. Kosinski, D. Stillwell, P. Kohli, Y. Bachrach, and T. Graepel, “Personality and website choice,” 2012.

- [136] J. Hu, H.-J. Zeng, H. Li, C. Niu, and Z. Chen, “Demographic prediction based on user’s browsing behavior,” in *Proceedings of the 16th international conference on World Wide Web*, 2007, pp. 151–160.
- [137] M. F. Mayer, *Rights of privacy. new york, law*, 1972.
- [138] W. Zelermeyer, *Invasion of privacy*. Syracuse University Press, 1959.
- [139] O. M. Ruebhausen and O. G. Brim Jr, “Privacy and behaviorial research,” *Colum. L. Rev.*, vol. 65, p. 1184, 1965.
- [140] J. Holvast, “History of Privacy,” en, in *The Future of Identity in the Information Society*, V. Matyas, S. Fischer-Hubner, D. CvrCek, and P. Svenda, Eds., ser. IFIP Advances in Information and Communication Technology, Berlin, Heidelberg: Springer, 2009, pp. 13–42, ISBN: 978-3-642-03315-5. DOI: [10.1007/978-3-642-03315-5_2](https://doi.org/10.1007/978-3-642-03315-5_2).
- [141] L. E. Daniel and L. E. Daniel, “Chapter 33 - Cellular System Evidence and Call Detail Records,” en, in *Digital Forensics for Legal Professionals*, L. E. Daniel and L. E. Daniel, Eds., Boston: Syngress, Jan. 2012, pp. 225–237, ISBN: 978-1-59749-643-8. DOI: [10.1016/B978-1-59749-643-8.00033-X](https://doi.org/10.1016/B978-1-59749-643-8.00033-X). [Online]. Available: <https://www.sciencedirect.com/science/article/pii/B978159749643800033X> (visited on 04/14/2021).
- [142] M. I. Jordan and T. M. Mitchell, “Machine learning: Trends, perspectives, and prospects,” en, *Science*, vol. 349, no. 6245, pp. 255–260, Jul. 2015, Publisher: American Association for the Advancement of Science Section: Review, ISSN: 0036-8075, 1095-9203. DOI: [10.1126/science.aaa8415](https://doi.org/10.1126/science.aaa8415). [Online]. Available: <https://science.sciencemag.org/content/349/6245/255> (visited on 04/14/2021).
- [143] O. Trelles, P. Prins, M. Snir, and R. C. Jansen, “Big data, but are we ready?” en, *Nature Reviews Genetics*, vol. 12, no. 3, pp. 224–224, Mar. 2011, Number: 3 Publisher: Nature Publishing Group, ISSN: 1471-0064. DOI: [10.1038/nrg2857-c1](https://doi.org/10.1038/nrg2857-c1). [Online]. Available: <https://www.nature.com/articles/nrg2857-c1> (visited on 04/14/2021).
- [144] S. Leonelli, “Scientific Research and Big Data,” in *The Stanford Encyclopedia of Philosophy*, E. N. Zalta, Ed., Summer 2020, Metaphysics Research Lab, Stanford University, 2020.

- [145] S. Lai, E. zu Erbach-Schoenberg, C. Pezzulo, N. W. Ruktanonchai, A. Sorichetta, J. Steele, T. Li, C. A. Dooley, and A. J. Tatem, “Exploring the use of mobile phone data for national migration statistics,” *Palgrave communications*, vol. 5, no. 1, pp. 1–10, 2019.
- [146] A. Pentland, T. Choudhury, N. Eagle, and P. Singh, “Human dynamics: Computation for organizations,” *Pattern Recognition Letters*, vol. 26, no. 4, pp. 503–511, 2005.
- [147] L. Deng, J. Gao, and C. Vuppapapati, “Building a big data analytics service framework for mobile advertising and marketing,” in *2015 IEEE First International Conference on Big Data Computing Service and Applications*, IEEE, 2015, pp. 256–266.
- [148] R. J. Bolton, D. J. Hand, *et al.*, “Statistical fraud detection: A review,” *Statistical science*, vol. 17, no. 3, pp. 235–255, 2002.
- [149] M. A. Utset, “Digital surveillance and preventive policing,” *Conn. L. Rev.*, vol. 49, p. 1453, 2016.
- [150] J. E. Steele, P. R. Sundsøy, C. Pezzulo, V. A. Alegana, T. J. Bird, J. Blumenstock, J. Bjelland, K. Engo-Monsen, Y.-A. de Montjoye, A. M. Iqbal, K. N. Hadiuzzaman, X. Lu, E. Wetter, A. J. Tatem, and L. Bengtsson, “Mapping poverty using mobile phone and satellite data,” *Journal of The Royal Society Interface*, vol. 14, no. 127, p. 20160690, Feb. 2017, ISSN: 1742-5689, 1742-5662. DOI: [10.1098/rsif.2016.0690](https://doi.org/10.1098/rsif.2016.0690). [Online]. Available: <http://rsif.royalsocietypublishing.org/content/14/127/20160690> (visited on 09/28/2018).
- [151] J. Blumenstock, G. Cadamuro, and R. On, “Predicting poverty and wealth from mobile phone metadata,” *Science*, vol. 350, no. 6264, pp. 1073–1076, Nov. 2015, ISSN: 0036-8075, 1095-9203. DOI: [10.1126/science.aac4420](https://doi.org/10.1126/science.aac4420). [Online]. Available: <http://science.sciencemag.org/content/350/6264/1073> (visited on 06/27/2018).
- [152] G. Li, L. Yu, W. S. Ng, W. Wu, and S. T. Goh, “Predicting Home and Work Locations Using Public Transport Smart Card Data by Spectral Analysis,” in *2015 IEEE 18th International Conference on Intelligent Transportation Systems*, Gran Canaria, Spain: IEEE, Sep. 2015, pp. 2788–2793, ISBN: 978-1-4673-6596-3. DOI: [10.1109/ITSC.2015](https://doi.org/10.1109/ITSC.2015).

445. [Online]. Available: <http://ieeexplore.ieee.org/document/7313540/> (visited on 09/28/2018).
- [153] D. Ashbrook and T. Starner, “Learning significant locations and predicting user movement with GPS,” in *Proceedings. Sixth International Symposium on Wearable Computers*, Seattle, WA, USA: IEEE, 2002, pp. 101–108, ISBN: 978-0-7695-1816-9. DOI: [10.1109/ISWC.2002.1167224](https://doi.org/10.1109/ISWC.2002.1167224). [Online]. Available: <http://ieeexplore.ieee.org/document/1167224/> (visited on 09/28/2018).
- [154] S. Isaacman, R. Becker, R. Caceres, S. Kobourov, M. Martonosi, J. Rowland, and A. Varshavsky, “Identifying Important Places in People’s Lives from Cellular Network Data,” in *Pervasive Computing*, K. Lyons, J. Hightower, and E. M. Huang, Eds., ser. Lecture Notes in Computer Science, vol. 6696, Springer Berlin Heidelberg, 2011, pp. 133–151, ISBN: 978-3-642-21726-5. DOI: [10.1007/978-3-642-21726-5_9](https://doi.org/10.1007/978-3-642-21726-5_9).
- [155] J. Mahmud, J. Nichols, and C. Drews, “Home Location Identification of Twitter Users,” *ACM Trans. Intell. Syst. Technol.*, vol. 5, no. 3, 47:1–47:21, Jul. 2014, ISSN: 2157-6904. DOI: [10.1145/2528548](https://doi.org/10.1145/2528548). [Online]. Available: <http://doi.acm.org/10.1145/2528548> (visited on 09/28/2018).
- [156] R. Li, S. Wang, H. Deng, R. Wang, and K. C.-C. Chang, “Towards Social User Profiling: Unified and Discriminative Influence Model for Inferring Home Locations,” in *Proceedings of the 18th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, New York, NY, USA: ACM, 2012, pp. 1023–1031, ISBN: 978-1-4503-1462-6. DOI: [10.1145/2339530.2339692](https://doi.org/10.1145/2339530.2339692). [Online]. Available: <http://doi.acm.org/10.1145/2339530.2339692> (visited on 09/28/2018).
- [157] E. Cho, S. A. Myers, and J. Leskovec, “Friendship and Mobility: User Movement in Location-based Social Networks,” in *Proceedings of the 17th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, New York, NY, USA: ACM, 2011, pp. 1082–1090, ISBN: 978-1-4503-0813-7. DOI: [10.1145/2020408.2020579](https://doi.org/10.1145/2020408.2020579). [Online]. Available: <http://doi.acm.org/10.1145/2020408.2020579> (visited on 09/28/2018).

- [158] D. Monsivais, A. Ghosh, K. Bhattacharya, R. I. M. Dunbar, and K. Kaski, "Tracking urban human activity from mobile phone calling patterns," *PLoS Computational Biology*, vol. 13, no. 11, Nov. 2017, ISSN: 1553-734X. DOI: [10.1371/journal.pcbi.1005824](https://doi.org/10.1371/journal.pcbi.1005824). [Online]. Available: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5697809/> (visited on 09/28/2018).
- [159] D. Monsivais, K. Bhattacharya, A. Ghosh, R. I. M. Dunbar, and K. Kaski, "Seasonal and geographical impact on human resting periods," *Scientific Reports*, vol. 7, Sep. 2017, ISSN: 2045-2322. DOI: [10.1038/s41598-017-11125-z](https://doi.org/10.1038/s41598-017-11125-z). [Online]. Available: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5587566/> (visited on 09/28/2018).
- [160] D. Kondor, B. Hashemian, Y.-A. de Montjoye, and C. Ratti, "Towards matching user mobility traces in large-scale datasets," *IEEE Transactions on Big Data*, pp. 1–1, 2018, ISSN: 2332-7790. DOI: [10.1109/TBDATA.2018.2871693](https://doi.org/10.1109/TBDATA.2018.2871693).
- [161] S. Hasan, X. Zhan, and S. V. Ukkusuri, "Understanding Urban Human Activity and Mobility Patterns Using Large-scale Location-based Data from Online Social Media," in *Proceedings of the 2Nd ACM SIGKDD International Workshop on Urban Computing*, ser. UrbComp '13, New York, NY, USA: ACM, 2013, 6:1–6:8, ISBN: 978-1-4503-2331-4. DOI: [10.1145/2505821.2505823](https://doi.org/10.1145/2505821.2505823). [Online]. Available: <http://doi.acm.org/10.1145/2505821.2505823> (visited on 09/28/2018).
- [162] R. Ahas, A. Aasa, S. Silm, and M. Tiru, "Daily rhythms of suburban commuters' movements in the Tallinn metropolitan area: Case study with mobile positioning data," *Transportation Research Part C: Emerging Technologies*, Information/Communication Technologies and Travel Behaviour, vol. 18, no. 1, pp. 45–54, Feb. 2010, ISSN: 0968-090X. DOI: [10.1016/j.trc.2009.04.011](https://doi.org/10.1016/j.trc.2009.04.011). [Online]. Available: <http://www.sciencedirect.com/science/article/pii/S0968090X09000400> (visited on 09/28/2018).
- [163] B. Felbo, P. Sundsøy, A. Pentland, S. Lehmann, and Y.-A. d. Montjoye, "Modeling the Temporal Nature of Human Behavior for Demographics Prediction," en, in *Machine Learning and Knowledge Discovery in Databases*, ser. Lecture Notes in Computer Science, vol. 10536, Springer, Cham, Sep. 2017, pp. 140–152, ISBN: 978-3-319-71272-

- 7 978-3-319-71273-4. DOI: [10.1007/978-3-319-71273-4_12](https://doi.org/10.1007/978-3-319-71273-4_12). [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-319-71273-4_12 (visited on 06/27/2018).
- [164] Y.-A. de Montjoye, J. Quoidbach, F. Robic, and A. S. Pentland, “Predicting personality using novel mobile phone-based metrics,” in *International conference on social computing, behavioral-cultural modeling, and prediction*, Springer, 2013, pp. 48–55. DOI: [10.1007/978-3-642-37210-0_6](https://doi.org/10.1007/978-3-642-37210-0_6).
- [165] J.-P. Onnela, J. Saramäki, J. Hyvönen, G. Szabó, D. Lazer, K. Kaski, J. Kertész, and A.-L. Barabási, “Structure and tie strengths in mobile communication networks,” *Proceedings of the national academy of sciences*, vol. 104, no. 18, pp. 7332–7336, 2007. DOI: [10.1073/pnas.0610245104](https://doi.org/10.1073/pnas.0610245104).
- [166] J.-P. Onnela, J. Saramäki, J. Hyvönen, G. Szabó, M. A. De Menezes, K. Kaski, A.-L. Barabási, and J. Kertész, “Analysis of a large-scale weighted network of one-to-one human communication,” *New journal of physics*, vol. 9, no. 6, p. 179, 2007. DOI: [10.1088/1367-2630/9/6/179](https://doi.org/10.1088/1367-2630/9/6/179).
- [167] C. Krumme, A. Llorente, M. Cebrian, E. Moro, *et al.*, “The predictability of consumer visitation patterns,” *Scientific reports*, vol. 3, p. 1645, 2013. DOI: [10.1038/srep01645](https://doi.org/10.1038/srep01645).
- [168] European Parliament and Council of European Union. (2016). “Regulation (eu) 2016/679,” [Online]. Available: <https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:32016R0679&from=EN> (visited on 11/12/2020).
- [169] W. Sheldon, “The concept of the negative,” *The Philosophical Review*, vol. 11, no. 5, pp. 485–496, 1902.
- [170] E. J. Williamson, A. J. Walker, K. Bhaskaran, S. Bacon, C. Bates, C. E. Morton, H. J. Curtis, A. Mehrkar, D. Evans, P. Inglesby, *et al.*, “Factors associated with covid-19-related death using opensafely,” *Nature*, vol. 584, no. 7821, pp. 430–436, 2020.
- [171] Y.-A. De Montjoye, E. Shmueli, S. S. Wang, and A. S. Pentland, “Openpds: Protecting the privacy of metadata through safeanswers,” *PloS one*, vol. 9, no. 7, e98790, 2014.

- [172] P. Francis, S. P. Eide, and R. Munz, “Diffix: High-utility database anonymization,” in *Annual Privacy Forum*, Springer, 2017, pp. 141–158.
- [173] A. Narayanan and V. Shmatikov, “Robust De-anonymization of Large Sparse Datasets,” IEEE, May 2008, pp. 111–125, ISBN: 978-0-7695-3168-7. DOI: [10.1109/SP.2008.33](https://doi.org/10.1109/SP.2008.33). [Online]. Available: <http://ieeexplore.ieee.org/document/4531148/> (visited on 11/03/2017).
- [174] Y.-A. de Montjoye, C. A. Hidalgo, M. Verleysen, and V. D. Blondel, “Unique in the crowd: The privacy bounds of human mobility,” *Scientific reports*, vol. 3, p. 1376, 2013.
- [175] A. Di Luzio, A. Mei, and J. Stefa, “Consensus robustness and transaction de-anonymization in the ripple currency exchange system,” in *2017 IEEE 37th International Conference on Distributed Computing Systems (ICDCS)*, IEEE, 2017, pp. 140–150.
- [176] Y.-A. de Montjoye, L. Radaelli, V. K. Singh, *et al.*, “Unique in the shopping mall: On the reidentifiability of credit card metadata,” *Science*, vol. 347, no. 6221, pp. 536–539, 2015.
- [177] A. Lavrenovs and K. Podins, “Privacy violations in riga open data public transport system,” in *2016 IEEE 4th Workshop on Advances in Information, Electronic and Electrical Engineering (AIEEE)*, IEEE, 2016, pp. 1–6.
- [178] Y. Erlich, T. Shor, I. Pe’er, and S. Carmi, “Identity inference of genomic data using long-range familial searches,” *Science*, vol. 362, no. 6415, pp. 690–694, 2018, ISSN: 0036-8075. DOI: [10.1126/science.aau4832](https://doi.org/10.1126/science.aau4832). [Online]. Available: <https://science.sciencemag.org/content/362/6415/690>.
- [179] A. Ceca, M. Mamei, and F. Zambonelli, “Re-identification and information fusion between anonymized cdr and social network data,” *Journal of Ambient Intelligence and Humanized Computing*, vol. 7, no. 1, pp. 83–96, 2016.
- [180] D. Sanchez, S. Martinez, and J. Domingo-Ferrer, “Comment on ”Unique in the shopping mall: On the reidentifiability of credit card metadata”,,” *Science*, vol. 351, no. 6279, pp. 1274–1274, Mar. 2016, ISSN: 0036-8075, 1095-9203. DOI: [10.1126/science.aad9295](https://doi.org/10.1126/science.aad9295).

- [Online]. Available: <http://science.sciencemag.org/content/351/6279/1274.1> (visited on 09/28/2018).
- [181] K. El Emam. (2015). “On Re-identification: Not Really Unique in the Shopping Mall,” [Online]. Available: <https://web.archive.org/web/20181203123033/https://iapp.org/news/a/on-re-identification-not-really-unique-in-the-shopping-mall/> (visited on 12/03/2018).
- [182] D. Barth-Jones, K. El Emam, J. Bambauer, A. Cavoukian, and B. Malin, “Assessing data intrusion threats,” *Science*, vol. 348, no. 6231, pp. 194–195, 2015.
- [183] D. de Zayas, “Carpenter v. united states and the emerging expectation of privacy in data comprehensiveness applied to browsing history,” *Am. UL Rev.*, vol. 68, p. 2209, 2018.
- [184] S. Sagiroglu and G. Canbek, “Keyloggers: Increasing threats to computer security and privacy,” *IEEE technology and society magazine*, vol. 28, no. 3, pp. 10–17, 2009.
- [185] D. Kravets, “An Intentional Mistake: The Anatomy of Google’s Wi-Fi Sniffing Debacle — WIRED,” *Wired.com*, 2012. [Online]. Available: <https://www.wired.com/2012/05/google-wifi-fcc-investigation/> (visited on 06/14/2021).
- [186] S. J. Daniel, “A brief history of information privacy law,” *PROSKAUER ON Privacy: A Guide to Privacy and Data Security Law in the Information Age*, pp. 1–3, 2006.
- [187] A. A. Bushkin and S. I. Schaen, *The privacy act of 1974: A reference manual for compliance*. System Development Corporation McLean, Va., 1976.
- [188] European Union, *Directive 95/46/EC of the European Parliament and of the Council on the Protection of Individuals with Regard to the Processing of Personal Data and on the Free Movement of Such Data*, 1995.
- [189] J. Wakefield, “Europe seeks to limit use of ai in society,” *BBC*, 2021. [Online]. Available: <https://www.bbc.co.uk/news/technology-56745730>.
- [190] L. Taylor, L. Floridi, and B. Van der Sloot, *Group privacy: New challenges of data technologies*. Springer, 2016, vol. 126.

- [191] E. Pollman, “A corporate right to privacy,” *Minn. L. Rev.*, vol. 99, p. 27, 2014.
- [192] N. Ferguson and B. Schneier, *Practical cryptography*. Wiley New York, 2003, vol. 141.
- [193] J. E. Cohen, “What privacy is for,” *Harv. L. Rev.*, vol. 126, p. 1904, 2012.
- [194] L. Sweeney, “K-anonymity: A model for protecting privacy,” *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems*, vol. 10, no. 05, pp. 557–570, 2002.
- [195] C. C. Aggarwal, “On k-anonymity and the curse of dimensionality,” in *VLDB*, vol. 5, 2005, pp. 901–909.
- [196] A. Machanavajjhala, D. Kifer, J. Gehrke, and M. Venkatasubramanian, “L-diversity: Privacy beyond k-anonymity,” *ACM Transactions on Knowledge Discovery from Data (TKDD)*, vol. 1, no. 1, 3–es, 2007.
- [197] N. Li, T. Li, and S. Venkatasubramanian, “T-closeness: Privacy beyond k-anonymity and l-diversity,” in *2007 IEEE 23rd International Conference on Data Engineering*, IEEE, 2007, pp. 106–115.
- [198] Article 29 Data Protection Working Party. (2014). “Opinion 05/2014 on anonymisation techniques,” [Online]. Available: https://ec.europa.eu/justice/article-29/documentation/opinion-recommendation/files/2014/wp216_en.pdf.
- [199] J. Pei, J. Xu, Z. Wang, W. Wang, and K. Wang, “Maintaining K-Anonymity against Incremental Updates,” in *19th International Conference on Scientific and Statistical Database Management (SSDBM 2007)*, ISSN: 1551-6393, Jul. 2007, pp. 5–5. DOI: [10.1109/SSDBM.2007.16](https://doi.org/10.1109/SSDBM.2007.16).
- [200] Y. Wu, Z. Sun, and X. Wang, “Privacy Preserving k-Anonymity for Re-publication of Incremental Datasets,” in *2009 WRI World Congress on Computer Science and Information Engineering*, vol. 4, Mar. 2009, pp. 53–60. DOI: [10.1109/CSIE.2009.549](https://doi.org/10.1109/CSIE.2009.549).
- [201] A. Oehmichen, S. Jain, A. Gadotti, and Y.-A. de Montjoye, “Opal: High performance platform for large-scale privacy-preserving location data analytics,” in *2019 IEEE International Conference on Big Data (Big Data)*, IEEE, 2019, pp. 1332–1342.

- [202] A. Schultze, A. J. Walker, B. MacKenna, C. E. Morton, K. Bhaskaran, J. P. Brown, C. T. Rentsch, E. Williamson, H. Drysdale, R. Croker, *et al.*, “Risk of covid-19-related death among patients with chronic obstructive pulmonary disease or asthma prescribed inhaled corticosteroids: An observational cohort study using the opensafely platform,” *The Lancet Respiratory Medicine*, vol. 8, no. 11, pp. 1106–1120, 2020.
- [203] C. T. Rentsch, N. J. DeVito, B. MacKenna, C. E. Morton, K. Bhaskaran, J. P. Brown, A. Schultze, W. J. Hulme, R. Croker, A. J. Walker, *et al.*, “Effect of pre-exposure use of hydroxychloroquine on covid-19 mortality: A population-based cohort study in patients with rheumatoid arthritis or systemic lupus erythematosus using the opensafely platform,” *The Lancet Rheumatology*, vol. 3, no. 1, e19–e27, 2021.
- [204] A. Y. Wong, B. MacKenna, C. E. Morton, A. Schultze, A. J. Walker, K. Bhaskaran, J. P. Brown, C. T. Rentsch, E. Williamson, H. Drysdale, *et al.*, “Use of non-steroidal anti-inflammatory drugs and risk of death from covid-19: An opensafely cohort analysis based on two cohorts,” *Annals of the rheumatic diseases*, 2021.
- [205] K. Bhaskaran, C. T. Rentsch, B. MacKenna, A. Schultze, A. Mehrkar, C. J. Bates, R. M. Eggo, C. E. Morton, S. C. Bacon, P. Inglesby, *et al.*, “Hiv infection and covid-19 death: A population-based cohort analysis of uk primary care data and linked national death registrations within the opensafely platform,” *The Lancet HIV*, vol. 8, no. 1, e24–e32, 2021.
- [206] H. Forbes, C. E. Morton, S. Bacon, H. I. McDonald, C. Minassian, J. P. Brown, C. T. Rentsch, R. Mathur, A. Schultze, N. J. DeVito, *et al.*, “Association between living with children and outcomes from covid-19: Opensafely cohort study of 12 million adults in england,” *bmj*, vol. 372, 2021.
- [207] D. J. Grint, K. Wing, E. Williamson, H. I. McDonald, K. Bhaskaran, D. Evans, S. J. Evans, A. J. Walker, G. Hickman, E. Nightingale, *et al.*, “Case fatality risk of the sars-cov-2 variant of concern b. 1.1. 7 in england, 16 november to 5 february,” *Eurosurveillance*, vol. 26, no. 11, p. 2100256, 2021.

- [208] A. Schultze, C. Bates, J. Cockburn, B. MacKenna, E. Nightingale, H. J. Curtis, W. J. Hulme, C. E. Morton, R. Croker, S. Bacon, *et al.*, “Identifying care home residents in electronic health records-an opensafely short data report,” *Wellcome Open Research*, vol. 6, no. 90, p. 90, 2021.
- [209] R. Mathur, C. T. Rentsch, C. E. Morton, W. J. Hulme, A. Schultze, B. MacKenna, R. M. Eggo, K. Bhaskaran, A. Y. Wong, E. J. Williamson, *et al.*, “Ethnic differences in sars-cov-2 infection and covid-19-related hospitalisation, intensive care unit admission, and death in 17 million adults in england: An observational cohort study using the opensafely platform,” *The Lancet*, vol. 397, no. 10286, pp. 1711–1724, 2021.
- [210] K. H. Grantz, H. R. Meredith, D. A. Cummings, C. J. E. Metcalf, B. T. Grenfell, J. R. Giles, S. Mehta, S. Solomon, A. Labrique, N. Kishore, *et al.*, “The use of mobile phone data to inform analysis of covid-19 pandemic epidemiology,” *Nature communications*, vol. 11, no. 1, pp. 1–8, 2020.
- [211] R. Wilson, E. zu Erbach-Schoenberg, M. Albert, D. Power, S. Tudge, M. Gonzalez, S. Guthrie, H. Chamberlain, C. Brooks, C. Hughes, *et al.*, “Rapid and near real-time assessments of population displacement using mobile phone data following disasters: The 2015 nepal earthquake,” *PLoS currents*, vol. 8, 2016.
- [212] C. Dwork, A. Roth, *et al.*, “The algorithmic foundations of differential privacy,” *Foundations and Trends in Theoretical Computer Science*, vol. 9, no. 3-4, pp. 211–407, 2014.
- [213] Ú. Erlingsson, V. Pihur, and A. Korolova, “Rappor: Randomized aggregatable privacy-preserving ordinal response,” in *Proceedings of the 2014 ACM SIGSAC conference on computer and communications security*, 2014, pp. 1054–1067.
- [214] B. Ding, J. Kulkarni, and S. Yekhanin, “Collecting telemetry data privately,” *arXiv preprint arXiv:1712.01524*, 2017.
- [215] R. Rogers, S. Subramaniam, S. Peng, D. Durfee, S. Lee, S. K. Kancha, S. Sahay, and P. Ahammad, “Linkedin’s audience engagements api: A privacy preserving data analytics system at scale,” *arXiv preprint arXiv:2002.05839*, 2020.

- [216] G. Cormode, S. Jha, T. Kulkarni, N. Li, D. Srivastava, and T. Wang, “Privacy at scale: Local differential privacy in practice,” in *Proceedings of the 2018 International Conference on Management of Data*, 2018, pp. 1655–1658.
- [217] J. M. Abowd, “The us census bureau adopts differential privacy,” in *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2018, pp. 2867–2867.
- [218] A. R. Santos-Lozada, J. T. Howard, and A. M. Verdery, “How differential privacy will affect our understanding of health disparities in the united states,” *Proceedings of the National Academy of Sciences*, vol. 117, no. 24, pp. 13 405–13 412, 2020.
- [219] J. Tang, A. Korolova, X. Bai, X. Wang, and X. Wang, *Privacy loss in apple’s implementation of differential privacy on macos 10.12*, 2017. arXiv: [1709.02753](https://arxiv.org/abs/1709.02753) [cs.CR].
- [220] F. du Pin Calmon and N. Fawaz, “Privacy against statistical inference,” in *2012 50th annual Allerton conference on communication, control, and computing (Allerton)*, IEEE, 2012, pp. 1401–1408.
- [221] S. L. Garfinkel, J. M. Abowd, and S. Powazek, “Issues encountered deploying differential privacy,” in *Proceedings of the 2018 Workshop on Privacy in the Electronic Society*, 2018, pp. 133–137.
- [222] A. Gadotti, F. Houssiau, L. Rocher, B. Livshits, and Y.-A. De Montjoye, “When the signal is in the noise: Exploiting diffix’s sticky noise,” in *28th {USENIX} Security Symposium ({USENIX} Security 19)*, 2019, pp. 1081–1098.
- [223] Aircloack, *Diffix Vulnerability #4*, en, 2019. [Online]. Available: <https://aircloak.com/security/diffix-vulnerability-4/> (visited on 06/14/2021).
- [224] D. Lazer and J. Radford, “Data ex machina: Introduction to big data,” *Annual Review of Sociology*, vol. 43, pp. 19–39, 2017.
- [225] C. Riederer, Y. Kim, A. Chaintreau, N. Korula, and S. Lattanzi, “Linking users across domains with location data: Theory and validation,” in *Proceedings of the 25th International Conference on World Wide Web*, International World Wide Web Conferences Steering Committee, 2016, pp. 707–719.

- [226] R. Pellungrini, L. Pappalardo, F. Pratesi, and A. Monreale, “A Data Mining Approach to Assess Privacy Risk in Human Mobility Data,” *ACM Trans. Intell. Syst. Technol.*, vol. 9, no. 3, 31:1–31:27, Dec. 2017, ISSN: 2157-6904. DOI: [10.1145/3106774](https://doi.org/10.1145/3106774). [Online]. Available: <http://doi.acm.org/10.1145/3106774> (visited on 09/28/2018).
- [227] J. P. Achara, G. Acs, and C. Castelluccia, “On the Unicity of Smartphone Applications,” ACM Press, 2015, pp. 27–36, ISBN: 978-1-4503-3820-2. DOI: [10.1145/2808138.2808146](https://doi.org/10.1145/2808138.2808146). [Online]. Available: <http://dl.acm.org/citation.cfm?doid=2808138.2808146> (visited on 12/01/2017).
- [228] V. Sekara, E. Mones, and H. Jonsson, “Temporal limits of privacy in human behavior,” *arXiv preprint arXiv:1806.03615*, 2018.
- [229] Y. Xu, A. Belyi, I. Bojic, and C. Ratti, “Human mobility and socioeconomic status: Analysis of Singapore and Boston,” *Computers, Environment and Urban Systems*, vol. 72, pp. 51–67, Nov. 2018, ISSN: 0198-9715. DOI: [10.1016/j.compenvurbsys.2018.04.001](https://doi.org/10.1016/j.compenvurbsys.2018.04.001). [Online]. Available: <http://www.sciencedirect.com/science/article/pii/S0198971517304179> (visited on 12/03/2018).
- [230] C. Deußer, S. Passmann, and T. Strufe, “Browsing unicity: On the limits of anonymizing web tracking data,” in *2020 IEEE Symposium on Security and Privacy (SP)*, IEEE, 2020, pp. 777–790.
- [231] Y. Erlich, T. Shor, I. Peter, and S. Carmi, “Identity inference of genomic data using long-range familial searches,” *Science*, vol. 362, no. 6415, pp. 690–694, 2018, ISSN: 0036-8075. DOI: [10.1126/science.aau4832](https://doi.org/10.1126/science.aau4832). [Online]. Available: <https://science.sciencemag.org/content/362/6415/690>.
- [232] President’s Council of Advisers on Science and Technology, *Big data and privacy: A technological perspective*, 2014. [Online]. Available: https://bigdatawg.nist.gov/pdf/pcast_big_data_and_privacy_-_may_2014.pdf (visited on 06/14/2021).
- [233] M. Gramaglia and M. Fiore, “On the anonymizability of mobile traffic datasets,” *arXiv preprint arXiv:1501.00100*, 2014.

- [234] Y. Song, D. Dahlmeier, and S. Bressan, “Not so unique in the crowd: A simple and effective algorithm for anonymizing location data,” in *PIR@ SIGIR*, 2014.
- [235] A. Boutet, S. B. Mokhtar, and V. Primault, “Uniqueness assessment of human mobility on multi-sensor datasets,” 2016.
- [236] Vodafone. (2018). “Vodafone UK’s company history and achievements,” [Online]. Available: <https://web.archive.org/web/20181203120615/https://www.vodafone.co.uk/about-us/company-history/> (visited on 12/03/2018).
- [237] L. Rocher, J. M. Hendrickx, and Y.-A. De Montjoye, “Estimating the success of re-identifications in incomplete datasets using generative models,” *Nature communications*, vol. 10, no. 1, pp. 1–9, 2019.
- [238] L. Pappalardo and F. Simini, “Data-driven generation of spatio-temporal routines in human mobility,” *Data Mining and Knowledge Discovery*, vol. 32, no. 3, pp. 787–829, 2018. DOI: [10.1007/s10618-017-0548-4](https://doi.org/10.1007/s10618-017-0548-4).
- [239] M. C. Gonzalez, C. A. Hidalgo, and A.-L. Barabasi, “Understanding individual human mobility patterns,” *Nature*, vol. 453, no. 7196, p. 779, Jun. 2008, ISSN: 1476-4687. DOI: [10.1038/nature06958](https://doi.org/10.1038/nature06958). [Online]. Available: <https://www.nature.com/articles/nature06958> (visited on 12/12/2017).
- [240] L. Alessandretti, P. Sapiezynski, V. Sekara, S. Lehmann, and A. Baronchelli, “Evidence for a conserved quantity in human mobility,” *Nature Human Behaviour*, p. 1, Jun. 2018, ISSN: 2397-3374. DOI: [10.1038/s41562-018-0364-x](https://doi.org/10.1038/s41562-018-0364-x). [Online]. Available: <https://www.nature.com/articles/s41562-018-0364-x> (visited on 06/25/2018).
- [241] C. Song, T. Koren, P. Wang, and A.-L. Barabasi, “Modelling the scaling properties of human mobility,” *Nature Physics*, vol. 6, no. 10, pp. 818–823, Oct. 2010, ISSN: 1745-2481. DOI: [10.1038/nphys1760](https://doi.org/10.1038/nphys1760). [Online]. Available: <https://www.nature.com/articles/nphys1760> (visited on 09/14/2018).
- [242] S. Hasan, C. M. Schneider, S. V. Ukkusuri, and M. C. Gonzalez, “Spatiotemporal Patterns of Urban Human Mobility,” *Journal of Statistical Physics*, vol. 151, no. 1, pp. 304–

- 318, Apr. 2013, ISSN: 1572-9613. DOI: [10.1007/s10955-012-0645-0](https://doi.org/10.1007/s10955-012-0645-0). [Online]. Available: <https://doi.org/10.1007/s10955-012-0645-0> (visited on 09/28/2018).
- [243] B. Delaunay, “Sur la sphere vide. a la memoire de georges voronoi,” *Bulletin de l’Academie des Sciences de l’URSS. Classe des sciences mathematiques et na*, pp. 793–800, 6 1934.
- [244] L. N. Vaserstein, “Markov processes over denumerable products of spaces, describing large systems of automata,” *Problemy Peredachi Informatsii*, vol. 5, no. 3, pp. 64–72, 1969.
- [245] Centre for Humanitarian Data of the United Nations Office for the Coordination of Humanitarian Affairs. (2019). “Guidance note series on data responsibility on humanitarian action. note 1: Statistical disclosure control,” [Online]. Available: https://centre.humdata.org/wp-content/uploads/2019/07/guidance_note_sdc.pdf.
- [246] D. J. Mir, S. Isaacman, R. Cáceres, M. Martonosi, and R. N. Wright, “Dp-where: Differentially private modeling of human mobility,” in *2013 IEEE international conference on big data*, IEEE, 2013, pp. 580–588.
- [247] M. C. Gonzalez, C. A. Hidalgo, and A.-L. Barabasi, “Understanding individual human mobility patterns,” *nature*, vol. 453, no. 7196, p. 779, 2008. DOI: [10.1038/nature06958](https://doi.org/10.1038/nature06958).
- [248] L. Alessandretti, P. Sapiezynski, V. Sekara, S. Lehmann, and A. Baronchelli, “Evidence for a conserved quantity in human mobility,” *Nature human behaviour*, vol. 2, no. 7, pp. 485–491, 2018.
- [249] E. Graells-Garrido, O. Peredo, and J. Garcia, “Sensing urban patterns with antenna mappings: The case of santiago, chile,” *Sensors*, vol. 16, no. 7, p. 1098, 2016.
- [250] L. Chen, A. Mislove, and C. Wilson, “An empirical analysis of algorithmic pricing on amazon marketplace,” in *Proceedings of the 25th international conference on World Wide Web*, 2016, pp. 1339–1349.
- [251] S. Kelkar, “Post-truth and the search for objectivity: Political polarization and the remaking of knowledge production,” *Engaging Science, Technology, and Society*, vol. 5, pp. 86–106, 2019.

- [252] C. Pattie and R. Johnston, “Talking with one voice? Conversation networks and political polarisation,” en, *The British Journal of Politics and International Relations*, vol. 18, no. 2, pp. 482–497, May 2016, Publisher: SAGE Publications, ISSN: 1369-1481. DOI: [10.1177/1369148115620989](https://doi.org/10.1177/1369148115620989). [Online]. Available: <https://doi.org/10.1177/1369148115620989> (visited on 06/14/2021).
- [253] Y. Kim, “Does Disagreement Mitigate Polarization? How Selective Exposure and Disagreement Affect Political Polarization,” en, *Journalism & Mass Communication Quarterly*, vol. 92, no. 4, pp. 915–937, Dec. 2015, Publisher: SAGE Publications Inc, ISSN: 1077-6990. DOI: [10.1177/1077699015596328](https://doi.org/10.1177/1077699015596328). [Online]. Available: <https://doi.org/10.1177/1077699015596328> (visited on 06/14/2021).
- [254] C. A. Bail, L. P. Argyle, T. W. Brown, J. P. Bumpus, H. Chen, M. B. F. Hunzaker, J. Lee, M. Mann, F. Merhout, and A. Volfovsky, “Exposure to opposing views on social media can increase political polarization,” *Proceedings of the National Academy of Sciences*, vol. 115, no. 37, pp. 9216–9221, 2018, ISSN: 0027-8424. DOI: [10.1073/pnas.1804840115](https://doi.org/10.1073/pnas.1804840115). [Online]. Available: <https://www.pnas.org/content/115/37/9216>.

Appendices

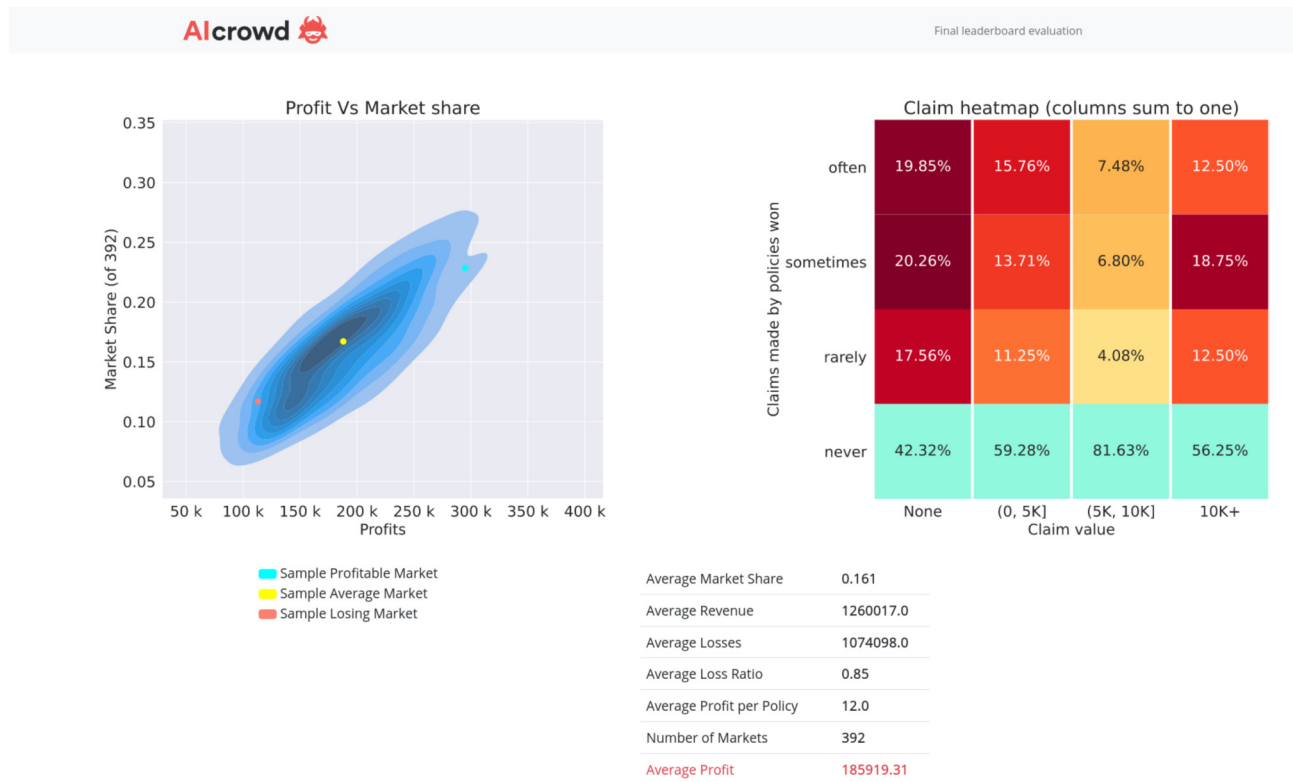
Appendix A

Insurance markets & machine learning

A case-study in simulating a market with real experts

A.1 Full participant feedback report example

In figure [A.1](#) a full example report made available to privately to each participant can be seen beginning from the next page.



For a description on how the tables and plots are computed, refer to [this post on discourse](#).

Policy

Policies won	year	pol_no_claims_discount	pol_coverage	pol_duration	pol_sit_duration	pol_pay_freq	pol_payd	pol_usage
often: 37.5 - 100.0%	5.00	0.01	Max (61%)	15.93	6.97	Yearly (41%)	No (95%)	WorkPrivate (64%)
sometimes: 7.4 - 37.5%	5.00	0.02	Max (48%)	15.65	7.37	Yearly (38%)	No (95%)	WorkPrivate (68%)
rarely: 0.3 - 7.4%	5.00	0.03	Max (33%)	14.66	7.16	Yearly (36%)	No (95%)	WorkPrivate (70%)
never:	5.00	0.04	Max (87%)	14.73	6.21	Yearly (34%)	No (97%)	WorkPrivate (64%)
in sample profitable market	5.00	0.02	Max (60%)	15.57	6.78	Yearly (40%)	No (95%)	WorkPrivate (65%)
in sample average market	5.00	0.01	Max (48%)	15.49	6.82	Yearly (38%)	No (95%)	WorkPrivate (67%)
in sample losing market	5.00	0.01	Max (60%)	15.05	6.89	Yearly (43%)	No (96%)	WorkPrivate (64%)

Driver

Policies won	drv_sex1	drv_age1	drv_age_lic1	drv_drv2	drv_sex2	drv_age2	drv_age_lic2
often: 37.5 - 100.0%	M (59%)	60.17	37.36	No (65%)	F (57%)	53.72	31.37
sometimes: 7.4 - 37.5%	M (60%)	59.17	36.67	No (69%)	F (62%)	51.48	29.55
rarely: 0.3 - 7.4%	M (63%)	58.01	35.96	No (71%)	F (62%)	49.57	28.11
never:	M (60%)	58.37	36.48	No (65%)	F (63%)	50.25	28.45
in sample profitable market	M (59%)	59.40	36.78	No (65%)	F (59%)	52.79	30.74
in sample average market	M (60%)	58.90	36.36	No (66%)	F (56%)	52.46	30.20
in sample losing market	M (61%)	60.03	37.10	No (63%)	F (58%)	54.03	31.29

Vehicle

Policies won	vh_make_model	vh_age	vh_fuel	vh_type	vh_speed	vh_value	vh_weight
often: 37.5 - 100.0%	rthsjevgdimkygk (9%)	14.17	Gasoline (57%)	Tourism (91%)	167.09	15743.08	1028.43
sometimes: 7.4 - 37.5%	rthsjevgdimkygk (9%)	16.79	Gasoline (58%)	Tourism (88%)	167.00	16065.81	1031.03
rarely: 0.3 - 7.4%	rthsjevgdimkygk (7%)	18.18	Gasoline (51%)	Tourism (84%)	165.03	16343.69	1037.64
never:	rthsjevgdimkygk (6%)	10.01	Diesel (68%)	Tourism (93%)	176.87	19858.77	1182.46
in sample profitable market	rthsjevgdimkygk (10%)	13.79	Gasoline (53%)	Tourism (90%)	168.44	16213.59	1047.75
in sample average market	rthsjevgdimkygk (9%)	14.60	Gasoline (57%)	Tourism (90%)	165.74	15626.33	1005.24
in sample losing market	rthsjevgdimkygk (9%)	14.03	Gasoline (55%)	Tourism (88%)	164.66	15735.73	1008.35

Location

Policies won	population	town_surface_area
often: 37.5 - 100.0%	546.94	188.55
sometimes: 7.4 - 37.5%	530.25	206.20
rarely: 0.3 - 7.4%	524.97	220.87
never:	618.73	203.68
in sample profitable market	540.99	189.95
in sample average market	562.58	194.56
in sample losing market	521.13	174.41

Financials By Conversion Rate

Policies won	claim_frequency	premiums	conversion_rate	profit per policy	profit_margin
often: 37.5 - 100.0%	0.07	83.70	0.65	8.35	0.20
sometimes: 7.4 - 37.5%	0.06	73.60	0.18	1.79	0.23
rarely: 0.3 - 7.4%	0.06	68.72	0.03	0.35	0.24
never:	0.12	234.95	0.00	0.00	0.37

Financials In Sample Markets

Policies won	claim_frequency	premiums	conversion_rate	profit	revenue	loss	market_share	profit_margin
in sample profitable market	0.07	85.78	0.53	294943.53	1864025.92	1569082.39	0.23	0.20
in sample average market	0.07	76.03	0.61	187937.55	1208356.16	1020418.61	0.17	0.21
in sample losing market	0.08	84.27	0.67	113035.62	937040.84	824005.22	0.12	0.20

Financials Of Policies Lost In Sample Markets

Policies lost	premiums (mean)	premiums (median)	premiums (std)	distance to winner (mean)
in sample profitable market	112.65	115.70	69.99	48.24
in sample average market	108.75	109.77	64.35	48.56
in sample losing market	105.30	103.49	63.98	46.29

Figure A.1: Example of a full participant feedback report.

A.2 Ethics approval

**Imperial College
London**

**Science Engineering Technology
Research Ethics Committee**
Imperial College London
Room 221
Medical School Building
St Marys Campus
London
W2 1PG
Tel: +44 (0)207 594 1862

researchethicscommittee@imperial.ac.uk

03 December 2020

Dear Yves-Alexandre de Montjoye

Study Title: Insurance Pricing Game
SETREC number: 20IC5995

The above study was approved by your Head of Department on 13/05/20 and by the Research Governance and Integrity Team on 03/12/20.

Under the Science Engineering Technology Research Ethics Committee process, a study that has been reviewed by the Research Governance and Integrity Team and Head of Division/Department, where no significant ethical issues have been identified in the protocol or ethics application, can be given RGIT approval without requiring it to go to full committee.

Documents

The documents reviewed were:

- Application form v3 01/10/2020
- Protocol v3 01/10/2020
- AI_Crowd Consent
- Data Authorisation Letter 21/04/2020
- Advertising Flyer v1
- Advertising Website v1

Yours sincerely,

Ruth Nicholson,
Head of Research Governance and Integrity,
Imperial College London

Imperial College of Science, Technology and Medicine

Figure A.2: Ethics approval form obtained for the insurance pricing game is copied and displayed here.

Appendix B

Privacy & reidentification risk

A case-study on the effect of dataset size

B.1 A naive unicity model: *the combinatorial approach*

A naive approach to model unicity is to approach it as a purely combinatorial problem with every time and location pair being equally likely to appear in a trace.

Given N trajectories of length λ hours, the model assigns to each of the λ hours a particular location $\mathcal{L}_i$ from the $|\mathcal{L}|$ possible locations uniformly at random.

Given p randomly picked points $A_p^{D_i}$ from a trajectory D_i , the probability of D_i being uniquely identifiable given $A_p^{D_i}$ is equal to the probability of all other $N - 1$ trajectories not containing all of $A_p^{D_i}$:

$$\mathbb{P}[A_p^{D_i} \not\subseteq D_j \ \forall j \neq i] = \prod_{i \neq j}^N \left(1 - \mathbb{P}[A_p^{D_i} \subseteq D_j]\right) \quad (\text{B.1})$$

The probability $\mathbb{P}[A_p^{D_i} \subseteq D_j]$ can be computed using the fact that the probability of each of the $|\mathcal{L}|$ locations appearing in $A_p^{(i)}$ is equal:

$$\mathbb{P}[A_p^{D_i} \not\subseteq D_j] = \prod_k^p \mathbb{P}[\mathcal{L}_k \in A_p^{D_i}] = \left(\frac{1}{|\mathcal{L}|}\right)^p \quad (\text{B.2})$$

The definition of our unicity estimate is the average of equation B.1 over all trajectories. Since the trajectories are independently generated, this equation indeed measures the unicity of the combinatorial model. Using the expression from equation B.2 we then have:

$$\begin{aligned} \varepsilon_p^{comb}(N) &= \prod_{i \neq j}^N \left(1 - \mathbb{P}[A_p^{D_i} \subseteq D_j]\right) \\ &= \prod_{i \neq j}^N \left(1 - \frac{1}{|\mathcal{L}|^p}\right) \\ &= \left(1 - \frac{1}{|\mathcal{L}|^p}\right)^{N-1} \end{aligned} \quad (\text{B.3})$$

where equation B.3 gives a simple form for the naive model.

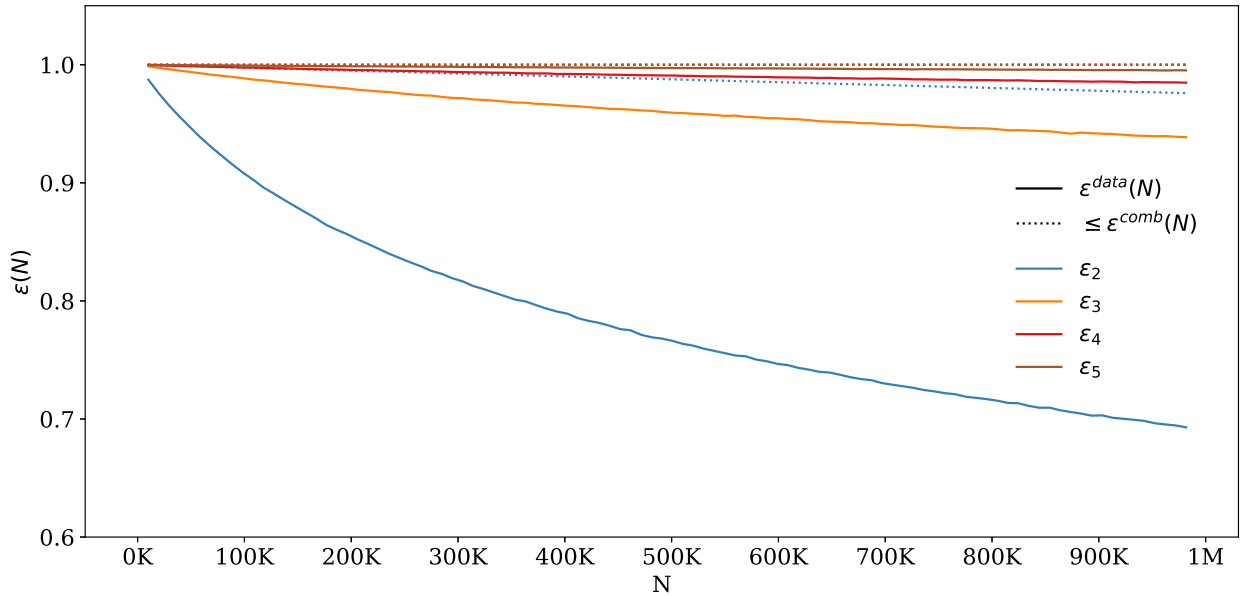


Figure B.1: The combinatorial unicity model (dotted lines) greatly overestimates empirical unicity (solid lines).

The combinatorial unicity model estimates are compared to empirical unicity in Figure B.1. It

is clear that the combinatorial model greatly overestimates unicity values with the dotted lines hugging the region of $\varepsilon \approx 1$ for nearly all values of p .

B.2 Convergence of the unicity model

The unicity model extracts a set of 3 distributions (collectively called R_n when computed from data of size (n)) to model the decrease of unicity. I now show how estimates of unicity converge as a function of the size of the data set n from which the distributions are extracted with the following approach:

1. I instantiate the unicity model using input distributions R_n with increasing n as sampled uniformly from the data set of 1M trajectories.
2. I use each instantiation to obtain unicity values for data sets ranging from 10K to 1M trajectories. This gives a vector of unicity values.
3. Finally, I compare each of these unicity vectors, to the vector obtained by a model instantiated using the entire data (i.e. $n = 1M$) using the [RMSE](#) of the vectors.

Figure [B.2](#) shows that for varying values of n (the horizontal axis), the instantiations of the unicity model quickly converge to the model with $n = 1M$. Indeed, the rate at which the input distributions converge is such that even with only $n = 2.5K$ ($\approx 0.0025\%$ of the data) the results of the unicity models are largely the same as a unicity model that used all 1M trajectories as its input (stable [RMSE](#) values). Finally, lines of best fit are also included showing almost horizontal lines across the board.

B.3 Estimating uniqueness by subsampling a larger dataset

El Emam [\[181\]](#) proposed a method to estimate the uniqueness of a population-size (N) dataset given its unicity $\varepsilon(m)$ for a smaller population size m : generating a dataset D of size N where a

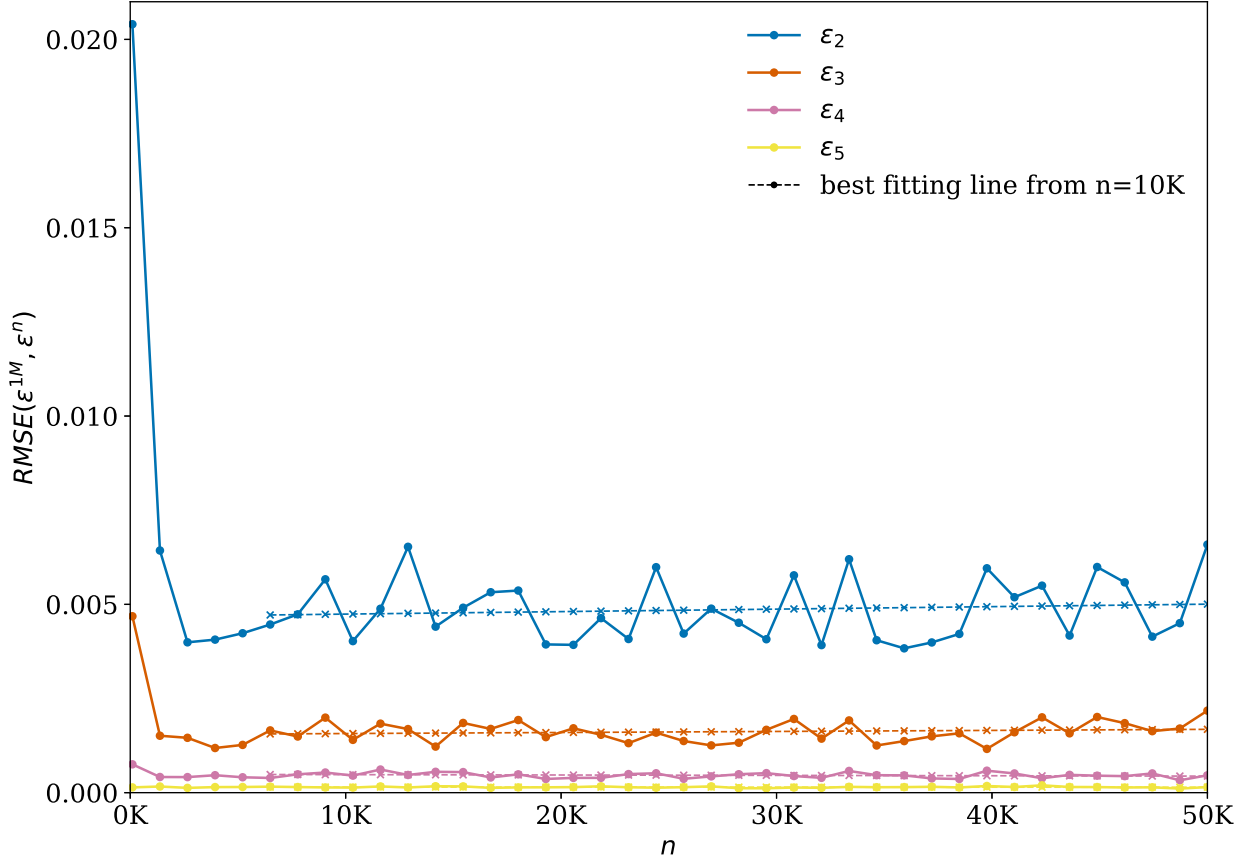


Figure B.2: Convergence of the unicity model as the size of the sample (n) from which the input distributions are extracted increases. In solid, the RMSE (vertical axis) of unicity models instantiated with distributions extracted from varying number of trajectories (horizontal axis) is displayed. The dotted line is the best fitting line to each learning curve starting from $n=10,000$.

fraction α of records are unique, and all other records are identical to exactly one other record. The parameter α is chosen such that the expected estimated uniqueness on a sample of size m , which we denote by $\nu_D(m)$, is equal to the empirical unicity.

By applying this method to prior results [36, main text] ($m = 10^6$, $\varepsilon_4(m) = 0.9$), El Emam estimates that the real population uniqueness for a population of size $N = 22 \cdot 10^6$ is of about 1%. This is used as a basis to argue that the unicity of a sample strongly overestimates the uniqueness of population-scale data, a claim this paper refutes.

In the remainder of this section, we prove that El Emam’s method computes a worst-case lower bound on the uniqueness of a population given a sample.

For this, we first show that for any dataset D of size N , the sample uniqueness as a function

of the sample size m , $\nu_D(m)$, is a convex decreasing function of m . We then show that for the dataset created by El Emam's method, D_α , sample uniqueness $\nu_{D_2}(m)$ is an *affine* function of m . Hence, the uniqueness estimated with El Emam's method will be lower or equal to the uniqueness of *any* other dataset for all (with $m' > m$ including for $m' = N$).

Let $D = (d_1, \dots, d_N)$ a population-scale dataset of size $N > 1$, with records d_i in some arbitrary set $\mathcal{D}$. A *matching* relation $\mathcal{M} : D \times D \rightarrow \{0, 1\}$ is defined over records in D , that outputs 1 iff two records are identical. This relation defines k equivalence classes $\mathcal{C}_1, \dots, \mathcal{C}_k$ of sizes $n_1, \dots, n_k$, such that $i, j \in \mathcal{C}_l \Leftrightarrow \mathcal{M}(d_i, d_j) = 1$. We show below that this list of class sizes $(n_1, \dots, n_k)$ uniquely characterises the behavior of the uniqueness in a sample of size $m < N$.

A dataset $D' \subset D$ of size $m \leq N$ (the sample dataset) is sampled uniformly without replacement from D . For each record d_i in D , we define the binary random variable $S_i = I\{d_i \in D'\}$. S_i follows a Bernoulli distribution with parameter $\frac{m}{N}$, and all Y_i satisfy the following equation: $\sum_{i=1}^N Y_i = m$. From this, we define the *count per class* variable $N_j = \sum_{i \in \mathcal{C}_j} S_i$ for $j \in \{1, \dots, k\}$.

Using these notations, the sample uniqueness for a sample size $m \leq N$ is defined as the expected fraction of records of D' that are unique in D' , i.e. that are the only records of their class in D' :

$$\nu_D(m) = \mathbb{E} \left[\frac{1}{m} \sum_{i=1}^k I\{N_i = 1\} \right] = \frac{1}{m} \sum_{i=1}^k \mathbb{P}[N_i = 1]$$

Denote by $\binom{a}{b}$ the number of subsets of size a of a set of size b ("a choose b"). The probability that $N_i = 1$ for some class of size n_i is equal to the fraction of samples of size m from the dataset (of size N) that have exactly 1 record from the class:

$$\mathbb{P}[N_i = 1] = \frac{\binom{1}{n_i} \cdot \binom{m-1}{N-n_i}}{\binom{m}{N}} = \frac{n_i \cdot \frac{(N-n_i)!}{(m-1)! (N-n_i-m+1)!}}{\frac{N!}{m! (N-m)!}} = n_i \cdot m \cdot \frac{(N-n_i)!}{N!} \cdot \prod_{j=0}^{n_i-2} (N-m-j)$$

We then use this expression to compute the sample uniqueness:

$$\nu_D(m) = \frac{1}{m} \sum_{i=1}^k m n_i \frac{(N-n_i)!}{N!} \prod_{j=0}^{n_i-2} (N-m-j) = \sum_{i=1}^k n_i \frac{(N-n_i)!}{N!} \cdot \prod_{j=0}^{n_i-2} (N-m-j)$$

We show that ν_D is a convex function of m , for $m \in [1, N]$, for any dataset D of size N . For

this, observe that $\nu_D(m)$ is a linear combination of the convex function $\hat{f}_{n_i}(m)$ as defined in Lemma 1, for $i = 1, \dots, k$, with nonnegative linear coefficients. We show in Lemma 1 that $\hat{f}_{n_i}$ is a convex function of m for all n_i , and thus, $\nu_D(m)$ is a convex, decreasing, function of m .

Lemma 1. *Let $n_i \in \mathbb{N}_0$, $N > n_i$. The function $\hat{f}_{n_i}$ defined as:*

$$\hat{f}_{n_i} : [0, N] \rightarrow \mathbb{R} : m \mapsto \begin{cases} \prod_{j=0}^{n_i-2} (N - m - j) & \text{if } m \leq N - n_i + 2 \\ 0 & \text{if } m > N - n_i + 2 \end{cases}$$

is a convex, decreasing, non-negative function of m .

Proof. First, Define the function $f_{n_i} : \mathbb{R}^+ \rightarrow \mathbb{R} : m \mapsto \prod_{j=0}^{n_i-2} (N - m - j)$. We show that $\forall m$ such that $N \geq m + n_i - 2$, the first derivative is non-positive, and the second derivative of f_{n_i} is non-negative:

$$\begin{cases} \frac{df_{n_i}}{dm}(m) = (-1) \cdot \sum_{l=0}^{n_i-2} \prod_{j \neq l} (N - m - j) \leq 0 \\ \frac{d^2 f_{n_i}}{dm^2}(m) = \sum_{l=0}^{n_i-2} \sum_{j \neq l} \prod_{s \neq j, s \neq l} (N - m - s) \geq 0 \end{cases}$$

and thus f_{n_i} is decreasing and convex over $[0, m + n_i - 2]$. From this, one gets that $\hat{f}_{n_i}$ is also a decreasing non-negative function over $[0, N]$. Further, since $f(N - n_i + 2) = 0$, this function is continuous.

We now show that $\hat{f}_{n_i}$ satisfies the definition of convexity over $[0, N]$: $\forall x, y \in \mathbb{R}^+$ with $x \leq y$, and $\forall \lambda \in [0, 1]$, $\hat{f}_{n_i}(\lambda x + (1 - \lambda)y) \leq \lambda \hat{f}_{n_i}(x) + (1 - \lambda)\hat{f}_{n_i}(y)$. Denote $z = \lambda x + (1 - \lambda)y$, and $\kappa = N - n_i + 2$, and observe that:

- Either $x \leq y \leq \kappa$, and the inequality holds by convexity of f_{n_i} over $[0, \kappa]$;
- Either $\kappa \leq x \leq y$, and the inequality holds since $\hat{f}_{n_i}(x) = \hat{f}_{n_i}(z) = \hat{f}_{n_i}(y) = 0$.
- Either $x < \kappa \leq z \leq y$, and the inequality holds since $\hat{f}_{n_i}(x) > 0$ and $\hat{f}_{n_i}(y) = \hat{f}_{n_i}(z) > 0$.
- Either $x \leq z < \kappa \leq y$. Then, there exists $\mu \in [0, 1]$ such that $z = \mu x + (1 - \mu)\kappa$. By convexity of f_{n_i} , we have $\hat{f}_{n_i}(z) \leq \mu \hat{f}_{n_i}(x) + (1 - \mu)\hat{f}_{n_i}(\kappa) = \mu \hat{f}_{n_i}(x)$. Further, observe that $z = \mu x + (1 - \mu)\kappa = \lambda x + (1 - \lambda)y \geq \lambda x + (1 - \lambda)\kappa$, and thus $\lambda \geq \mu$. Combined together,

this implies $\hat{f}_{n_i}(z) \leq \mu \hat{f}_{n_i}(x) \leq \lambda \hat{f}_{n_i}(x) = \lambda \hat{f}_{n_i}(x) + (1 - \lambda) \hat{f}_{n_i}(y)$, since $\hat{f}_{n_i}(x) > 0$ and $\hat{f}_{n_i}(y) = 0$.

□

We then compute the sample uniqueness of the dataset used by El Emam's method, which we call D_α . By construction, this dataset has $\alpha N \in \mathbb{N}$ classes of size $n_i = 1$, $i = 1, \dots, \alpha N$, and $\frac{(1-\alpha)N}{2}$ classes of size $n_i = 2$, $i = \alpha N + 1, \dots, k$. We thus get:

$$\nu_{D_\alpha}(m) = \alpha N \cdot \frac{1}{N} + \frac{(1-\alpha)N}{2} \cdot 2 \frac{N-m}{N(N-1)} = \alpha + (1-\alpha) \left(1 - \frac{m-1}{N-1}\right)$$

which is an affine function of m .

Let (m^*, ε^*) an empirical measurement of uniqueness on a dataset of size m^* , assumed to be a uniform sample from a sample of size N . We show that if there exists an α for which $\nu_{D_\alpha}(m^*) = \varepsilon^*$, then *any* other dataset D of size N such that $\nu_D(m^*) = \varepsilon^*$ will have higher sample uniqueness for any $m' \geq m^*$: $\nu_D(m') \geq \nu_{D_\alpha}(m')$. This is a direct consequence of the convexity of $\nu_D(m)$ and linearity of $\nu_{D_\alpha}(m)$. Indeed, convexity for $\nu_D(m)$ implies that, $\forall m' \geq m^*$ (using the definition with $x = m'$, $y = 1$ and $\lambda' = \frac{m^*-1}{m'-1} \leq 1$):

$$\varepsilon^* = \nu_D(m^*) \leq \lambda' \cdot \nu_D(m') + (1 - \lambda') \cdot \nu_D(1)$$

which in turn entails, since $\nu_D(1) = 1$:

$$\nu_D(m') \geq \frac{\varepsilon^* - 1 + \lambda'}{\lambda'} = \frac{m' - 1}{m^* - 1} (\varepsilon^* - 1) + 1$$

Observe that choosing $\alpha^* = 1 + \frac{N-1}{m^*-1} (\varepsilon^* - 1)$ in El Emam's method, if $\alpha \leq 1$, is such that $\nu_{D_{\alpha^*}}(m^*) = \varepsilon^*$. If $\alpha > 1$, then El Emam's method cannot be used. We then get, for all $m' \geq m^*$:

$$\nu_{D_{\alpha^*}}(m') = \frac{m' - 1}{m^* - 1} (\varepsilon^* - 1) + 1$$

And thus, for *any* dataset D of size N , if α exists, we have that, $\forall m' \geq m^*$:

$$\nu_{D_{\alpha^*}}(m') \leq \nu_D(m')$$